

ImPRESS Statistical Analysis Mock-up report Report

Baseline Review for the Newly Diagnosed Glioblastoma Cohort

ImPRESS Statistical Team

27 March 2026

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Introduction

The ImPRESS study is a prospective investigation of losartan as an adjunctive therapy for patients with glioblastoma. According to Protocol Version 7.0 (17-Dec-2024), the trial collects comprehensive demographic, clinical, imaging, and quality-of-life information prior to treatment initiation. The statistical objectives for the interim analyses focus on characterising baseline status, neurological function, and tumour burden before randomisation to evaluate comparability across planned treatment arms.

This report summarises the currently available baseline data with emphasis on clinical status, imaging assessments, and prior treatments that may influence subsequent outcomes. The material is intended to provide early insight into data quality and cohort composition ahead of formal efficacy or safety analyses.

Randomisation Status

The present dataset is built on sham randomisation generated from the most recent export (see `_targets.R` workflow). No true treatment assignments have been unlocked. All findings in this document should therefore be interpreted as preliminary and descriptive only. Final efficacy comparisons will not be conducted until the database is locked and genuine randomisation codes are released.

Cohort Scope

The results herein are restricted to participants in cohort code 2 as configured in `config/cfg.yml`. Subjects from other cohorts are excluded from all summaries, figures, and tables in this draft. A combined-cohort analysis can be delivered once recruitment and cleaning are complete for all strata.

Baseline Characteristics

This section summarises baseline characteristics for the newly diagnosed glioblastoma cohort using the `adb1` ADaM dataset.

Demographics and Baseline characteristics

	all obs
Age (years)	
n	54
Mean (SD)	52.6 (10.2)
Median	55.0
Min - Max	26.0 - 69.0
Sex	
n	54
Female	29 (53.7%)
Male	25 (46.3%)
Cohort	
n	54

Demographics and Baseline characteristics

	all obs
Newly diagnosed glioblastoma	54 (100%)
Country	
n	54
NOR	54 (100%)
Childbearing potential?	
n	29
No	16 (55.2%)
Yes	13 (44.8%)
Diagnosis	
n	54
Newly diagnosed glioblastoma	54 (100%)
Time from surgery to randomisation (days)	
n	54
Mean (SD)	27.4 (7.3)
Median	25.5
Min - Max	13.0 - 48.0
IDH1/2 mutation	
n	54
Negative	47 (87%)
Positive	7 (13%)
MGMT status	
n	54
Negative	30 (55.6%)
Positive	24 (44.4%)
Pregnancy test result	
n	13
Negative	13 (100%)
ECOG Score	
n	54
Mean (SD)	0.3 (0.5)
Median	0.0
Min - Max	0.0 - 2.0
ECOG Score (categorical)	
n	54

Demographics and Baseline characteristics

	all obs
ECOG 0	40 (74.1%)
ECOG 1	13 (24.1%)
ECOG 2	1 (1.9%)
Karnofsky Performance Scale	
n	54
Mean (SD)	95.0 (8.0)
Median	100.0
Min - Max	60.0 - 100.0
Karnofsky Performance Scale (categorical)	
n	54
KPS 100	34 (63%)
KPS 60	1 (1.9%)
KPS 80	4 (7.4%)
KPS 90	15 (27.8%)
Total NANO score	
n	54
Mean (SD)	1.0 (1.3)
Median	0.0
Min - Max	0.0 - 4.0
EORTC QLQ-C30 score	
n	53
Mean (SD)	56.3 (8.6)
Median	54.0
Min - Max	41.0 - 77.0
EORTC QLQ-BN20 score	
n	53
Mean (SD)	30.8 (6.6)
Median	30.0
Min - Max	20.0 - 49.0
Systolic blood pressure	
n	54
Mean (SD)	130.0 (15.4)
Median	128.5
Min - Max	106.0 - 180.0

Demographics and Baseline characteristics

	all obs
Diastolic blood pressure	
n	54
Mean (SD)	81.9 (7.1)
Median	82.0
Min - Max	67.0 - 97.0
Pulse Rate	
n	54
Mean (SD)	80.6 (15.6)
Median	78.0
Min - Max	54.0 - 114.0
Weight (kg)	
n	54
Mean (SD)	78.8 (13.2)
Median	78.3
Min - Max	48.2 - 116.0
BMI	
n	51
Mean (SD)	25.0 (4.2)
Median	23.9
Min - Max	18.6 - 39.9
Height (cm)	
n	51
Mean (SD)	176.5 (8.8)
Median	175.0
Min - Max	158.0 - 194.0
Has the patient received chemotherapy	
n	54
No	54 (100%)
Has the patient received immunotherapy	
n	54
No	54 (100%)
Has the patient received radiation therapy to the brain?	
n	54
No	54 (100%)

Demographics and Baseline characteristics

	all obs
CNS surgery	
n	54
No	3 (5.6%)
Yes	51 (94.4%)
Steroids at time of MRI	
n	54
No	42 (77.8%)
Yes	12 (22.2%)
Mean relative cerebral blood flow (PTA-ROI)	
n	54
Mean (SD)	1.5 (0.4)
Median	1.4
Min - Max	0.7 - 2.4
Mean relative cerebral blood flow (ROI2)	
n	47
Mean (SD)	1.1 (0.5)
Median	1.0
Min - Max	0.2 - 2.2
Mean relative solid stress (PTA-ROI)	
n	41
Mean (SD)	0.8 (0.1)
Median	0.8
Min - Max	0.5 - 1.1
Mean relative solid stress (ROI2)	
n	32
Mean (SD)	0.9 (0.2)
Median	0.9
Min - Max	0.5 - 1.4
Mean relative apparent diffusion coefficient	
n	52
Mean (SD)	1.4 (0.2)
Median	1.4
Min - Max	1.0 - 2.0
Mean relative fractional anisotropy	

Demographics and Baseline characteristics

	all obs
n	52
Mean (SD)	0.5 (0.1)
Median	0.5
Min - Max	0.3 - 1.0
Mean relative restricted spectrum imaging	
n	50
Mean (SD)	0.5 (0.2)
Median	0.5
Min - Max	0.2 - 1.1
Mean DSC K2 leakage	
n	54
Mean (SD)	0.2 (1.6)
Median	0.4
Min - Max	-3.9 - 4.5
Mean relative cerebral blood	
n	54
Mean (SD)	1.7 (0.5)
Median	1.7
Min - Max	1.0 - 2.8
Mean relative transit time	
n	54
Mean (SD)	1.4 (0.3)
Median	1.4
Min - Max	0.9 - 2.3
Total volume of contrast agent enhancing lesion on T1-weighted MRI in cm3	
n	54
Mean (SD)	6.3 (6.5)
Median	4.3
Min - Max	0.0 - 26.3
Total volume of pathologic signal on T2/FLAIR-weighted MRI in cm3	
n	54
Mean (SD)	13.5 (14.3)
Median	8.5
Min - Max	0.1 - 57.4

Demographics and Baseline characteristics

	all obs
Mean tissue elasticity	
n	41
Mean (SD)	0.8 (0.1)
Median	0.8
Min - Max	0.5 - 1.2
Mean tissue relative viscosity	
n	41
Mean (SD)	0.9 (0.2)
Median	0.9
Min - Max	0.0 - 1.1
Mean tissue stiffness	
n	41
Mean (SD)	0.8 (0.1)
Median	0.8
Min - Max	0.5 - 1.1
Mean tissue viscosity	
n	41
Mean (SD)	0.7 (0.2)
Median	0.7
Min - Max	-0.0 - 1.1
Mean relative DSC extravasation	
n	28
Mean (SD)	1.2 (0.8)
Median	1.0
Min - Max	0.7 - 5.1
Mean relative direction value	
n	28
Mean (SD)	0.4 (1.4)
Median	0.3
Min - Max	-2.7 - 5.4
Mean relative vessel caliber	
n	28
Mean (SD)	1.0 (0.3)
Median	1.0

Demographics and Baseline characteristics

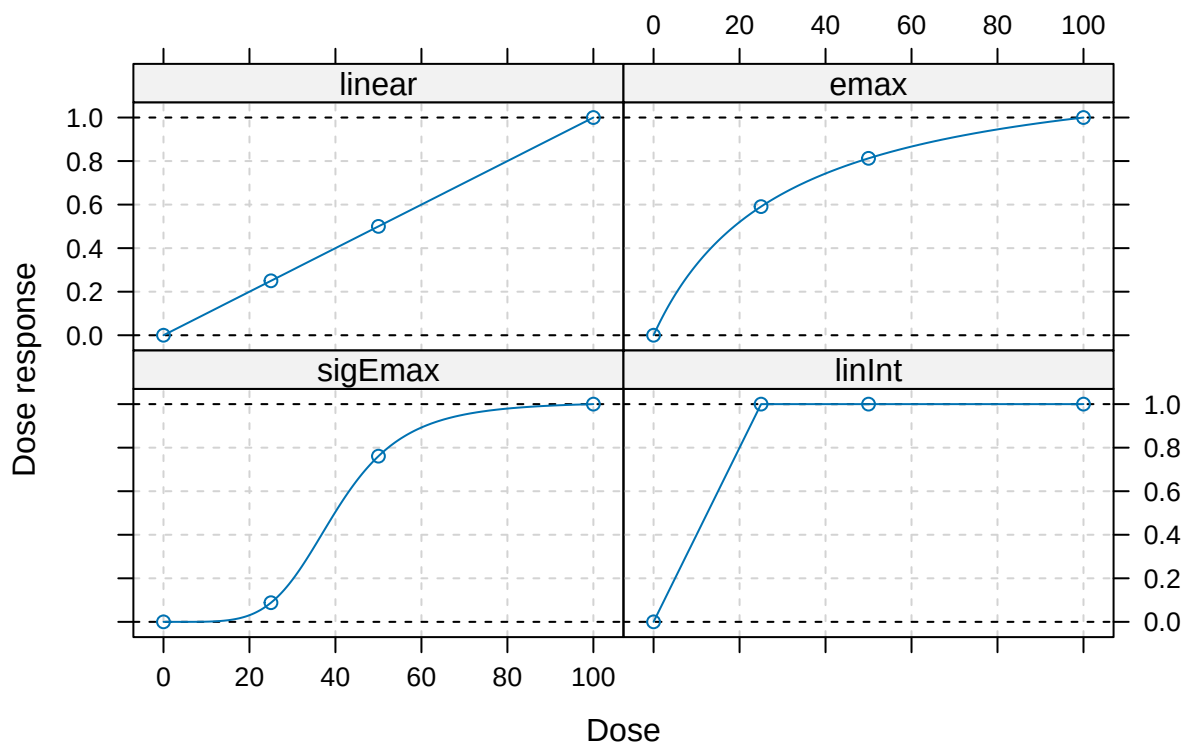
	all obs
Min - Max	0.5 - 1.9
Mean relative vessel size index	
n	28
Mean (SD)	1.4 (0.6)
Median	1.2
Min - Max	0.6 - 2.6
Mean relative vortex area	
n	28
Mean (SD)	1.0 (0.1)
Median	1.0
Min - Max	0.8 - 1.1

Primary Efficacy Analyses (MRI)

The primary variable is analysed with the MCP-Mod methodology as described in the protocol and SAP. We pre-specify four candidate dose-response models (linear, Emax, sigmoid Emax, logistic) and derive optimal contrasts using the estimated variance-covariance from the mixed model. The MCP step tests for a dose-response signal across candidate models with multiplicity control; if significant, the Mod step selects the best-fitting model (by AIC) and reports ED values. Results below are grouped by MRI endpoint (primary, key secondary, sensitivity). All analyses are based on data up to and including Day 43 (Week 6).

Candidate dose-response models

Figure 1: Pre-specified candidate dose-response models



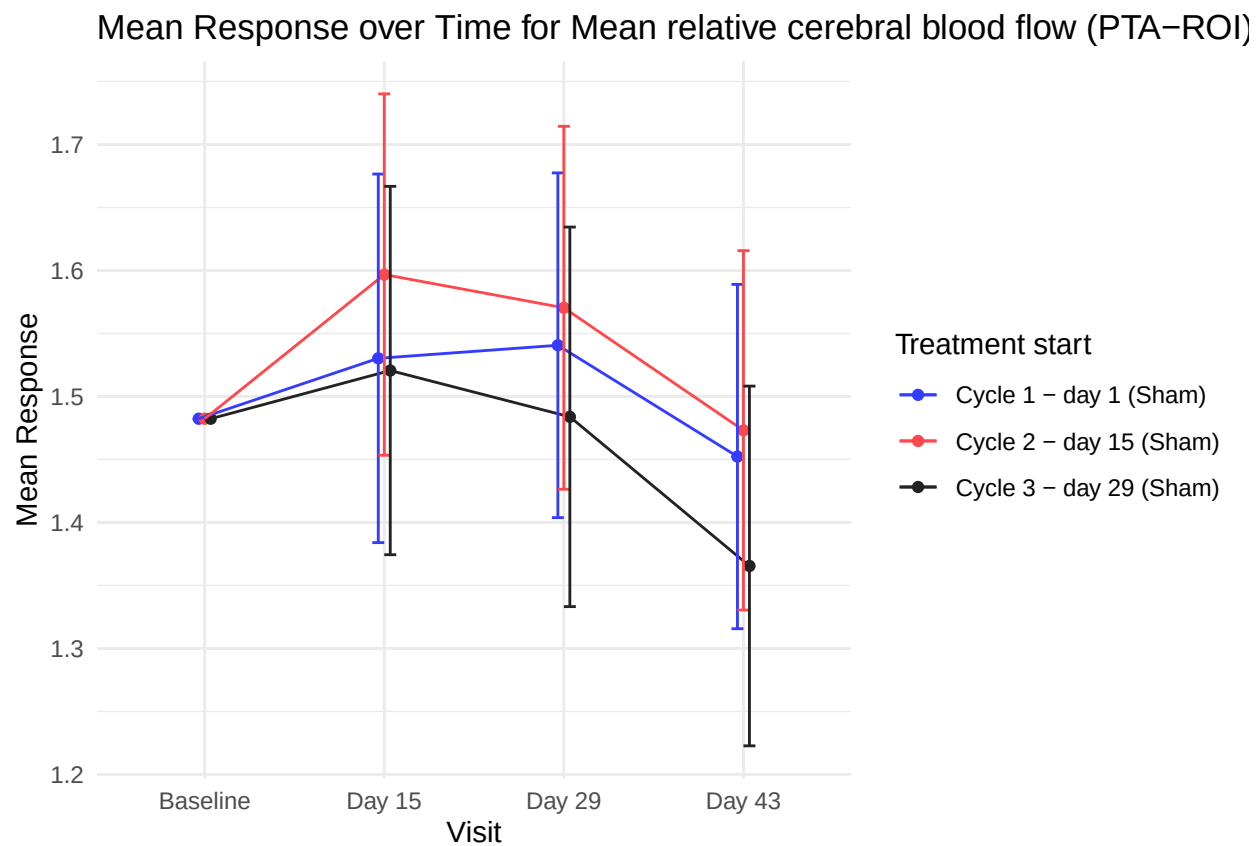
Mean relative cerebral blood flow (PTA-ROI)

The primary analysis follows the MCP-Mod methodology as detailed in the SAP, using pre-specified candidate dose-response models.

A total of 54 subjects were included in the analysis for this endpoint.

The four candidate models (linear, piecewise linear, Emax, sigmoid Emax) are shown in Figure 1.

Line plot



Optimal contrasts

Treatment	linear	emax	sigEmax	linInt
0 mg	-0.645	-0.809	-0.603	-0.864
25 mg	-0.176	0.036	-0.352	0.284
50 mg	0.081	0.236	0.311	0.242
100 mg	0.739	0.537	0.644	0.338

MCP-Mod tests

Candidate model	t-statistic	Adjusted p-value
sigEmax	2.45	0.015
linear	1.90	0.054
emax	1.05	0.24
linInt	-0.10	0.69

Fitted model parameters (AIC/ED)

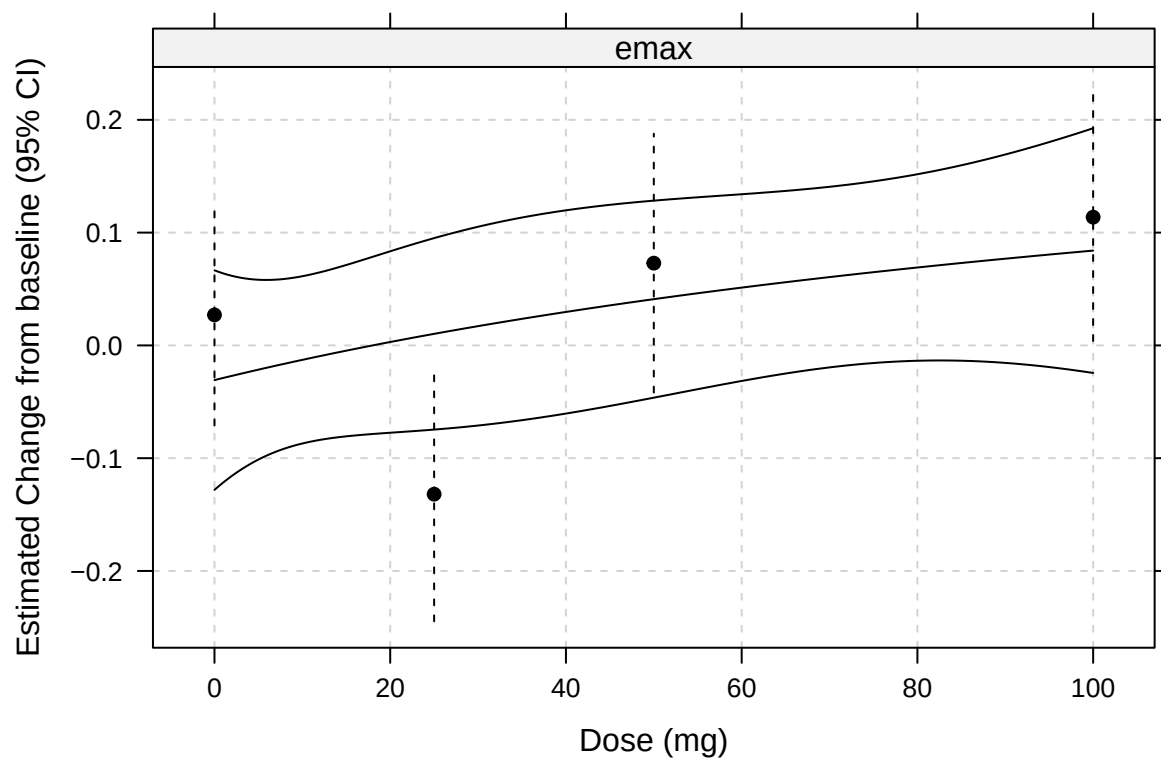
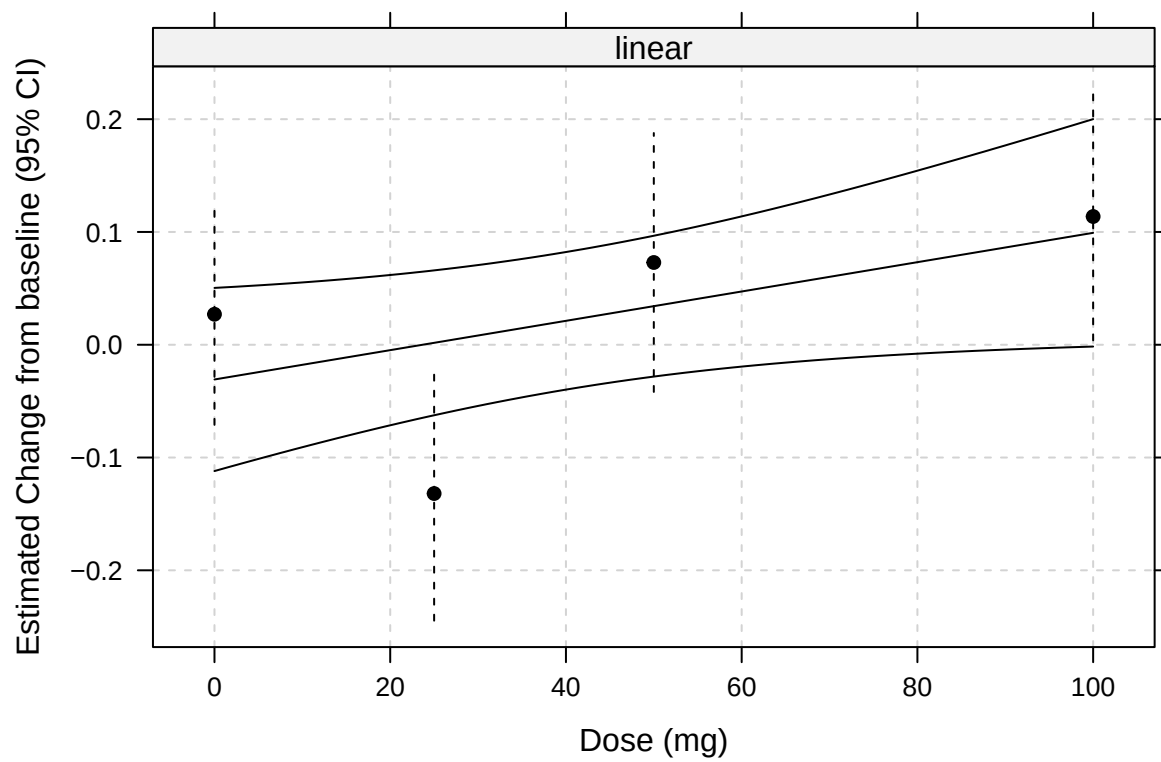
Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
linInt	8.00	12.50	17.50	22.50

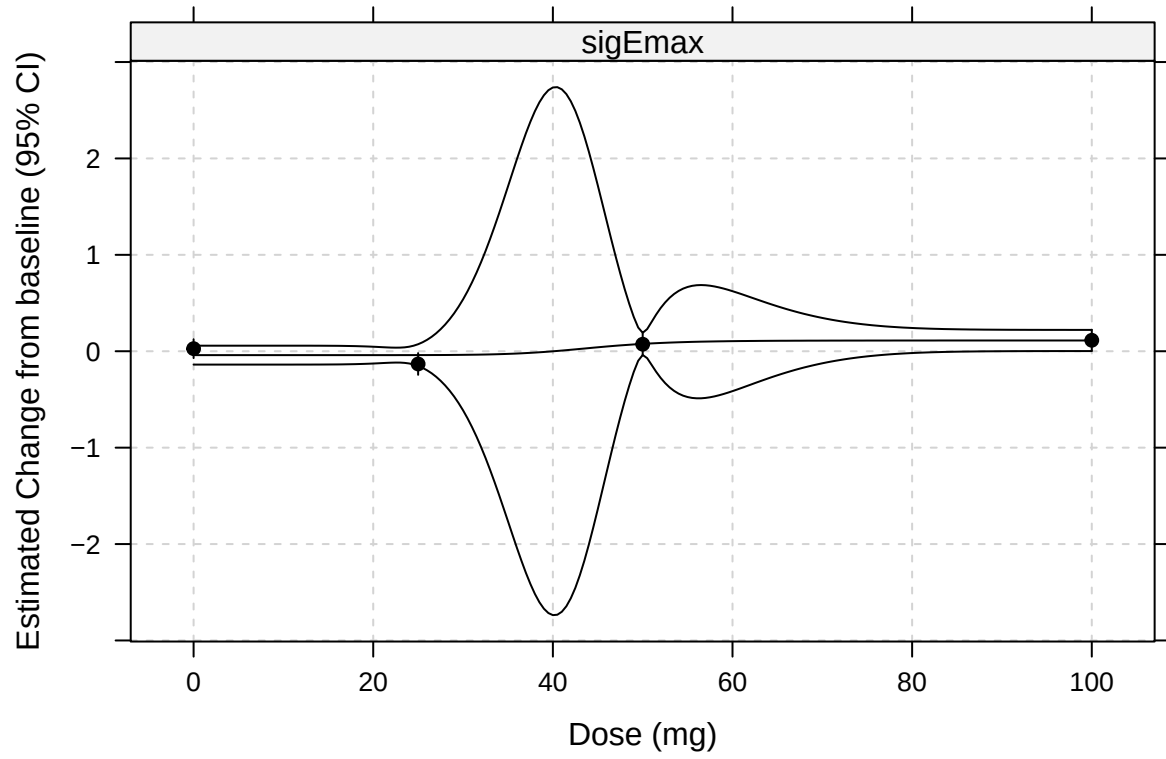
Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
linear	12.00	50.00	70.00	90.00
sigEmax	12.79	44.37	48.29	55.26
emax	14.81	37.50	58.33	84.37

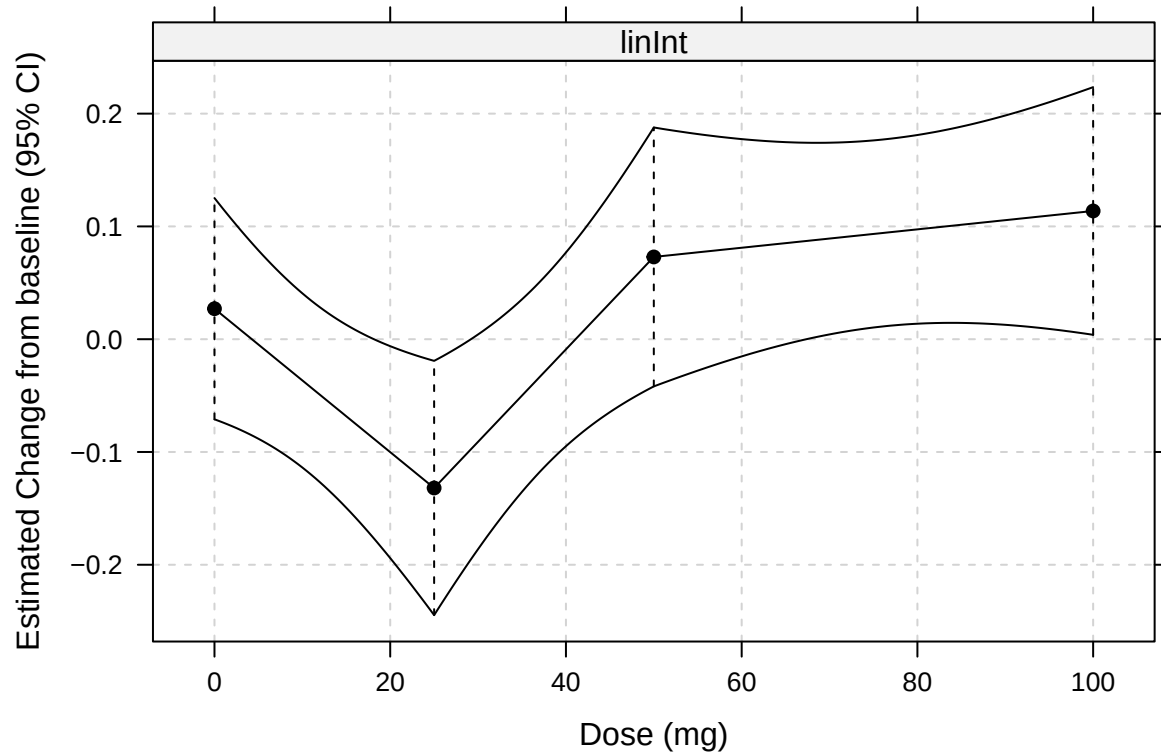
Estimated means by dose

Treatment	Estimated Change from Baseline (95% CI)
0 mg	0.03 (-0.07 to 0.13)
25 mg	-0.13 (-0.25 to -0.02)
50 mg	0.07 (-0.04 to 0.19)
100 mg	0.11 (0.00 to 0.22)

Fitted dose-response curves







Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	0.00 (-0.12 to 0.11)

Mean relative cerebral blood flow (ROI2)

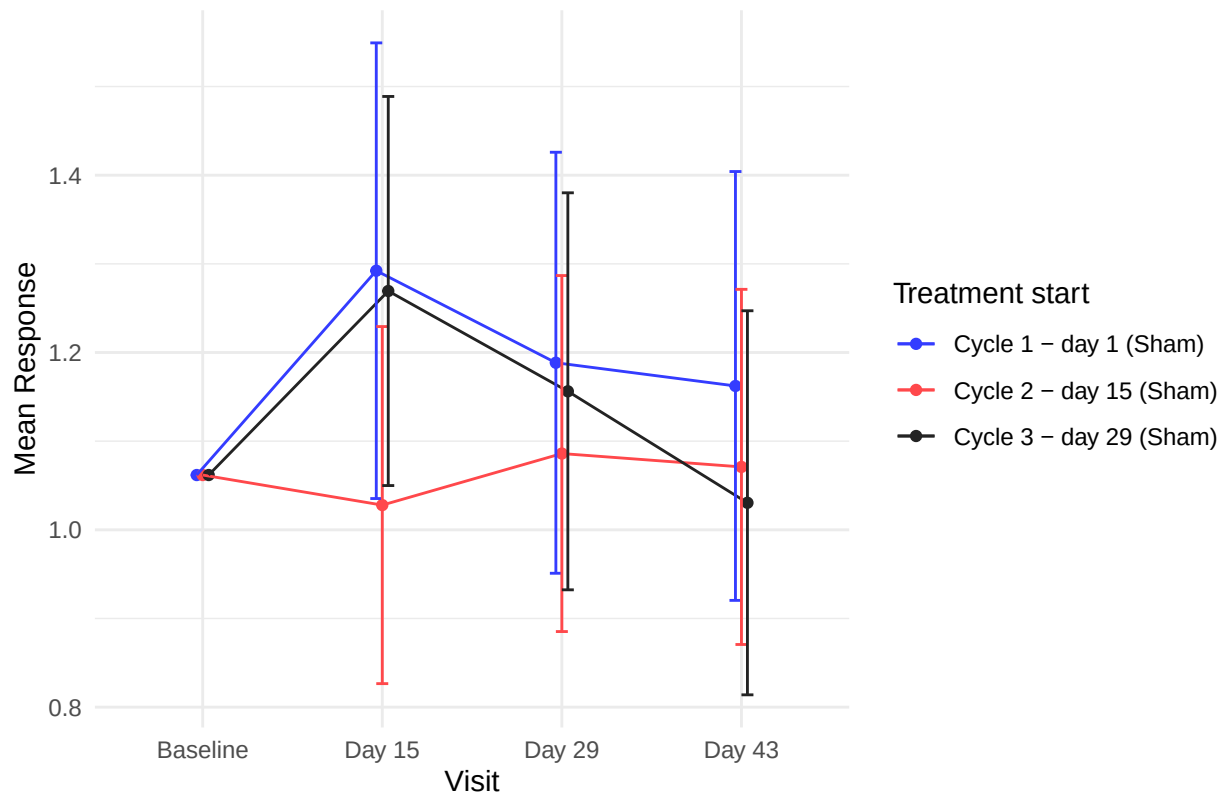
The primary analysis follows the MCP-Mod methodology as detailed in the SAP, using pre-specified candidate dose-response models.

A total of 54 subjects were included in the analysis for this endpoint.

The four candidate models (linear, piecewise linear, Emax, sigmoid Emax) are shown in Figure 1.

Line plot

Mean Response over Time for Mean relative cerebral blood flow (ROI2)



Optimal contrasts

Treatment	linear	emax	sigEmax	linInt
0 mg	-0.668	-0.819	-0.620	-0.864
25 mg	-0.154	0.063	-0.331	0.301
50 mg	0.102	0.234	0.312	0.231
100 mg	0.720	0.521	0.639	0.332

MCP-Mod tests

Candidate model	t-statistic	Adjusted p-value
linear	2.12	0.033
sigEmax	1.77	0.071
emax	1.27	0.17
linInt	0.52	0.44

Fitted model parameters (AIC/ED)

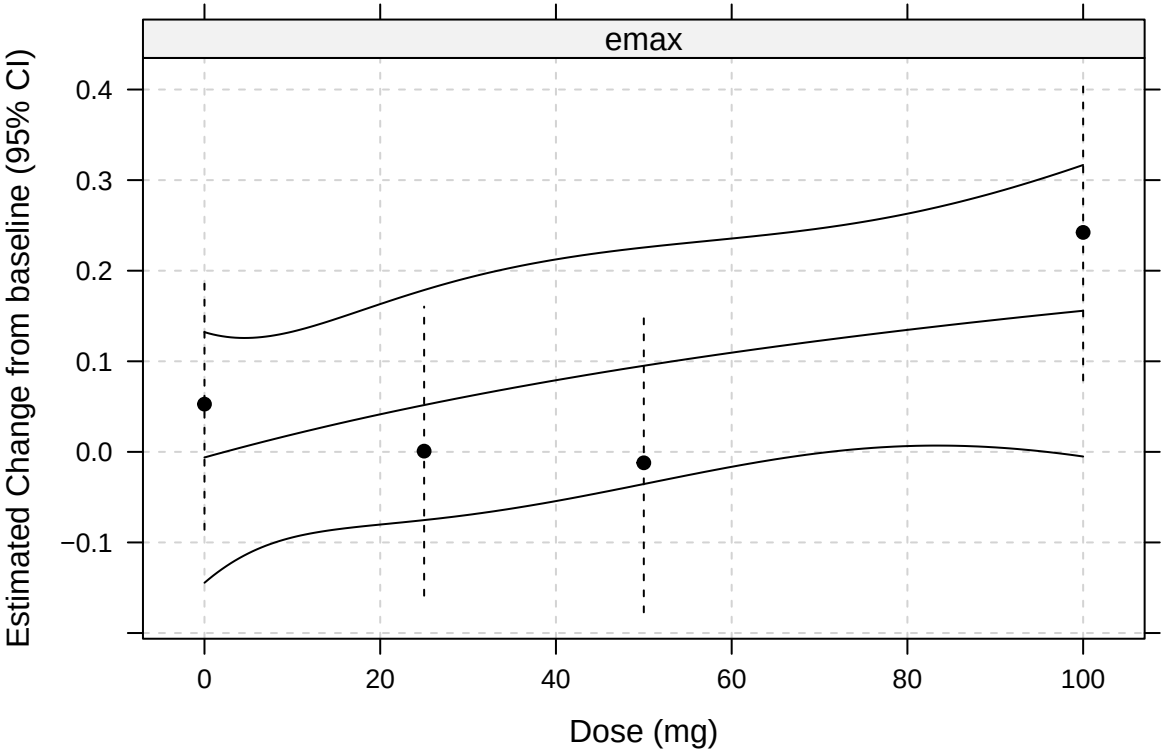
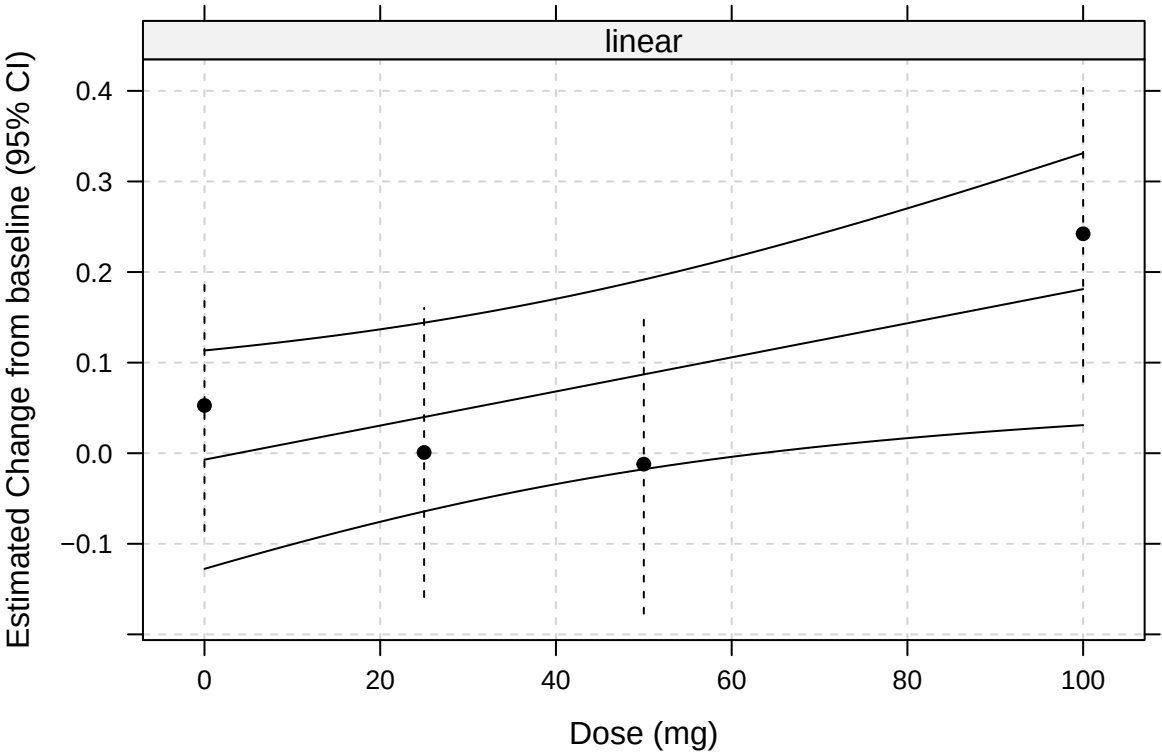
Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
linear	7.51	50.00	70.00	90.00

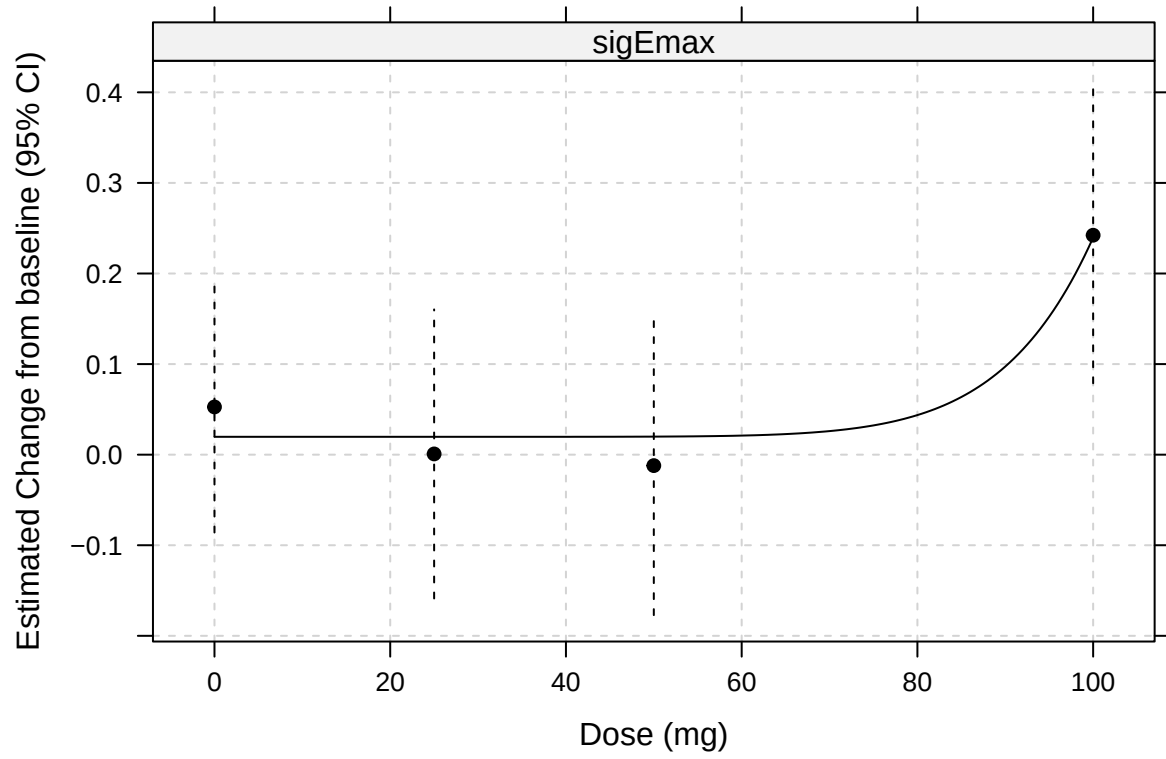
Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
linInt	8.00	81.37	88.82	96.27
sigEmax	8.54	93.22	96.44	98.93
emax	10.64	37.50	58.33	84.37

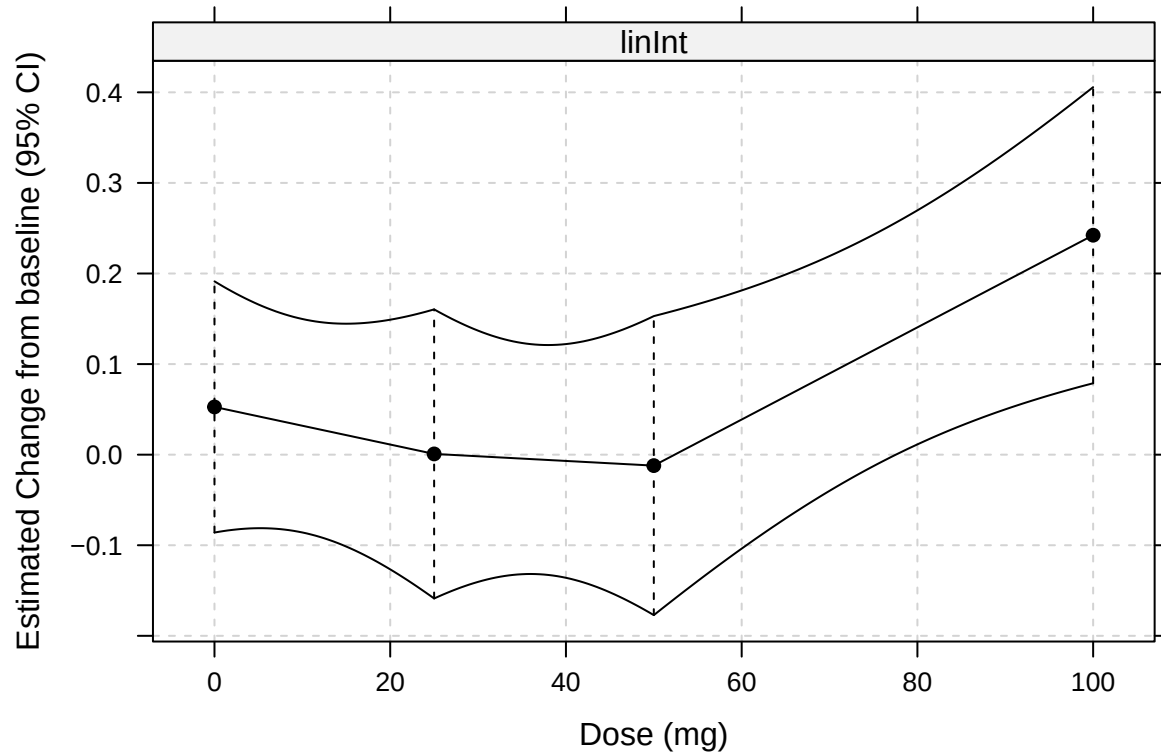
Estimated means by dose

Treatment	Estimated Change from Baseline (95% CI)
0 mg	0.05 (-0.09 to 0.19)
25 mg	0.00 (-0.16 to 0.16)
50 mg	-0.01 (-0.18 to 0.16)
100 mg	0.24 (0.08 to 0.41)

Fitted dose-response curves







Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	0.04 (-0.11 to 0.18)

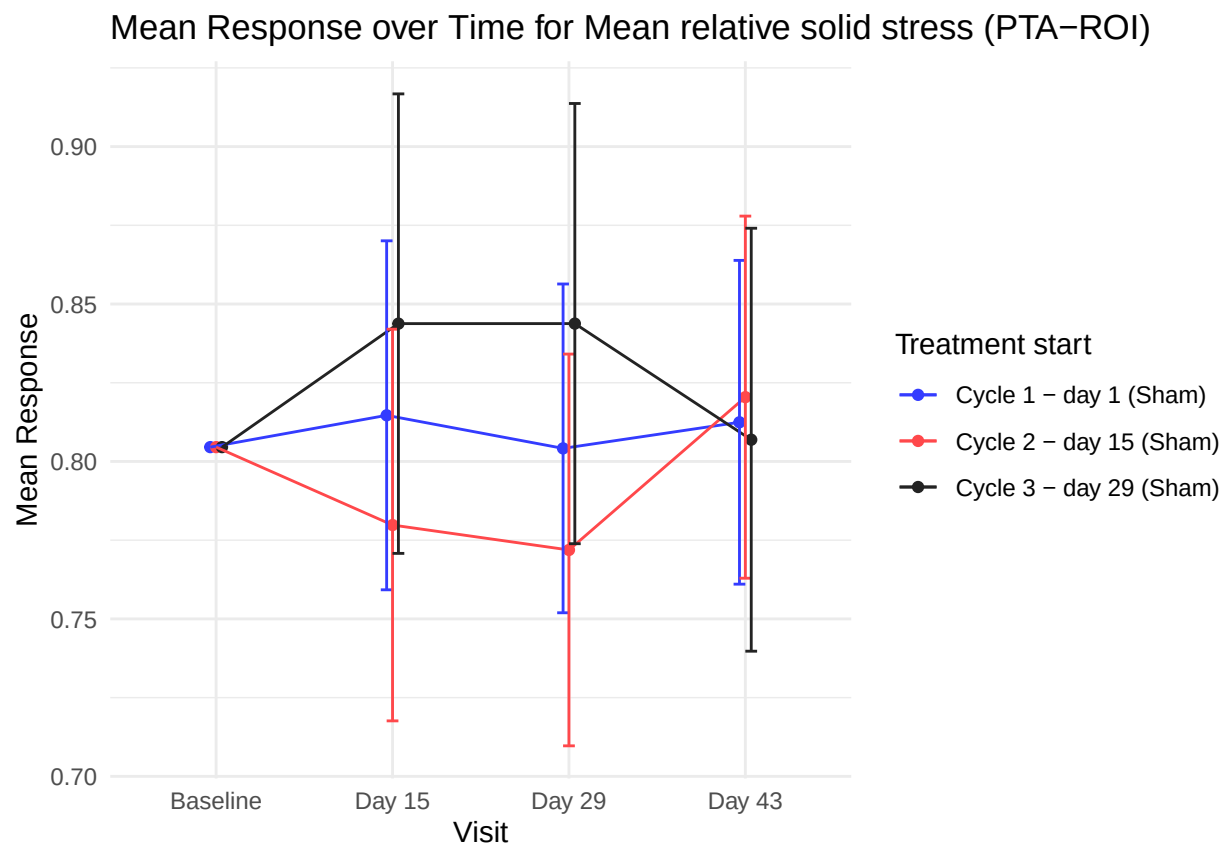
Mean relative solid stress (PTA-ROI)

The primary analysis follows the MCP-Mod methodology as detailed in the SAP, using pre-specified candidate dose-response models.

A total of 54 subjects were included in the analysis for this endpoint.

The four candidate models (linear, piecewise linear, Emax, sigmoid Emax) are shown in Figure 1.

Line plot



Optimal contrasts

Treatment	linear	emax	sigEmax	linInt
0 mg	-0.661	-0.799	-0.635	-0.858
25 mg	-0.142	0.036	-0.289	0.249
50 mg	0.068	0.197	0.255	0.212
100 mg	0.734	0.567	0.669	0.397

MCP-Mod tests

Candidate model	t-statistic	Adjusted p-value
sigEmax	0.03	0.63
linear	-0.14	0.69
emax	-0.52	0.82
linInt	-0.87	0.9

Fitted model parameters (AIC/ED)

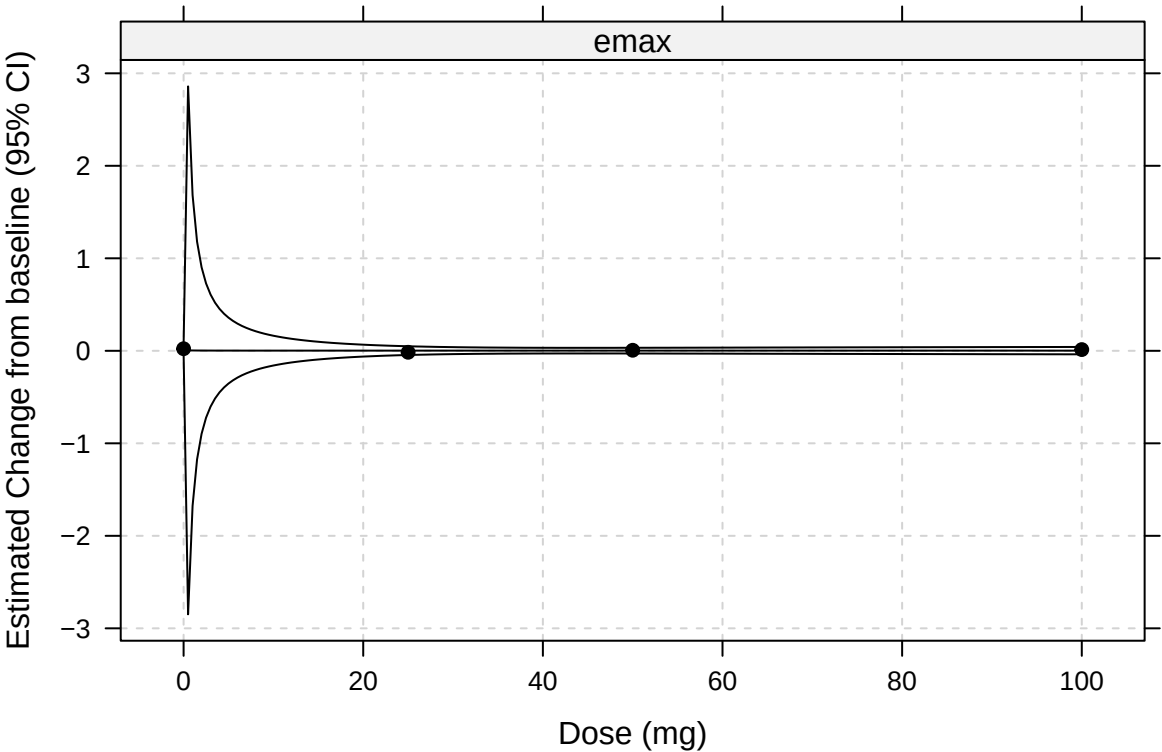
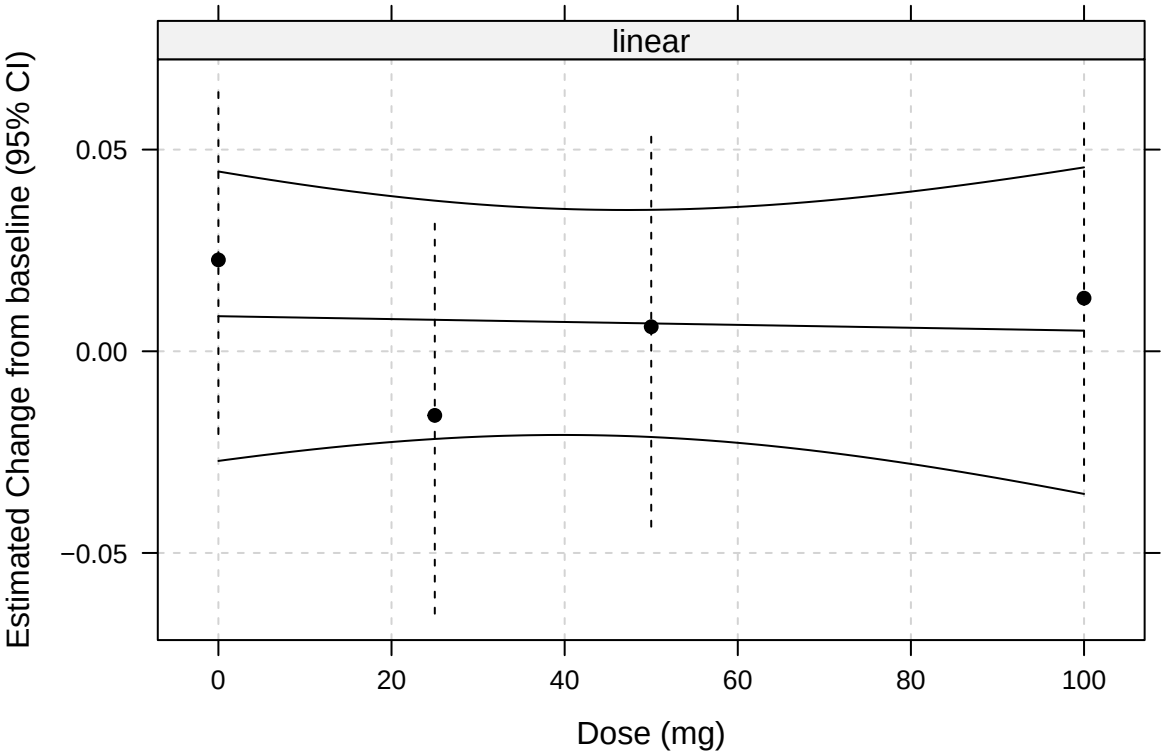
Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
linear	5.61	50.00	70.00	90.00

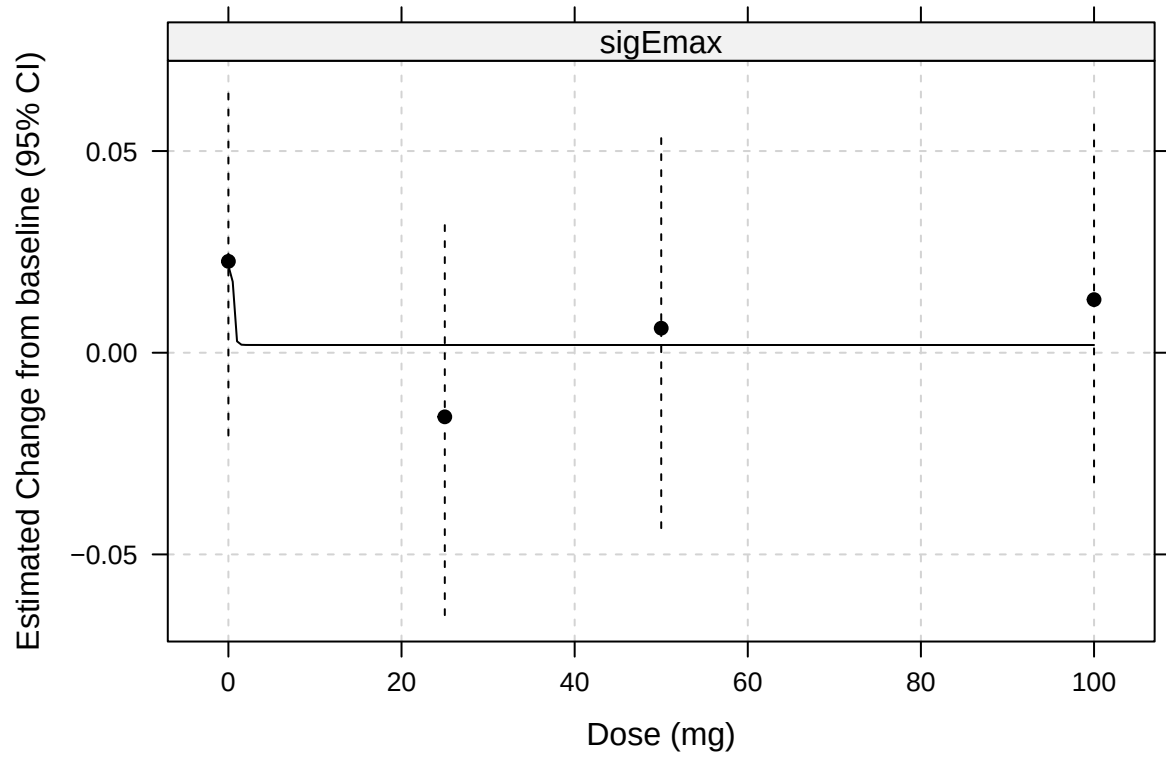
Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
emax	6.88	0.10	0.23	0.89
linInt	8.00	12.50	17.50	22.50
sigEmax	8.87	0.62	0.71	0.88

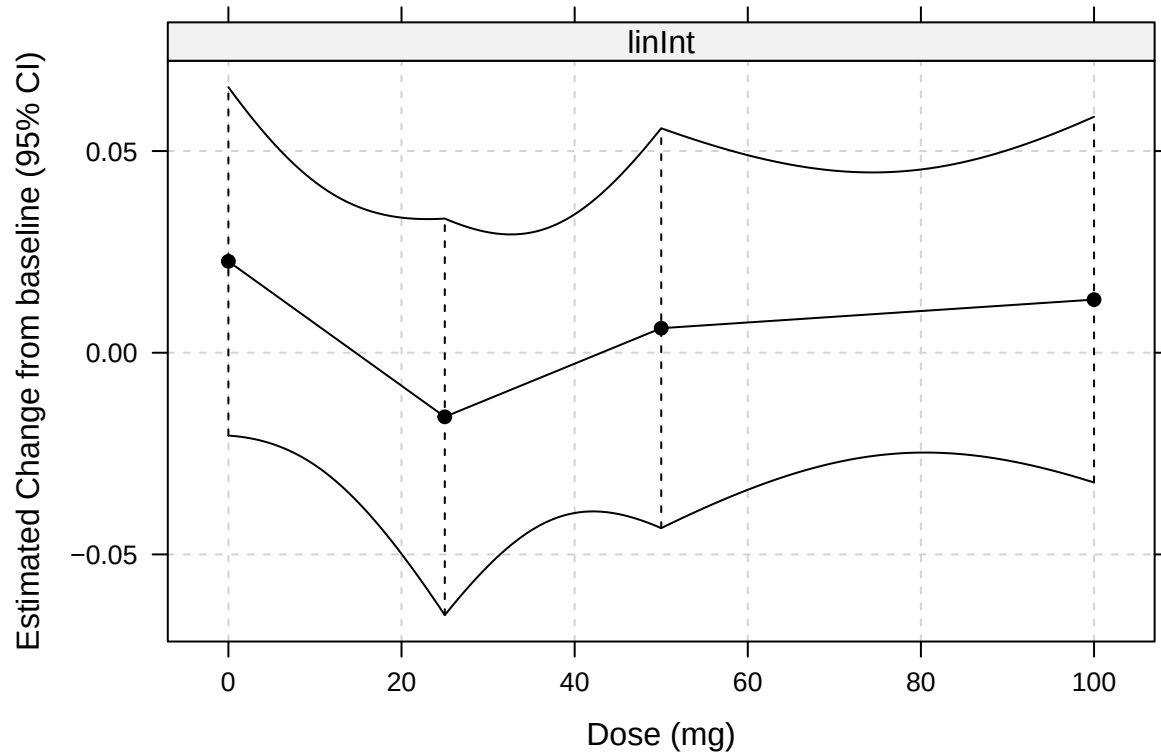
Estimated means by dose

Treatment	Estimated Change from Baseline (95% CI)
0 mg	0.02 (-0.02 to 0.07)
25 mg	-0.02 (-0.07 to 0.03)
50 mg	0.01 (-0.04 to 0.06)
100 mg	0.01 (-0.03 to 0.06)

Fitted dose-response curves







Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.02 (-0.06 to 0.03)

Mean relative solid stress (ROI2)

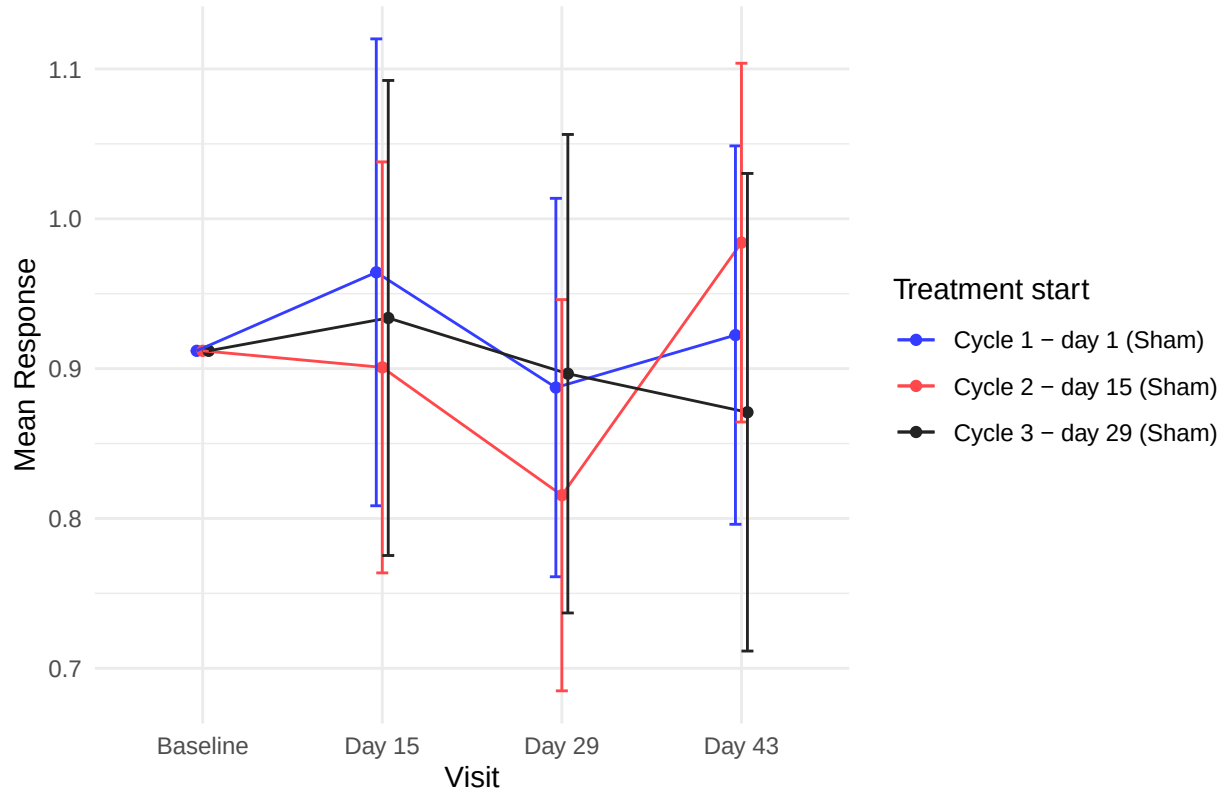
The primary analysis follows the MCP-Mod methodology as detailed in the SAP, using pre-specified candidate dose-response models.

A total of 54 subjects were included in the analysis for this endpoint.

The four candidate models (linear, piecewise linear, Emax, sigmoid Emax) are shown in Figure 1.

Line plot

Mean Response over Time for Mean relative solid stress (ROI2)



Optimal contrasts

Treatment	linear	emax	sigEmax	linInt
0 mg	-0.588	-0.771	-0.572	-0.859
25 mg	-0.233	-0.062	-0.387	0.193
50 mg	0.048	0.249	0.303	0.292
100 mg	0.773	0.583	0.657	0.374

MCP-Mod tests

Candidate model	t-statistic	Adjusted p-value
linear	1.12	0.21
sigEmax	0.63	0.39
emax	0.18	0.58
linInt	-0.51	0.82

Fitted model parameters (AIC/ED)

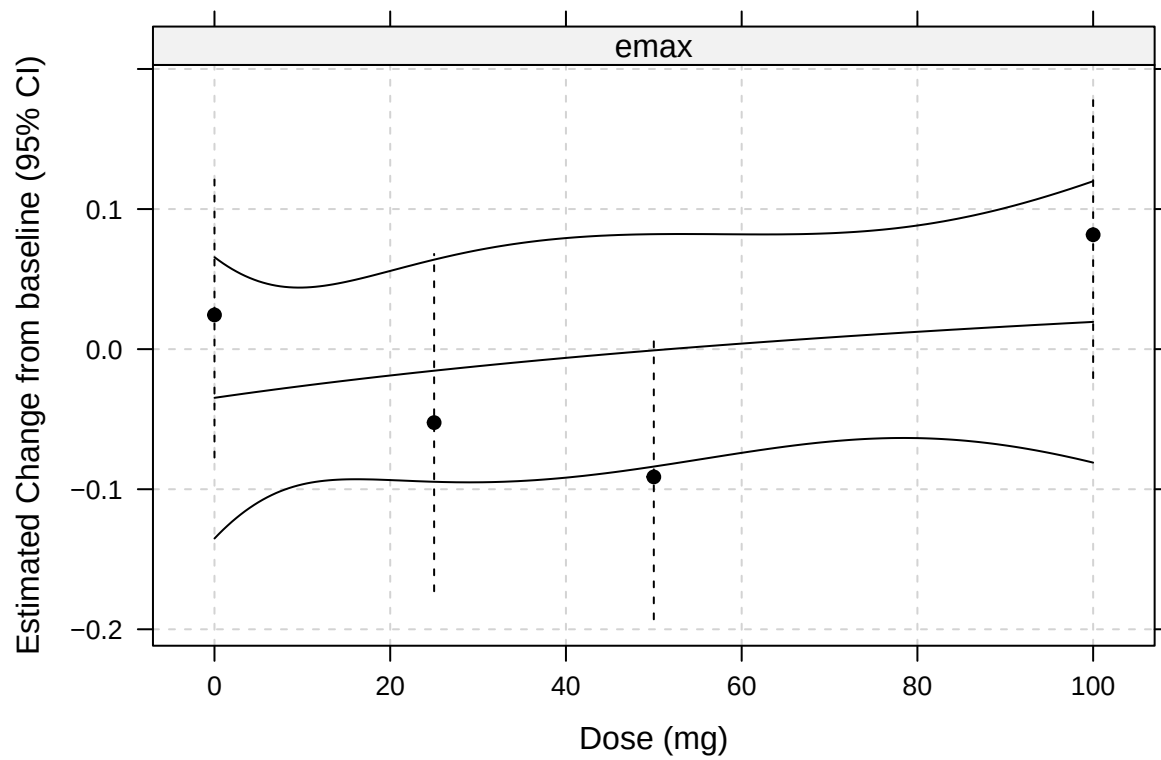
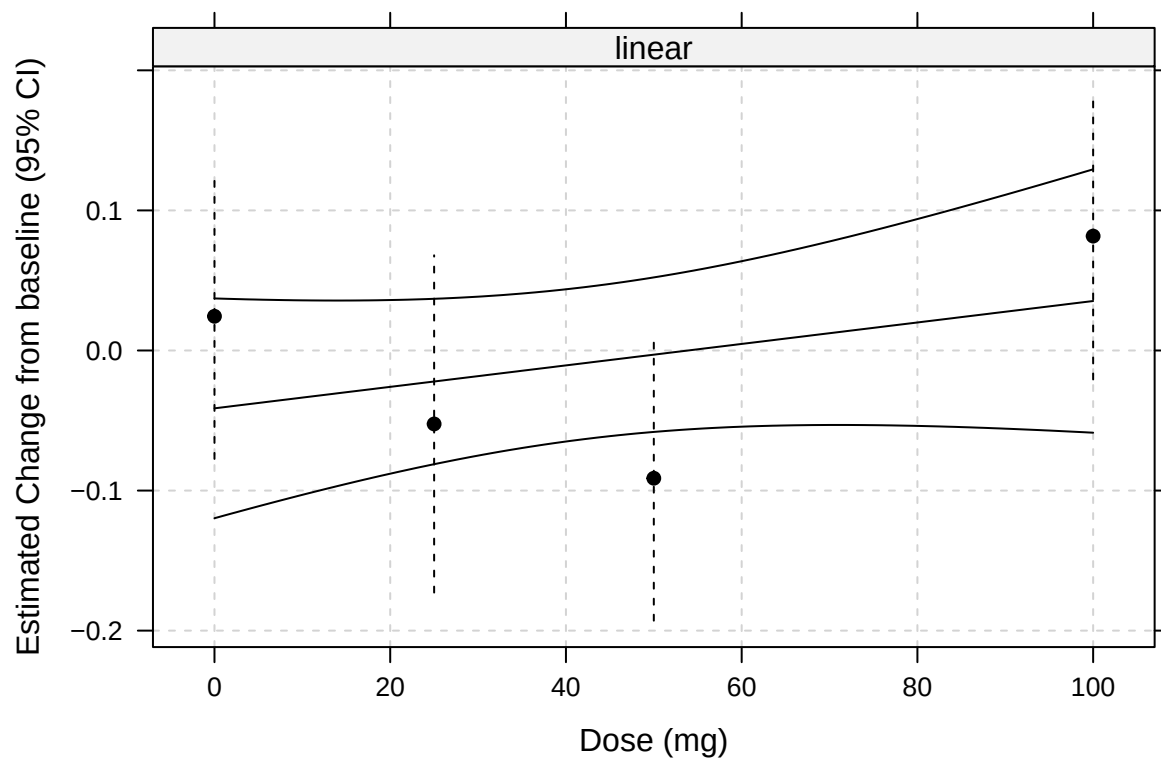
Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
linInt	8.00	18.81	27.63	42.54

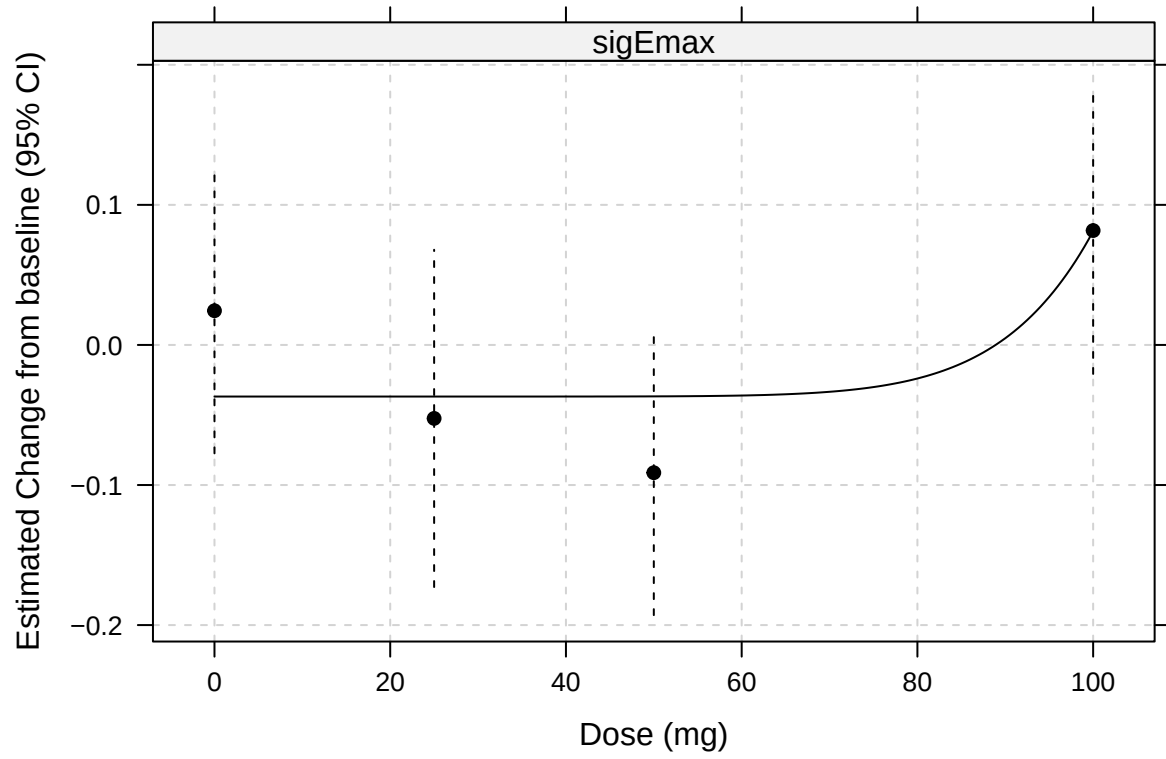
Model	AIC-value	ED ₅₀	ED ₇₀	ED ₉₀
linear	9.26	50.00	70.00	90.00
sigEmax	10.39	93.22	96.44	98.93
emax	11.91	37.50	58.33	84.37

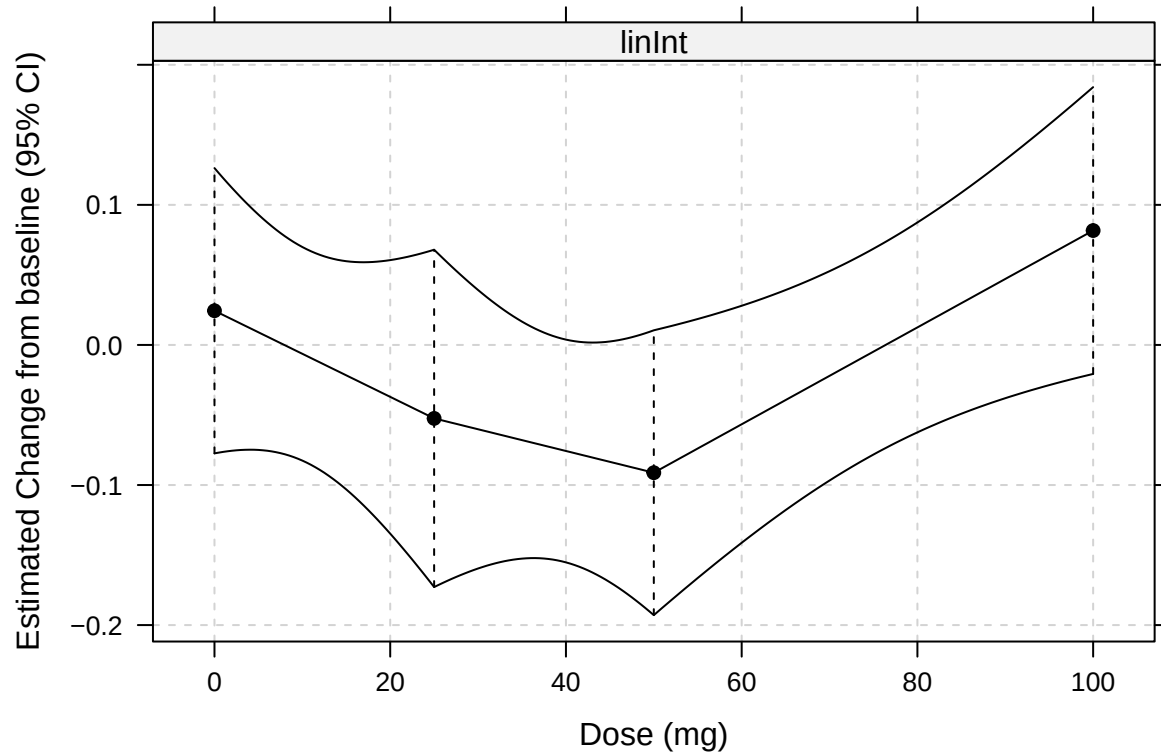
Estimated means by dose

Treatment	Estimated Change from Baseline (95% CI)
0 mg	0.02 (-0.08 to 0.13)
25 mg	-0.05 (-0.18 to 0.07)
50 mg	-0.09 (-0.20 to 0.01)
100 mg	0.08 (-0.02 to 0.19)

Fitted dose-response curves







Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.03 (-0.16 to 0.10)

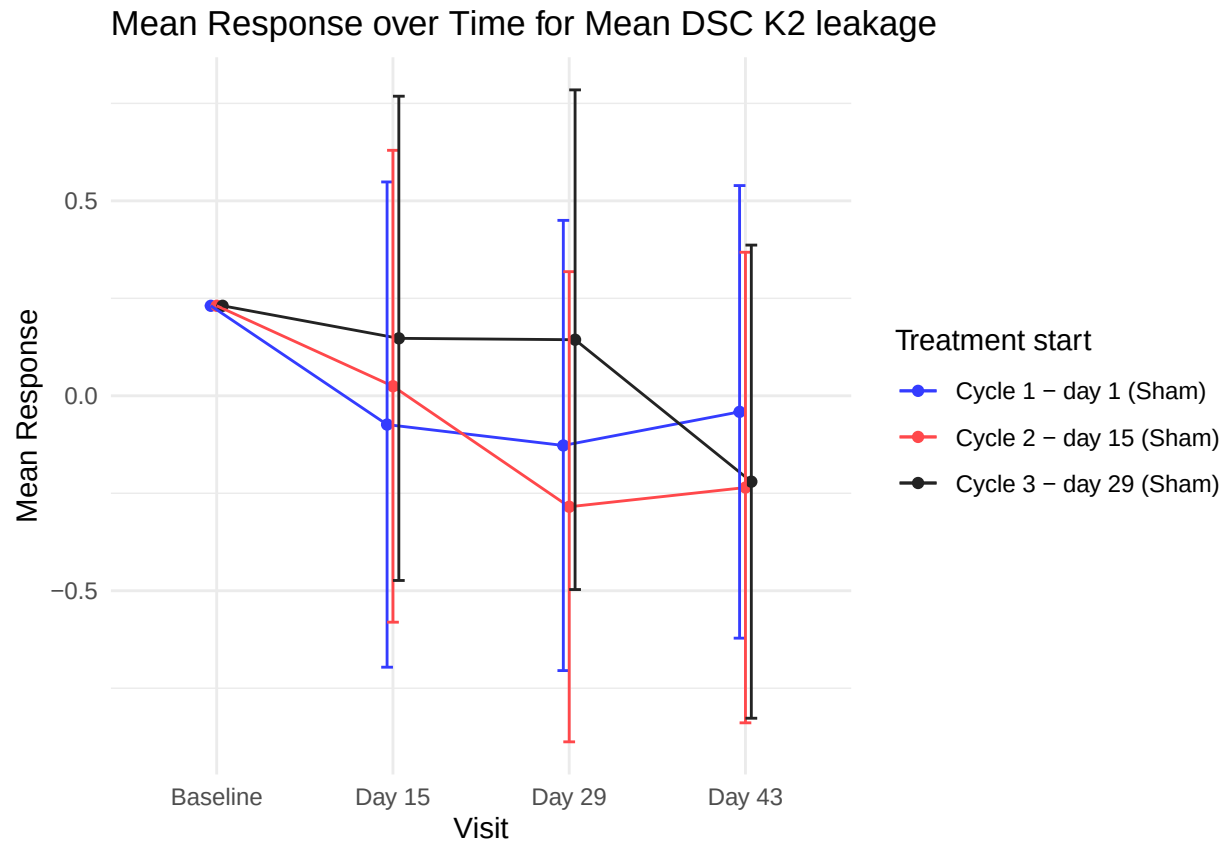
Other MRI Endpoints up to Day 43

For remaining MRI endpoints, we present estimated means by dose, contrasts vs 0 mg, and a plot with confidence intervals.

Mean DSC K2 leakage

A total of 53 subjects were included in the analysis for this endpoint.

Line plot by treatment start



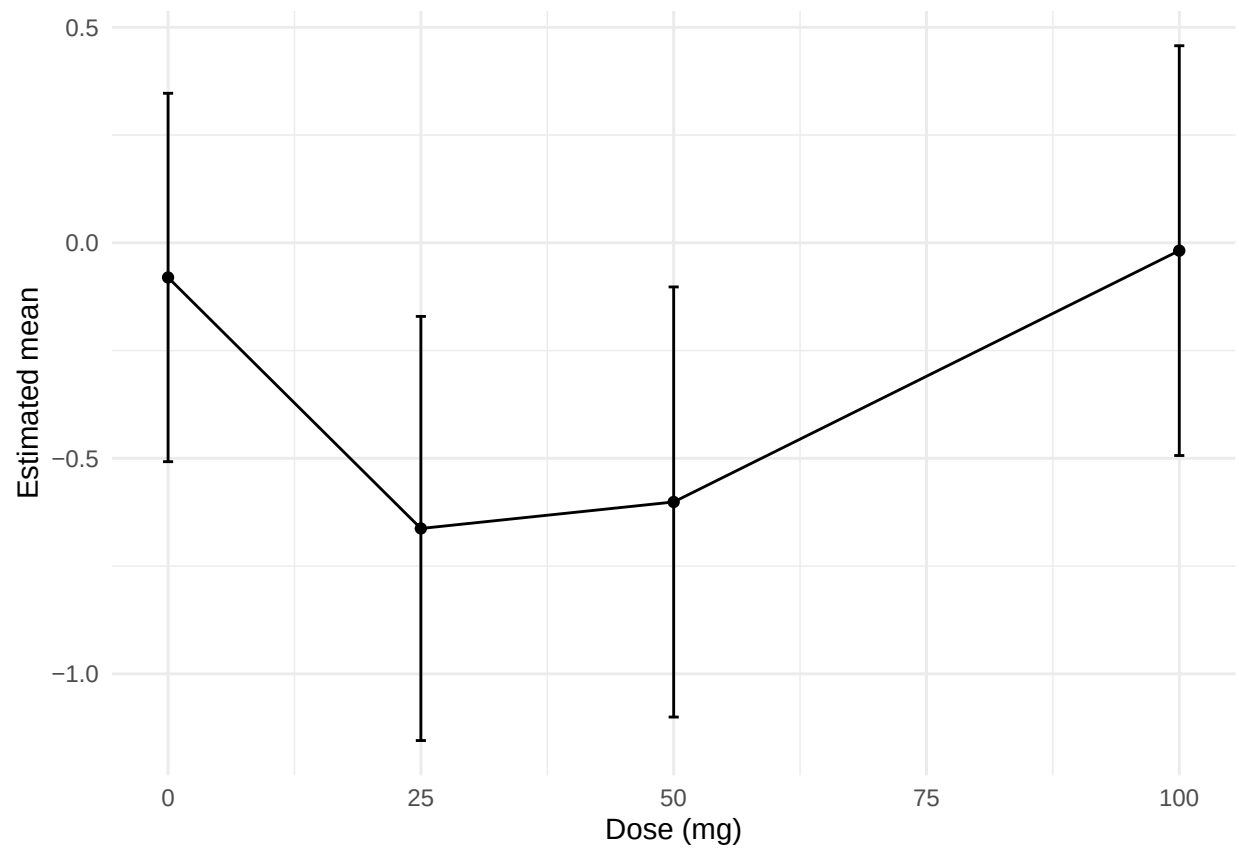
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	-0.08 (-0.51 to 0.35)
25 mg	-0.66 (-1.15 to -0.17)
50 mg	-0.60 (-1.10 to -0.10)
100 mg	-0.02 (-0.49 to 0.46)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	-0.58 (-1.34 to 0.17)
50 mg vs 0 mg	-0.52 (-1.30 to 0.25)
100 mg vs 0 mg	0.06 (-0.66 to 0.79)

Plot



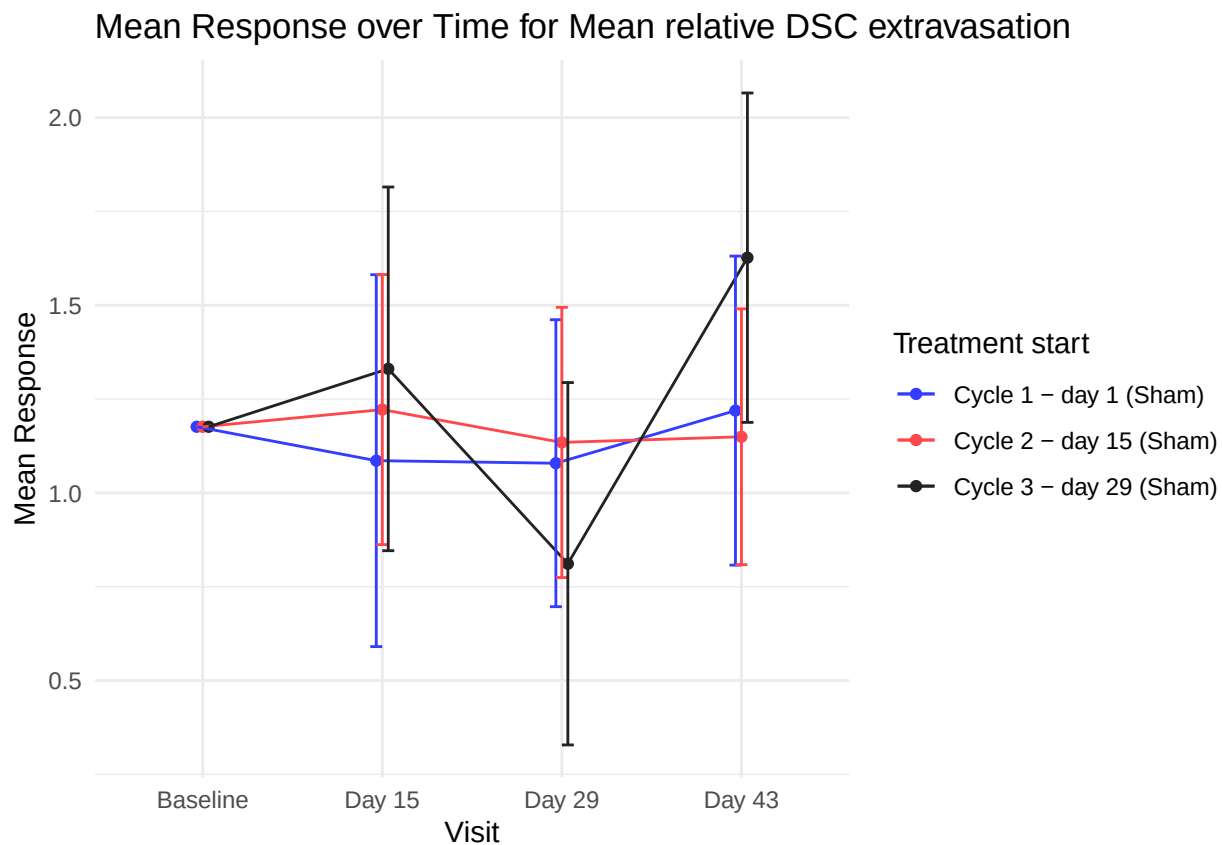
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.31 (-0.81 to 0.19)

Mean relative DSC extravasation

A total of 27 subjects were included in the analysis for this endpoint.

Line plot by treatment start



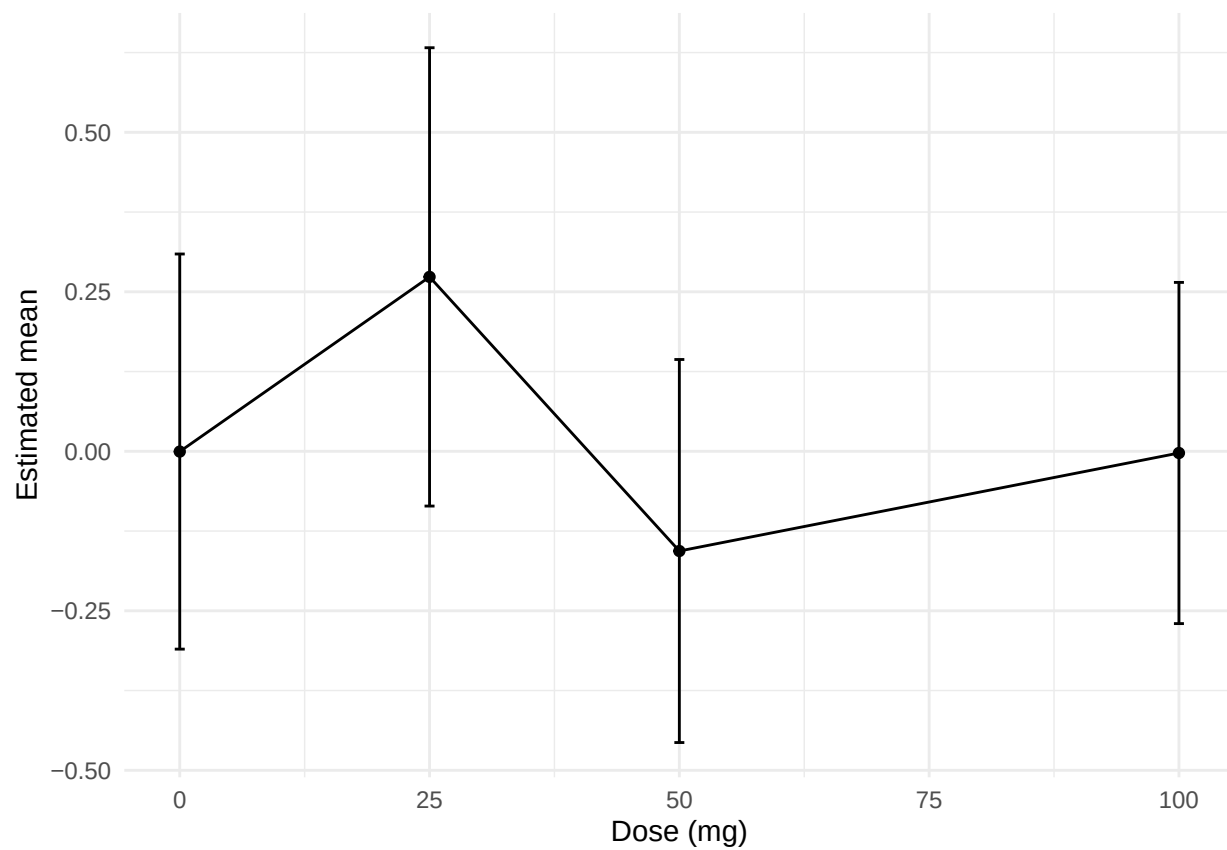
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	0.00 (-0.31 to 0.31)
25 mg	0.27 (-0.09 to 0.63)
50 mg	-0.16 (-0.46 to 0.14)
100 mg	0.00 (-0.27 to 0.26)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	0.27 (-0.33 to 0.88)
50 mg vs 0 mg	-0.16 (-0.72 to 0.41)
100 mg vs 0 mg	0.00 (-0.54 to 0.53)

Plot



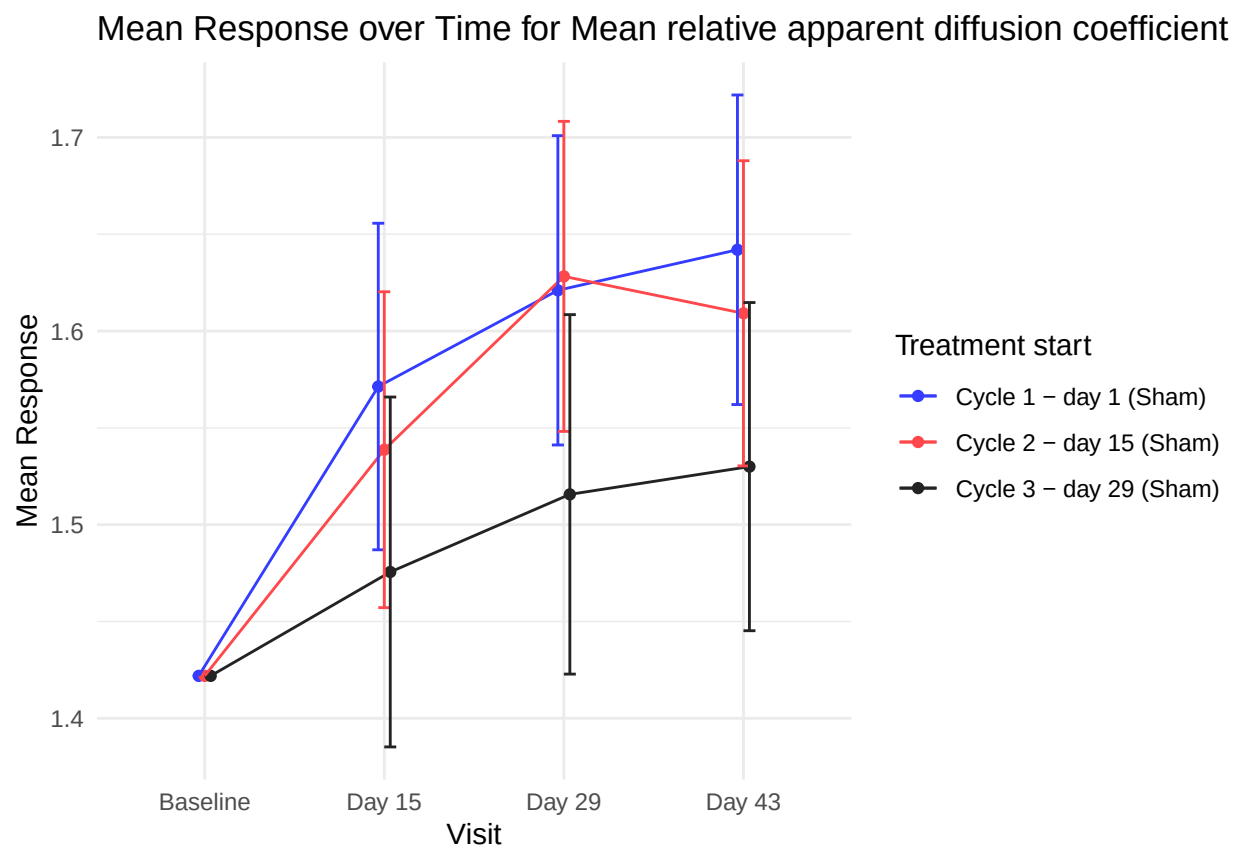
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	0.01 (-0.39 to 0.42)

Mean relative apparent diffusion coefficient

A total of 51 subjects were included in the analysis for this endpoint.

Line plot by treatment start



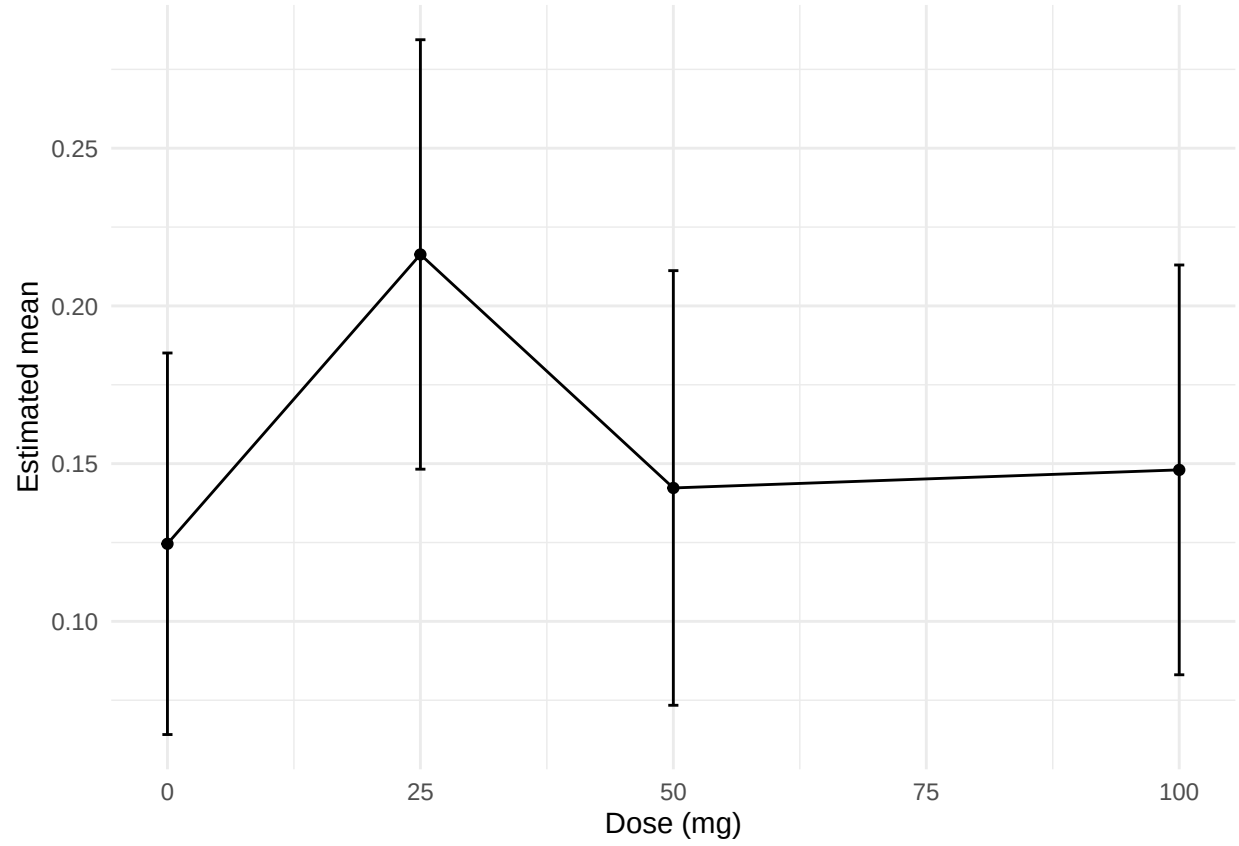
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	0.12 (0.06 to 0.19)
25 mg	0.22 (0.15 to 0.28)
50 mg	0.14 (0.07 to 0.21)
100 mg	0.15 (0.08 to 0.21)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	0.09 (-0.01 to 0.19)
50 mg vs 0 mg	0.02 (-0.08 to 0.12)
100 mg vs 0 mg	0.02 (-0.07 to 0.12)

Plot



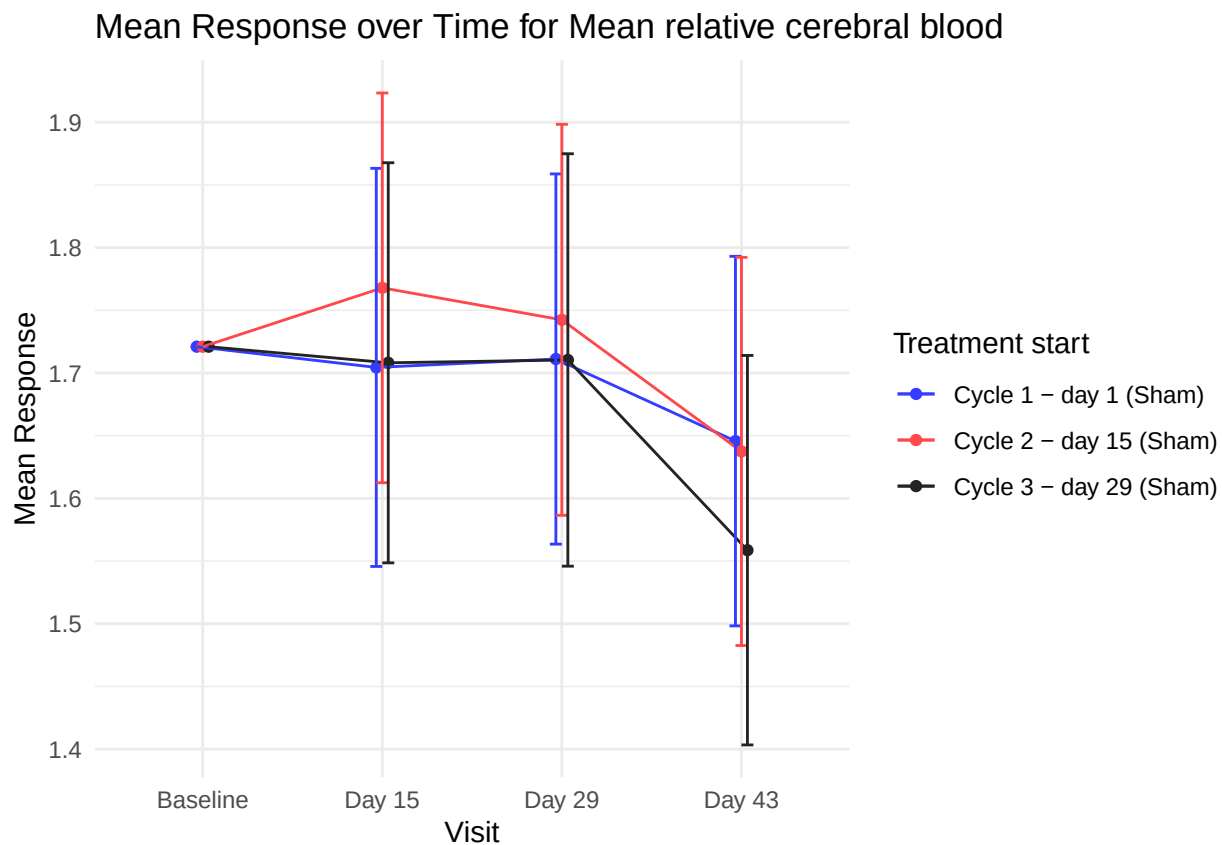
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	0.04 (-0.02 to 0.11)

Mean relative cerebral blood

A total of 53 subjects were included in the analysis for this endpoint.

Line plot by treatment start



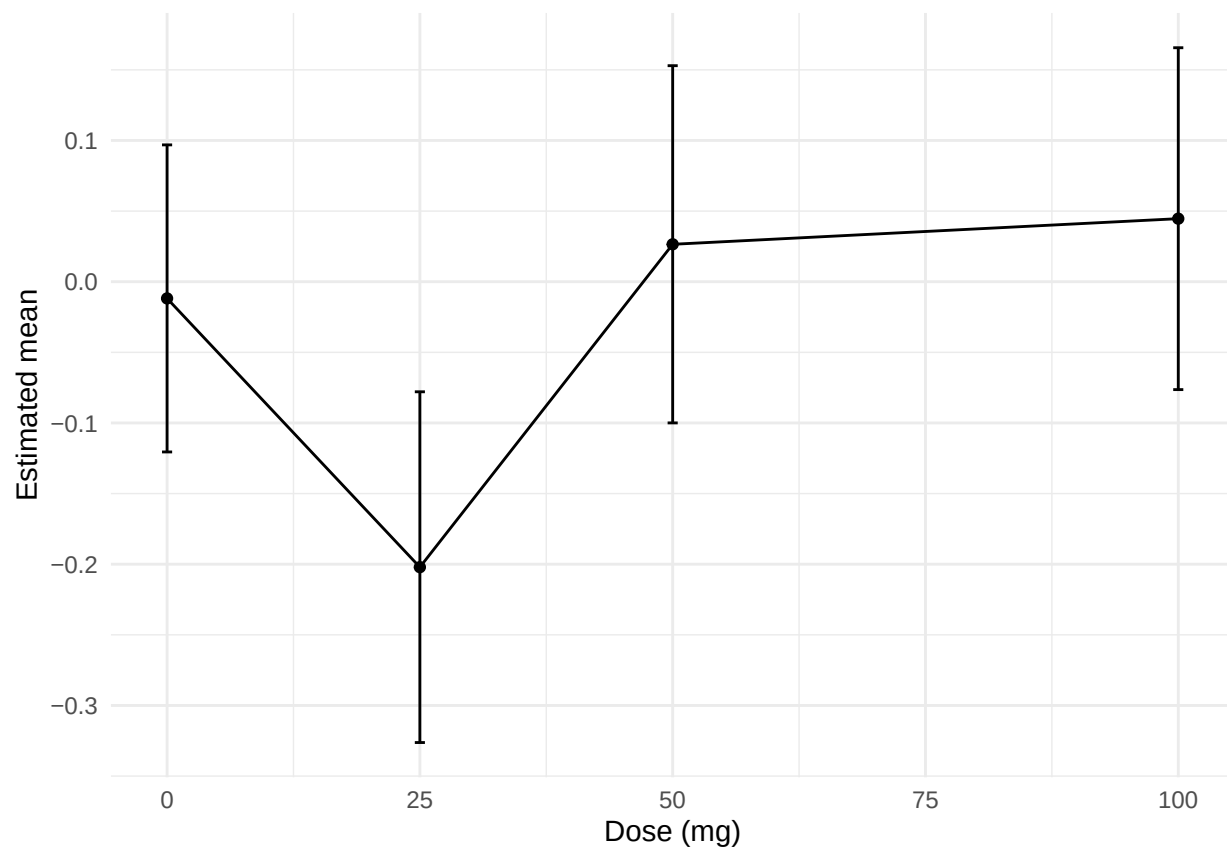
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	-0.01 (-0.12 to 0.10)
25 mg	-0.20 (-0.33 to -0.08)
50 mg	0.03 (-0.10 to 0.15)
100 mg	0.04 (-0.08 to 0.17)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	-0.19 (-0.38 to 0.00)
50 mg vs 0 mg	0.04 (-0.16 to 0.24)
100 mg vs 0 mg	0.06 (-0.13 to 0.24)

Plot



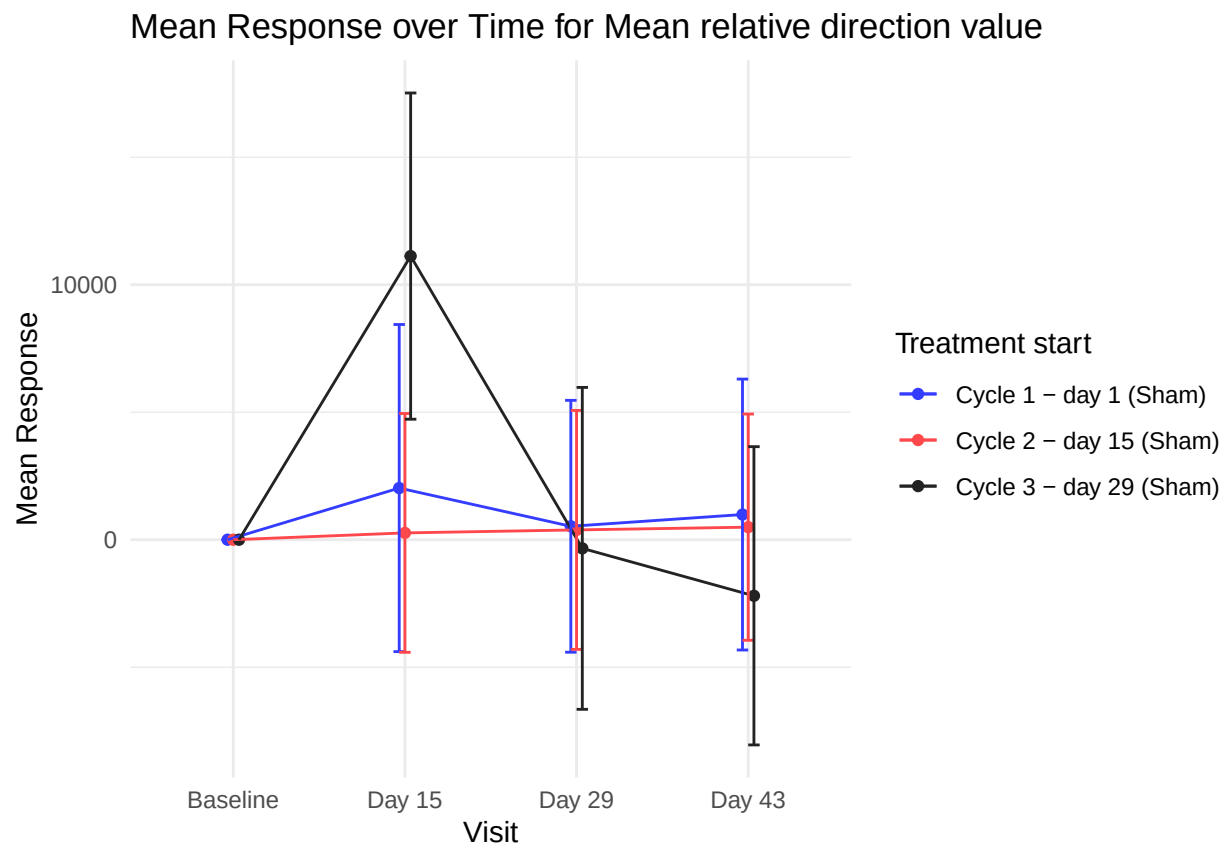
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.03 (-0.16 to 0.10)

Mean relative direction value

A total of 27 subjects were included in the analysis for this endpoint.

Line plot by treatment start



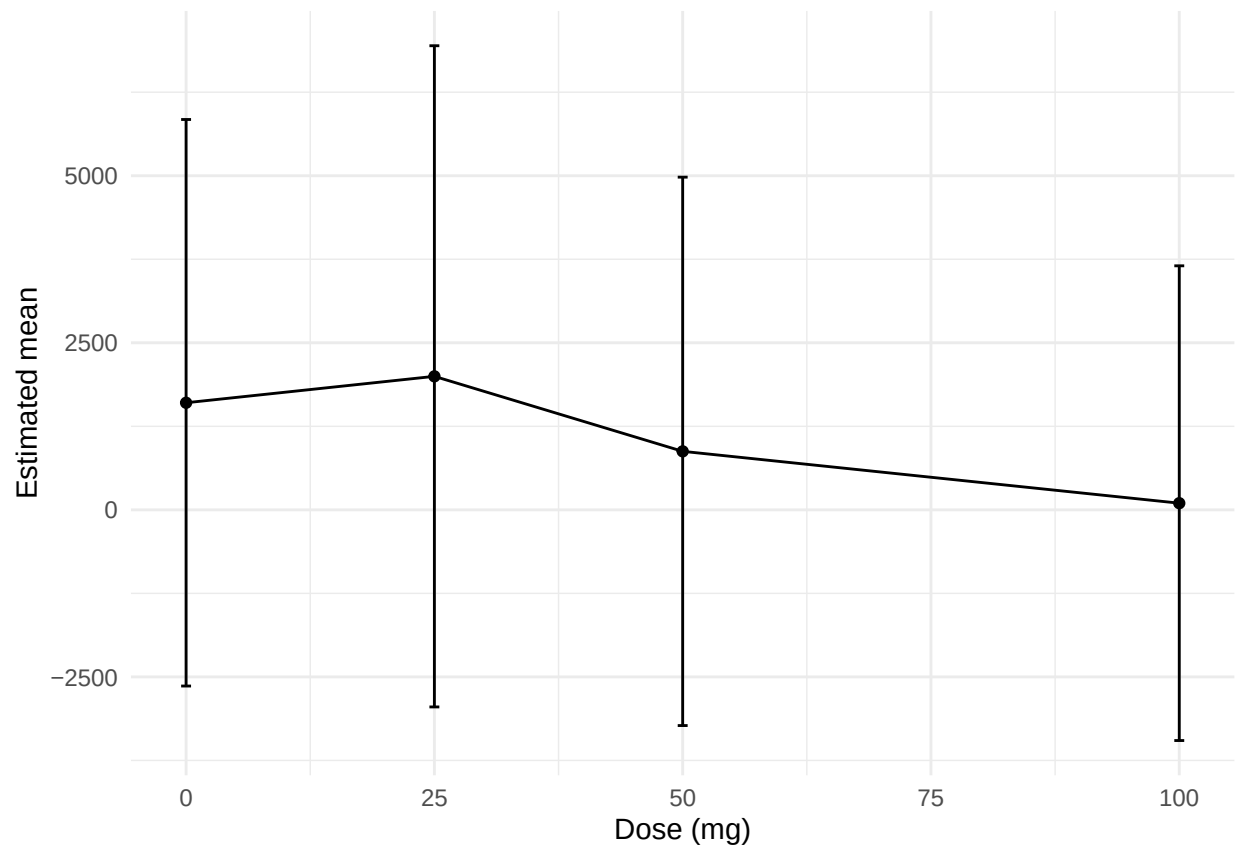
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	1602.43 (-2636.48 to 5841.34)
25 mg	1997.84 (-2949.72 to 6945.39)
50 mg	875.35 (-3227.90 to 4978.60)
100 mg	99.63 (-3452.41 to 3651.67)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	395.41 (-8045.52 to 8836.34)
50 mg vs 0 mg	-727.08 (-8496.49 to 7042.33)
100 mg vs 0 mg	-1502.80 (-8648.91 to 5643.31)

Plot



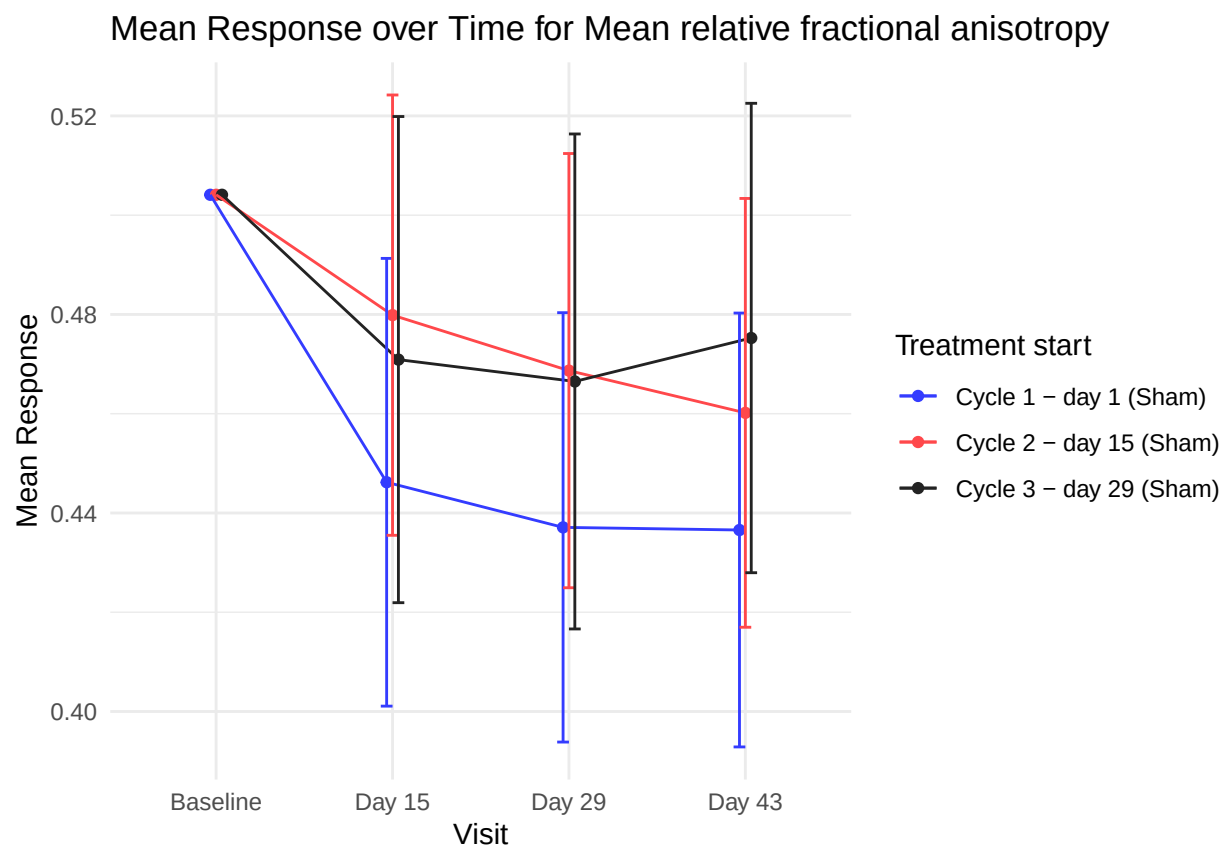
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-899.49 (-6255.58 to 4456.61)

Mean relative fractional anisotropy

A total of 51 subjects were included in the analysis for this endpoint.

Line plot by treatment start



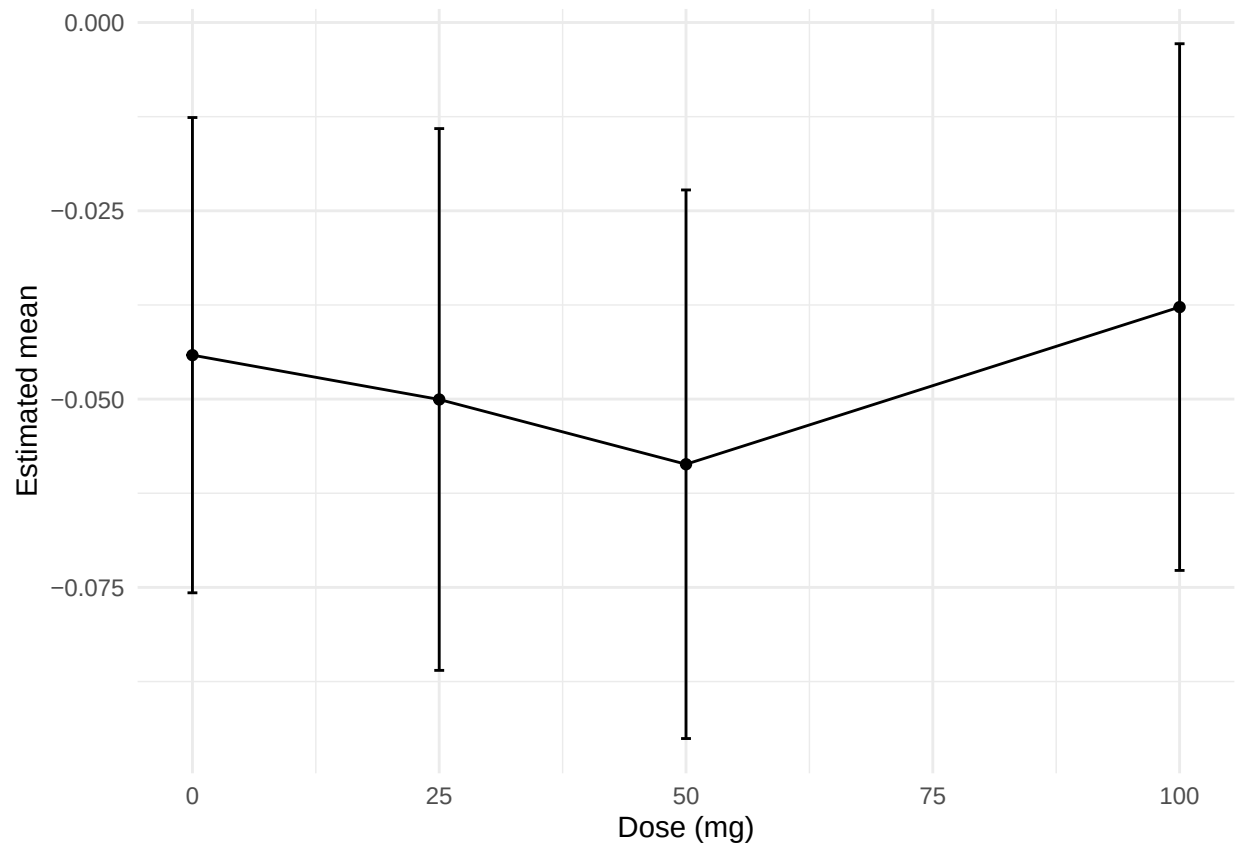
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	-0.04 (-0.08 to -0.01)
25 mg	-0.05 (-0.09 to -0.01)
50 mg	-0.06 (-0.10 to -0.02)
100 mg	-0.04 (-0.07 to 0.00)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	-0.01 (-0.06 to 0.04)
50 mg vs 0 mg	-0.01 (-0.07 to 0.04)
100 mg vs 0 mg	0.01 (-0.04 to 0.05)

Plot



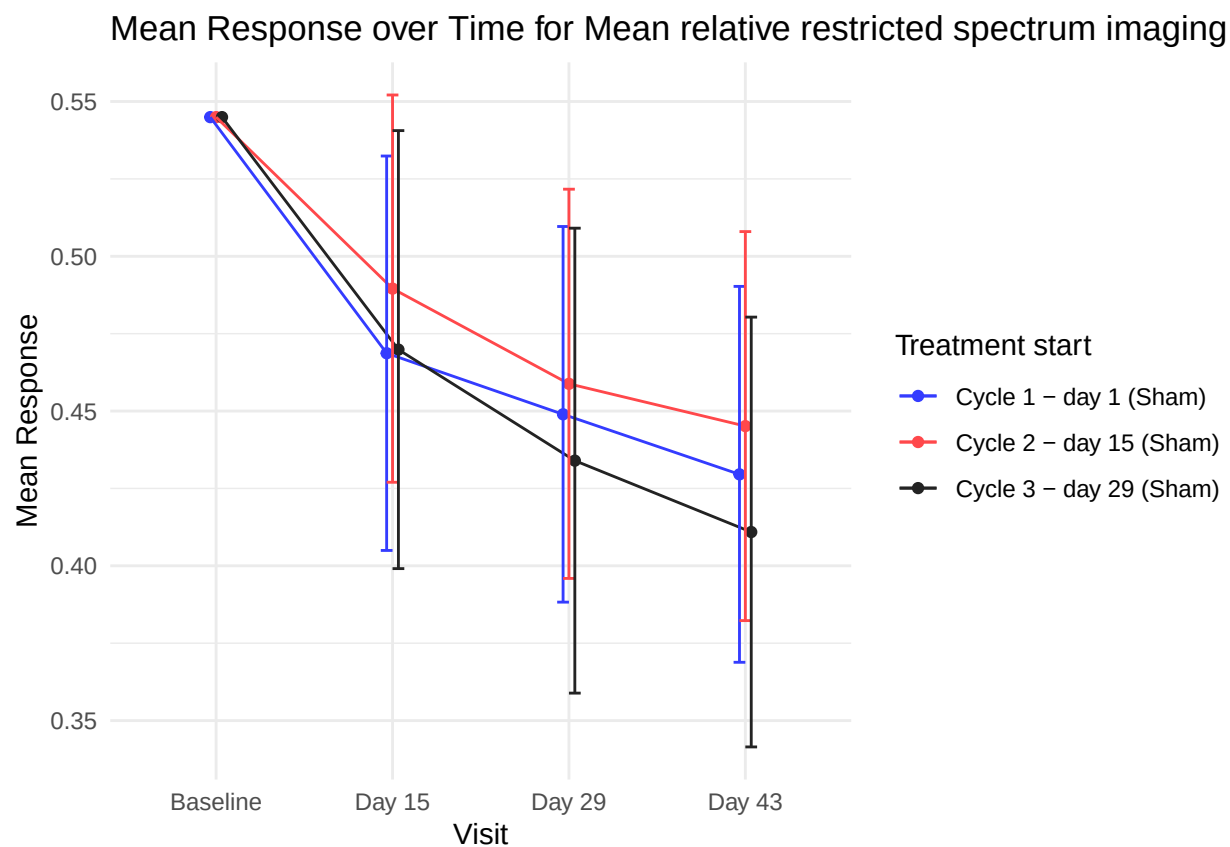
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	0.00 (-0.03 to 0.03)

Mean relative restricted spectrum imaging

A total of 49 subjects were included in the analysis for this endpoint.

Line plot by treatment start



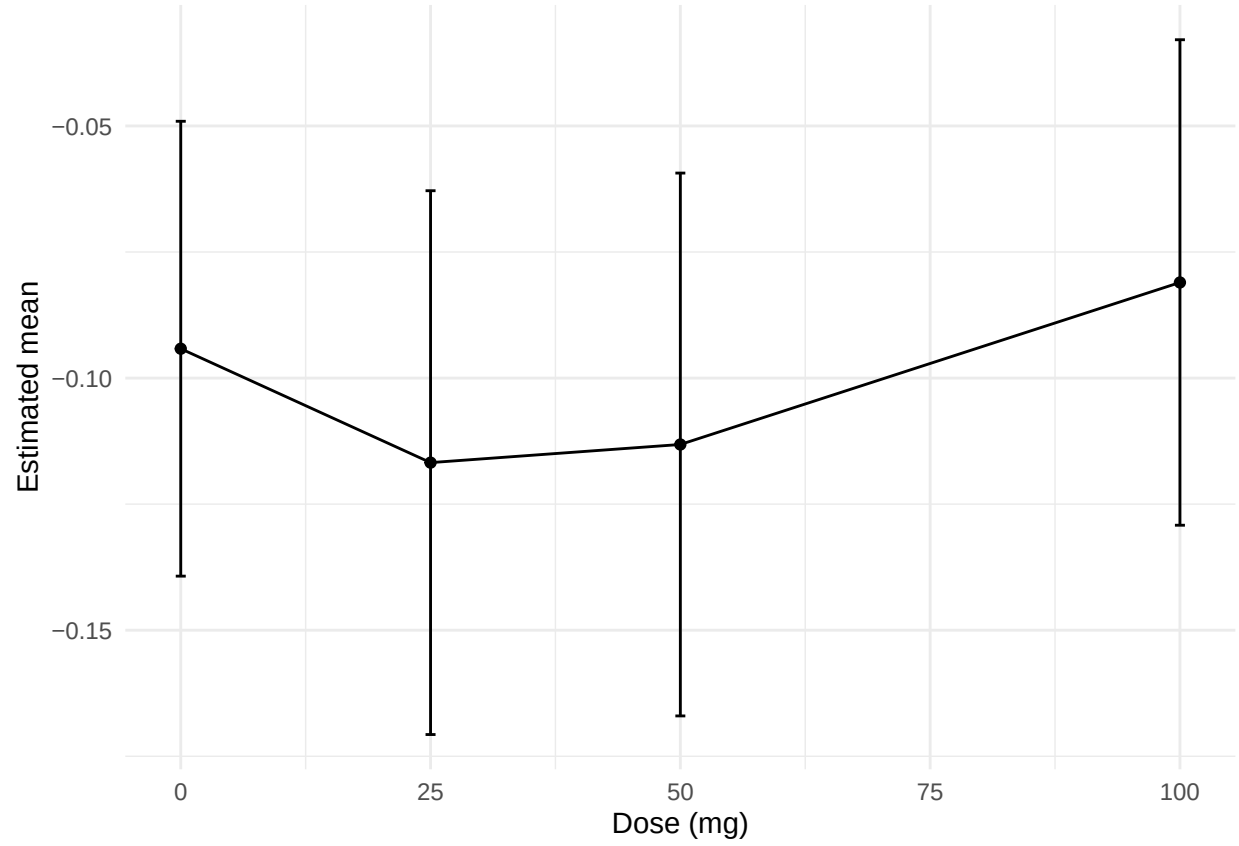
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	-0.09 (-0.14 to -0.05)
25 mg	-0.12 (-0.17 to -0.06)
50 mg	-0.11 (-0.17 to -0.06)
100 mg	-0.08 (-0.13 to -0.03)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	-0.02 (-0.10 to 0.05)
50 mg vs 0 mg	-0.02 (-0.09 to 0.06)
100 mg vs 0 mg	0.01 (-0.05 to 0.08)

Plot



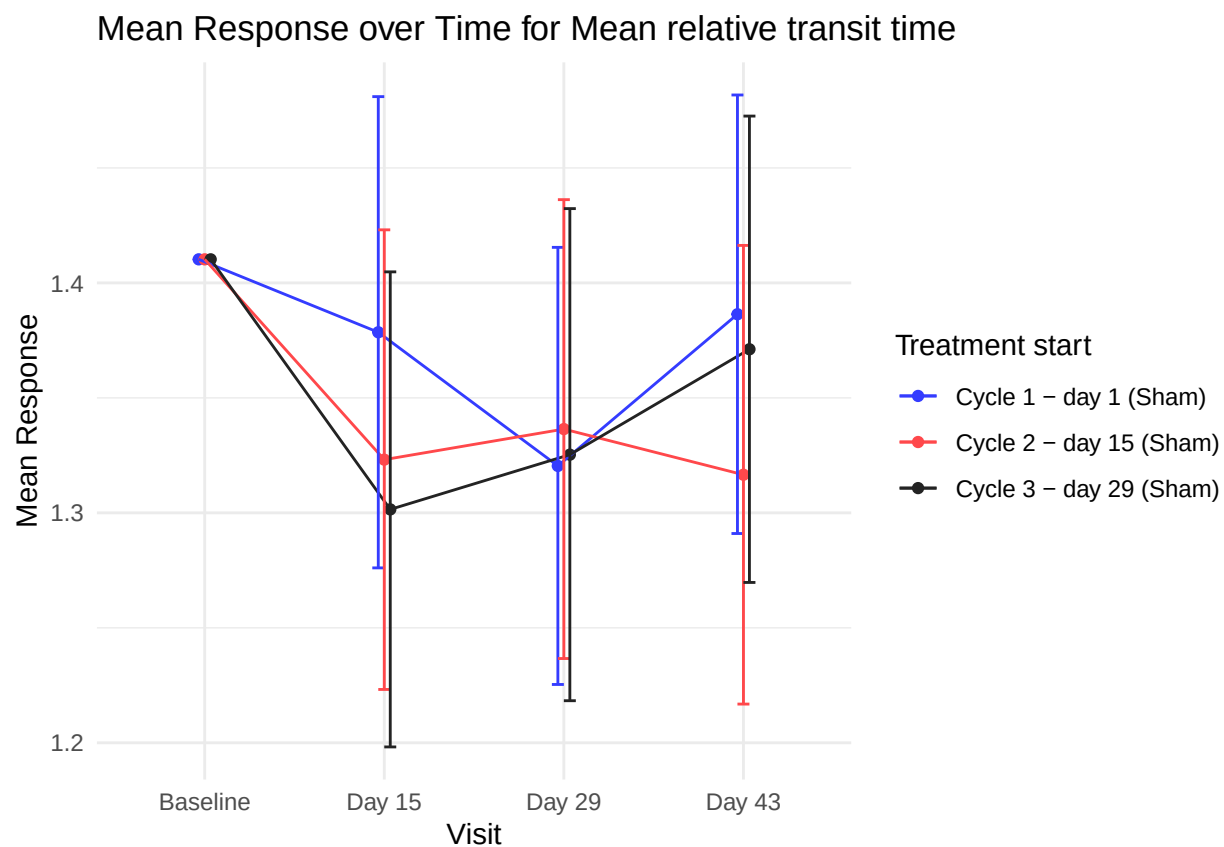
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.01 (-0.05 to 0.04)

Mean relative transit time

A total of 53 subjects were included in the analysis for this endpoint.

Line plot by treatment start



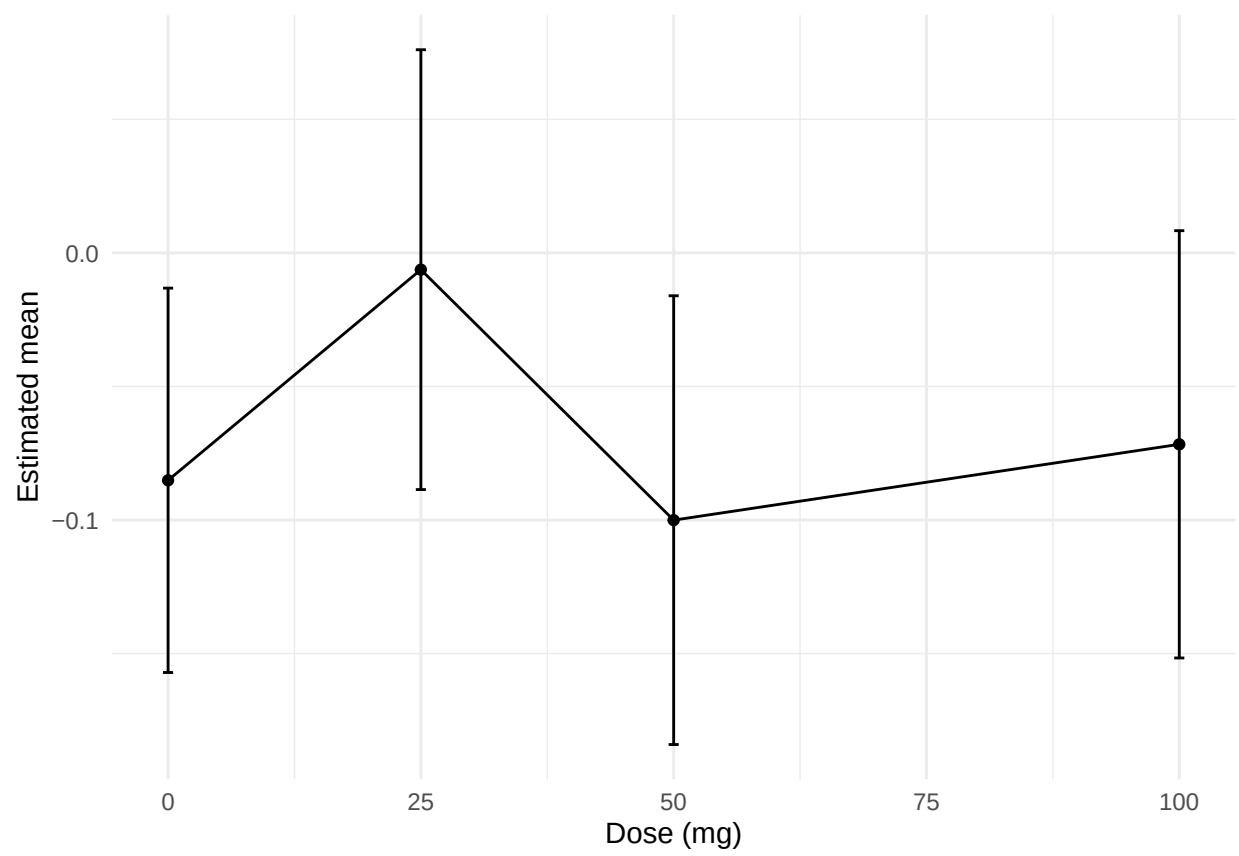
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	-0.09 (-0.16 to -0.01)
25 mg	-0.01 (-0.09 to 0.08)
50 mg	-0.10 (-0.18 to -0.02)
100 mg	-0.07 (-0.15 to 0.01)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	0.08 (-0.05 to 0.21)
50 mg vs 0 mg	-0.01 (-0.15 to 0.12)
100 mg vs 0 mg	0.01 (-0.11 to 0.14)

Plot



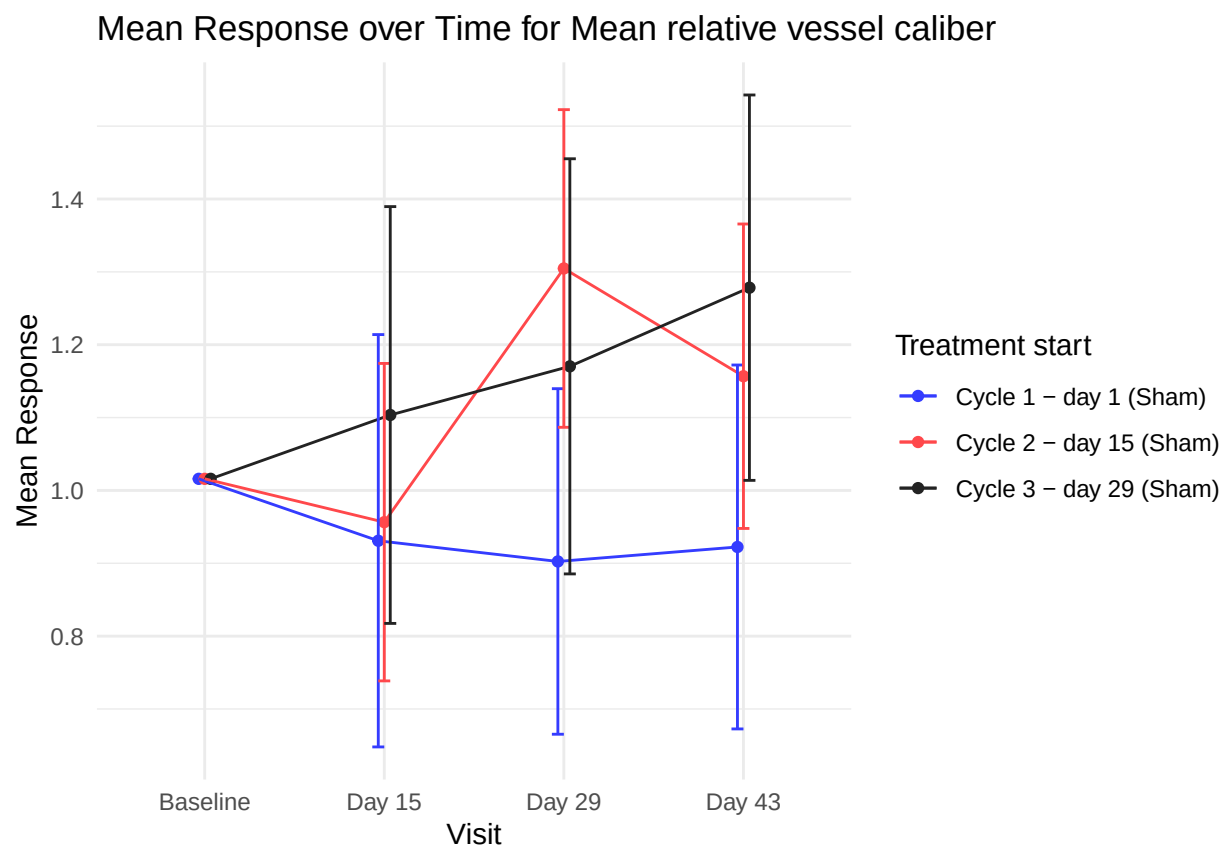
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	0.03 (-0.06 to 0.11)

Mean relative vessel caliber

A total of 27 subjects were included in the analysis for this endpoint.

Line plot by treatment start



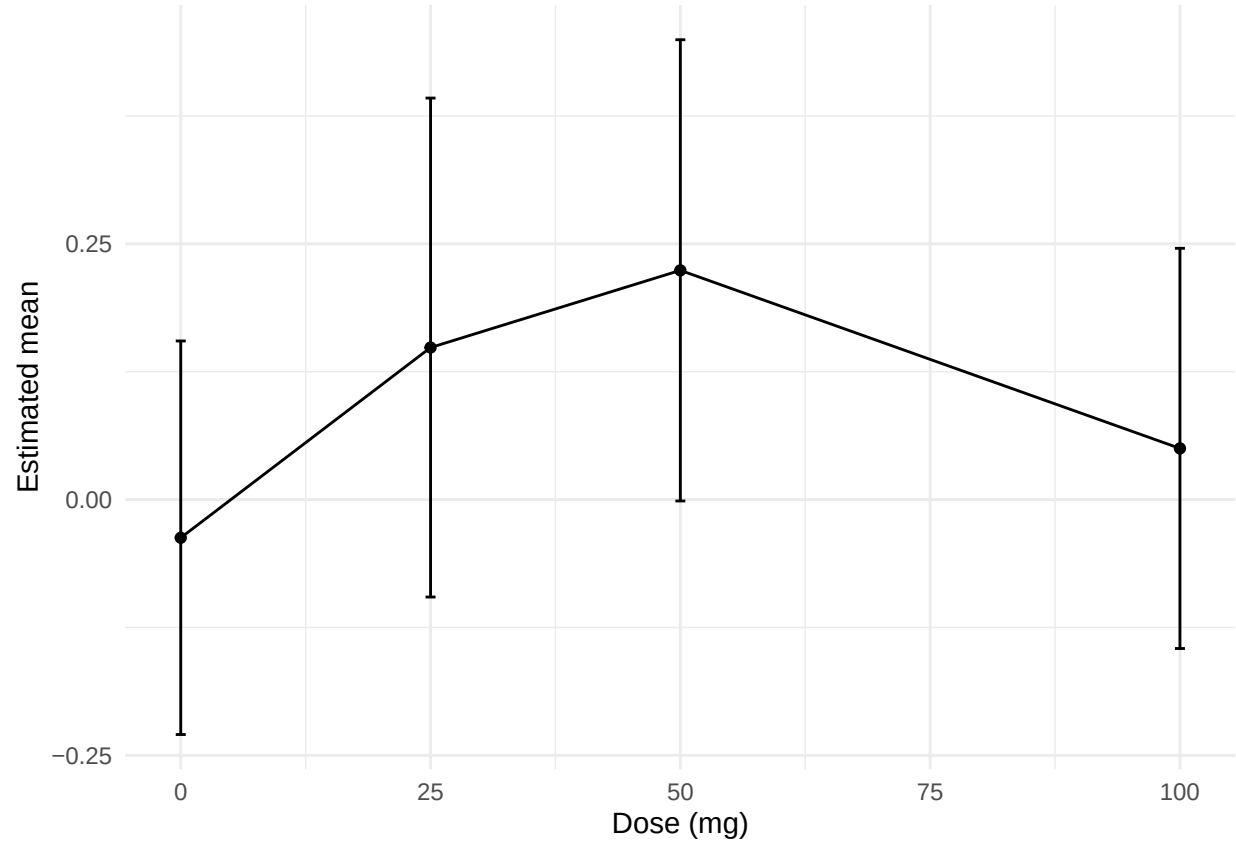
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	-0.04 (-0.23 to 0.16)
25 mg	0.15 (-0.10 to 0.39)
50 mg	0.22 (0.00 to 0.45)
100 mg	0.05 (-0.15 to 0.25)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	0.19 (-0.17 to 0.54)
50 mg vs 0 mg	0.26 (-0.09 to 0.62)
100 mg vs 0 mg	0.09 (-0.21 to 0.39)

Plot



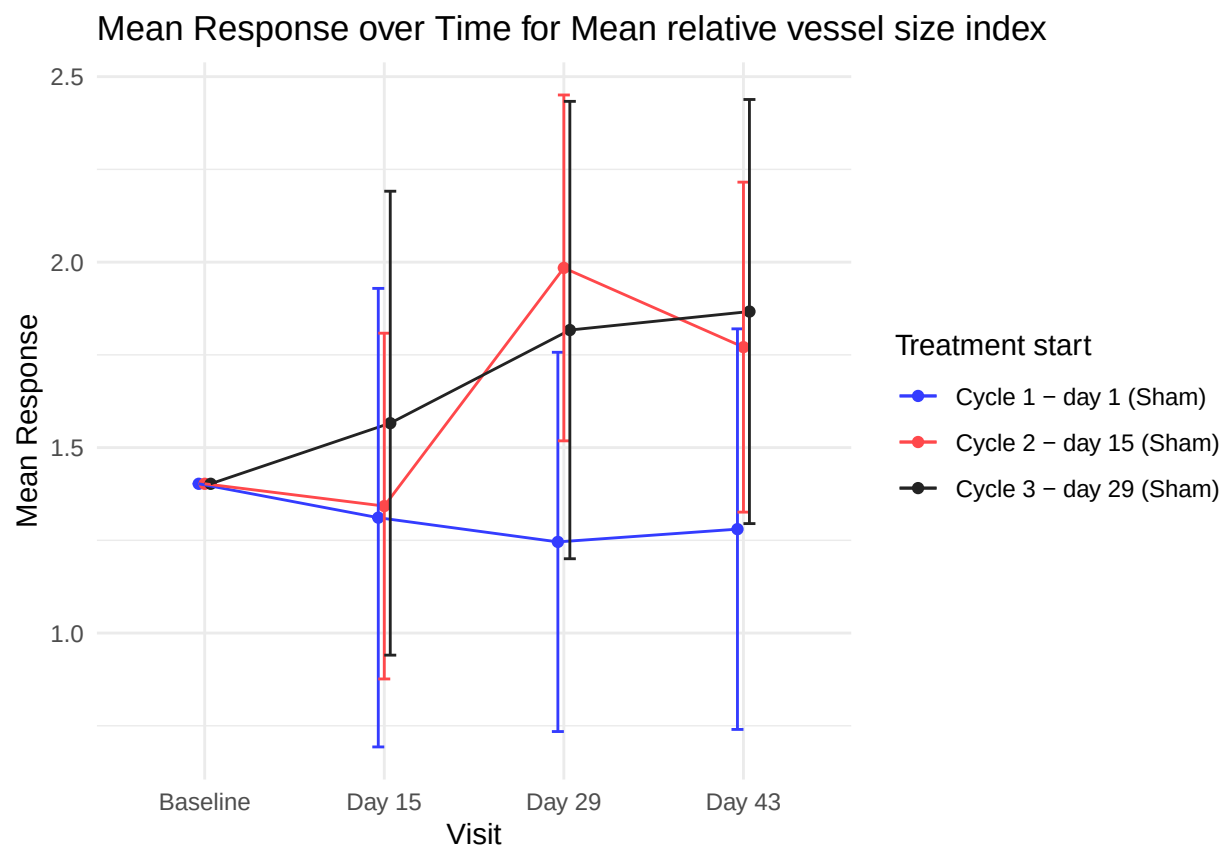
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	0.15 (-0.07 to 0.36)

Mean relative vessel size index

A total of 27 subjects were included in the analysis for this endpoint.

Line plot by treatment start



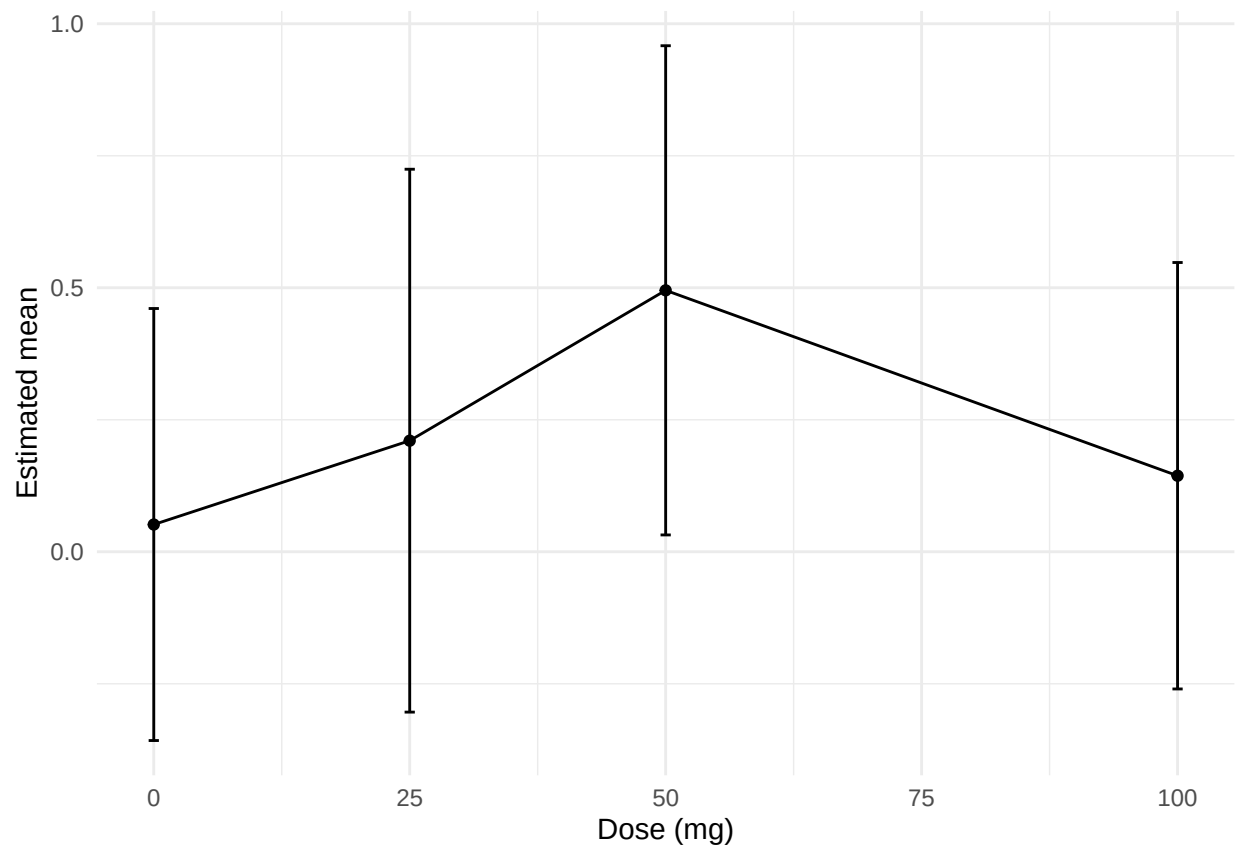
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	0.05 (-0.36 to 0.46)
25 mg	0.21 (-0.30 to 0.72)
50 mg	0.50 (0.03 to 0.96)
100 mg	0.14 (-0.26 to 0.55)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	0.16 (-0.62 to 0.94)
50 mg vs 0 mg	0.44 (-0.32 to 1.21)
100 mg vs 0 mg	0.09 (-0.57 to 0.75)

Plot



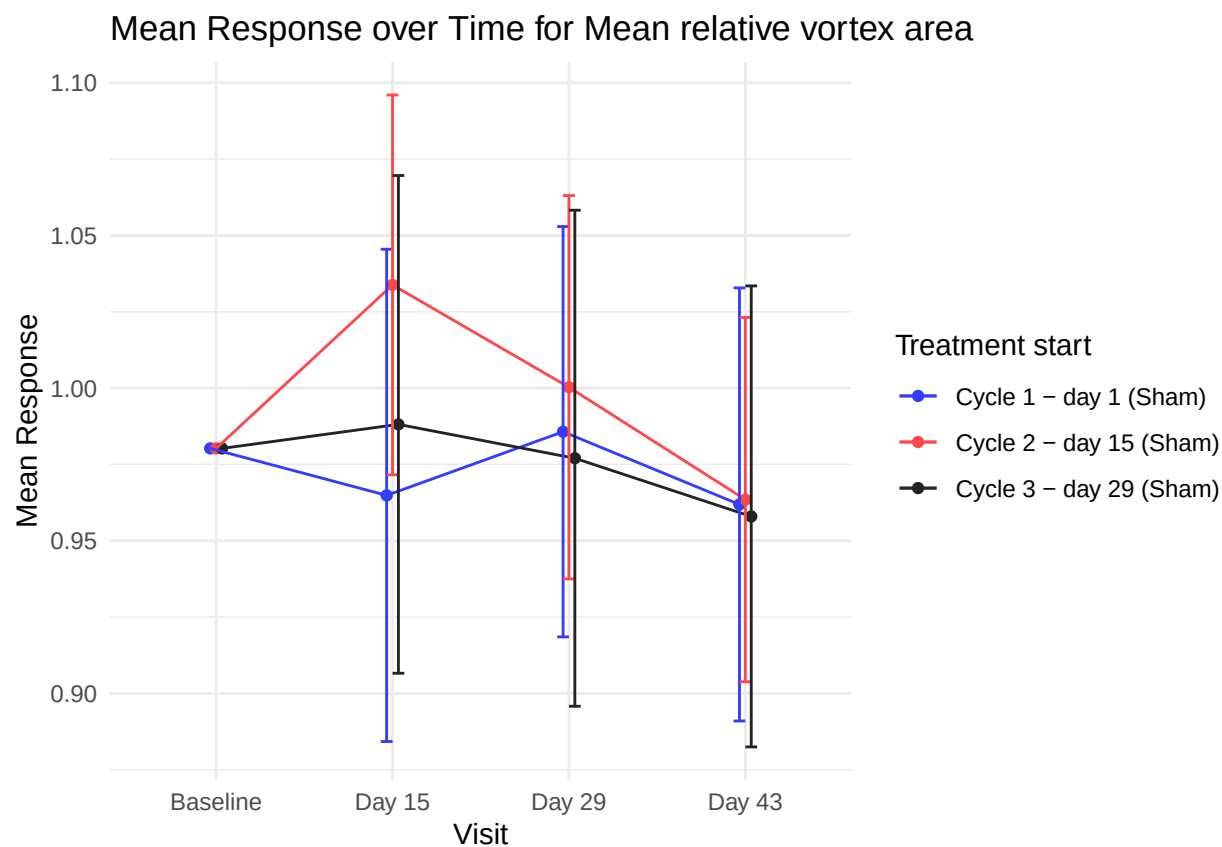
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	0.19 (-0.29 to 0.67)

Mean relative vortex area

A total of 27 subjects were included in the analysis for this endpoint.

Line plot by treatment start



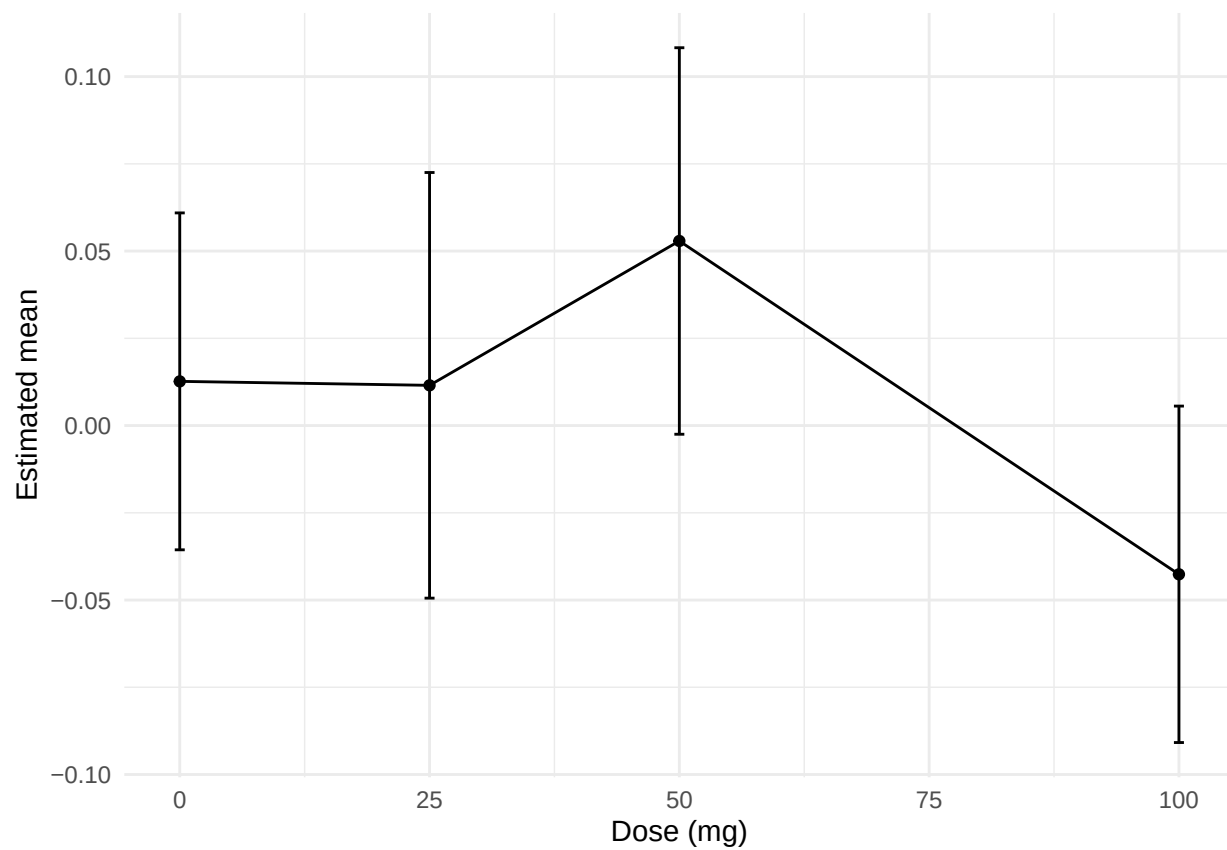
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	0.01 (-0.04 to 0.06)
25 mg	0.01 (-0.05 to 0.07)
50 mg	0.05 (0.00 to 0.11)
100 mg	-0.04 (-0.09 to 0.01)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	0.00 (-0.09 to 0.09)
50 mg vs 0 mg	0.04 (-0.05 to 0.13)
100 mg vs 0 mg	-0.06 (-0.13 to 0.02)

Plot



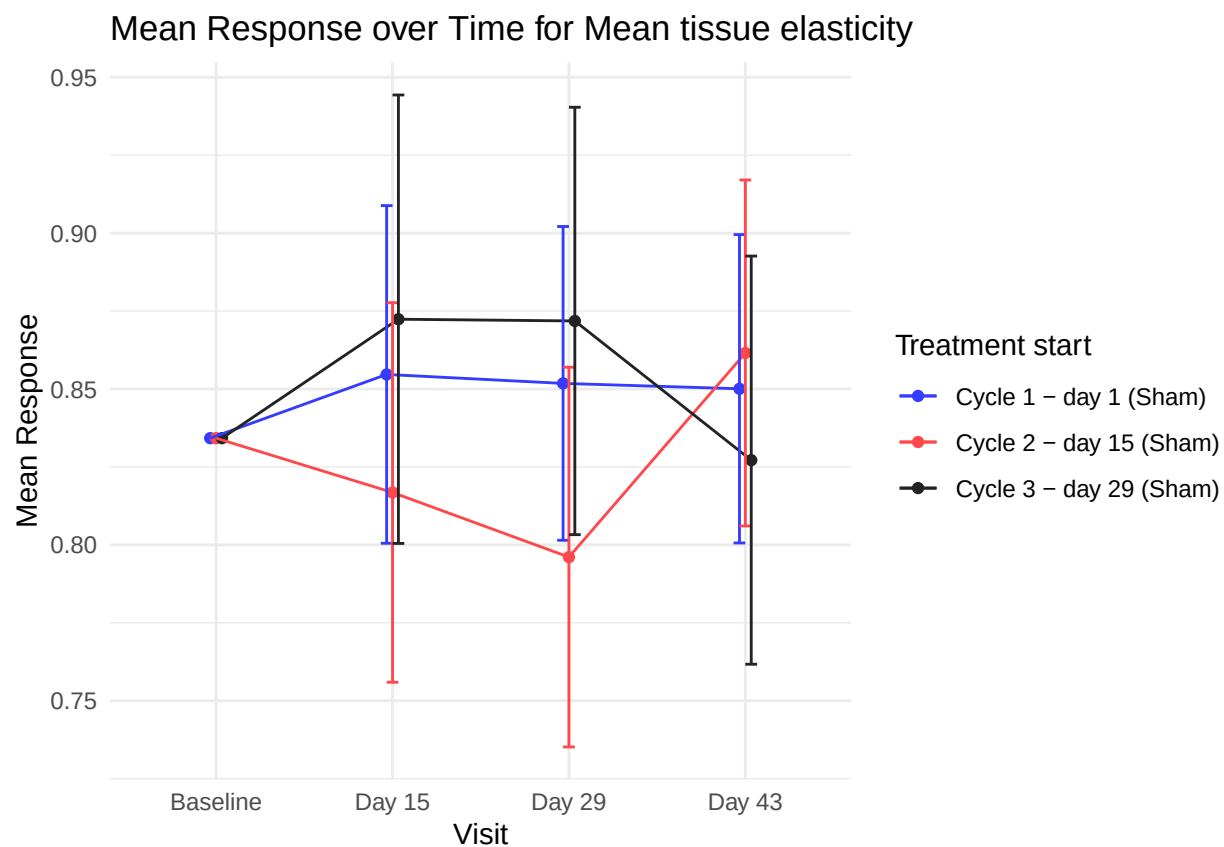
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.02 (-0.08 to 0.04)

Mean tissue elasticity

A total of 40 subjects were included in the analysis for this endpoint.

Line plot by treatment start



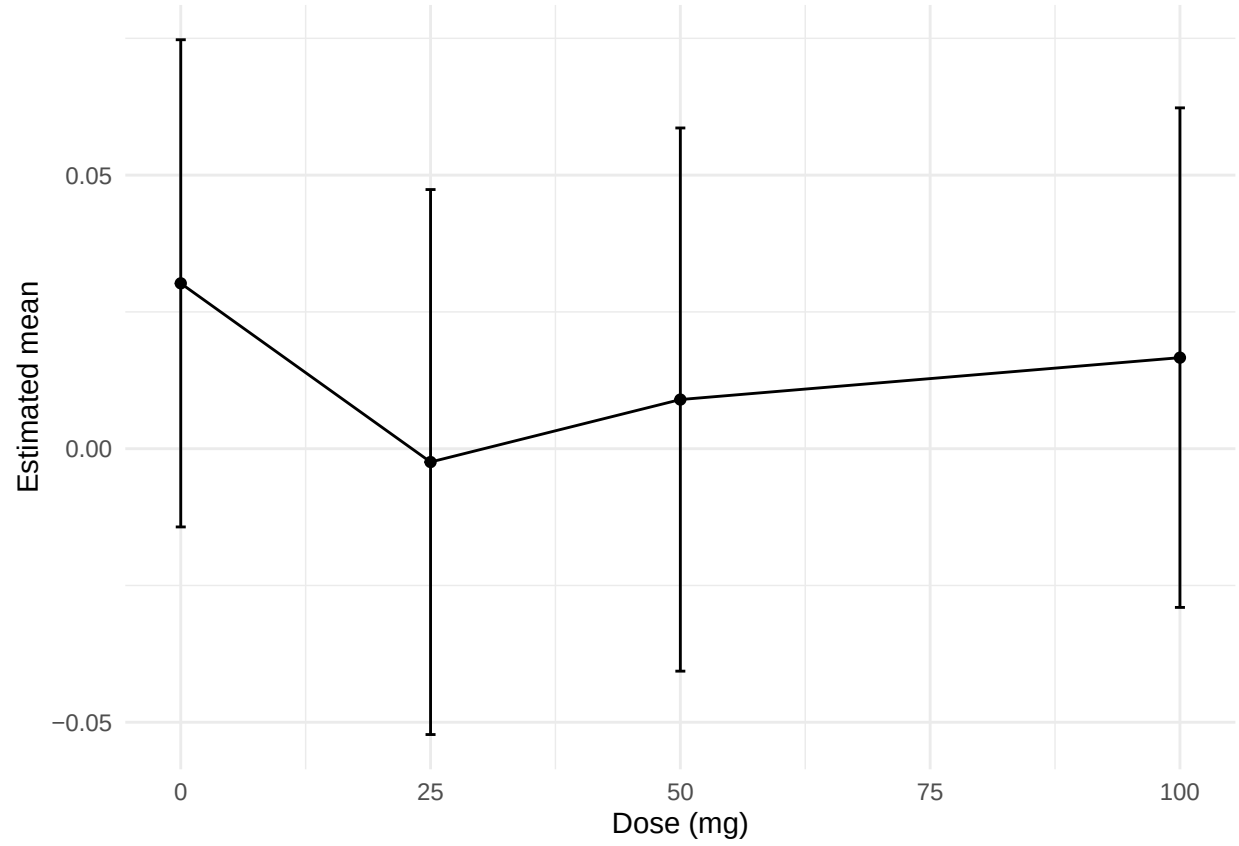
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	0.03 (-0.01 to 0.07)
25 mg	0.00 (-0.05 to 0.05)
50 mg	0.01 (-0.04 to 0.06)
100 mg	0.02 (-0.03 to 0.06)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	-0.03 (-0.11 to 0.04)
50 mg vs 0 mg	-0.02 (-0.10 to 0.06)
100 mg vs 0 mg	-0.01 (-0.08 to 0.05)

Plot



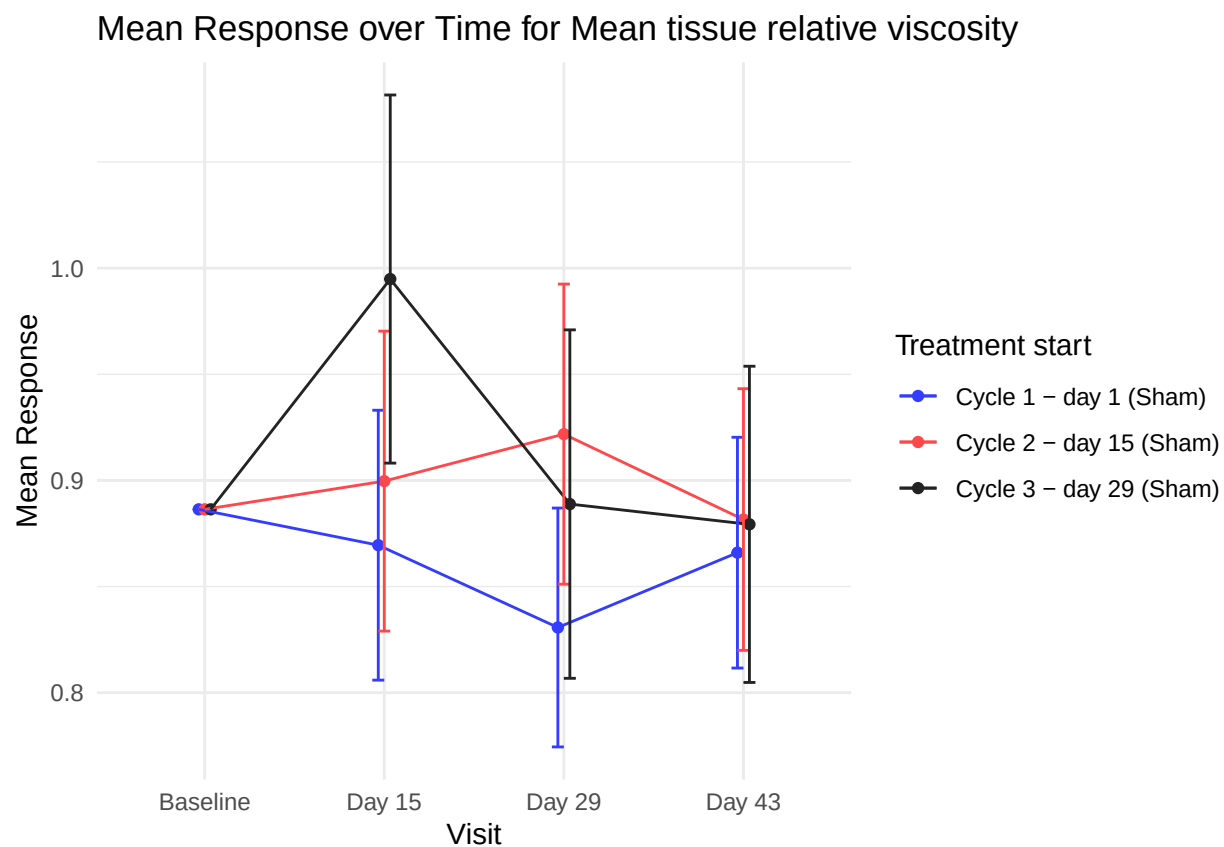
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.02 (-0.07 to 0.03)

Mean tissue relative viscosity

A total of 40 subjects were included in the analysis for this endpoint.

Line plot by treatment start



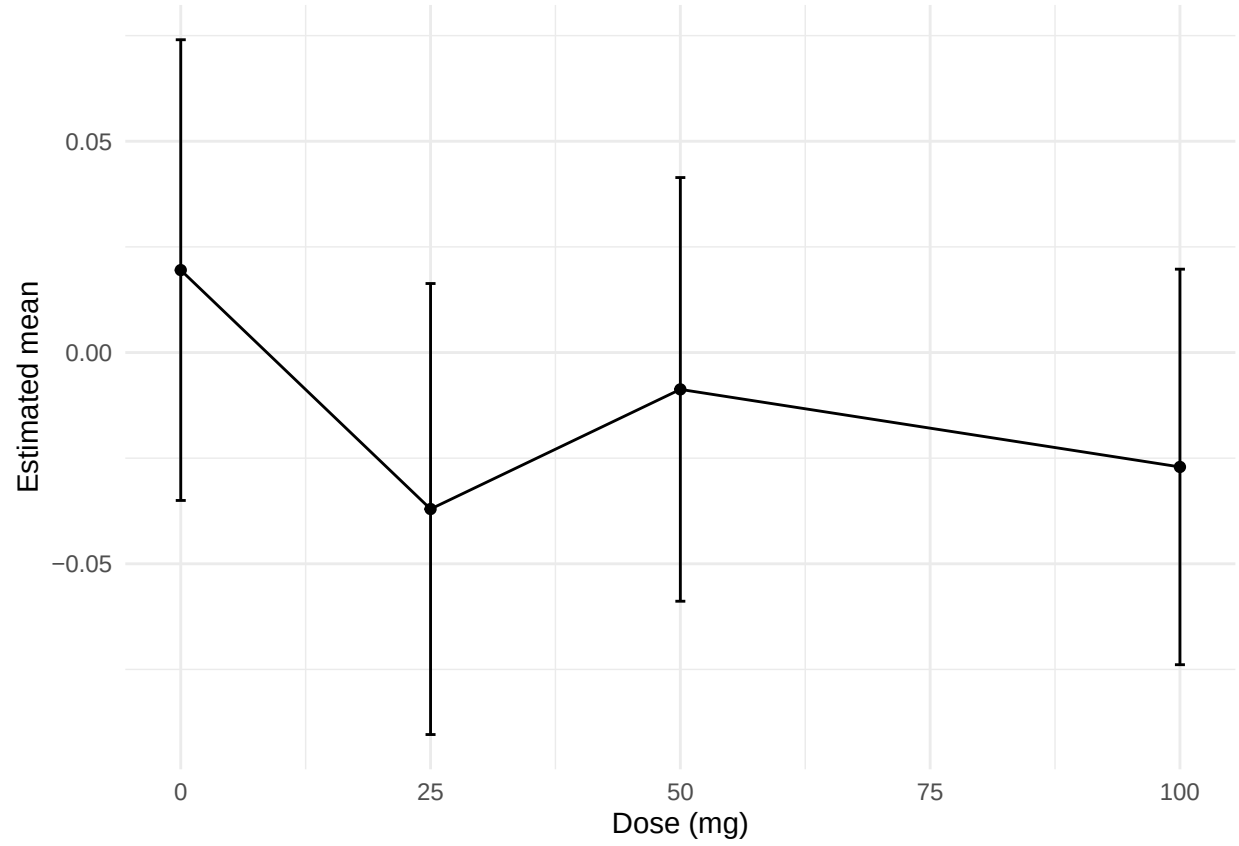
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	0.02 (-0.04 to 0.07)
25 mg	-0.04 (-0.09 to 0.02)
50 mg	-0.01 (-0.06 to 0.04)
100 mg	-0.03 (-0.07 to 0.02)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	-0.06 (-0.15 to 0.04)
50 mg vs 0 mg	-0.03 (-0.12 to 0.06)
100 mg vs 0 mg	-0.05 (-0.13 to 0.04)

Plot



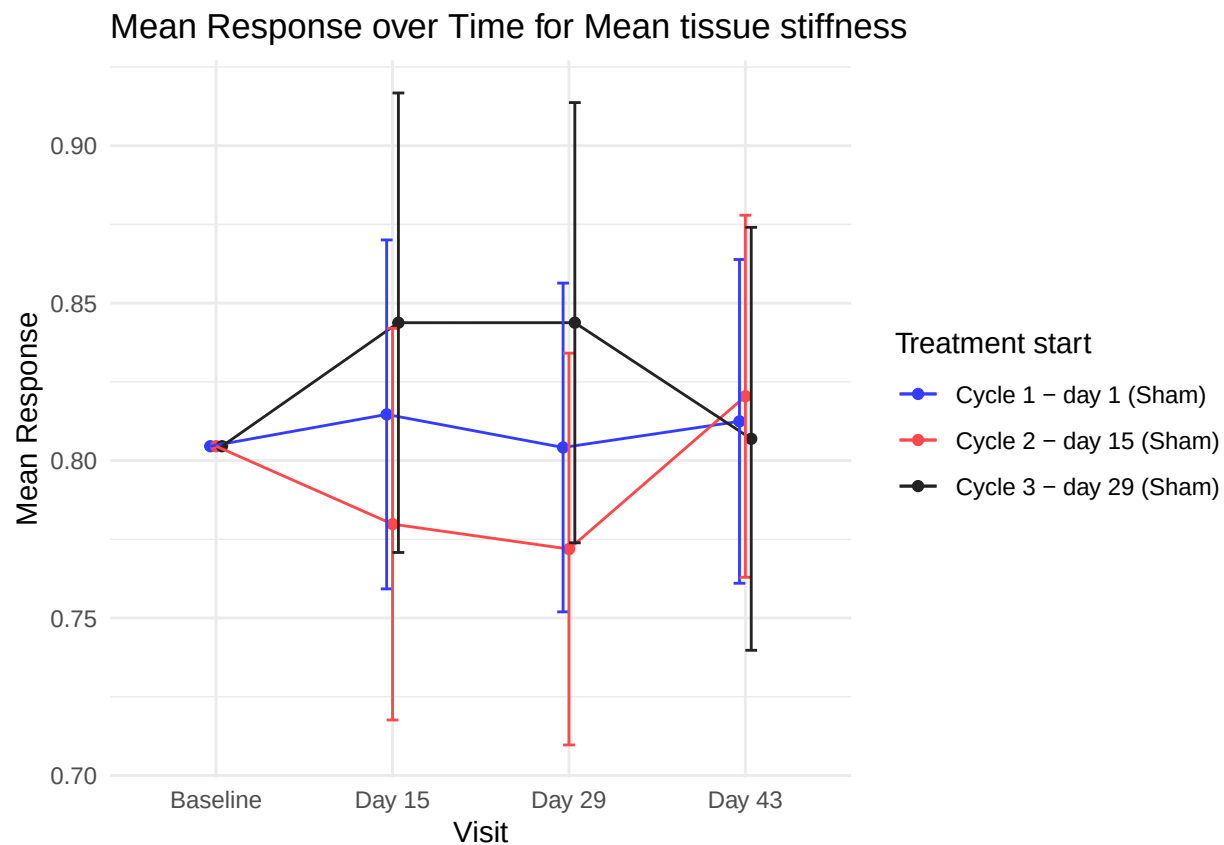
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.04 (-0.11 to 0.02)

Mean tissue stiffness

A total of 40 subjects were included in the analysis for this endpoint.

Line plot by treatment start



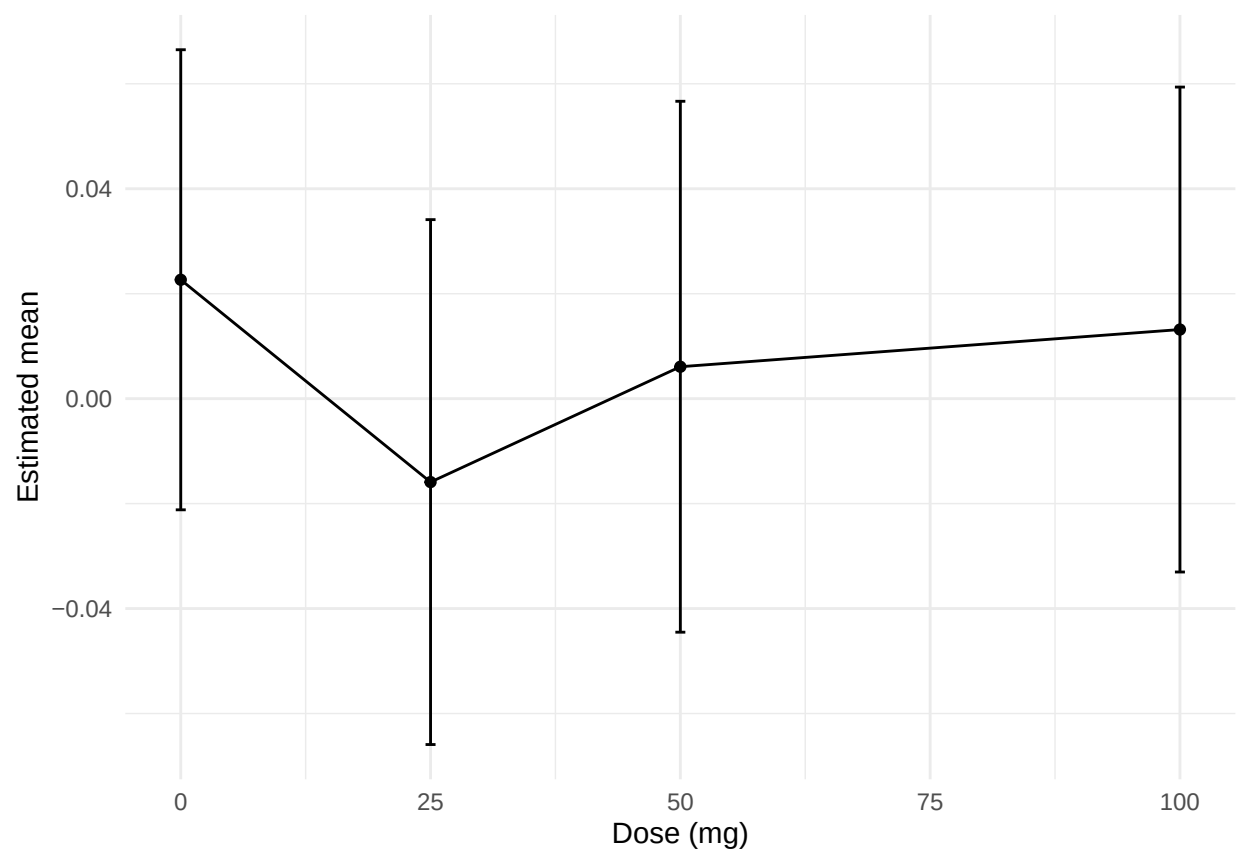
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	0.02 (-0.02 to 0.07)
25 mg	-0.02 (-0.07 to 0.03)
50 mg	0.01 (-0.04 to 0.06)
100 mg	0.01 (-0.03 to 0.06)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	-0.04 (-0.11 to 0.03)
50 mg vs 0 mg	-0.02 (-0.09 to 0.06)
100 mg vs 0 mg	-0.01 (-0.07 to 0.06)

Plot



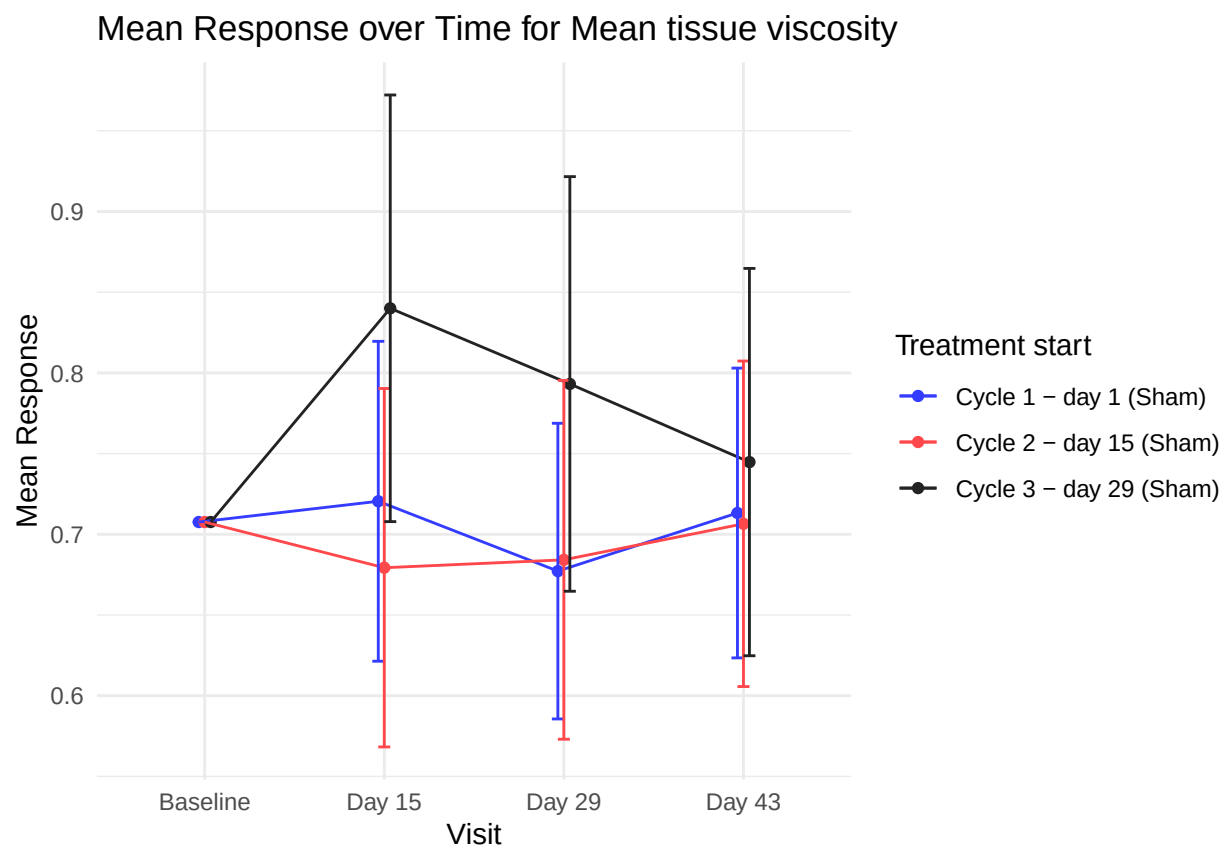
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.02 (-0.06 to 0.03)

Mean tissue viscosity

A total of 40 subjects were included in the analysis for this endpoint.

Line plot by treatment start



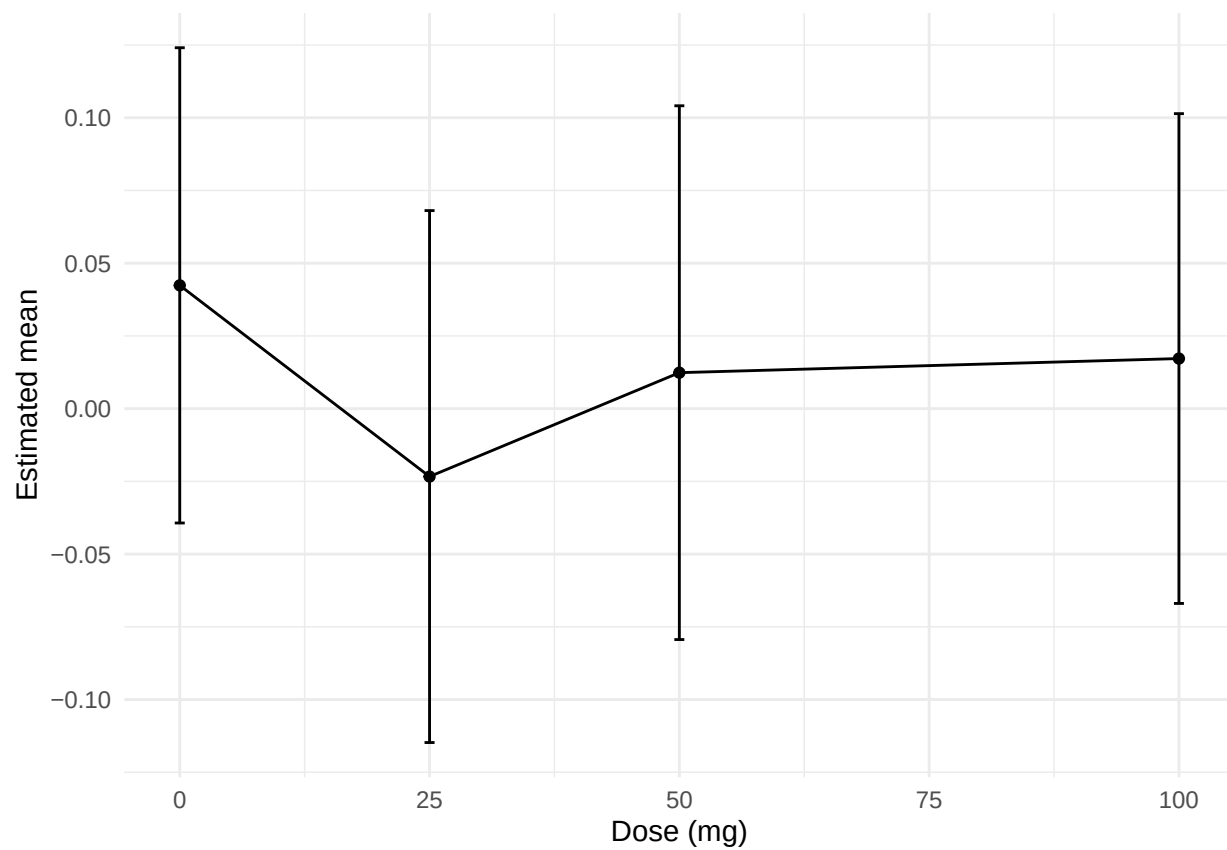
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	0.04 (-0.04 to 0.12)
25 mg	-0.02 (-0.11 to 0.07)
50 mg	0.01 (-0.08 to 0.10)
100 mg	0.02 (-0.07 to 0.10)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	-0.07 (-0.20 to 0.07)
50 mg vs 0 mg	-0.03 (-0.17 to 0.11)
100 mg vs 0 mg	-0.03 (-0.15 to 0.10)

Plot



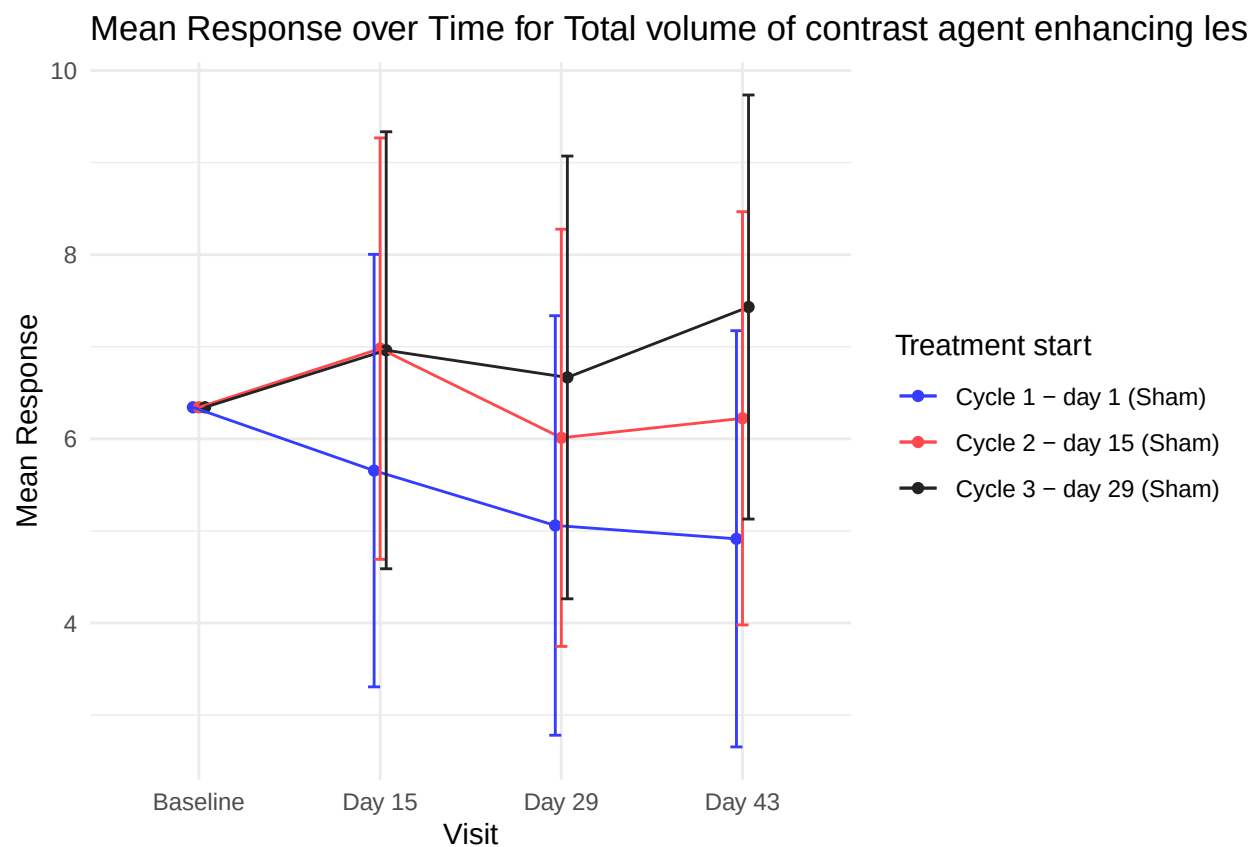
Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.04 (-0.13 to 0.05)

Total volume of contrast agent enhancing lesion on T1-weighted MRI in cm3

A total of 53 subjects were included in the analysis for this endpoint.

Line plot by treatment start



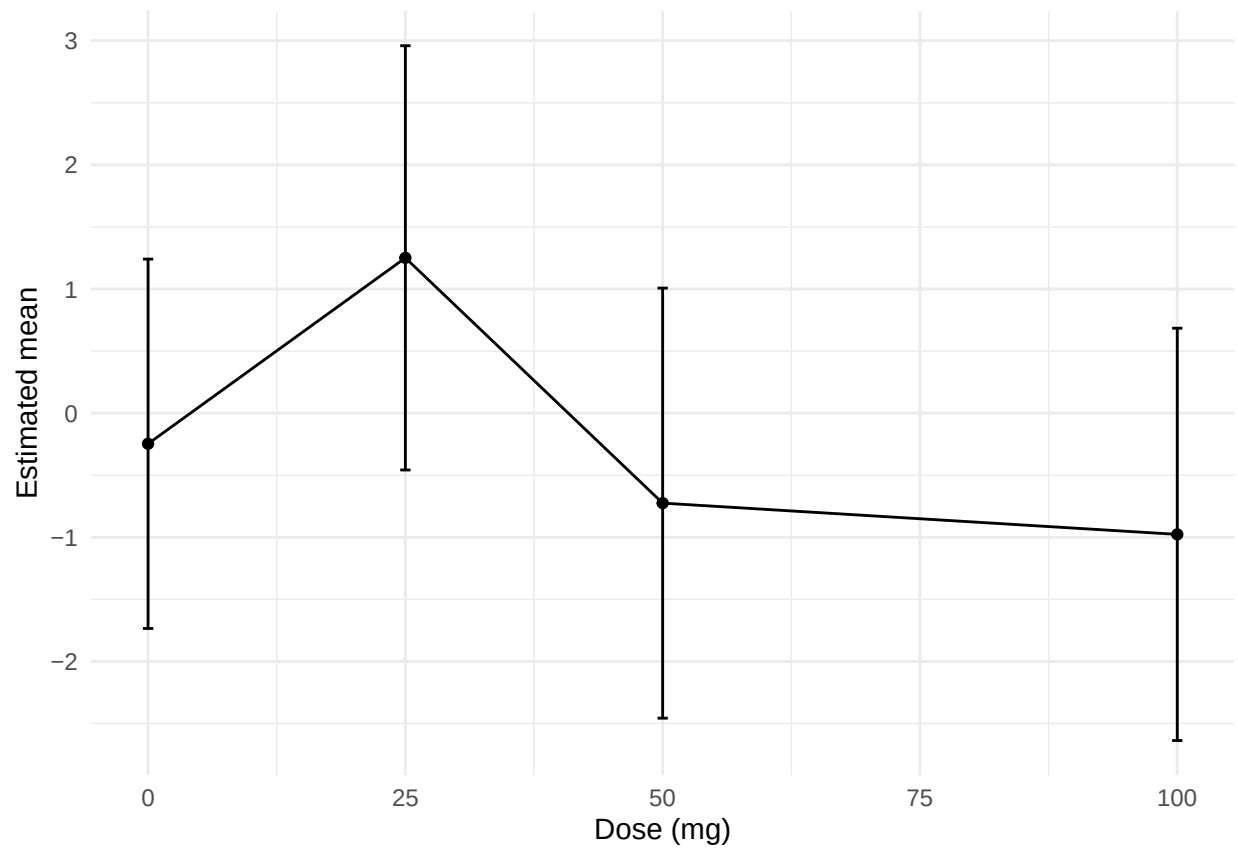
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	-0.25 (-1.73 to 1.24)
25 mg	1.25 (-0.46 to 2.96)
50 mg	-0.72 (-2.46 to 1.01)
100 mg	-0.98 (-2.64 to 0.68)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	1.50 (-0.69 to 3.69)
50 mg vs 0 mg	-0.48 (-2.73 to 1.77)
100 mg vs 0 mg	-0.73 (-2.76 to 1.30)

Plot



Supplementary analysis: any dose vs 0 mg

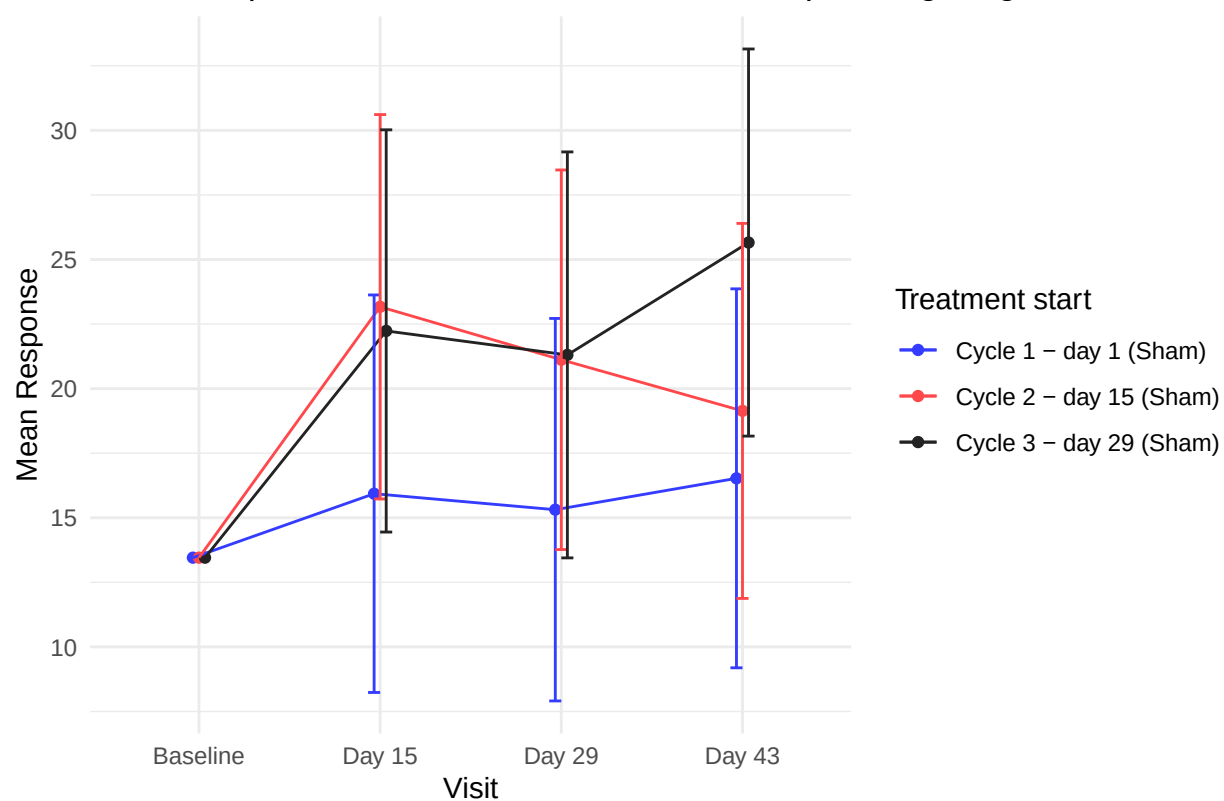
Comparison	Difference (95% CI)
Any dose vs 0 mg	0.03 (-1.31 to 1.38)

Total volume of pathologic signal on T2/FLAIR-weighted MRI in cm3

A total of 53 subjects were included in the analysis for this endpoint.

Line plot by treatment start

Mean Response over Time for Total volume of pathologic signal on T2/FLAIR



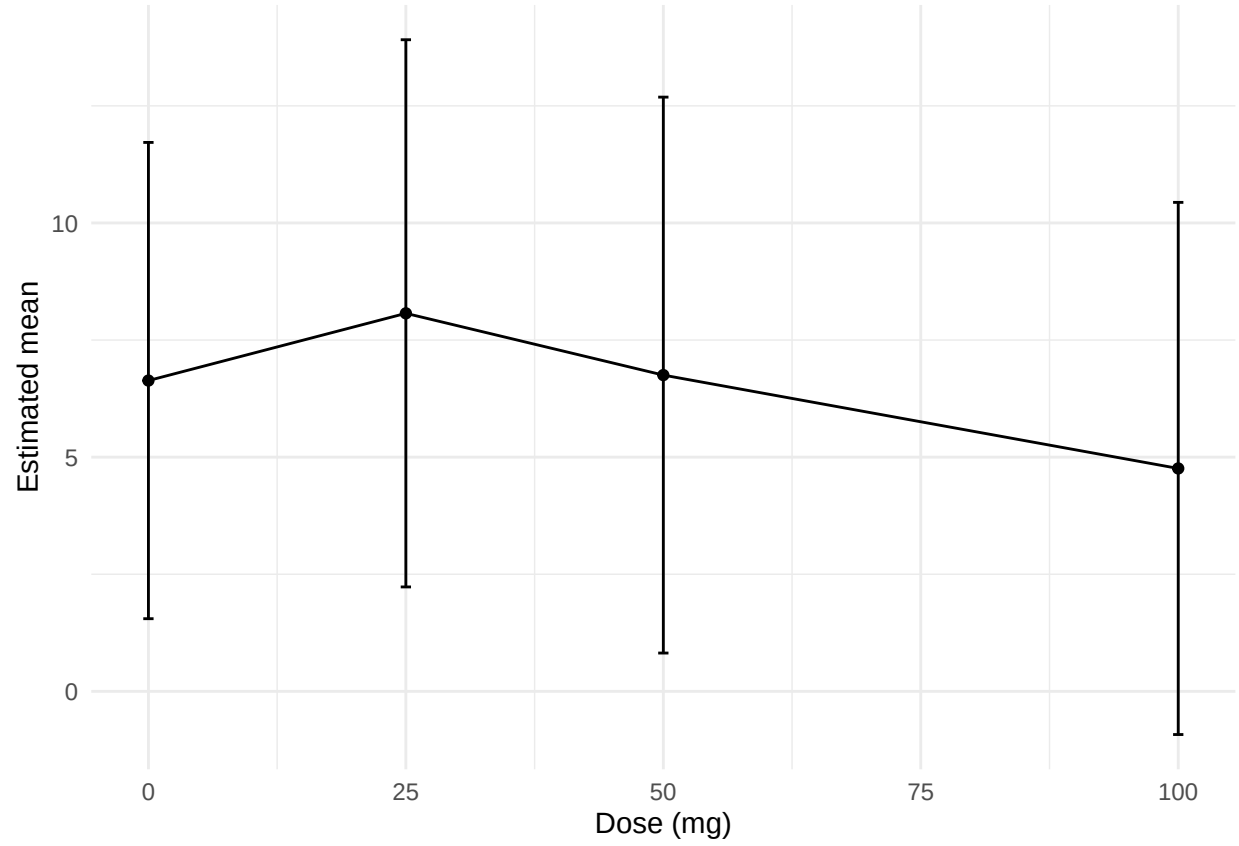
Means by dose

Dose	Estimated change from baseline (95% CI)
0 mg	6.64 (1.55 to 11.72)
25 mg	8.07 (2.23 to 13.91)
50 mg	6.75 (0.82 to 12.69)
100 mg	4.76 (-0.92 to 10.44)

Differences vs 0 mg

Comparison	Difference (95% CI)
25 mg vs 0 mg	1.44 (-6.37 to 9.24)
50 mg vs 0 mg	0.12 (-7.92 to 8.16)
100 mg vs 0 mg	-1.88 (-9.18 to 5.43)

Plot



Supplementary analysis: any dose vs 0 mg

Comparison	Difference (95% CI)
Any dose vs 0 mg	-0.24 (-5.03 to 4.54)

MRI Endpoints at Cycle 11 (Day 141)

For all MRI endpoints, we report estimated change from baseline at Cycle 11 (day 141), grouped by the extended treatment variable. The summaries include mean estimates, contrasts versus the reference group, and a plot with confidence intervals.

Mean DSC K2 leakage

A total of 47 subjects were included in the analysis for this endpoint.

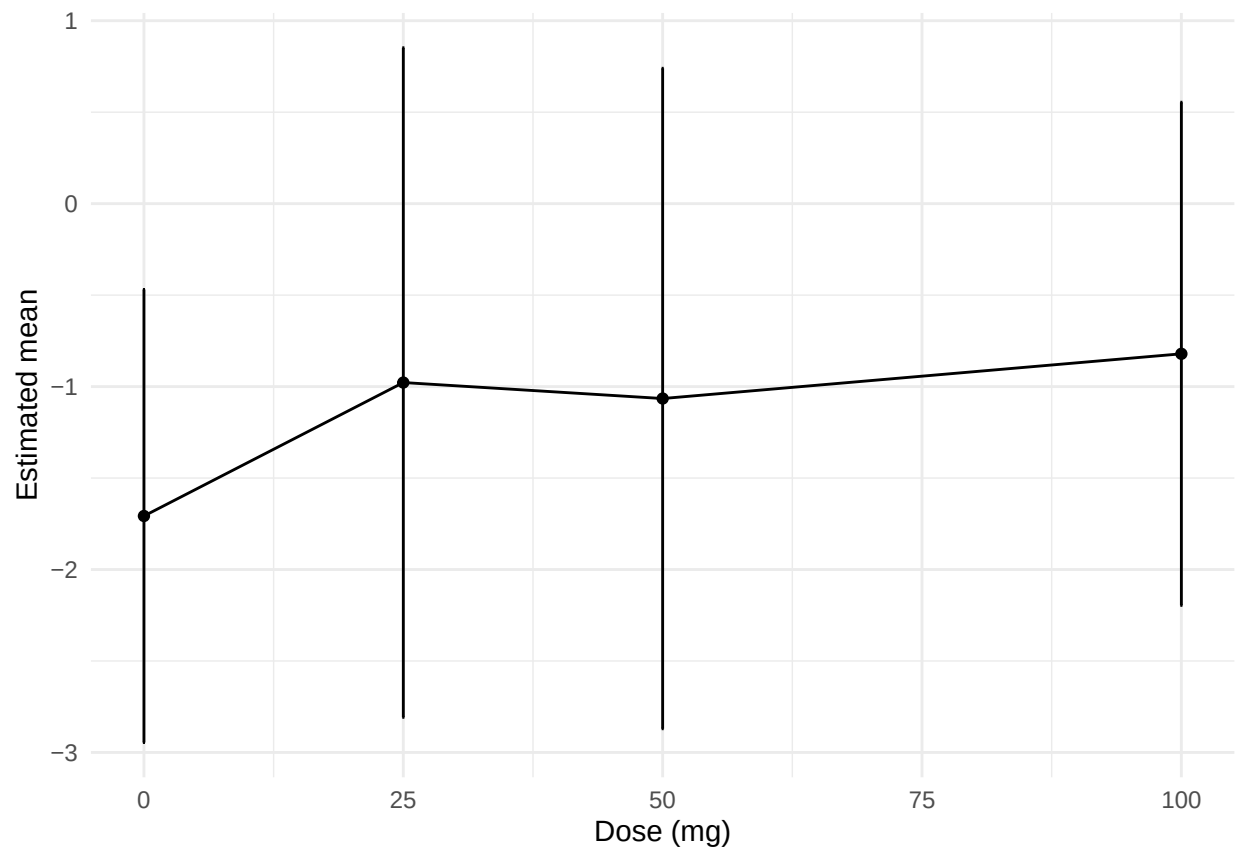
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-1.71 (-2.95 to -0.47)
losartan 25 mg	-0.98 (-2.81 to 0.85)
losartan 50 mg	-1.07 (-2.87 to 0.74)
losartan 100 mg	-0.82 (-2.20 to 0.55)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.73 (-1.95 to 3.41)
losartan 50 mg - losartan 0 mg	0.64 (-2.02 to 3.30)
losartan 100 mg - losartan 0 mg	0.89 (-1.36 to 3.13)

Plot



Mean relative DSC extravasation

A total of 20 subjects were included in the analysis for this endpoint.

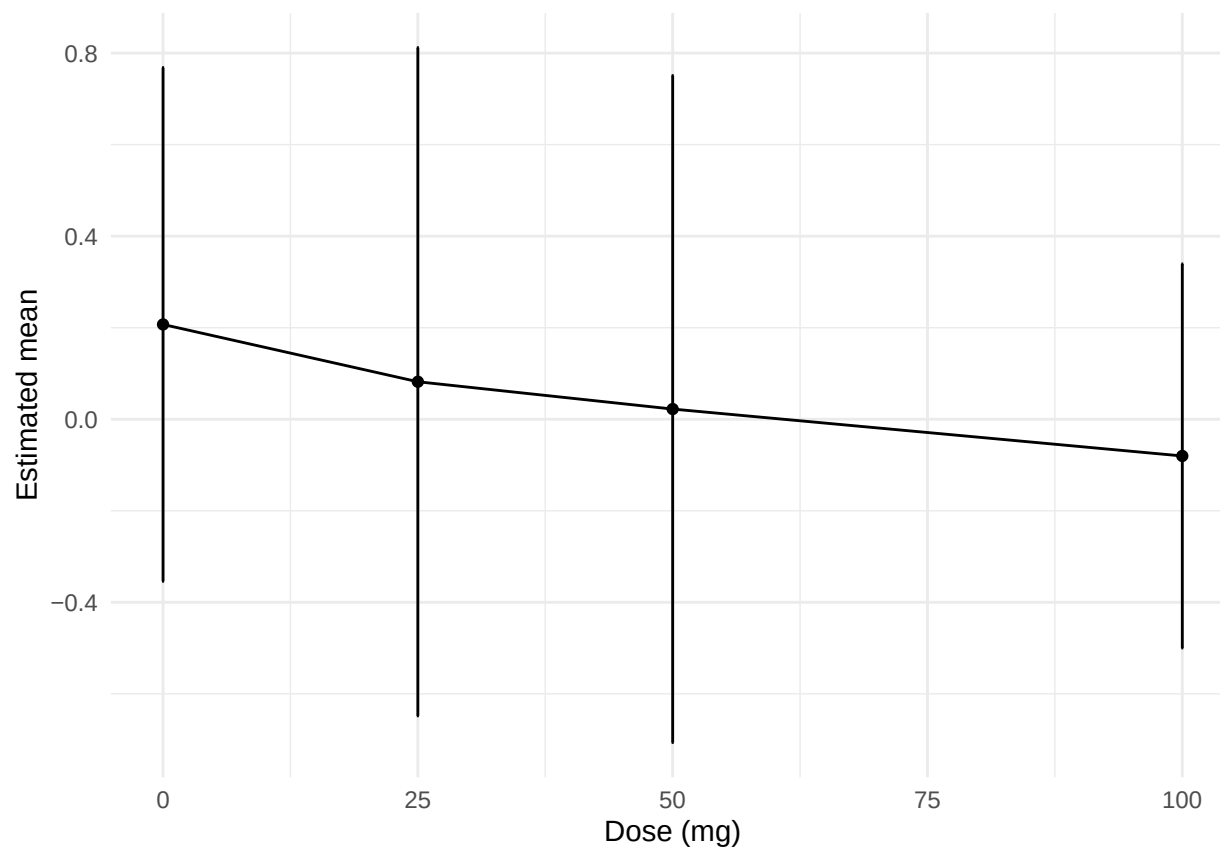
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	0.21 (-0.35 to 0.77)
losartan 25 mg	0.08 (-0.65 to 0.81)
losartan 50 mg	0.02 (-0.71 to 0.75)
losartan 100 mg	-0.08 (-0.50 to 0.34)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-0.13 (-1.27 to 1.02)
losartan 50 mg - losartan 0 mg	-0.18 (-1.32 to 0.95)
losartan 100 mg - losartan 0 mg	-0.29 (-1.15 to 0.57)

Plot



Mean relative apparent diffusion coefficient

A total of 43 subjects were included in the analysis for this endpoint.

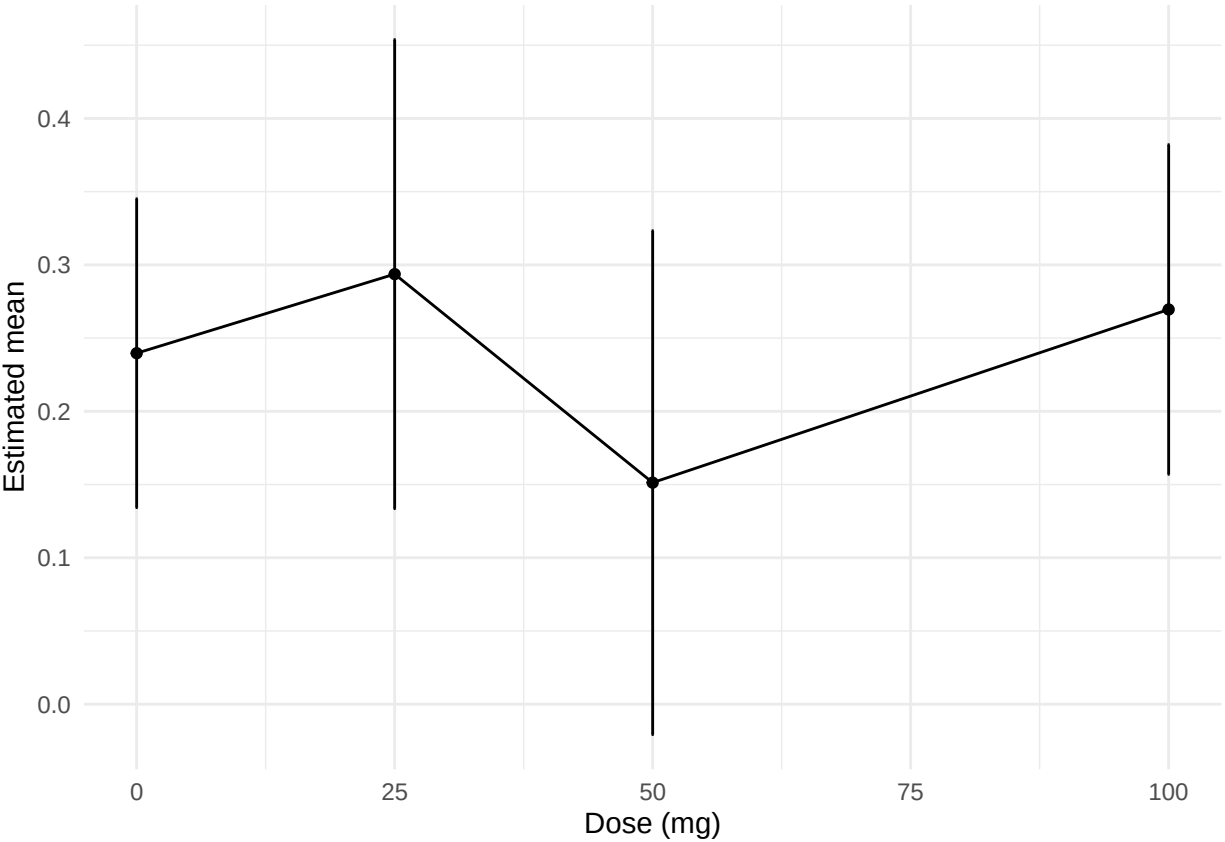
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	0.24 (0.13 to 0.35)
losartan 25 mg	0.29 (0.13 to 0.45)
losartan 50 mg	0.15 (-0.02 to 0.32)
losartan 100 mg	0.27 (0.16 to 0.38)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.05 (-0.18 to 0.29)
losartan 50 mg - losartan 0 mg	-0.09 (-0.33 to 0.16)
losartan 100 mg - losartan 0 mg	0.03 (-0.16 to 0.22)

Plot



Mean relative cerebral blood

A total of 47 subjects were included in the analysis for this endpoint.

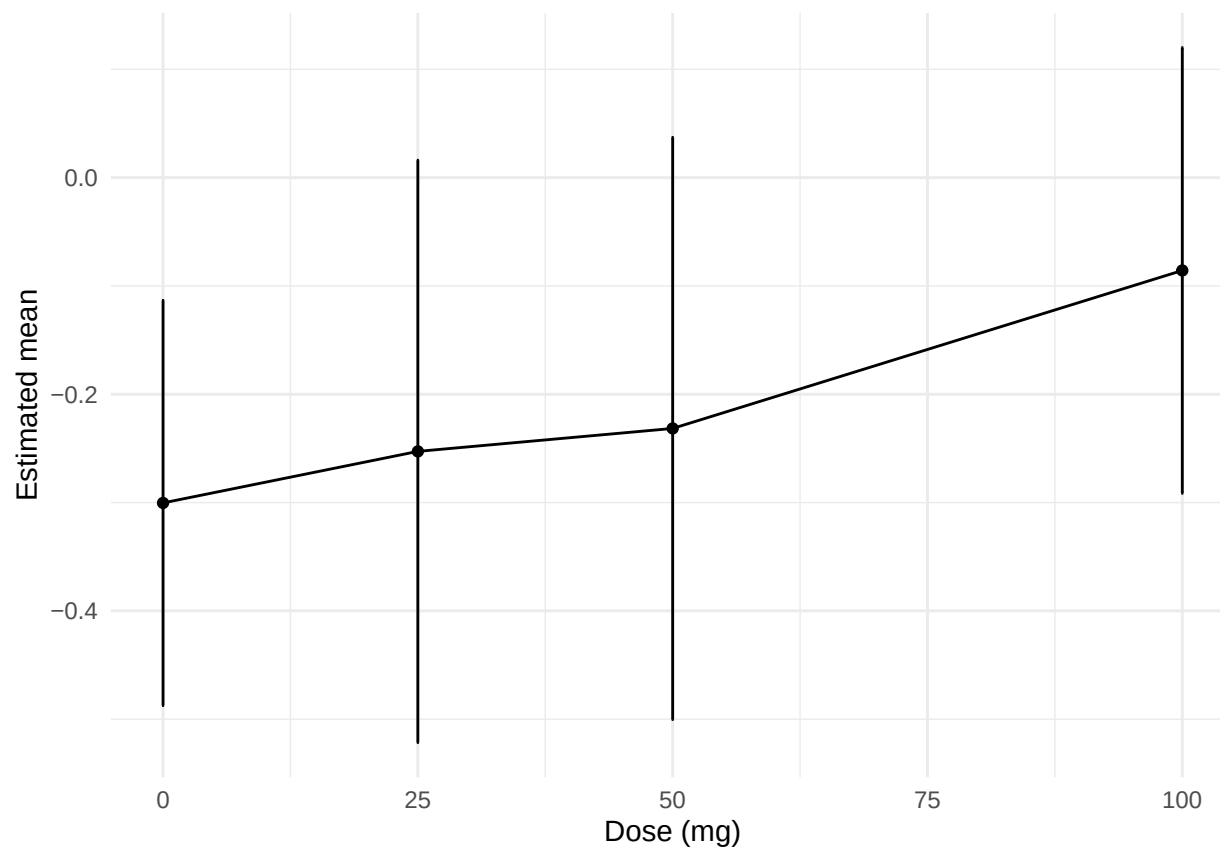
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.30 (-0.49 to -0.11)
losartan 25 mg	-0.25 (-0.52 to 0.02)
losartan 50 mg	-0.23 (-0.50 to 0.04)
losartan 100 mg	-0.09 (-0.29 to 0.12)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.05 (-0.35 to 0.45)
losartan 50 mg - losartan 0 mg	0.07 (-0.33 to 0.47)
losartan 100 mg - losartan 0 mg	0.21 (-0.13 to 0.56)

Plot



Mean relative cerebral blood flow (PTA-ROI)

A total of 47 subjects were included in the analysis for this endpoint.

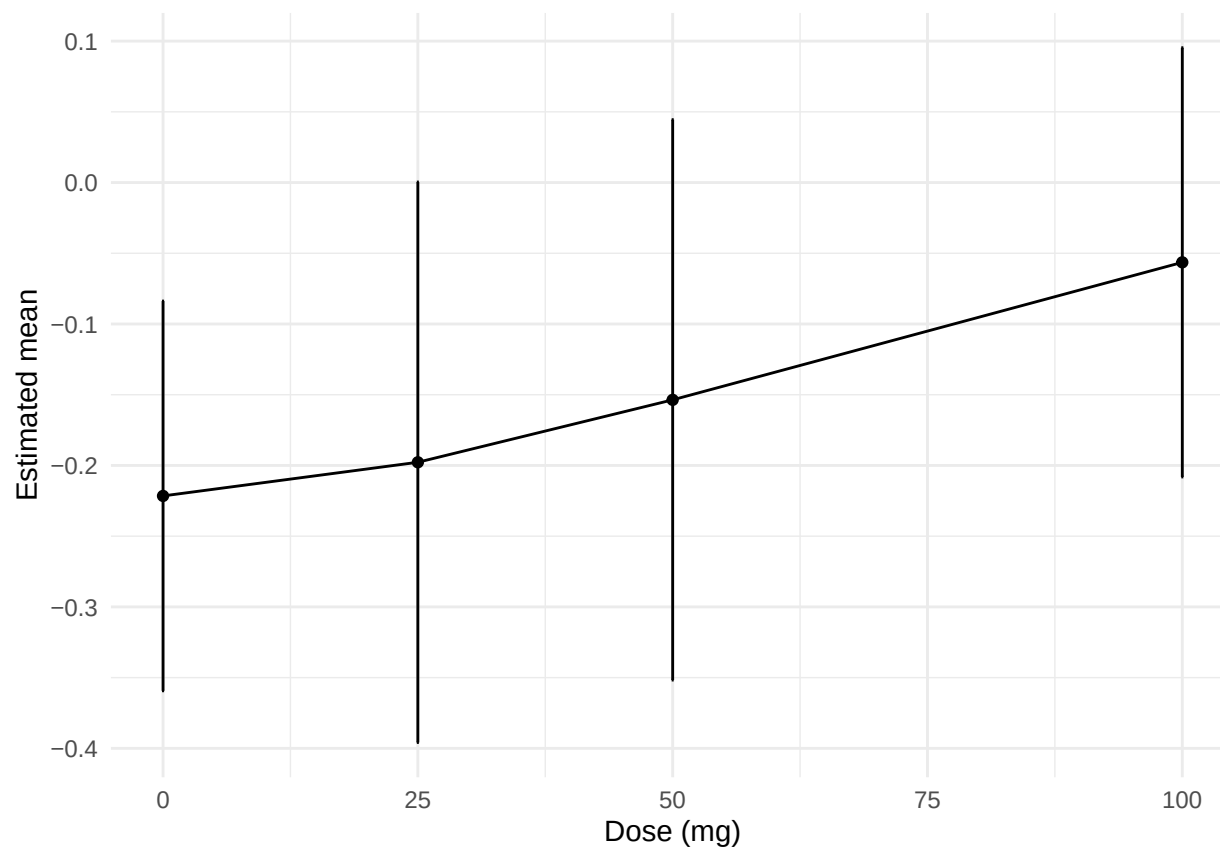
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.22 (-0.36 to -0.08)
losartan 25 mg	-0.20 (-0.40 to 0.00)
losartan 50 mg	-0.15 (-0.35 to 0.04)
losartan 100 mg	-0.06 (-0.21 to 0.10)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.02 (-0.27 to 0.32)
losartan 50 mg - losartan 0 mg	0.07 (-0.23 to 0.36)
losartan 100 mg - losartan 0 mg	0.17 (-0.09 to 0.42)

Plot



Mean relative cerebral blood flow (ROI2)

A total of 40 subjects were included in the analysis for this endpoint.

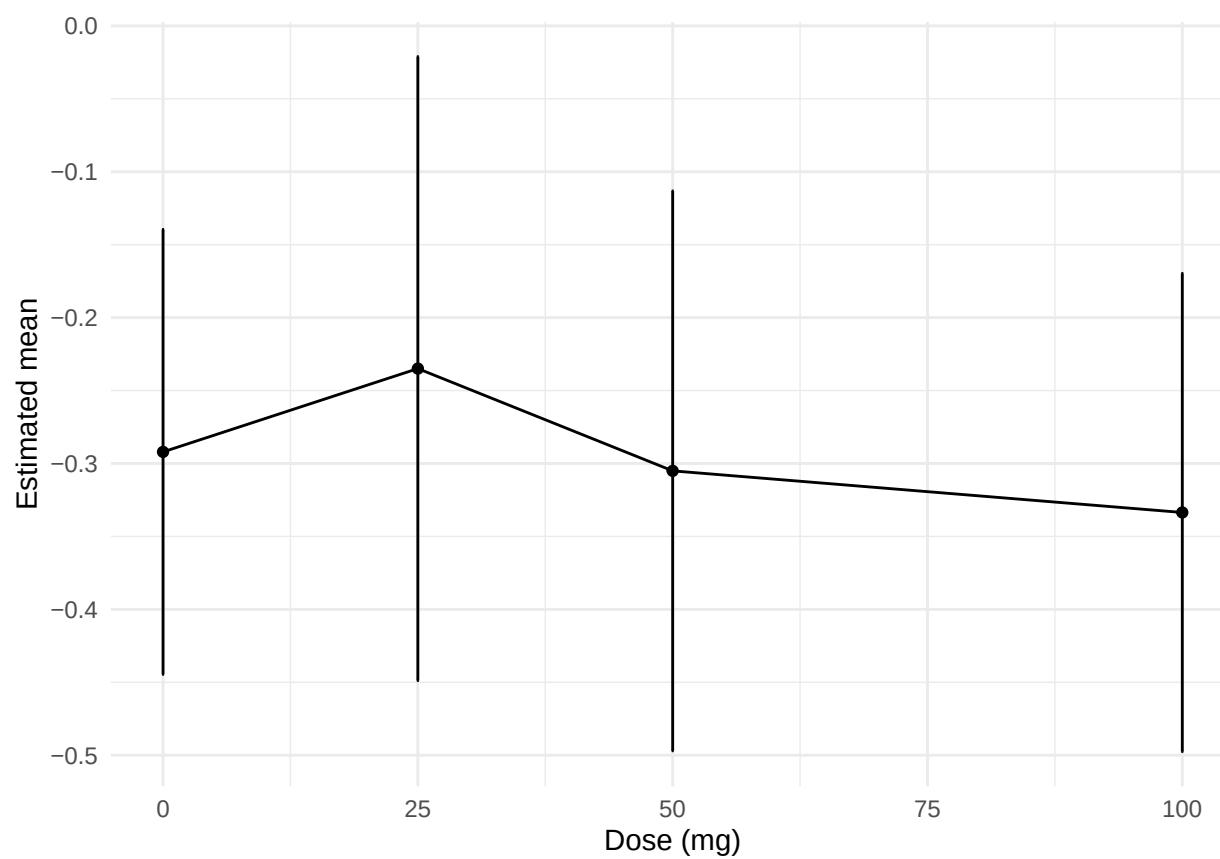
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.29 (-0.44 to -0.14)
losartan 25 mg	-0.23 (-0.45 to -0.02)
losartan 50 mg	-0.31 (-0.50 to -0.11)
losartan 100 mg	-0.33 (-0.50 to -0.17)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.06 (-0.28 to 0.39)
losartan 50 mg - losartan 0 mg	-0.01 (-0.31 to 0.29)
losartan 100 mg - losartan 0 mg	-0.04 (-0.32 to 0.23)

Plot



Mean relative direction value

A total of 20 subjects were included in the analysis for this endpoint.

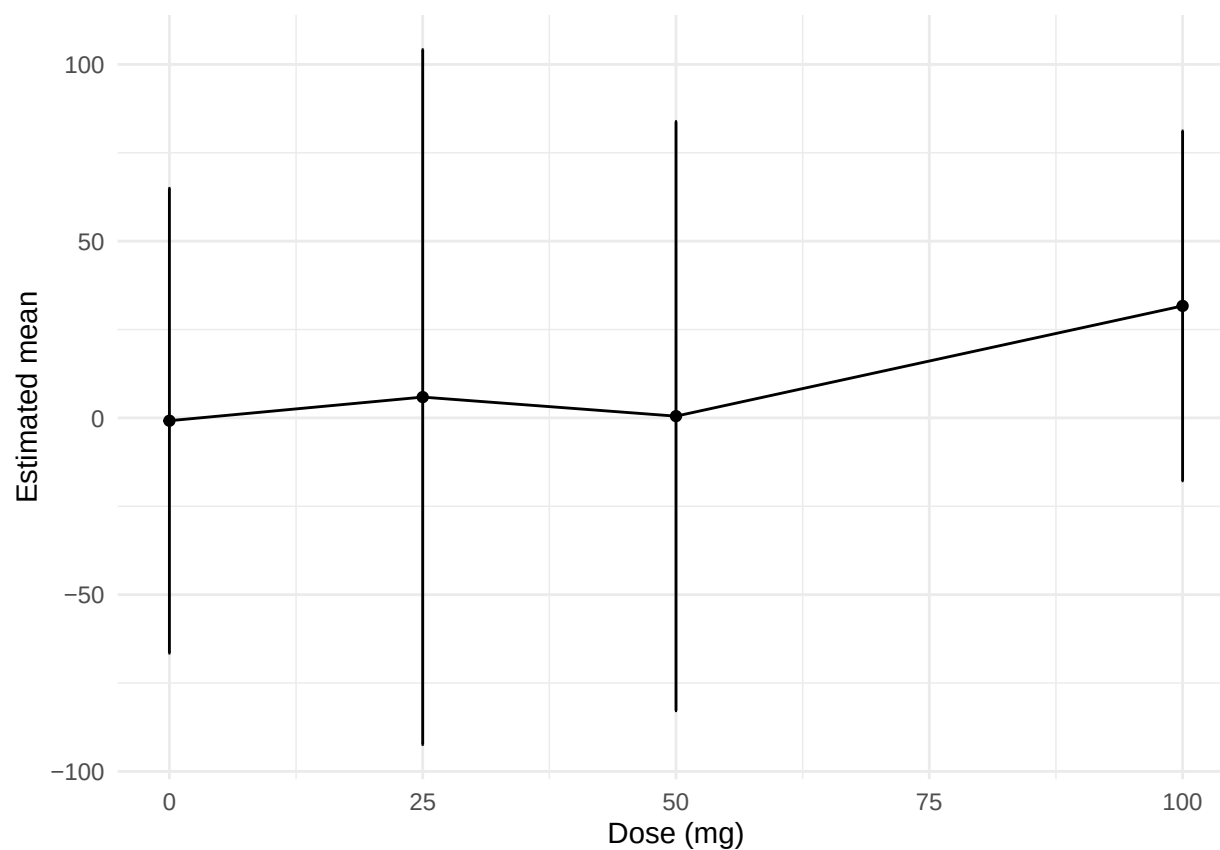
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.80 (-66.53 to 64.94)
losartan 25 mg	5.89 (-92.39 to 104.16)
losartan 50 mg	0.50 (-82.82 to 83.81)
losartan 100 mg	31.68 (-17.78 to 81.15)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	6.68 (-146.18 to 159.55)
losartan 50 mg - losartan 0 mg	1.30 (-130.42 to 133.02)
losartan 100 mg - losartan 0 mg	32.48 (-66.61 to 131.57)

Plot



Mean relative fractional anisotropy

A total of 40 subjects were included in the analysis for this endpoint.

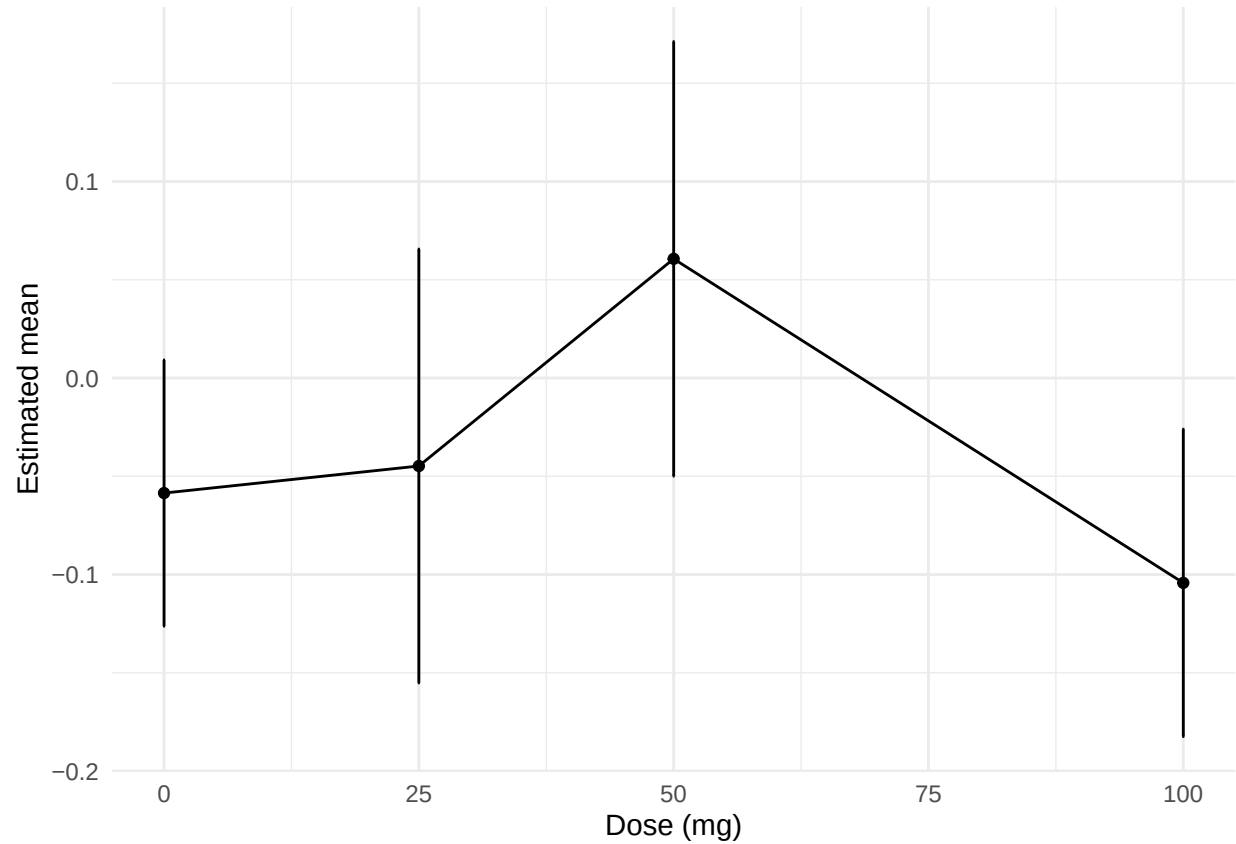
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.06 (-0.13 to 0.01)
losartan 25 mg	-0.04 (-0.16 to 0.07)
losartan 50 mg	0.06 (-0.05 to 0.17)
losartan 100 mg	-0.10 (-0.18 to -0.03)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.01 (-0.14 to 0.17)
losartan 50 mg - losartan 0 mg	0.12 (-0.04 to 0.28)
losartan 100 mg - losartan 0 mg	-0.05 (-0.17 to 0.08)

Plot



Mean relative restricted spectrum imaging

A total of 40 subjects were included in the analysis for this endpoint.

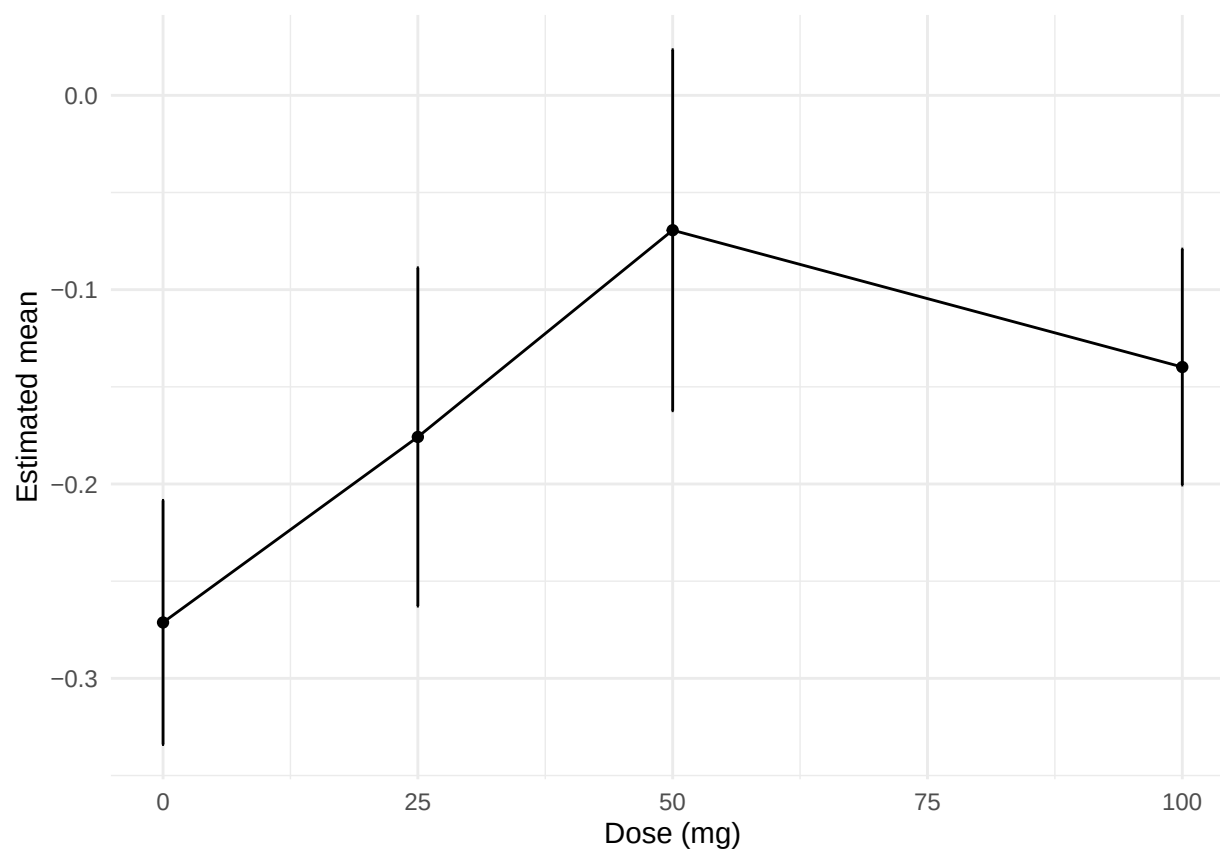
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.27 (-0.33 to -0.21)
losartan 25 mg	-0.18 (-0.26 to -0.09)
losartan 50 mg	-0.07 (-0.16 to 0.02)
losartan 100 mg	-0.14 (-0.20 to -0.08)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.10 (-0.03 to 0.23)
losartan 50 mg - losartan 0 mg	0.20 (0.07 to 0.34)
losartan 100 mg - losartan 0 mg	0.13 (0.03 to 0.24)

Plot



Mean relative solid stress (PTA-ROI)

A total of 29 subjects were included in the analysis for this endpoint.

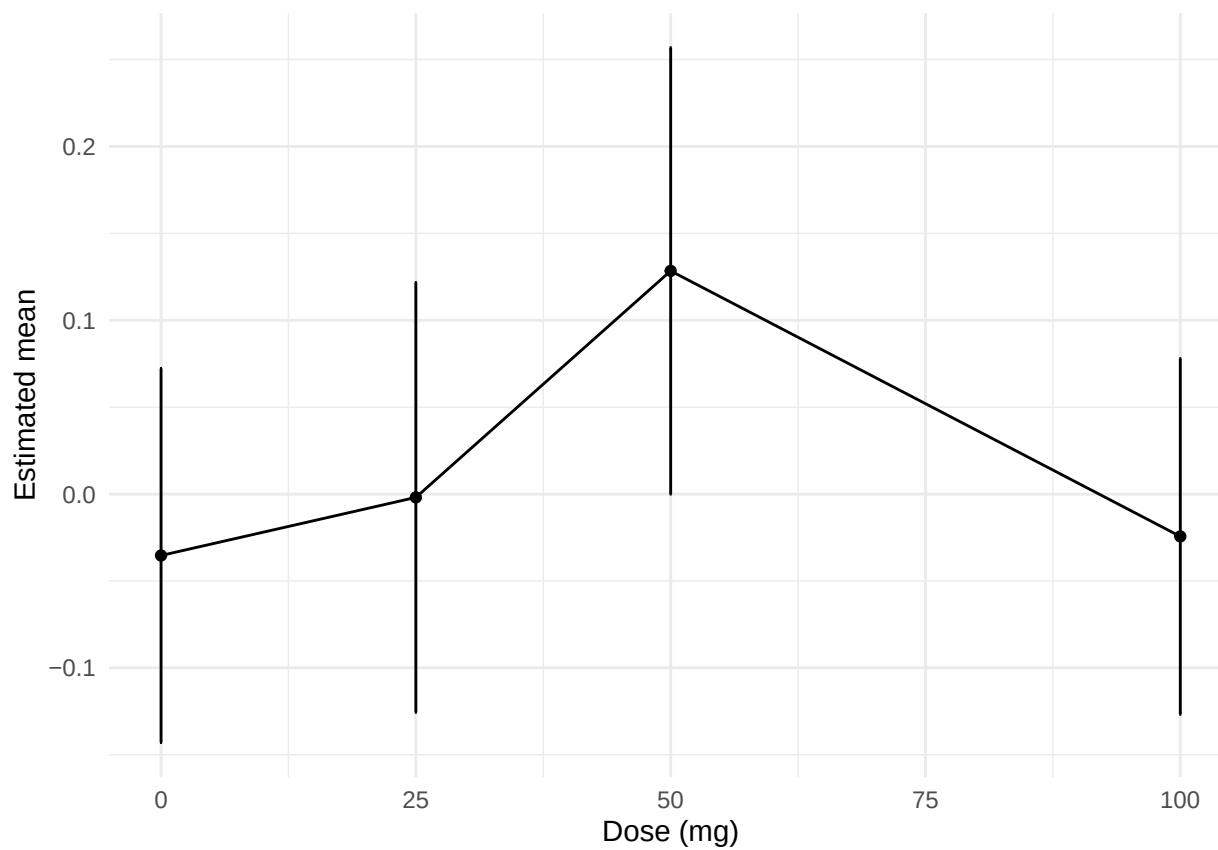
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.04 (-0.14 to 0.07)
losartan 25 mg	0.00 (-0.13 to 0.12)
losartan 50 mg	0.13 (0.00 to 0.26)
losartan 100 mg	-0.02 (-0.13 to 0.08)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.03 (-0.17 to 0.23)
losartan 50 mg - losartan 0 mg	0.16 (-0.04 to 0.37)
losartan 100 mg - losartan 0 mg	0.01 (-0.17 to 0.19)

Plot



Mean relative solid stress (ROI2)

A total of 23 subjects were included in the analysis for this endpoint.

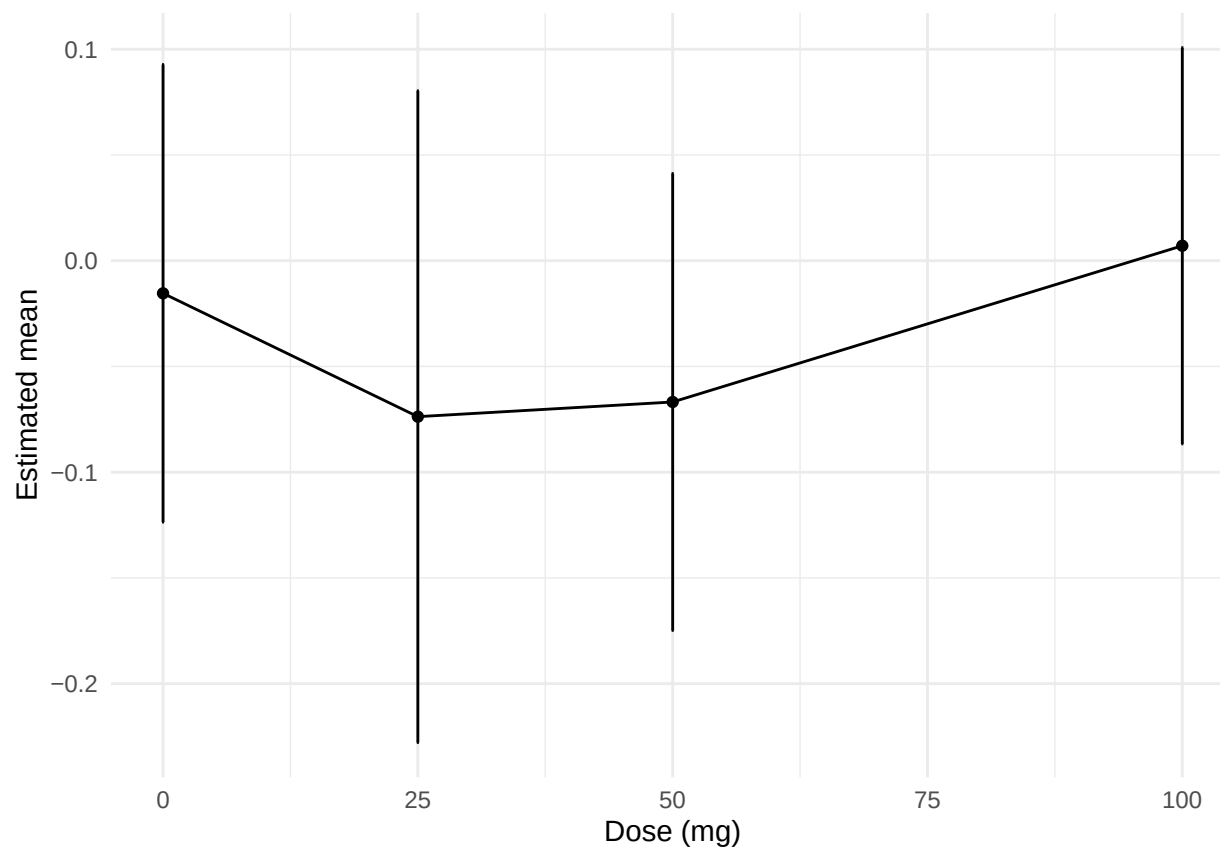
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.02 (-0.12 to 0.09)
losartan 25 mg	-0.07 (-0.23 to 0.08)
losartan 50 mg	-0.07 (-0.17 to 0.04)
losartan 100 mg	0.01 (-0.09 to 0.10)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-0.06 (-0.29 to 0.17)
losartan 50 mg - losartan 0 mg	-0.05 (-0.24 to 0.14)
losartan 100 mg - losartan 0 mg	0.02 (-0.15 to 0.20)

Plot



Mean relative transit time

A total of 47 subjects were included in the analysis for this endpoint.

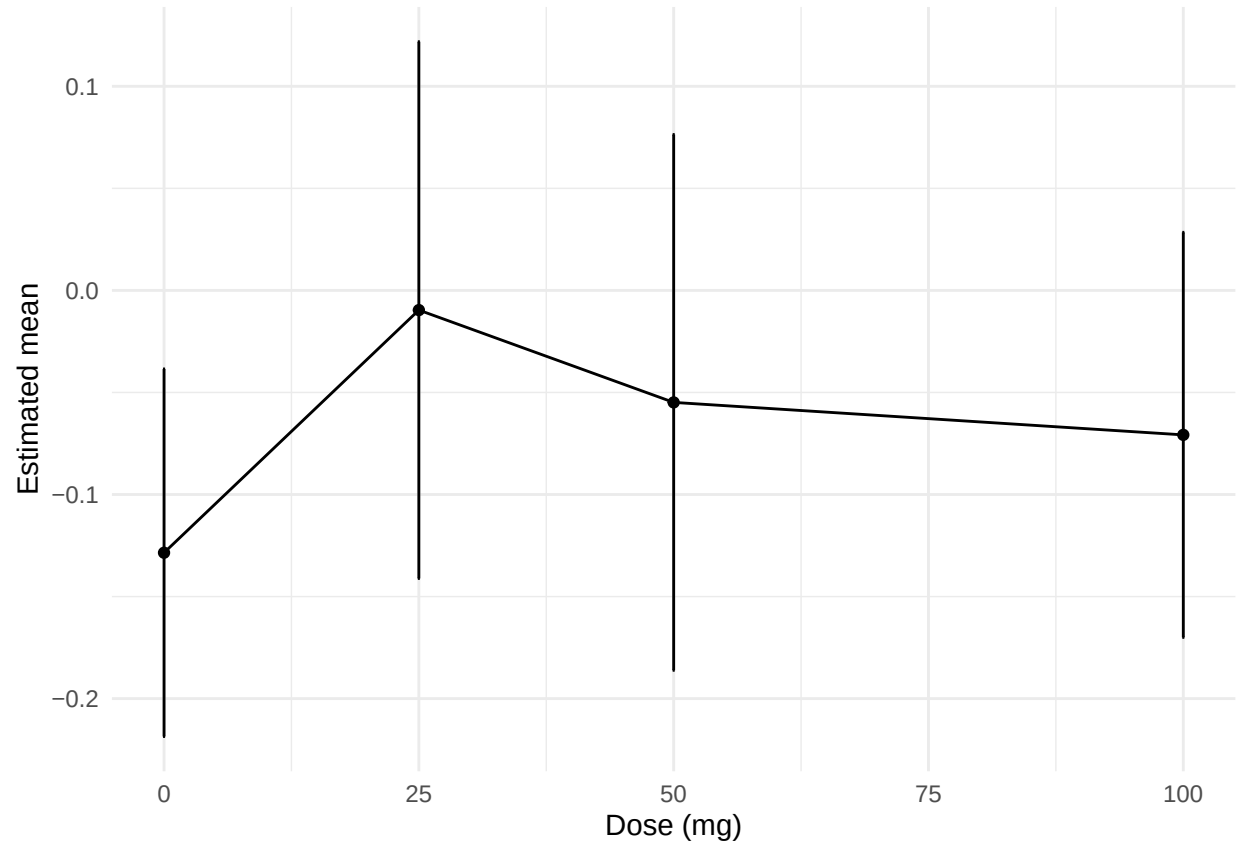
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.13 (-0.22 to -0.04)
losartan 25 mg	-0.01 (-0.14 to 0.12)
losartan 50 mg	-0.05 (-0.19 to 0.08)
losartan 100 mg	-0.07 (-0.17 to 0.03)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.12 (-0.07 to 0.31)
losartan 50 mg - losartan 0 mg	0.07 (-0.12 to 0.27)
losartan 100 mg - losartan 0 mg	0.06 (-0.10 to 0.22)

Plot



Mean relative vessel caliber

A total of 20 subjects were included in the analysis for this endpoint.

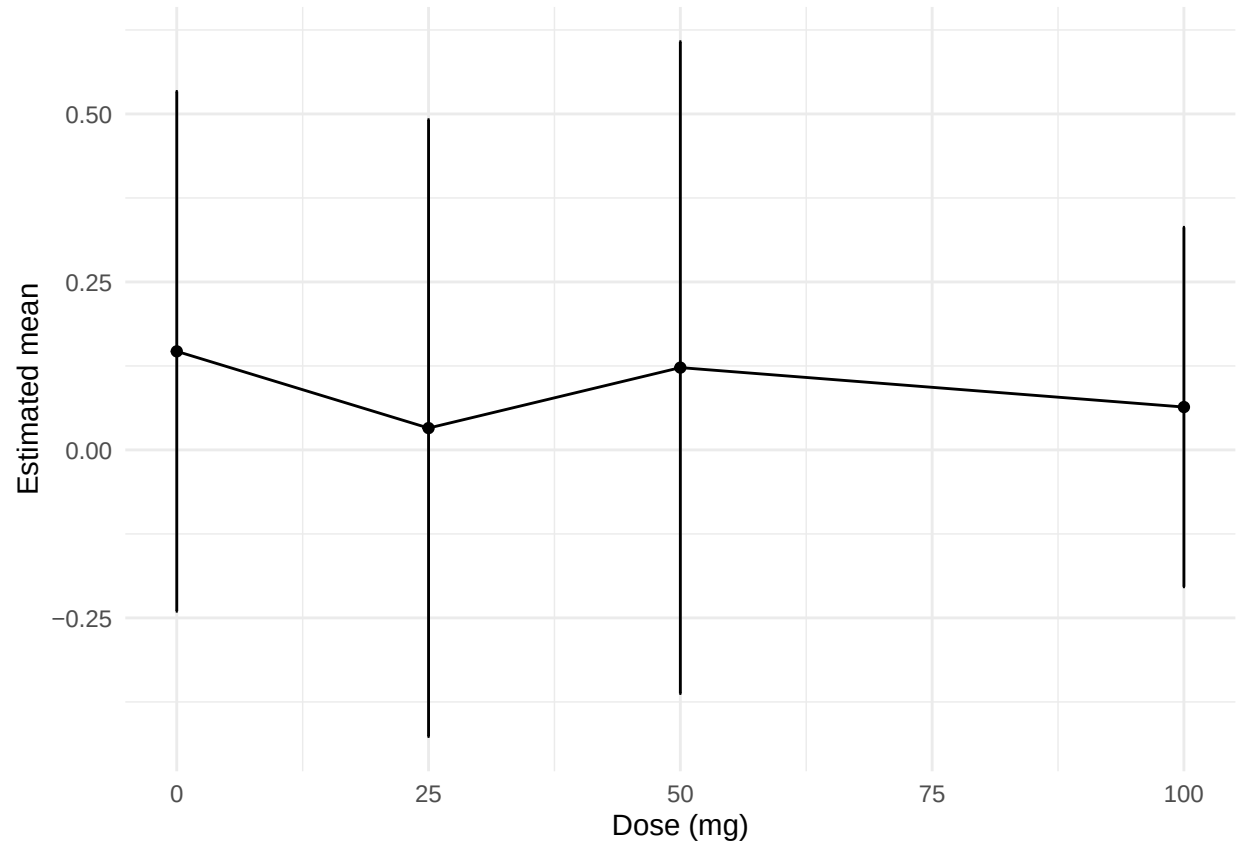
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	0.15 (-0.24 to 0.53)
losartan 25 mg	0.03 (-0.43 to 0.49)
losartan 50 mg	0.12 (-0.36 to 0.61)
losartan 100 mg	0.06 (-0.20 to 0.33)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-0.11 (-0.85 to 0.62)
losartan 50 mg - losartan 0 mg	-0.02 (-0.84 to 0.79)
losartan 100 mg - losartan 0 mg	-0.08 (-0.68 to 0.51)

Plot



Mean relative vessel size index

A total of 20 subjects were included in the analysis for this endpoint.

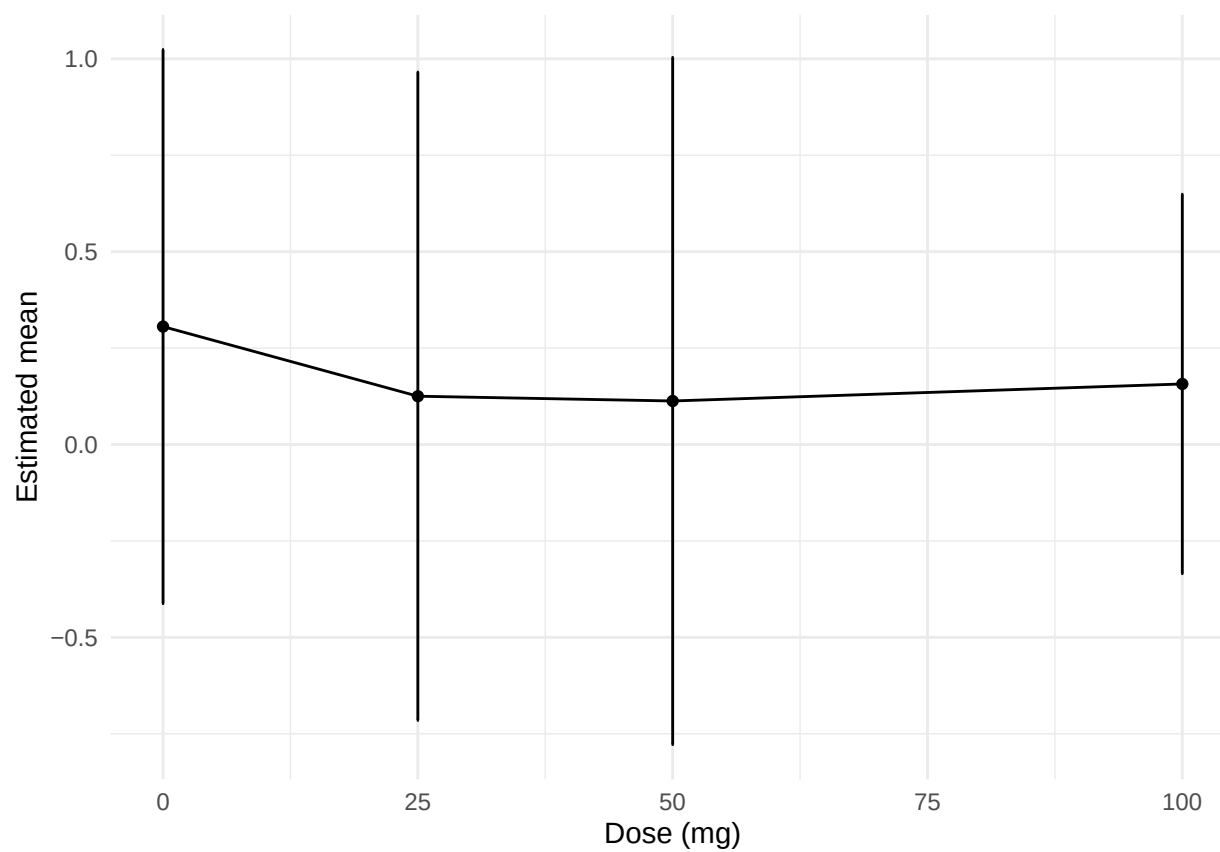
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	0.31 (-0.41 to 1.02)
losartan 25 mg	0.13 (-0.71 to 0.97)
losartan 50 mg	0.11 (-0.78 to 1.00)
losartan 100 mg	0.16 (-0.33 to 0.65)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-0.18 (-1.53 to 1.17)
losartan 50 mg - losartan 0 mg	-0.19 (-1.70 to 1.31)
losartan 100 mg - losartan 0 mg	-0.15 (-1.26 to 0.96)

Plot



Mean relative vortex area

A total of 20 subjects were included in the analysis for this endpoint.

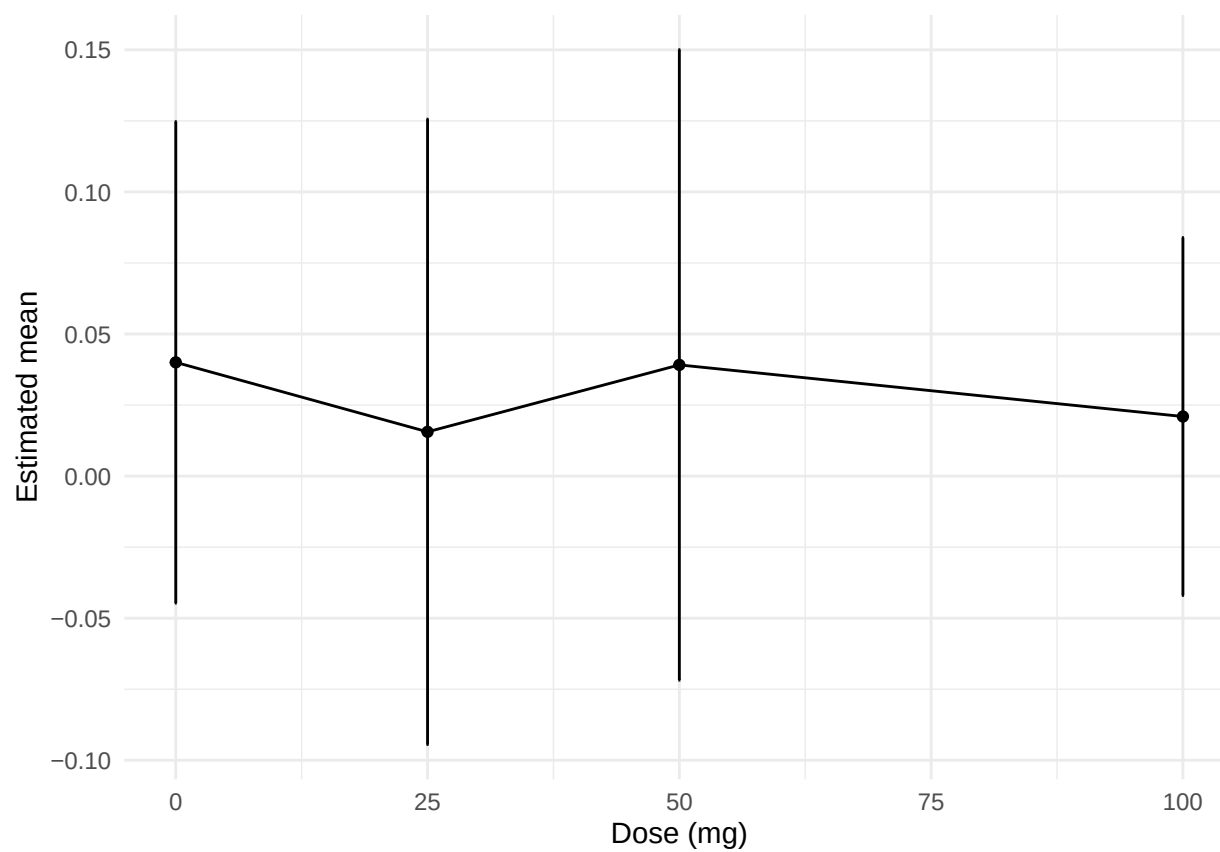
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	0.04 (-0.04 to 0.12)
losartan 25 mg	0.02 (-0.09 to 0.13)
losartan 50 mg	0.04 (-0.07 to 0.15)
losartan 100 mg	0.02 (-0.04 to 0.08)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-0.02 (-0.19 to 0.15)
losartan 50 mg - losartan 0 mg	0.00 (-0.17 to 0.17)
losartan 100 mg - losartan 0 mg	-0.02 (-0.15 to 0.11)

Plot



Mean tissue elasticity

A total of 29 subjects were included in the analysis for this endpoint.

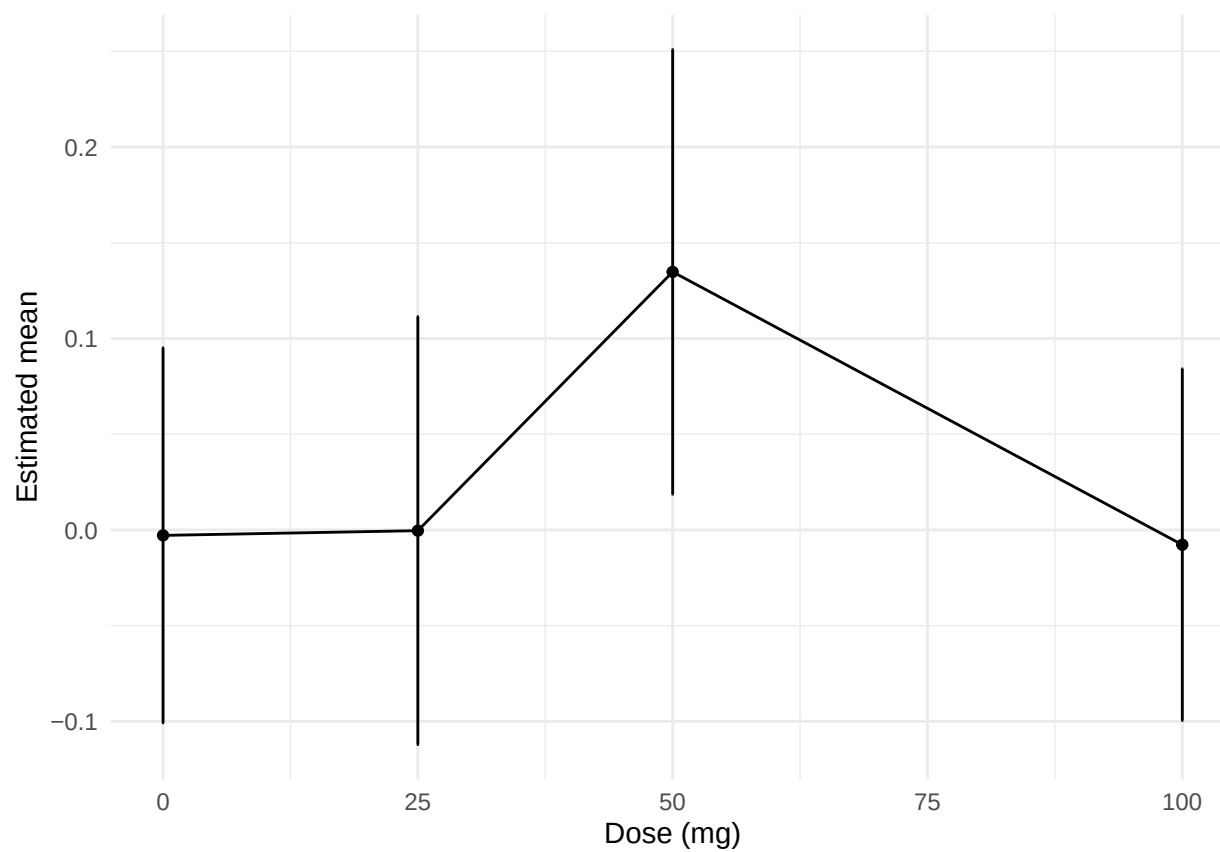
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	0.00 (-0.10 to 0.10)
losartan 25 mg	0.00 (-0.11 to 0.11)
losartan 50 mg	0.13 (0.02 to 0.25)
losartan 100 mg	-0.01 (-0.10 to 0.08)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.00 (-0.18 to 0.19)
losartan 50 mg - losartan 0 mg	0.14 (-0.05 to 0.33)
losartan 100 mg - losartan 0 mg	0.00 (-0.17 to 0.16)

Plot



Mean tissue relative viscosity

A total of 29 subjects were included in the analysis for this endpoint.

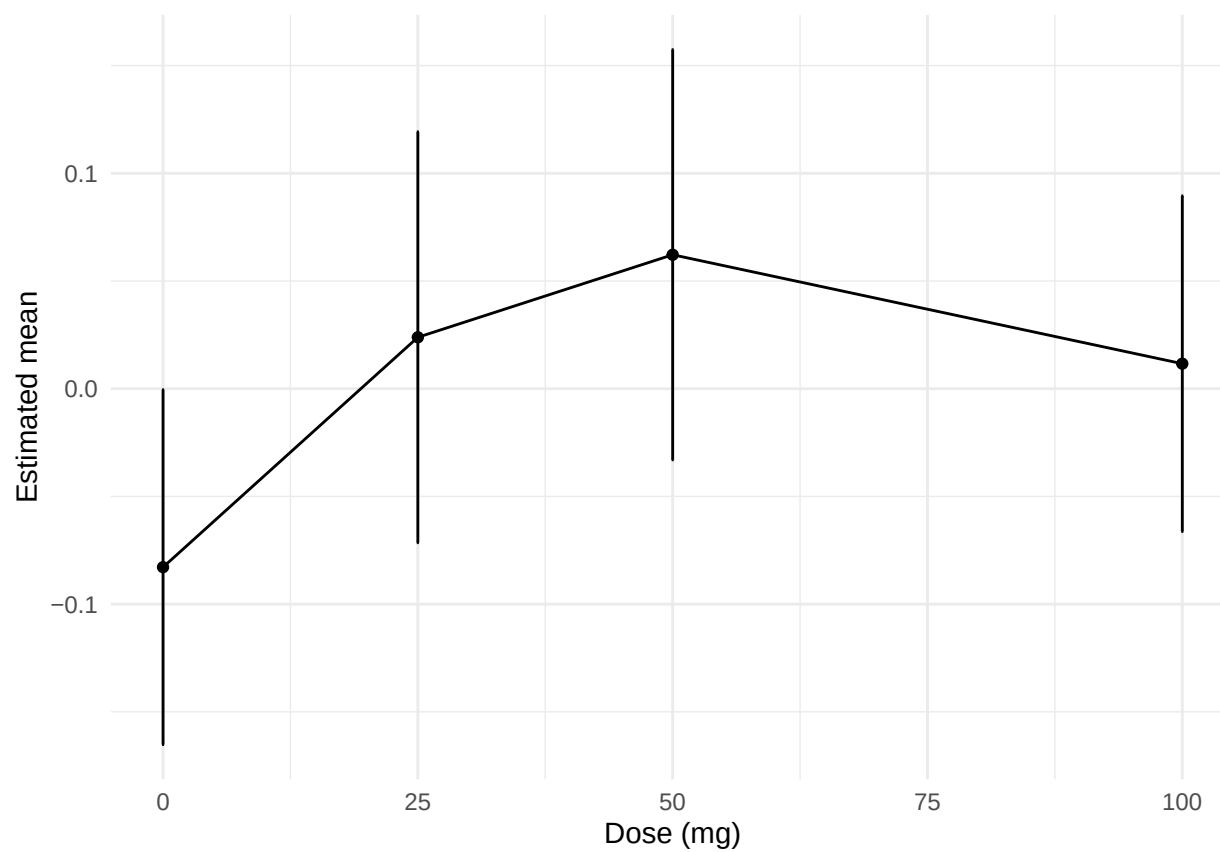
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.08 (-0.17 to 0.00)
losartan 25 mg	0.02 (-0.07 to 0.12)
losartan 50 mg	0.06 (-0.03 to 0.16)
losartan 100 mg	0.01 (-0.07 to 0.09)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.11 (-0.05 to 0.26)
losartan 50 mg - losartan 0 mg	0.15 (-0.01 to 0.30)
losartan 100 mg - losartan 0 mg	0.09 (-0.04 to 0.23)

Plot



Mean tissue stiffness

A total of 29 subjects were included in the analysis for this endpoint.

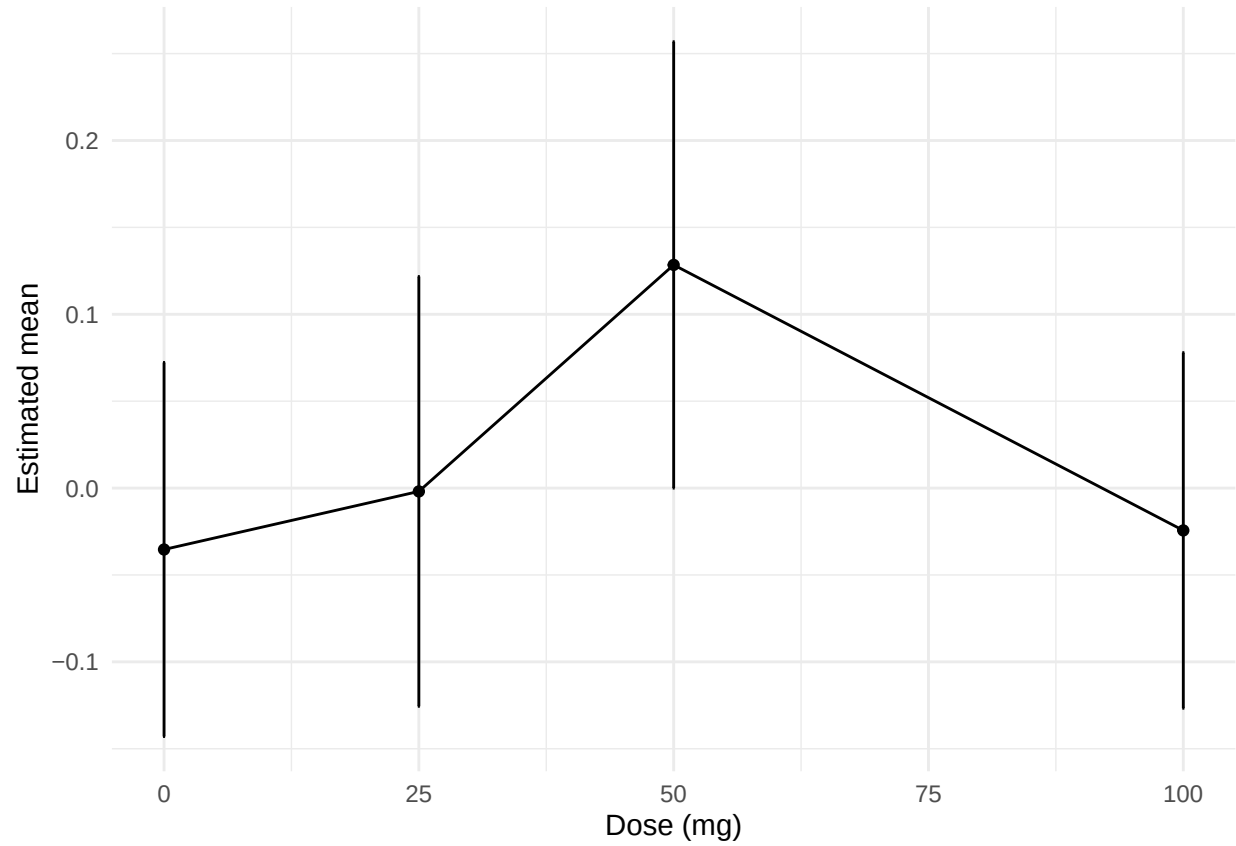
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.04 (-0.14 to 0.07)
losartan 25 mg	0.00 (-0.13 to 0.12)
losartan 50 mg	0.13 (0.00 to 0.26)
losartan 100 mg	-0.02 (-0.13 to 0.08)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.03 (-0.17 to 0.23)
losartan 50 mg - losartan 0 mg	0.16 (-0.04 to 0.37)
losartan 100 mg - losartan 0 mg	0.01 (-0.17 to 0.19)

Plot



Mean tissue viscosity

A total of 29 subjects were included in the analysis for this endpoint.

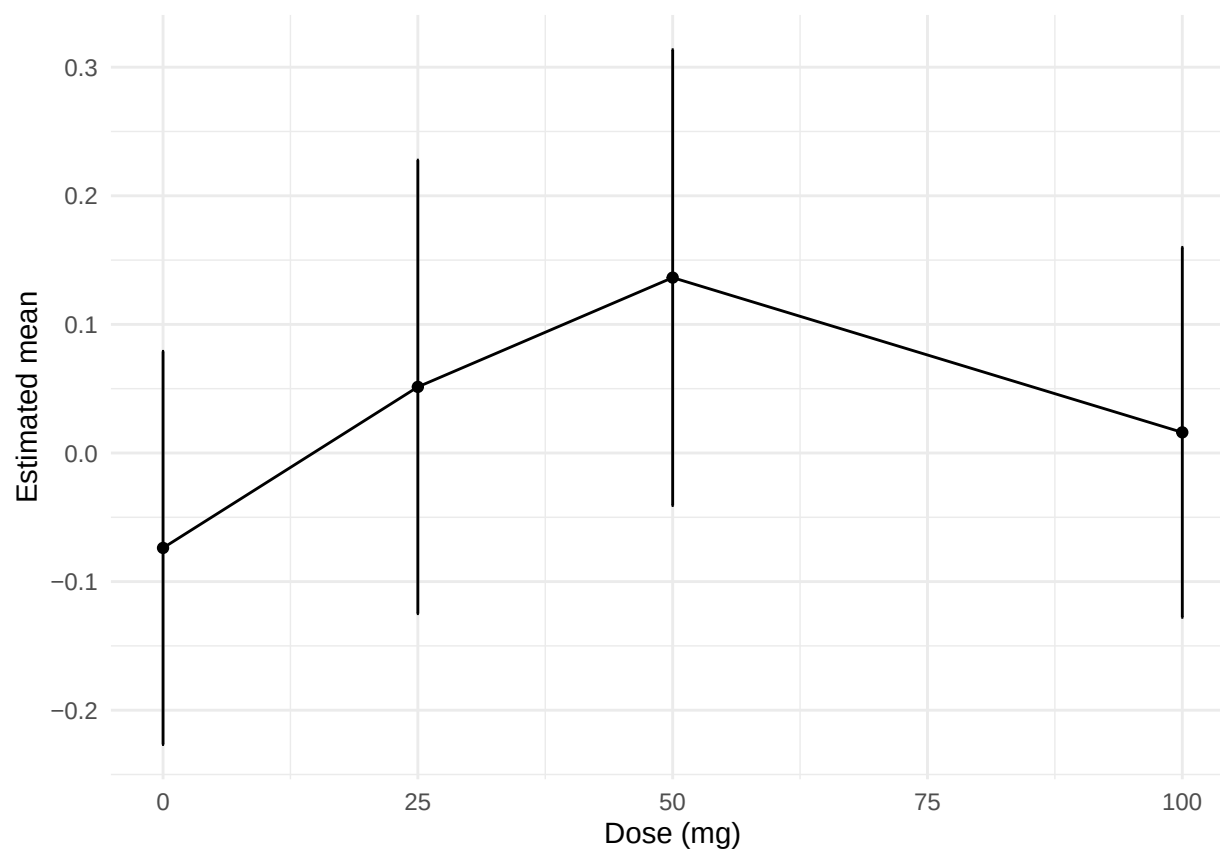
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	-0.07 (-0.23 to 0.08)
losartan 25 mg	0.05 (-0.12 to 0.23)
losartan 50 mg	0.14 (-0.04 to 0.31)
losartan 100 mg	0.02 (-0.13 to 0.16)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	0.13 (-0.16 to 0.41)
losartan 50 mg - losartan 0 mg	0.21 (-0.08 to 0.50)
losartan 100 mg - losartan 0 mg	0.09 (-0.17 to 0.35)

Plot



Total volume of contrast agent enhancing lesion on T1-weighted MRI in cm3

A total of 47 subjects were included in the analysis for this endpoint.

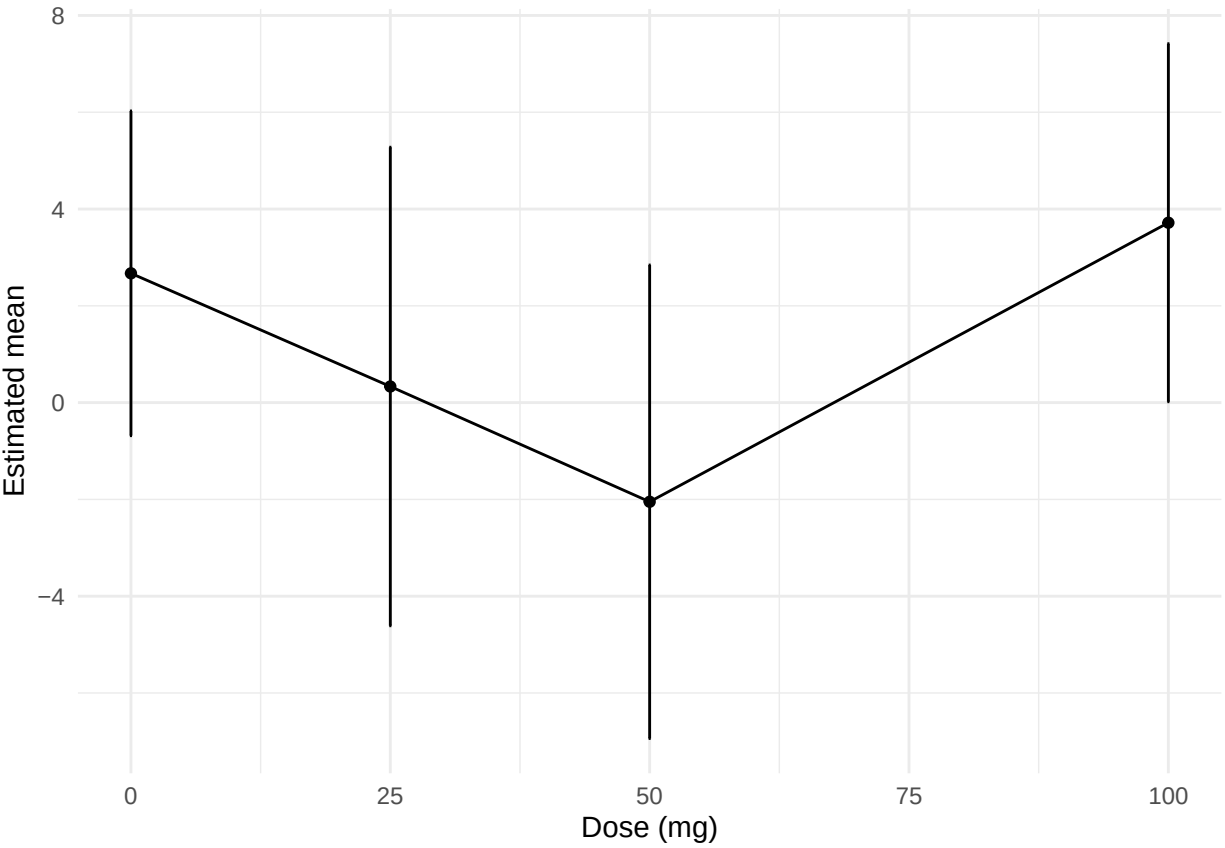
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	2.67 (-0.68 to 6.03)
losartan 25 mg	0.33 (-4.61 to 5.28)
losartan 50 mg	-2.05 (-6.94 to 2.84)
losartan 100 mg	3.72 (0.02 to 7.42)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-2.34 (-9.65 to 4.97)
losartan 50 mg - losartan 0 mg	-4.72 (-11.95 to 2.51)
losartan 100 mg - losartan 0 mg	1.05 (-4.99 to 7.08)

Plot



Total volume of pathologic signal on T2/FLAIR-weighted MRI in cm3

A total of 47 subjects were included in the analysis for this endpoint.

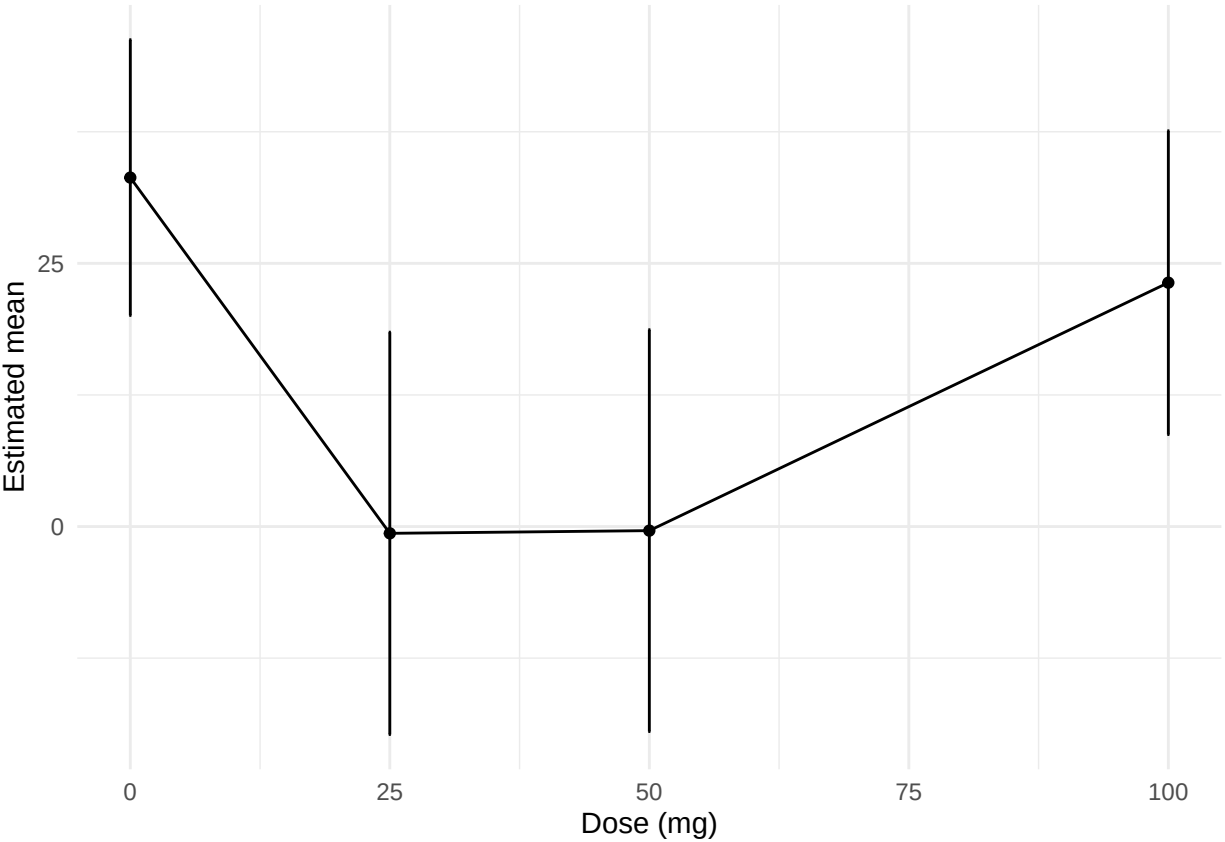
Means by treatment

Treatment	Estimated change from baseline (95% CI)
losartan 0 mg	33.15 (20.05 to 46.25)
losartan 25 mg	-0.65 (-19.76 to 18.46)
losartan 50 mg	-0.39 (-19.47 to 18.70)
losartan 100 mg	23.16 (8.74 to 37.59)

Differences vs reference

Comparison	Difference (95% CI)
losartan 25 mg - losartan 0 mg	-33.79 (-61.94 to -5.65)
losartan 50 mg - losartan 0 mg	-33.54 (-61.63 to -5.44)
losartan 100 mg - losartan 0 mg	-9.98 (-33.64 to 13.68)

Plot



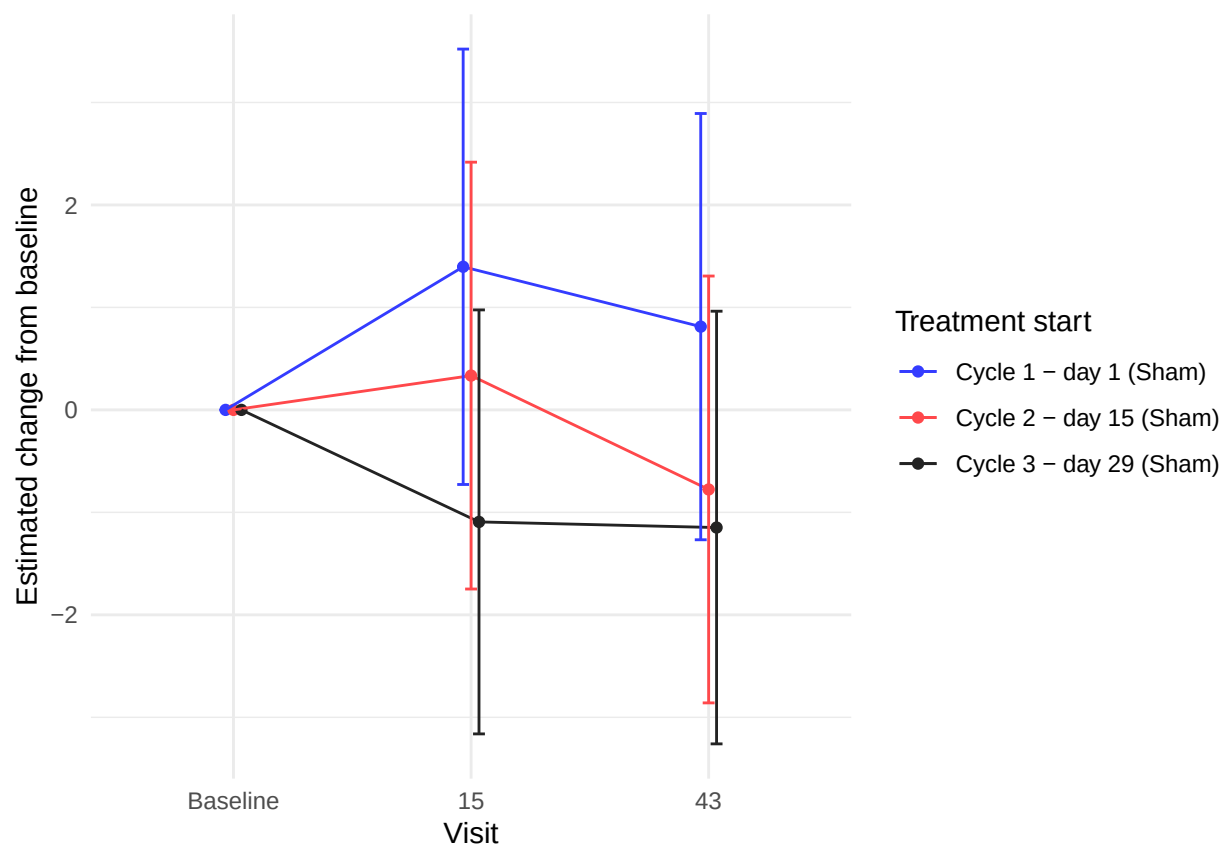
Neurologic Performance

Change from baseline in KPS, ECOG, and NANO scores at Days 15/43 and model-based estimates at Days 155/239 are reported below for the newly diagnosed glioblastoma cohort. Note that since the day 29 is not measured for any of the measures, the dose-response analyses are based on Days 15 and 43 only for the early timepoints. There might therefore be a limited ability to detect dose-response relationships at these early visits because we are lacking one contrast.

Karnofsky Performance Scale

A total of 54 subjects were included in the analysis for this endpoint.

Line plot by treatment start



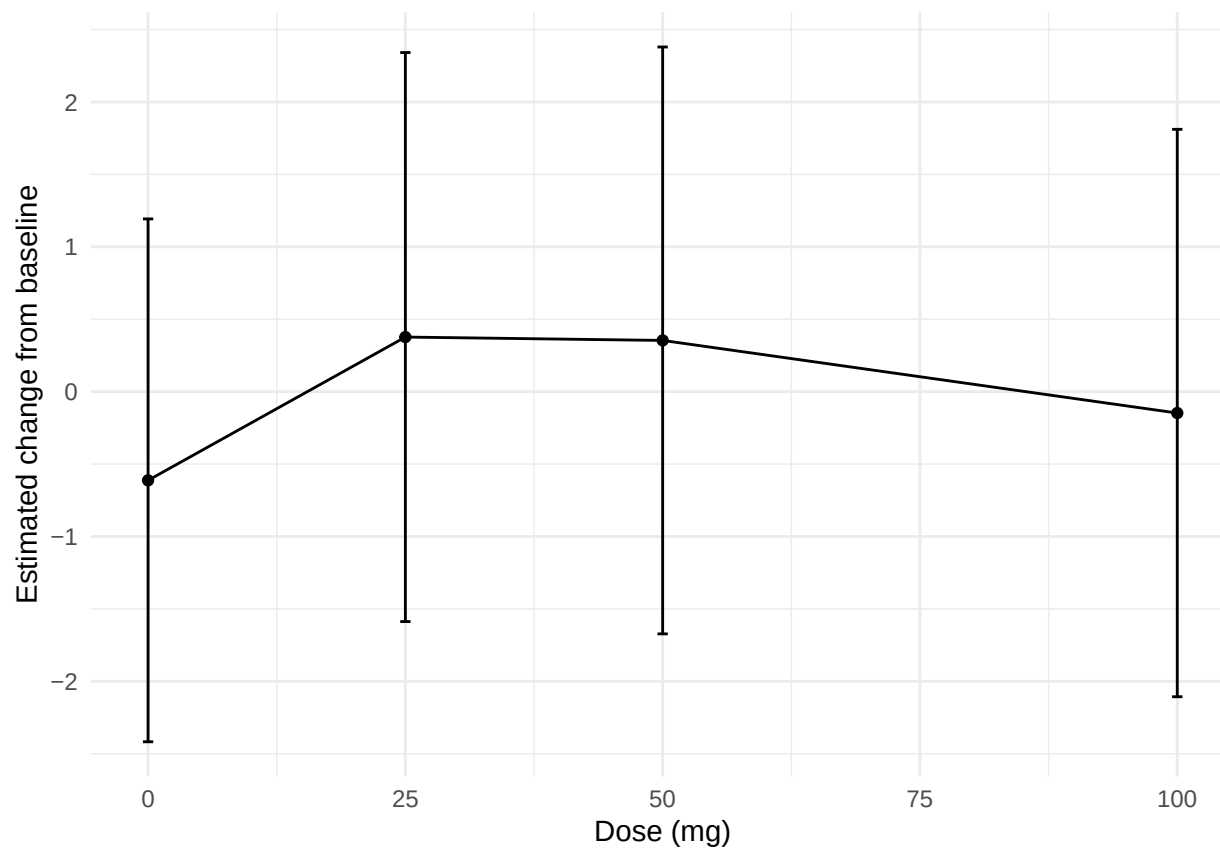
Days 15/43 (change from baseline)

Dose (mg)	Estimated change from baseline (95% CI)
0	-0.61 (-2.42 to 1.19)
25	0.38 (-1.59 to 2.34)
50	0.35 (-1.67 to 2.38)
100	-0.15 (-2.11 to 1.81)

Differences vs 0 mg

Comparison	Difference (95% CI)
losartan 25 mg vs losartan 0 mg	0.99 (-2.23 to 4.21)
losartan 50 mg vs losartan 0 mg	0.97 (-2.32 to 4.26)
losartan 100 mg vs losartan 0 mg	0.46 (-2.74 to 3.67)

Plot



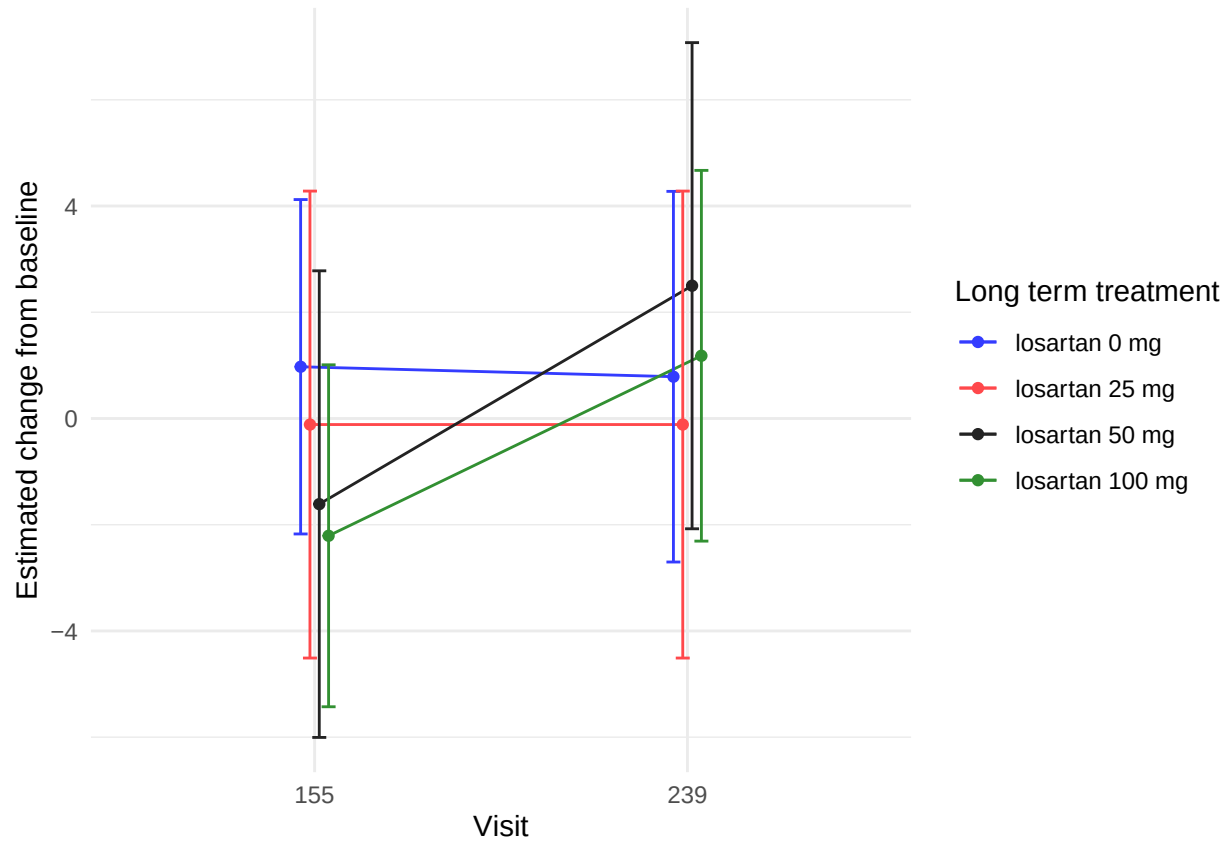
Days 155/239 (mixed model)

Visit	Treatment	Estimated change from baseline (95% CI)
155	losartan 0 mg	0.97 (-2.17 to 4.12)
155	losartan 25 mg	-0.11 (-4.51 to 4.28)
155	losartan 50 mg	-1.61 (-6.00 to 2.78)
155	losartan 100 mg	-2.21 (-5.43 to 1.01)
239	losartan 0 mg	0.79 (-2.70 to 4.28)
239	losartan 25 mg	-0.11 (-4.51 to 4.28)
239	losartan 50 mg	2.50 (-2.08 to 7.08)
239	losartan 100 mg	1.18 (-2.31 to 4.67)

Differences vs reference

Visit	Comparison	Difference (95% CI)
155	losartan 25 mg - losartan 0 mg	-1.09 (-7.63 to 5.45)
155	losartan 50 mg - losartan 0 mg	-2.59 (-9.11 to 3.93)
155	losartan 100 mg - losartan 0 mg	-3.18 (-8.61 to 2.25)
239	losartan 25 mg - losartan 0 mg	-0.90 (-7.69 to 5.89)
239	losartan 50 mg - losartan 0 mg	1.71 (-5.23 to 8.65)
239	losartan 100 mg - losartan 0 mg	0.39 (-5.56 to 6.35)

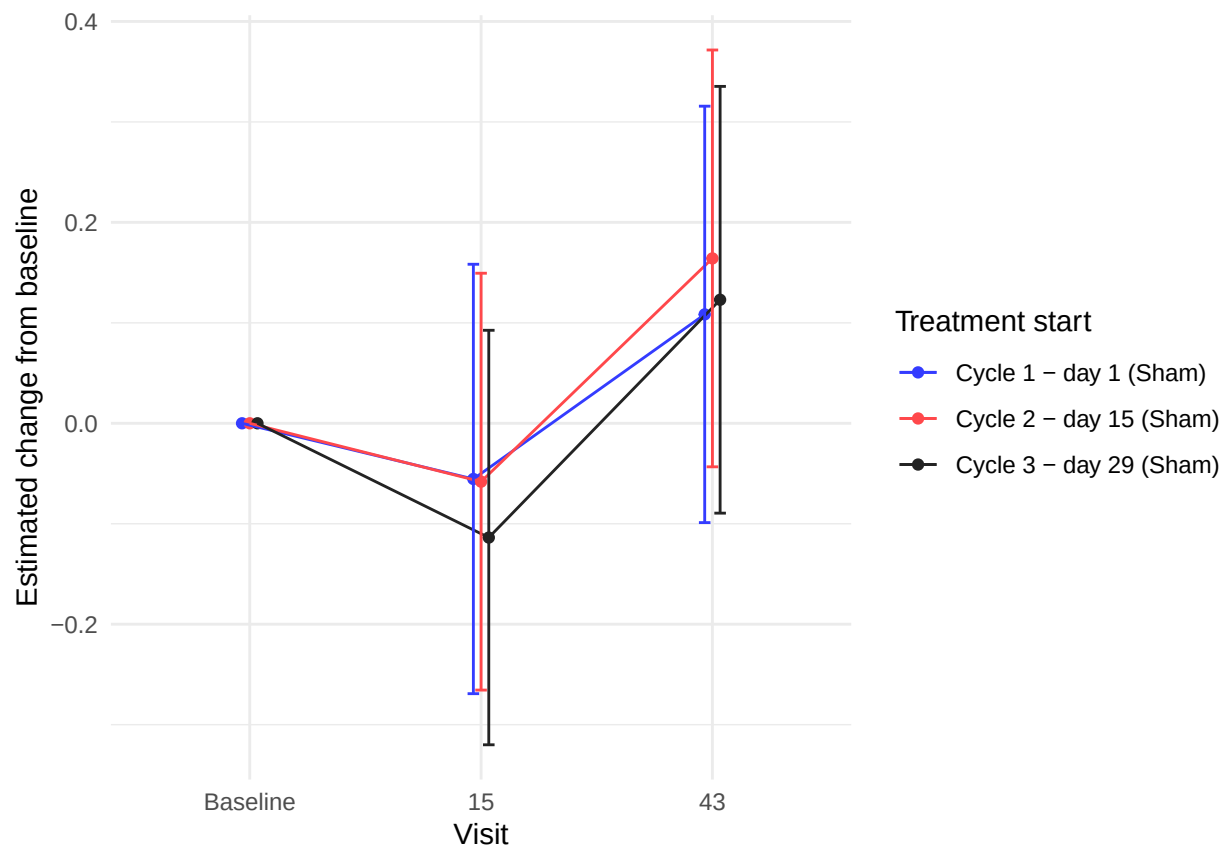
Plot



ECOG Score

A total of 54 subjects were included in the analysis for this endpoint.

Line plot by treatment start



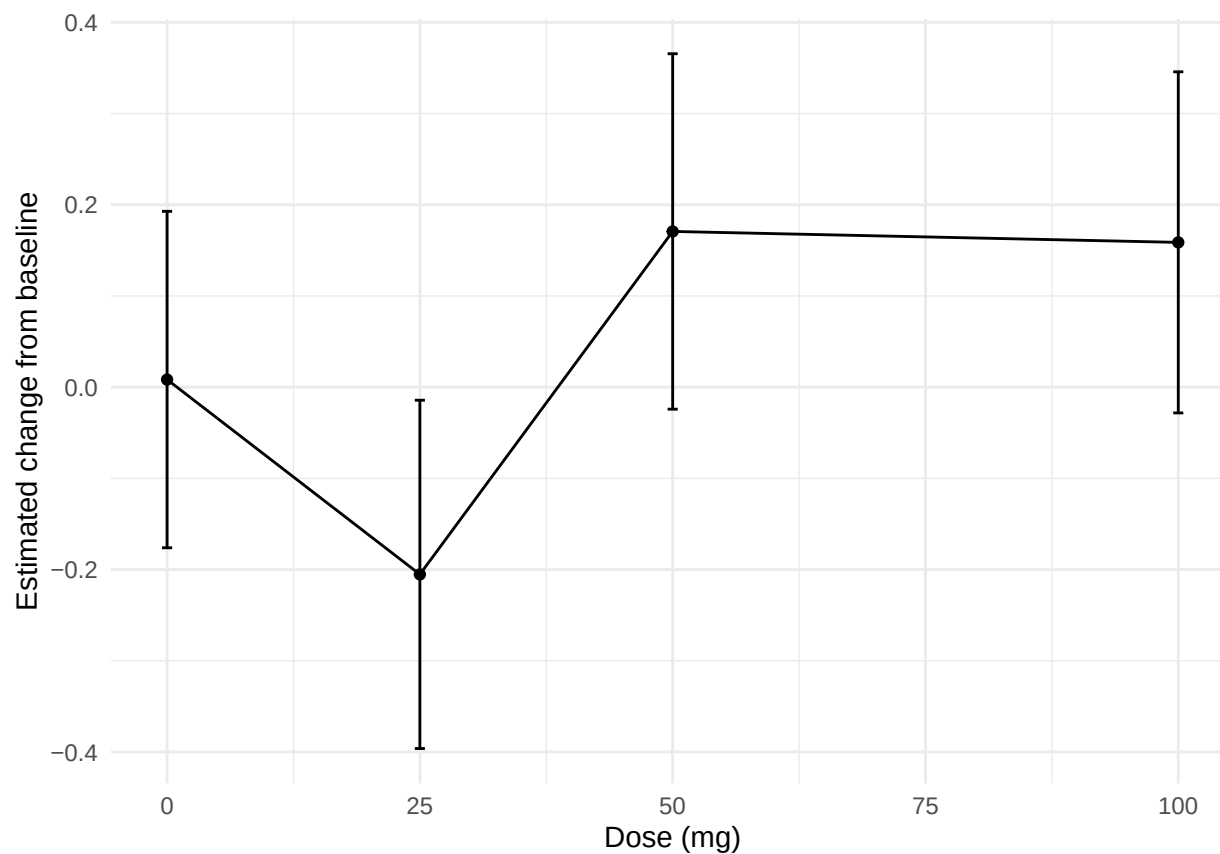
Days 15/43 (change from baseline)

Dose (mg)	Estimated change from baseline (95% CI)
0	0.01 (-0.18 to 0.19)
25	-0.21 (-0.40 to -0.01)
50	0.17 (-0.02 to 0.37)
100	0.16 (-0.03 to 0.35)

Differences vs 0 mg

Comparison	Difference (95% CI)
losartan 25 mg vs losartan 0 mg	-0.21 (-0.56 to 0.14)
losartan 50 mg vs losartan 0 mg	0.16 (-0.19 to 0.52)
losartan 100 mg vs losartan 0 mg	0.15 (-0.19 to 0.49)

Plot



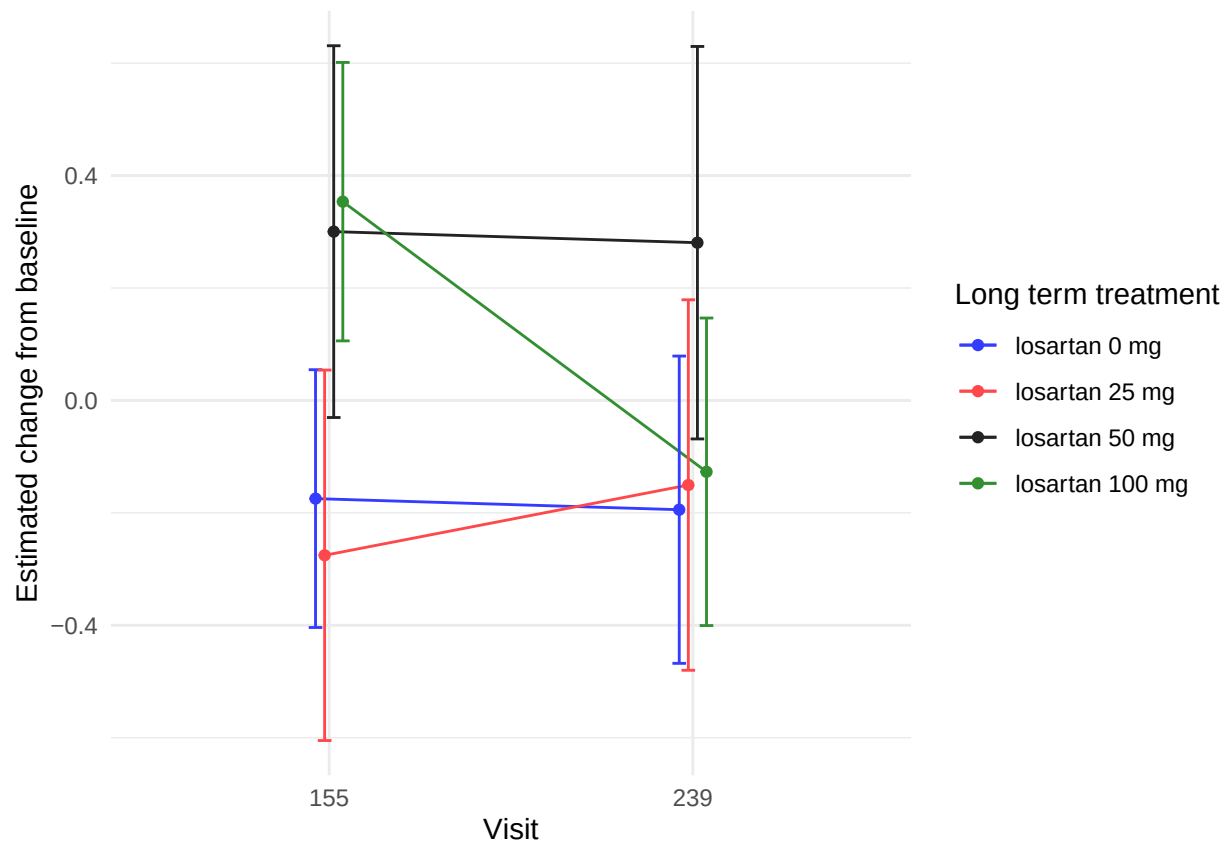
Days 155/239 (mixed model)

Visit	Treatment	Estimated change from baseline (95% CI)
155	losartan 0 mg	-0.17 (-0.40 to 0.05)
155	losartan 25 mg	-0.28 (-0.61 to 0.05)
155	losartan 50 mg	0.30 (-0.03 to 0.63)
155	losartan 100 mg	0.35 (0.11 to 0.60)
239	losartan 0 mg	-0.19 (-0.47 to 0.08)
239	losartan 25 mg	-0.15 (-0.48 to 0.18)
239	losartan 50 mg	0.28 (-0.07 to 0.63)
239	losartan 100 mg	-0.13 (-0.40 to 0.15)

Differences vs reference

Visit	Comparison	Difference (95% CI)
155	losartan 25 mg - losartan 0 mg	-0.10 (-0.59 to 0.38)
155	losartan 50 mg - losartan 0 mg	0.47 (-0.01 to 0.96)
155	losartan 100 mg - losartan 0 mg	0.53 (0.12 to 0.94)
239	losartan 25 mg - losartan 0 mg	0.04 (-0.47 to 0.56)
239	losartan 50 mg - losartan 0 mg	0.47 (-0.06 to 1.01)
239	losartan 100 mg - losartan 0 mg	0.07 (-0.40 to 0.54)

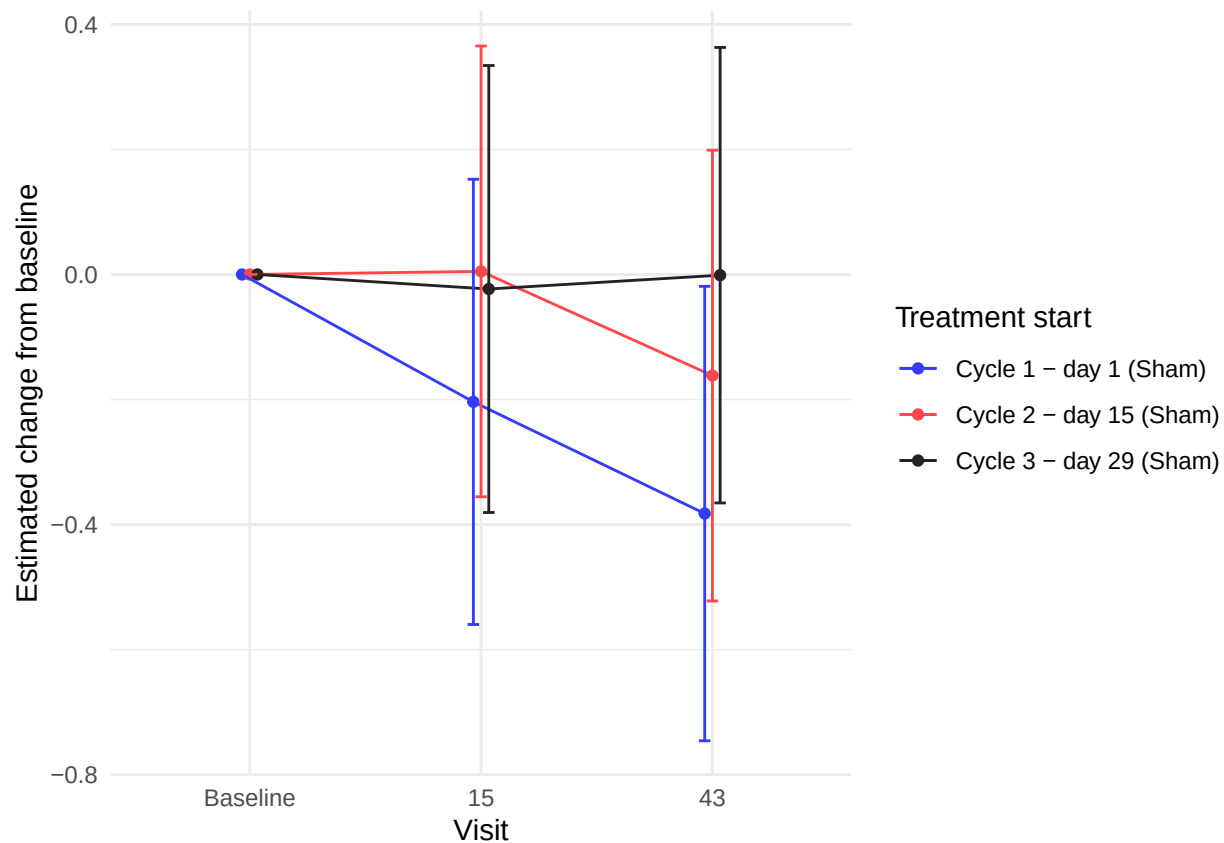
Plot



Total NANO score

A total of 54 subjects were included in the analysis for this endpoint.

Line plot by treatment start



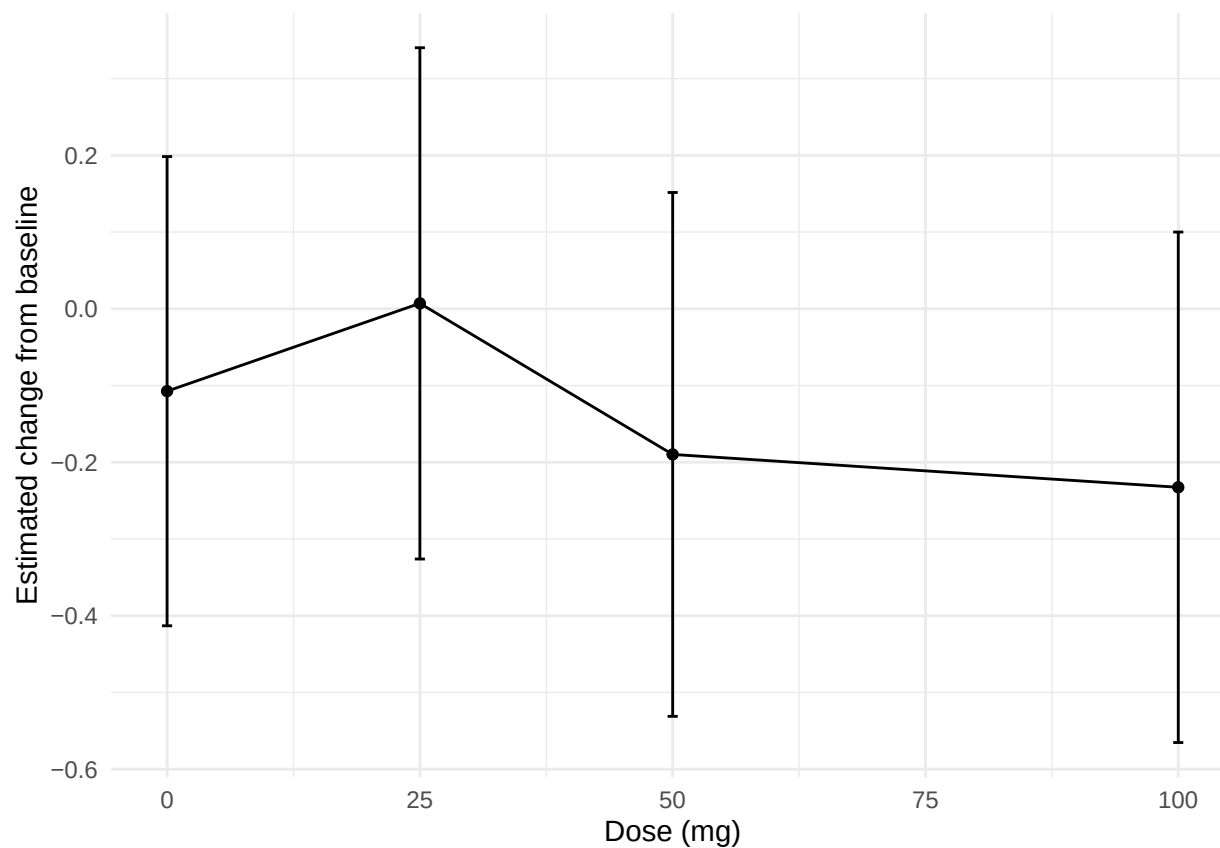
Days 15/43 (change from baseline)

Dose (mg)	Estimated change from baseline (95% CI)
0	-0.11 (-0.41 to 0.20)
25	0.01 (-0.33 to 0.34)
50	-0.19 (-0.53 to 0.15)
100	-0.23 (-0.57 to 0.10)

Differences vs 0 mg

Comparison	Difference (95% CI)
losartan 25 mg vs losartan 0 mg	0.11 (-0.42 to 0.65)
losartan 50 mg vs losartan 0 mg	-0.08 (-0.63 to 0.46)
losartan 100 mg vs losartan 0 mg	-0.13 (-0.66 to 0.41)

Plot



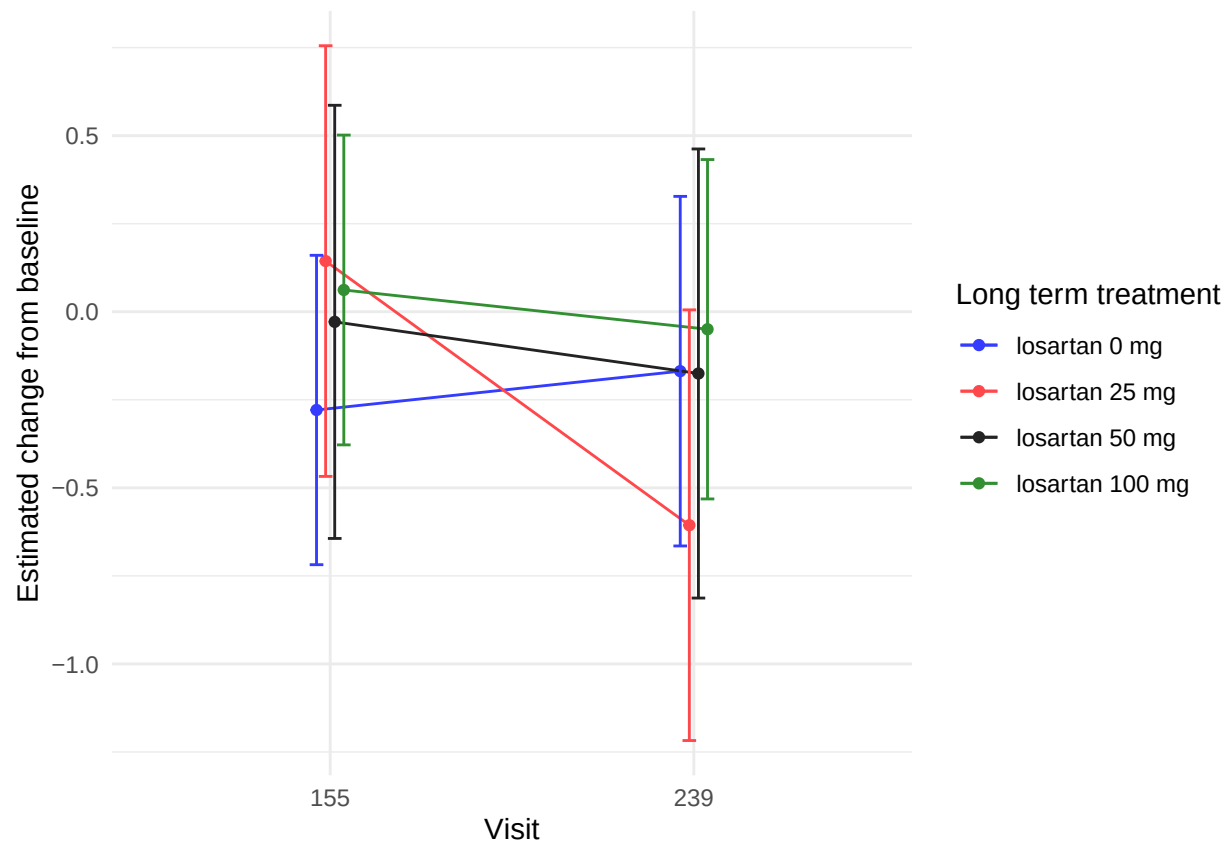
Days 155/239 (mixed model)

Visit	Treatment	Estimated change from baseline (95% CI)
155	losartan 0 mg	-0.28 (-0.72 to 0.16)
155	losartan 25 mg	0.14 (-0.47 to 0.76)
155	losartan 50 mg	-0.03 (-0.64 to 0.59)
155	losartan 100 mg	0.06 (-0.38 to 0.50)
239	losartan 0 mg	-0.17 (-0.67 to 0.33)
239	losartan 25 mg	-0.61 (-1.22 to 0.01)
239	losartan 50 mg	-0.18 (-0.81 to 0.46)
239	losartan 100 mg	-0.05 (-0.53 to 0.43)

Differences vs reference

Visit	Comparison	Difference (95% CI)
155	losartan 25 mg - losartan 0 mg	0.42 (-0.49 to 1.33)
155	losartan 50 mg - losartan 0 mg	0.25 (-0.66 to 1.16)
155	losartan 100 mg - losartan 0 mg	0.34 (-0.41 to 1.09)
239	losartan 25 mg - losartan 0 mg	-0.44 (-1.39 to 0.51)
239	losartan 50 mg - losartan 0 mg	-0.01 (-0.98 to 0.97)
239	losartan 100 mg - losartan 0 mg	0.12 (-0.72 to 0.95)

Plot



Quality of Life Outcomes

Model based estimates of change from baseline in QLQ-C30 and QLQ-BN20 scores at Days 155/239 are reported below for the newly diagnosed glioblastoma cohort. Note that even if the QoL outcomes were collected at Day 43 it is not possible to make a meaningful analysis of the effect of losartan as all participants received the treatment at that time point.

EORTC QLQ-C30 score

A total of 19 subjects were included in the analysis for this endpoint.

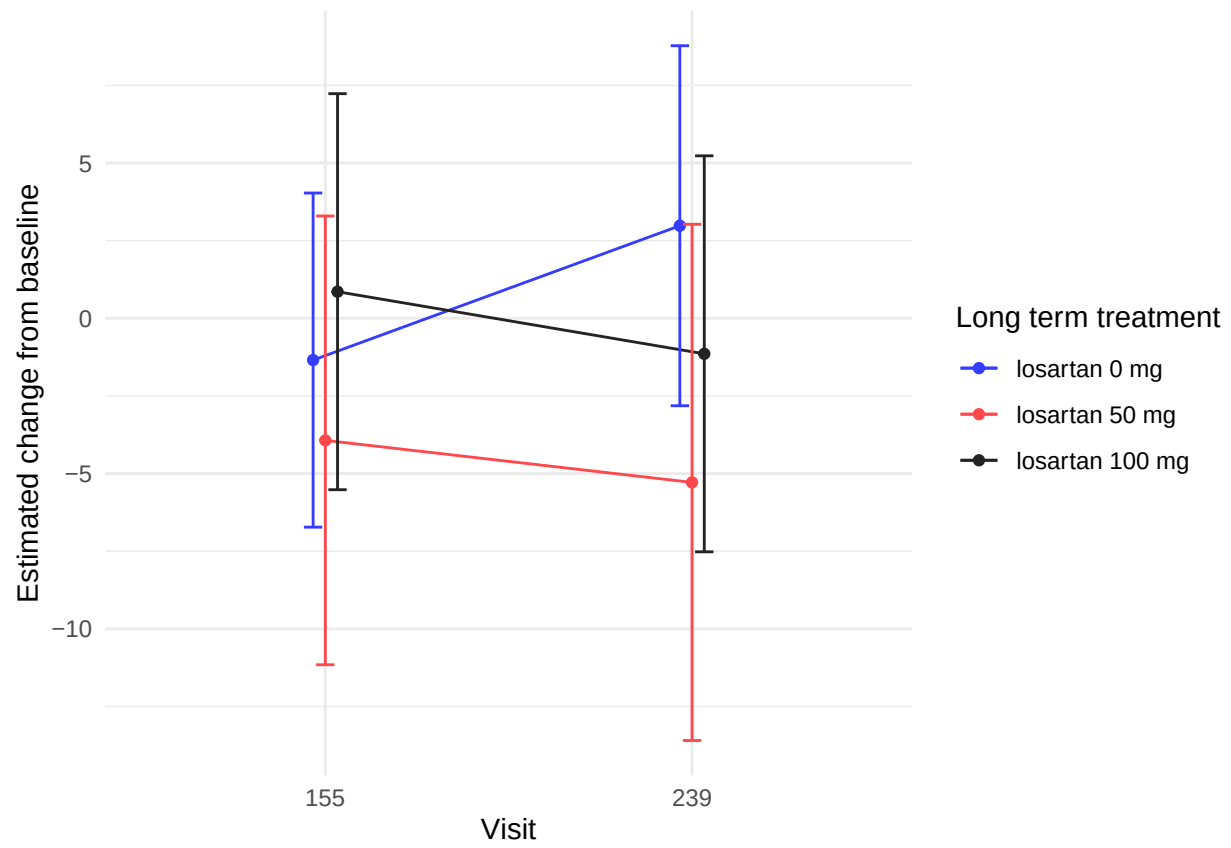
Days 155/239 (mixed model)

Visit	Treatment	Estimated change from baseline (95% CI)
155	losartan 0 mg	-1.35 (-6.72 to 4.03)
155	losartan 50 mg	-3.93 (-11.16 to 3.29)
155	losartan 100 mg	0.86 (-5.52 to 7.23)
239	losartan 0 mg	2.98 (-2.82 to 8.77)
239	losartan 50 mg	-5.28 (-13.60 to 3.03)
239	losartan 100 mg	-1.14 (-7.52 to 5.23)

Differences vs reference

Visit	Comparison	Difference (95% CI)
155	losartan 50 mg - losartan 0 mg	-2.59 (-12.91 to 7.74)
155	losartan 100 mg - losartan 0 mg	2.20 (-7.30 to 11.70)
239	losartan 50 mg - losartan 0 mg	-8.26 (-19.82 to 3.29)
239	losartan 100 mg - losartan 0 mg	-4.12 (-13.96 to 5.71)

Plot



EORTC QLQ-BN20 score

A total of 18 subjects were included in the analysis for this endpoint.

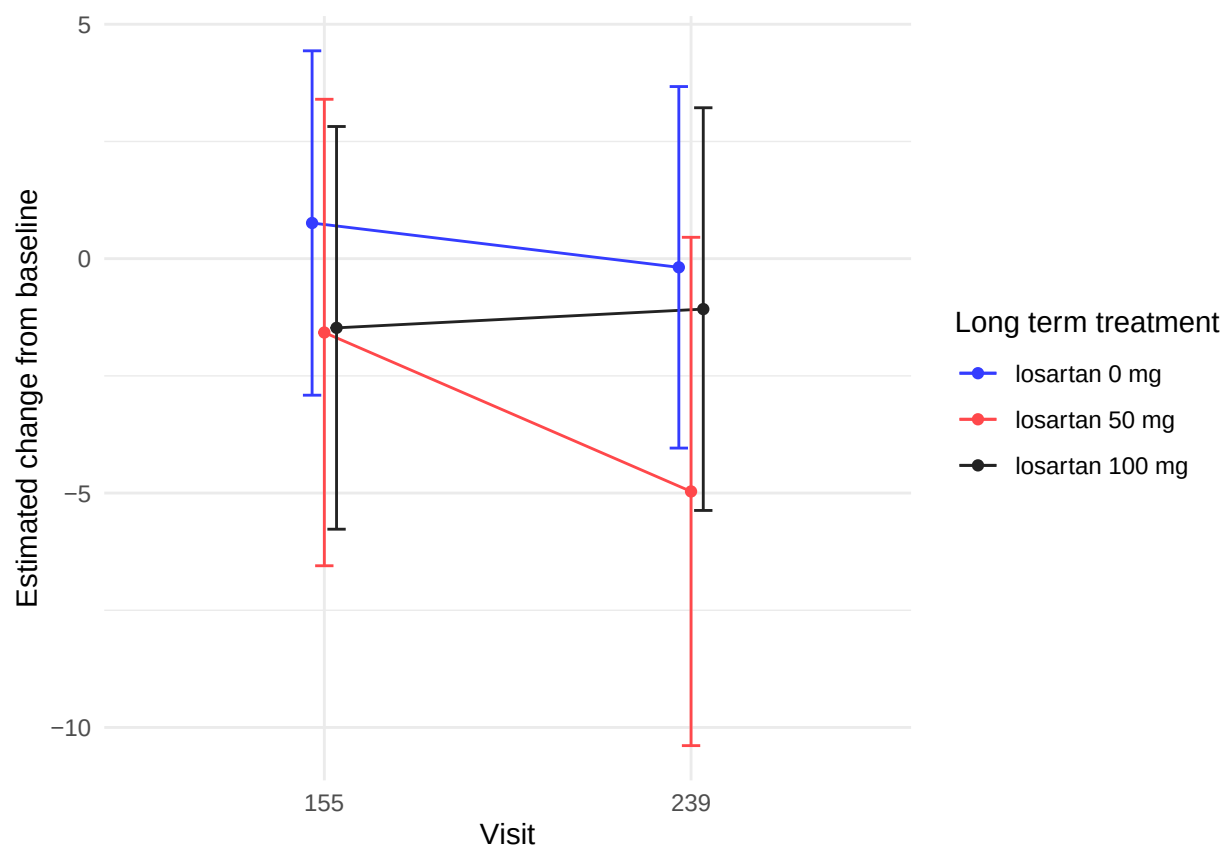
Days 155/239 (mixed model)

Visit	Treatment	Estimated change from baseline (95% CI)
155	losartan 0 mg	0.76 (-2.91 to 4.43)
155	losartan 50 mg	-1.58 (-6.55 to 3.40)
155	losartan 100 mg	-1.48 (-5.77 to 2.82)
239	losartan 0 mg	-0.18 (-4.04 to 3.67)
239	losartan 50 mg	-4.97 (-10.39 to 0.46)
239	losartan 100 mg	-1.08 (-5.37 to 3.22)

Differences vs reference

Visit	Comparison	Difference (95% CI)
155	losartan 50 mg - losartan 0 mg	-2.34 (-9.45 to 4.78)
155	losartan 100 mg - losartan 0 mg	-2.24 (-8.70 to 4.23)
239	losartan 50 mg - losartan 0 mg	-4.78 (-12.43 to 2.87)
239	losartan 100 mg - losartan 0 mg	-0.89 (-7.49 to 5.70)

Plot



Steroid Use

Binary steroid use is analysed with longitudinal logistic regression and population-averaged marginal risk differences for early (Days 15/29/43) and late (Days 155/239) visits in the newly diagnosed glioblastoma cohort.

Steroid use

A total of 54 subjects were included in the analysis.

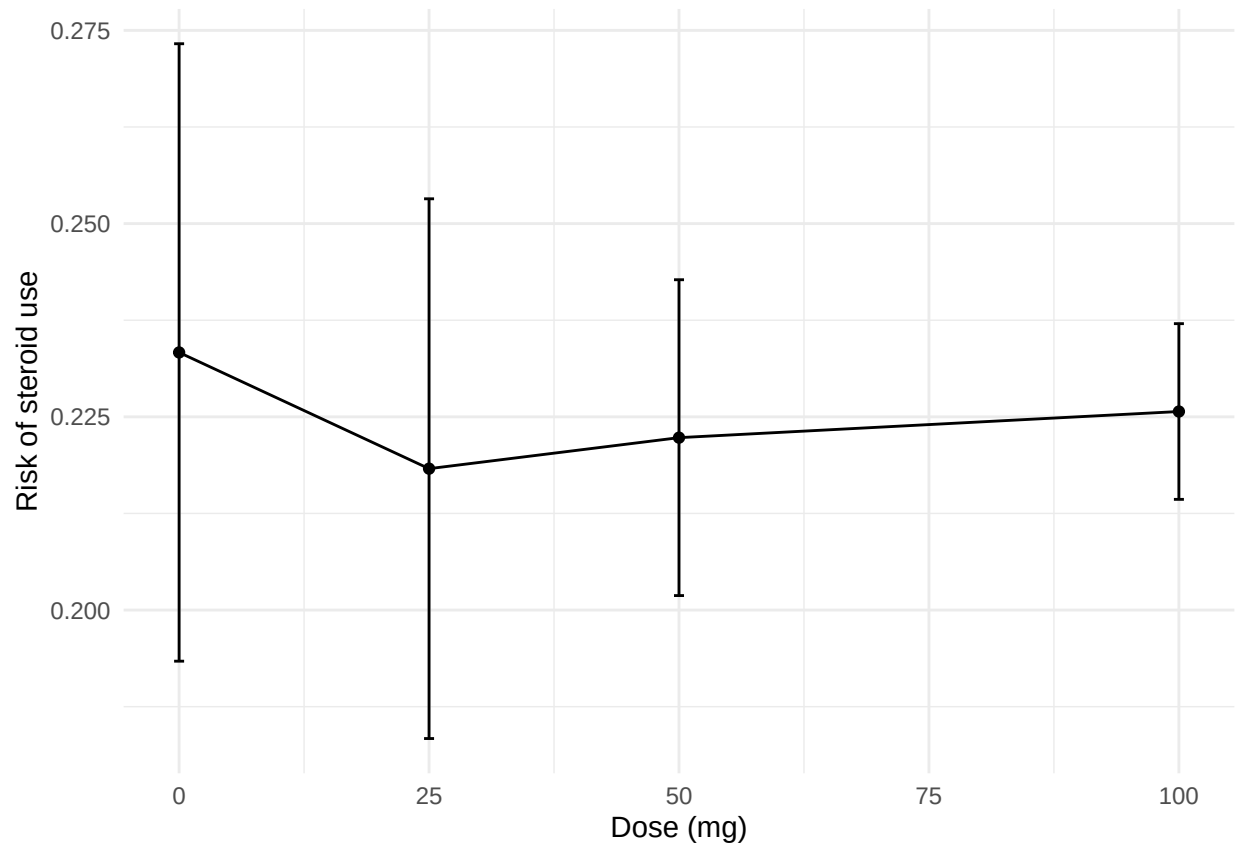
Days 15/29/43 (by dose)

Treatment	Risk (95% CI)
losartan 0 mg	0.233 (0.193 to 0.273)
losartan 25 mg	0.218 (0.183 to 0.253)
losartan 50 mg	0.222 (0.202 to 0.243)
losartan 100 mg	0.226 (0.214 to 0.237)

Risk differences vs 0 mg

Comparison	Risk difference (95% CI)
100 - 0	-0.008 (-0.048 to 0.033)
25 - 0	-0.015 (-0.082 to 0.051)
50 - 0	-0.011 (-0.065 to 0.043)

Plot



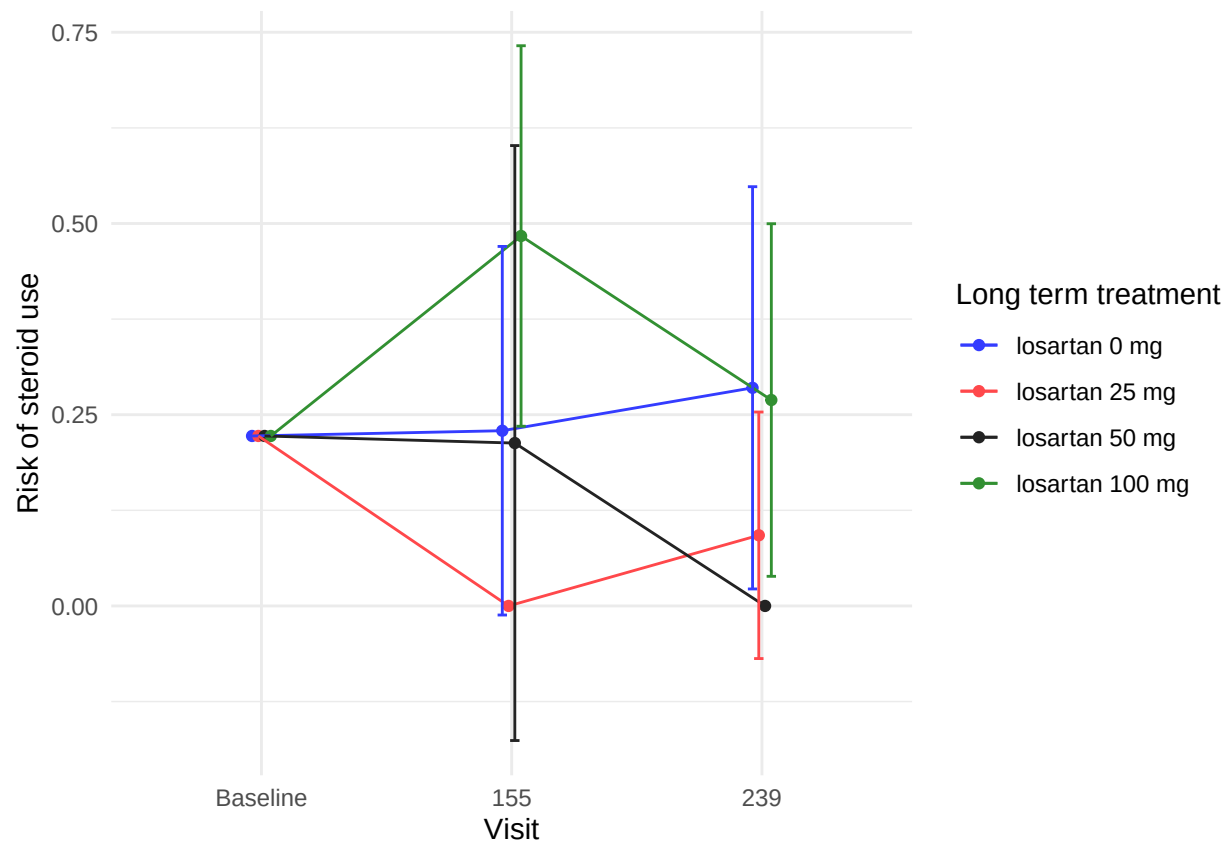
Days 155/239 (population-averaged risk)

Visit	Treatment	Risk (95% CI)
155	losartan 0 mg	0.229 (-0.012 to 0.470)
239	losartan 0 mg	0.285 (0.022 to 0.548)
155	losartan 25 mg	0.000 (0.000 to 0.000)
239	losartan 25 mg	0.092 (-0.069 to 0.254)
155	losartan 50 mg	0.213 (-0.176 to 0.602)
239	losartan 50 mg	0.000 (0.000 to 0.000)
155	losartan 100 mg	0.484 (0.235 to 0.732)
239	losartan 100 mg	0.269 (0.039 to 0.500)

Risk differences vs reference

Visit	Comparison	Risk difference (95% CI)
155	losartan 100 mg - losartan 0 mg	0.222 (-0.087 to 0.531)
155	losartan 25 mg - losartan 0 mg	-0.248 (-0.463 to -0.033)
155	losartan 50 mg - losartan 0 mg	-0.015 (-0.427 to 0.398)
239	losartan 100 mg - losartan 0 mg	-0.014 (-0.323 to 0.295)
239	losartan 25 mg - losartan 0 mg	-0.187 (-0.482 to 0.107)
239	losartan 50 mg - losartan 0 mg	-0.308 (-0.537 to -0.078)

Plot



RANO Responses

Ordinal RANO outcomes are analysed using proportional odds models within the configured cohort. For State of disease according to the RANO criteria, visits at Days 141 and 225 are included; for Best overall radiographic response, only Day 225 is included. Proportional odds ratios and confidence intervals are reported below.

State of disease according to the RANO criteria

Visit 141

Comparison	Propoprtional Odds Ratio (95% CI)
losartan 25 vs 0 mg	2.065 (0.359 to 12.269)
losartan 50 vs 0 mg	5.388 (0.964 to 36.797)
losartan 100 vs 0 mg	0.929 (0.206 to 4.084)

Visit 225

Comparison	Propoprtional Odds Ratio (95% CI)
losartan 25 vs 0 mg	8.166 (1.024 to 176.747)
losartan 50 vs 0 mg	1.556 (0.244 to 10.815)
losartan 100 vs 0 mg	3.111 (0.590 to 19.664)

Best overall radiographic response

Visit 225

Comparison	Propoprtional Odds Ratio (95% CI)
losartan 25 vs 0 mg	6.898 (0.811 to 90.991)
losartan 50 vs 0 mg	6.898 (0.730 to 101.826)
losartan 100 vs 0 mg	4.035 (0.676 to 31.150)

Biomarker Endpoints (Serum Cytokines / Angiogenic Factors)

Serum cytokine and angiogenic-factor concentrations from Bio-Plex 19-plex and 14-plex panels are analysed as exploratory biomarker endpoints. For each analyte, a linear mixed model with random subject intercept is fitted with change from Screening baseline as the response. Where the log-scale model fits better (lower AIC), the analysis is performed on the log-concentration scale and results are labelled accordingly.

Biomarker samples are collected at screening (baseline), Day 15 (Cycle 2), and Day 43 (Cycle 4) for the early analysis, and again at Cycle 17 (Day 225) and later for the long-term analysis. The early analysis uses the stepped-wedge treatment assignment; the long-term analysis uses the assigned long-term dose (**trt02p**).

A formal model selection and residual diagnostic assessment is available in a separate document ([impress_lb_model_fit.pdf](#)).

Fractalkine

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	0.025 (-0.236 to 0.287)
25 mg	0.119 (-0.149 to 0.388)
50 mg	0.110 (-0.180 to 0.400)
100 mg	-0.014 (-0.268 to 0.241)

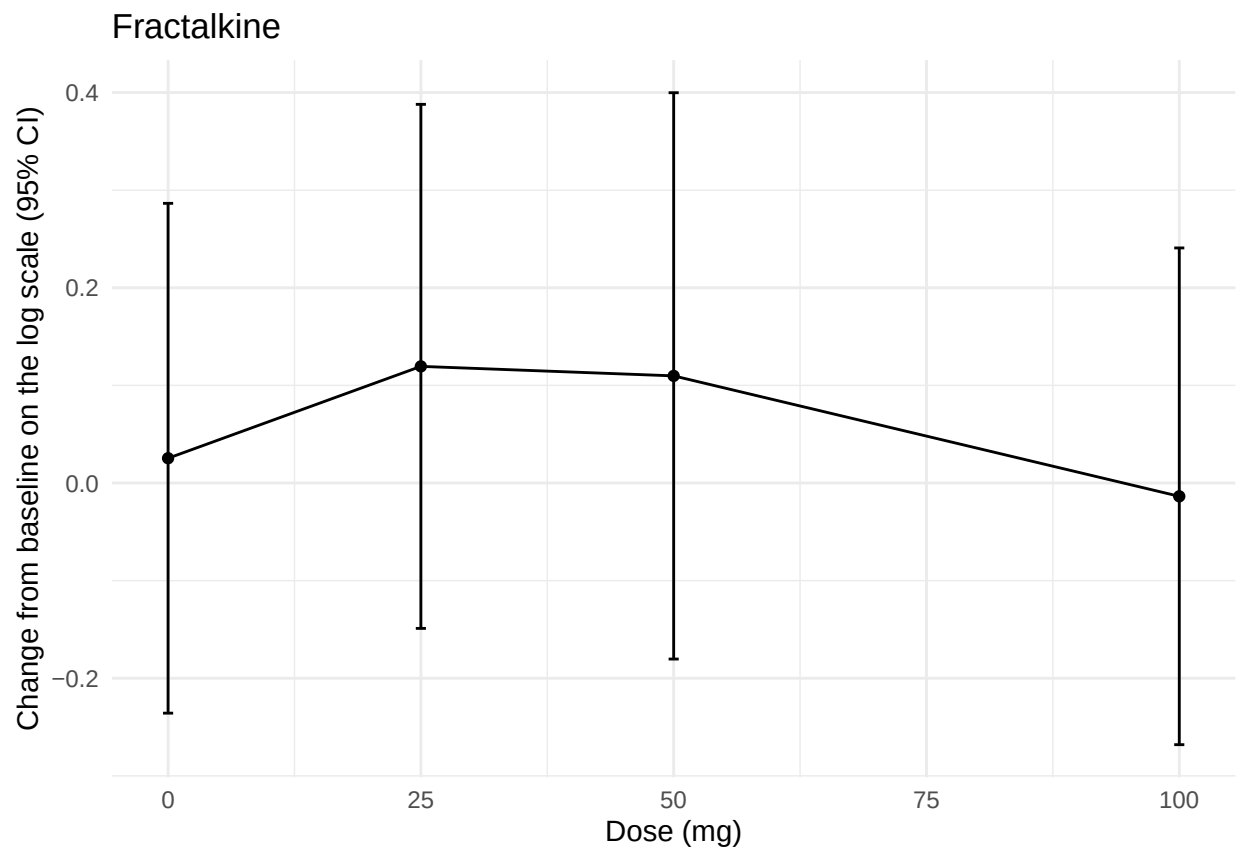
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	0.09 (-0.39 to 0.58)
losartan 50 mg vs losartan 0 mg0	0.08 (-0.42 to 0.58)
losartan 100 mg vs losartan 0 mg0	-0.04 (-0.51 to 0.43)

Model comparison (AIC)

Model	AIC	Preferred
Linear ($\text{chg} \sim \text{base} + \text{trtcd} + \text{avisitn}$)	1468.3	FALSE
Log-transformed ($\log(\text{value}) - \log(\text{base}) \sim \log(\text{base}) + \text{trtcd} + \text{avisitn}$)	223.3	TRUE

Plot



Granzyme B

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.176 (-0.370 to 0.018)
25 mg	-0.197 (-0.408 to 0.014)
50 mg	-0.027 (-0.249 to 0.195)
100 mg	-0.104 (-0.309 to 0.101)

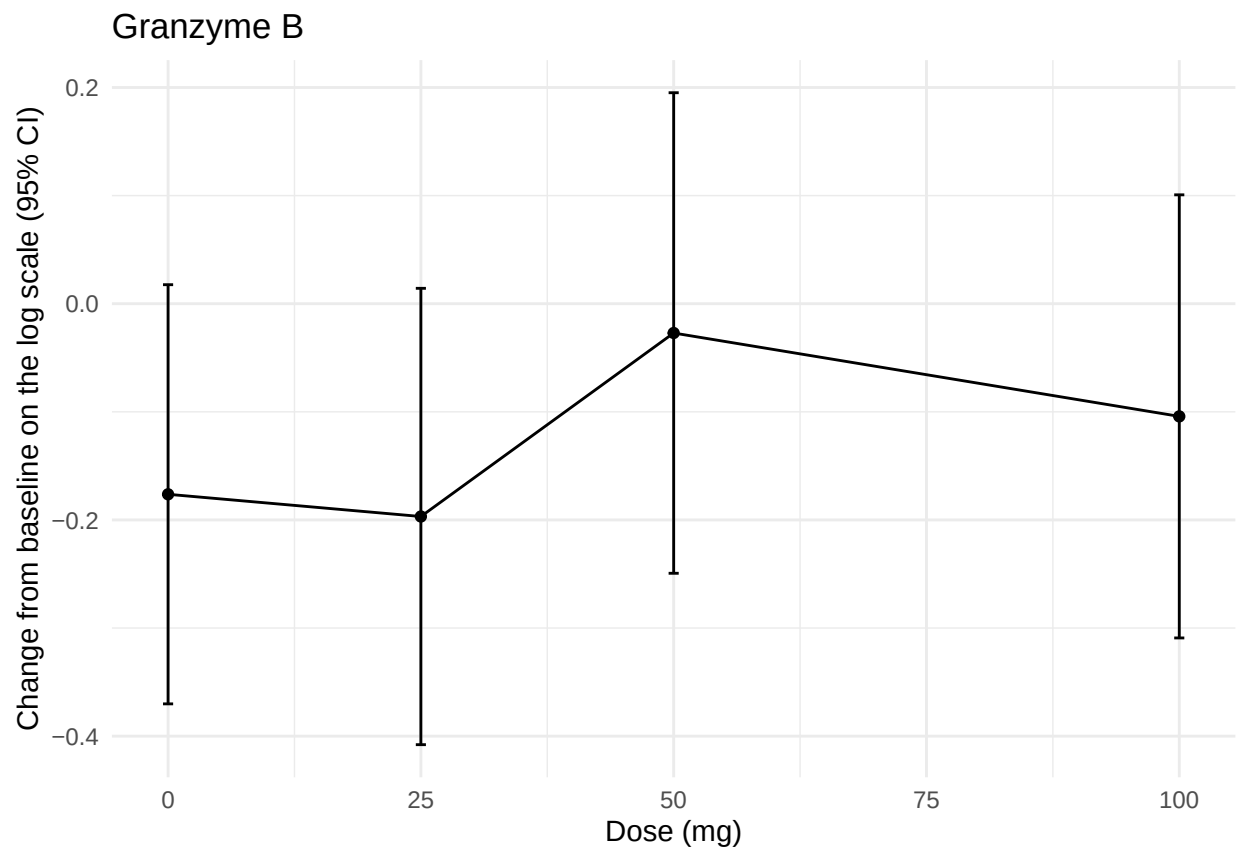
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg	-0.02 (-0.37 to 0.33)
losartan 50 mg vs losartan 0 mg	0.15 (-0.21 to 0.50)
losartan 100 mg vs losartan 0 mg	0.07 (-0.27 to 0.42)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	826.6	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	161.7	TRUE

Plot



IFN-g

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.212 (-0.553 to 0.129)
25 mg	-0.073 (-0.445 to 0.299)
50 mg	-0.161 (-0.553 to 0.230)
100 mg	-0.289 (-0.650 to 0.072)

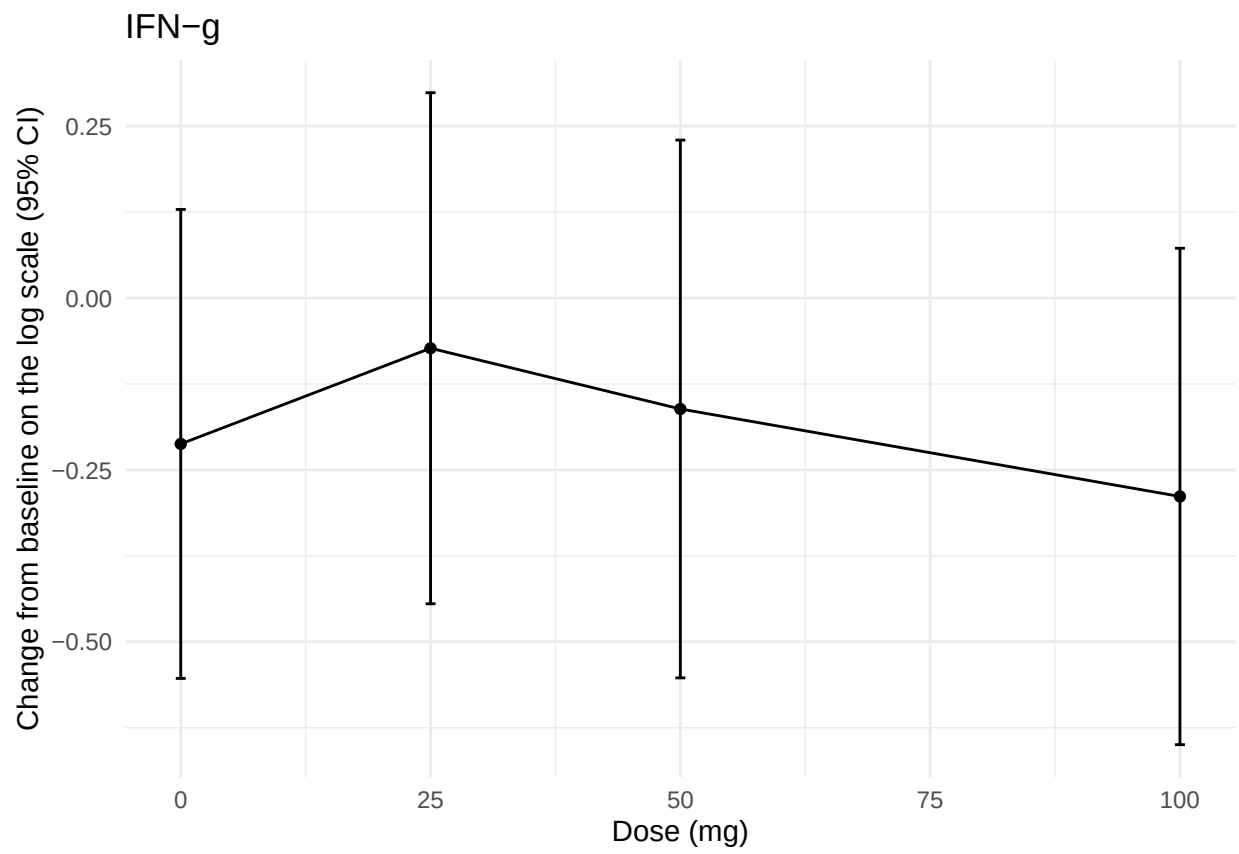
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	0.14 (-0.48 to 0.76)
losartan 50 mg vs losartan 0 mg0	0.05 (-0.58 to 0.68)
losartan 100 mg vs losartan 0 mg0	-0.08 (-0.68 to 0.53)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	626.2	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	272.0	TRUE

Plot



IL-1b

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.102 (-0.471 to 0.266)
25 mg	-0.039 (-0.458 to 0.381)
50 mg	-0.196 (-0.633 to 0.241)
100 mg	0.042 (-0.370 to 0.454)

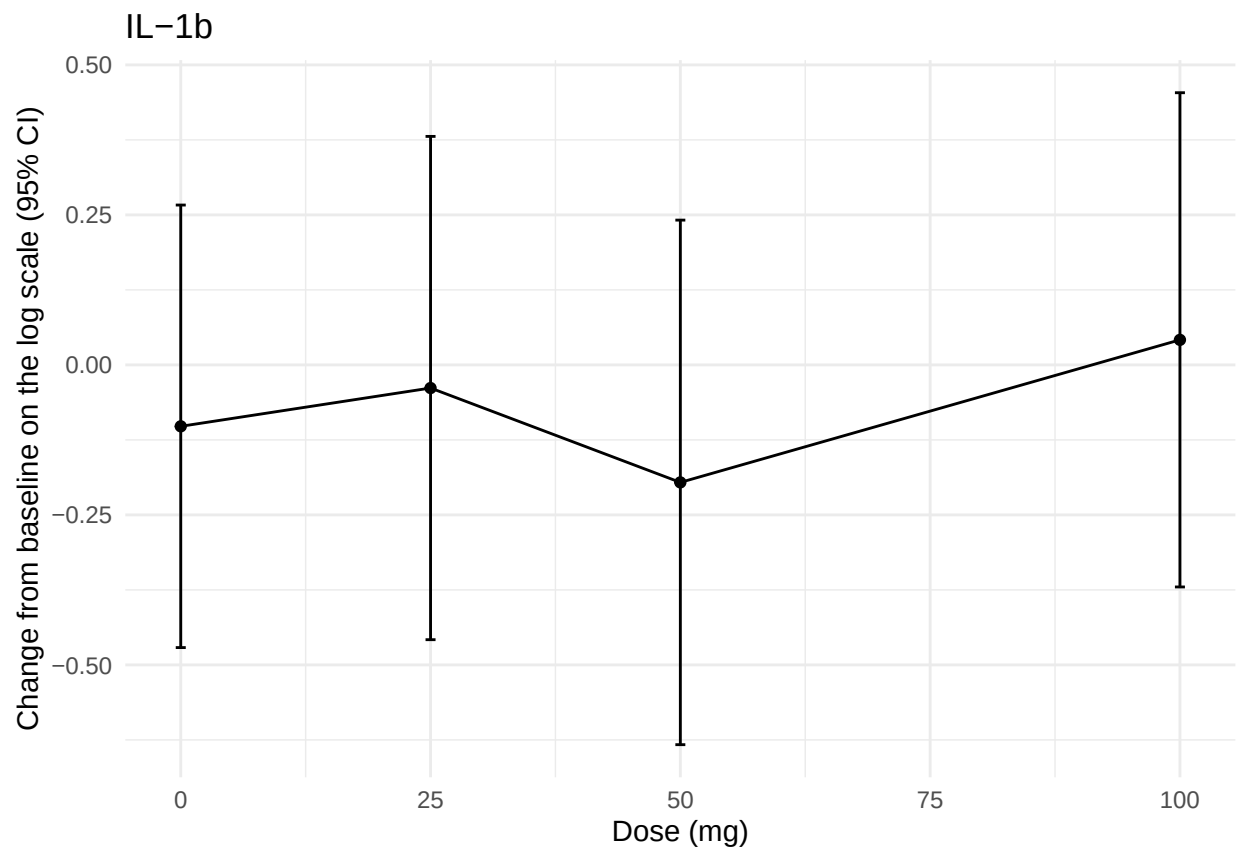
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	0.06 (-0.57 to 0.70)
losartan 50 mg vs losartan 0 mg0	-0.09 (-0.73 to 0.55)
losartan 100 mg vs losartan 0 mg0	0.14 (-0.48 to 0.77)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	511.2	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	288.0	TRUE

Plot



IL-6

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.369 (-0.916 to 0.178)
25 mg	0.024 (-0.553 to 0.601)
50 mg	-0.088 (-0.710 to 0.534)
100 mg	-0.131 (-0.682 to 0.420)

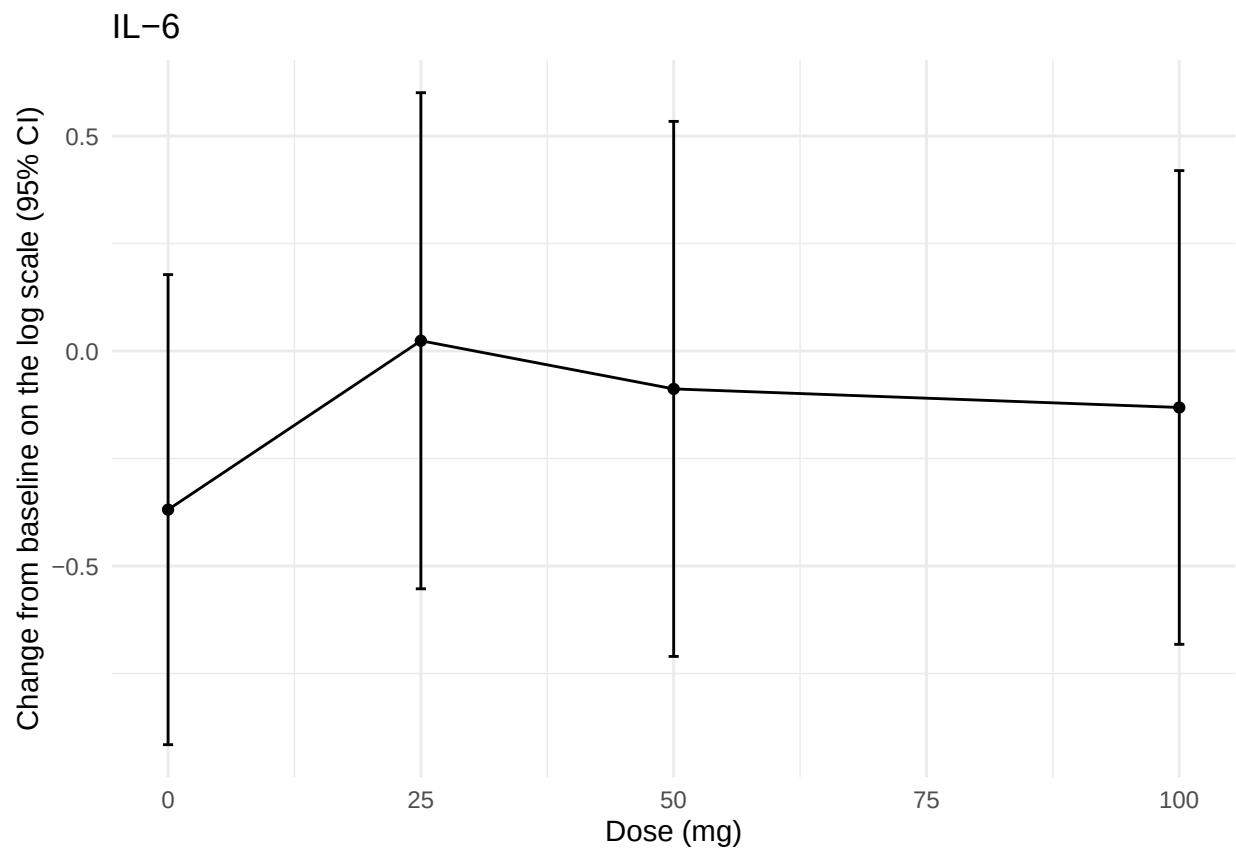
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg	0.39 (-0.62 to 1.40)
losartan 50 mg vs losartan 0 mg	0.28 (-0.76 to 1.32)
losartan 100 mg vs losartan 0 mg	0.24 (-0.74 to 1.21)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	390.6	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	364.0	TRUE

Plot



IL-8

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.056 (-0.171 to 0.059)
25 mg	-0.164 (-0.289 to -0.039)
50 mg	0.038 (-0.096 to 0.173)
100 mg	-0.108 (-0.230 to 0.013)

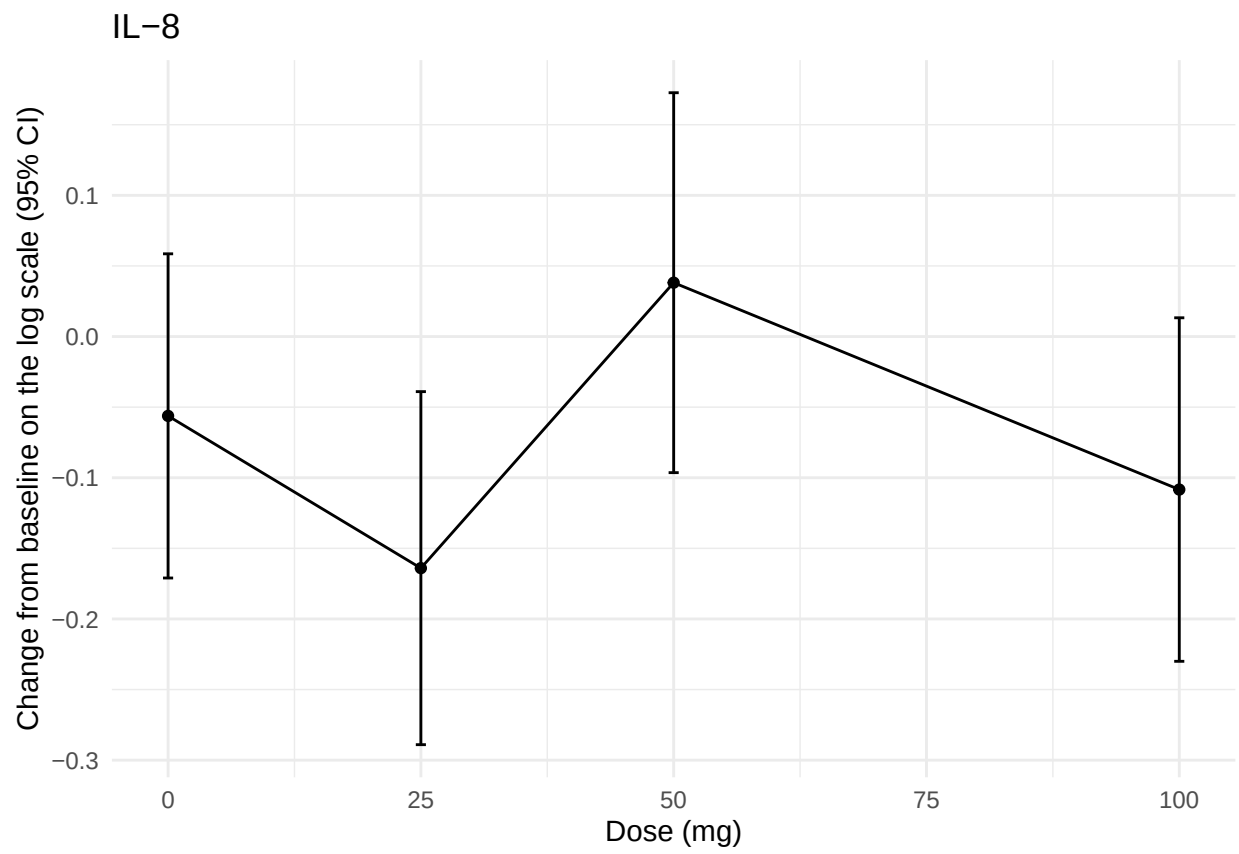
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	-0.11 (-0.32 to 0.10)
losartan 50 mg vs losartan 0 mg0	0.09 (-0.12 to 0.31)
losartan 100 mg vs losartan 0 mg0	-0.05 (-0.25 to 0.15)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	423.1	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	60.3	TRUE

Plot



MCP-1

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.174 (-0.409 to 0.062)
25 mg	-0.157 (-0.419 to 0.104)
50 mg	0.028 (-0.246 to 0.303)
100 mg	-0.085 (-0.340 to 0.170)

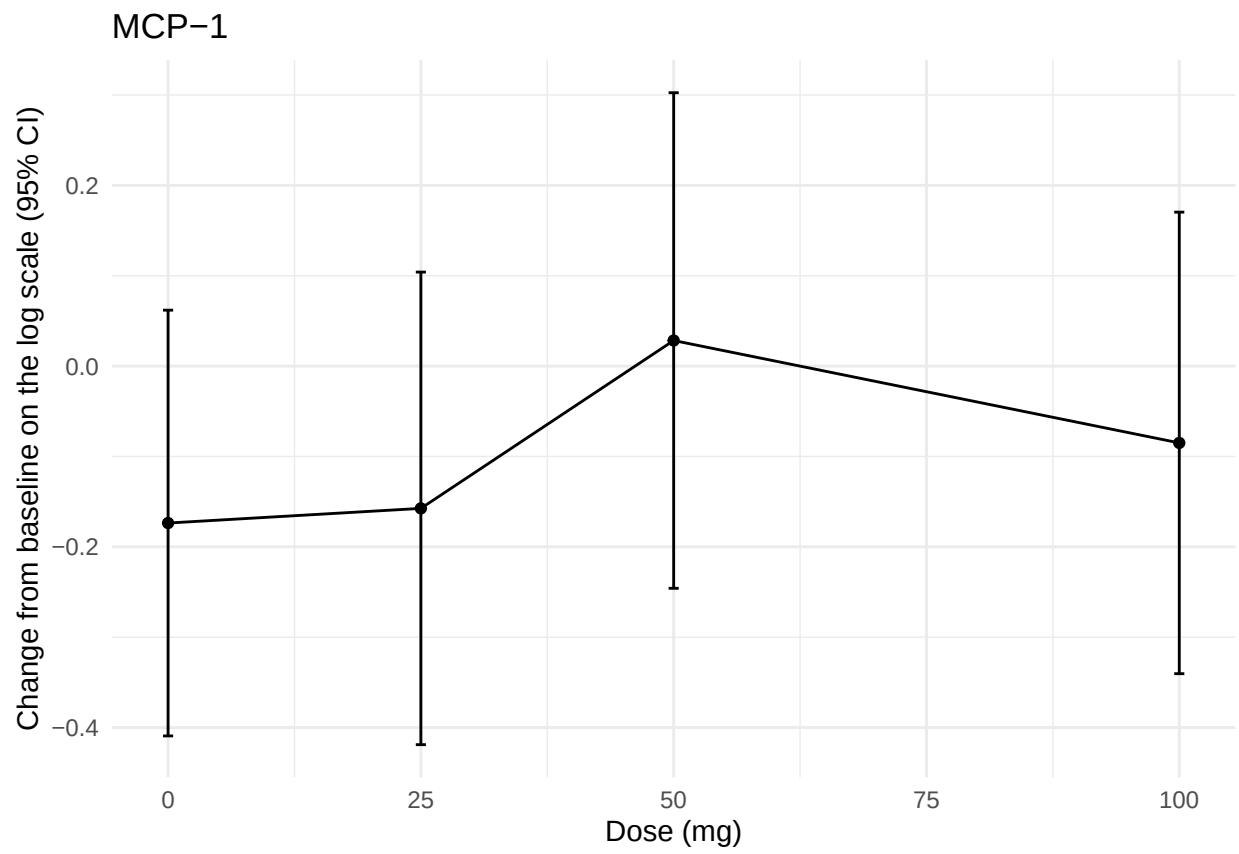
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	0.02 (-0.41 to 0.44)
losartan 50 mg vs losartan 0 mg0	0.20 (-0.22 to 0.63)
losartan 100 mg vs losartan 0 mg0	0.09 (-0.32 to 0.50)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	1148.1	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	199.4	TRUE

Plot



MMP-10

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	0.104 (0.008 to 0.200)
25 mg	0.039 (-0.066 to 0.145)
50 mg	0.015 (-0.095 to 0.125)
100 mg	0.045 (-0.057 to 0.146)

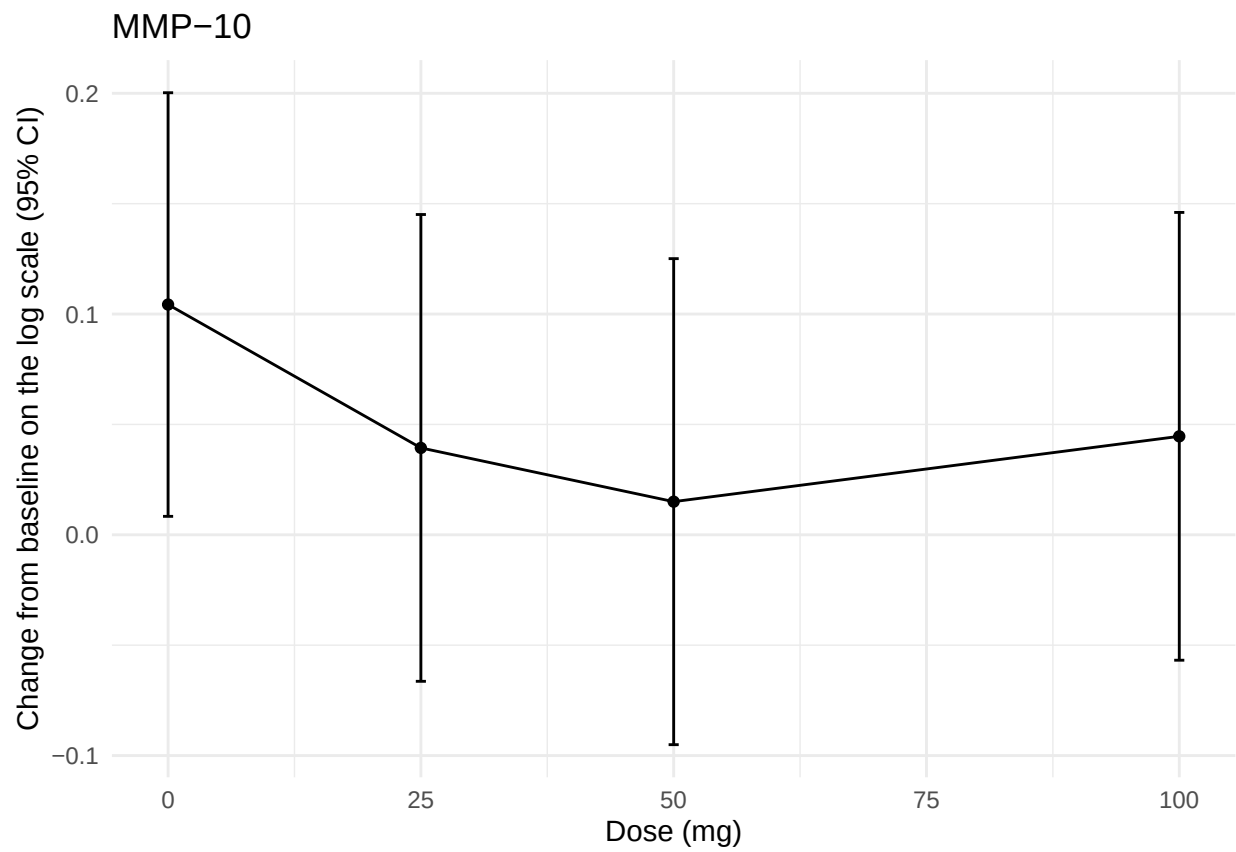
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	-0.06 (-0.24 to 0.11)
losartan 50 mg vs losartan 0 mg0	-0.09 (-0.26 to 0.09)
losartan 100 mg vs losartan 0 mg0	-0.06 (-0.23 to 0.11)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	1254.9	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	25.1	TRUE

Plot



PDGF-DD

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.301 (-0.536 to -0.067)
25 mg	-0.237 (-0.499 to 0.026)
50 mg	-0.296 (-0.573 to -0.020)
100 mg	-0.261 (-0.516 to -0.005)

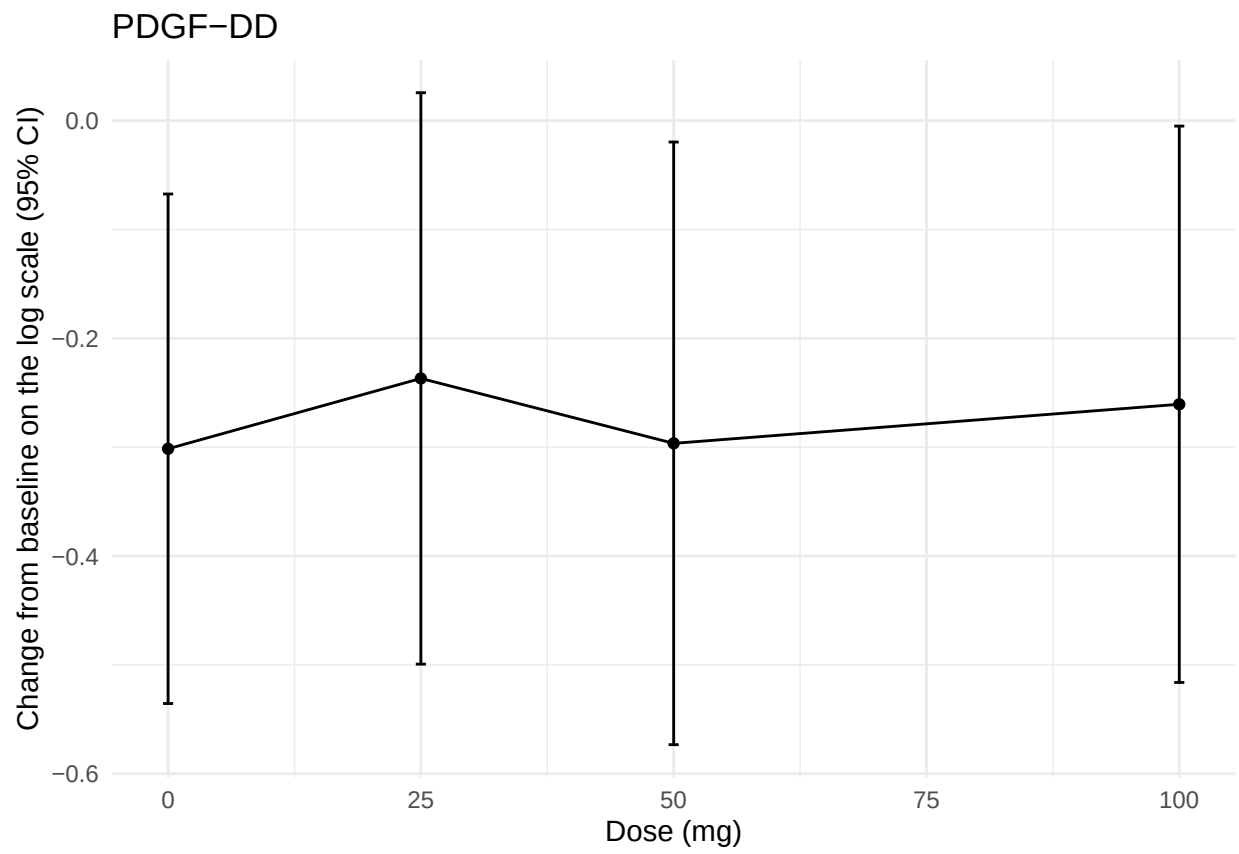
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	0.06 (-0.35 to 0.48)
losartan 50 mg vs losartan 0 mg0	0.00 (-0.42 to 0.43)
losartan 100 mg vs losartan 0 mg0	0.04 (-0.37 to 0.45)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	1109.7	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	198.7	TRUE

Plot



PECAM-1

A total of 52 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.482 (-0.789 to -0.174)
25 mg	-0.126 (-0.446 to 0.193)
50 mg	-0.373 (-0.714 to -0.032)
100 mg	0.074 (-0.237 to 0.384)

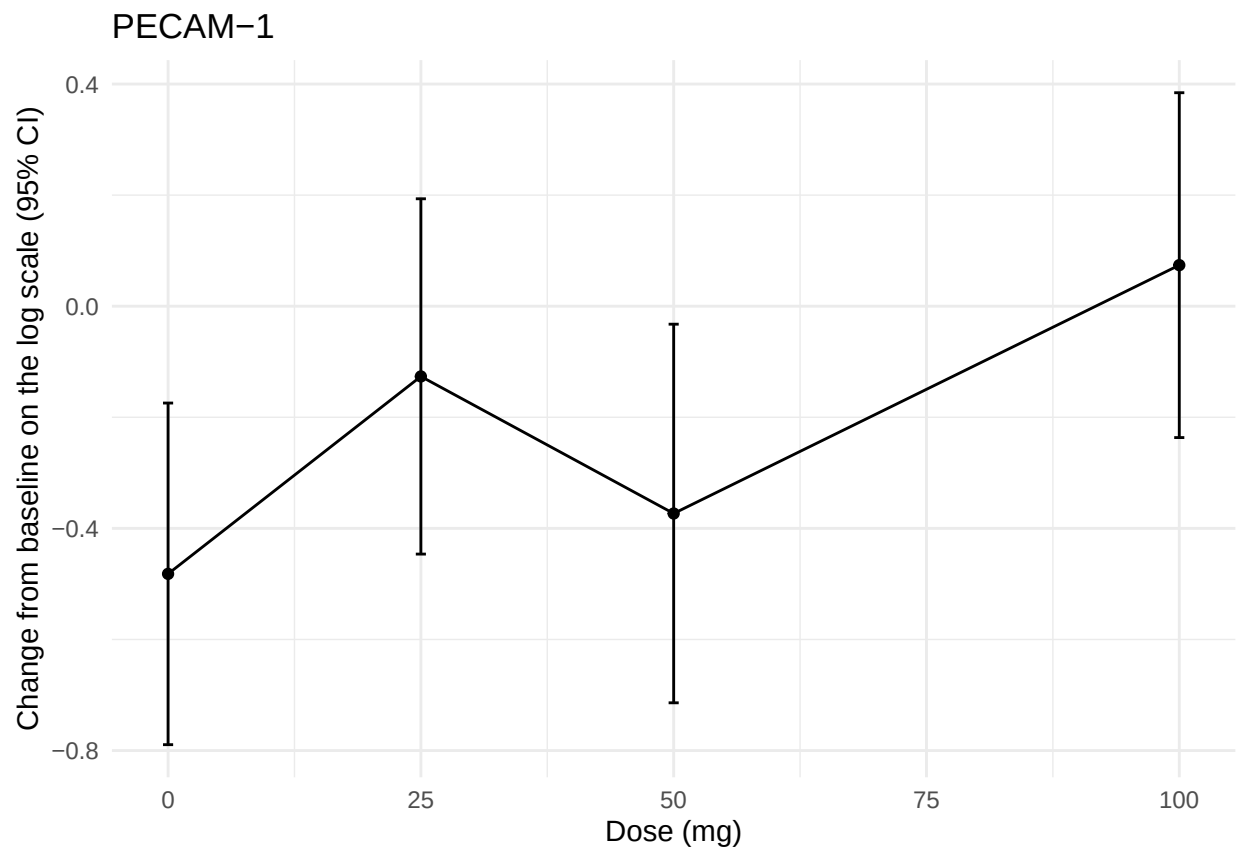
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	0.36 (-0.21 to 0.92)
losartan 50 mg vs losartan 0 mg0	0.11 (-0.47 to 0.68)
losartan 100 mg vs losartan 0 mg0	0.56 (0.01 to 1.11)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	2298.5	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	246.1	TRUE

Plot



TGF-alpha

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.072 (-0.395 to 0.251)
25 mg	0.129 (-0.220 to 0.479)
50 mg	0.045 (-0.321 to 0.411)
100 mg	0.125 (-0.213 to 0.463)

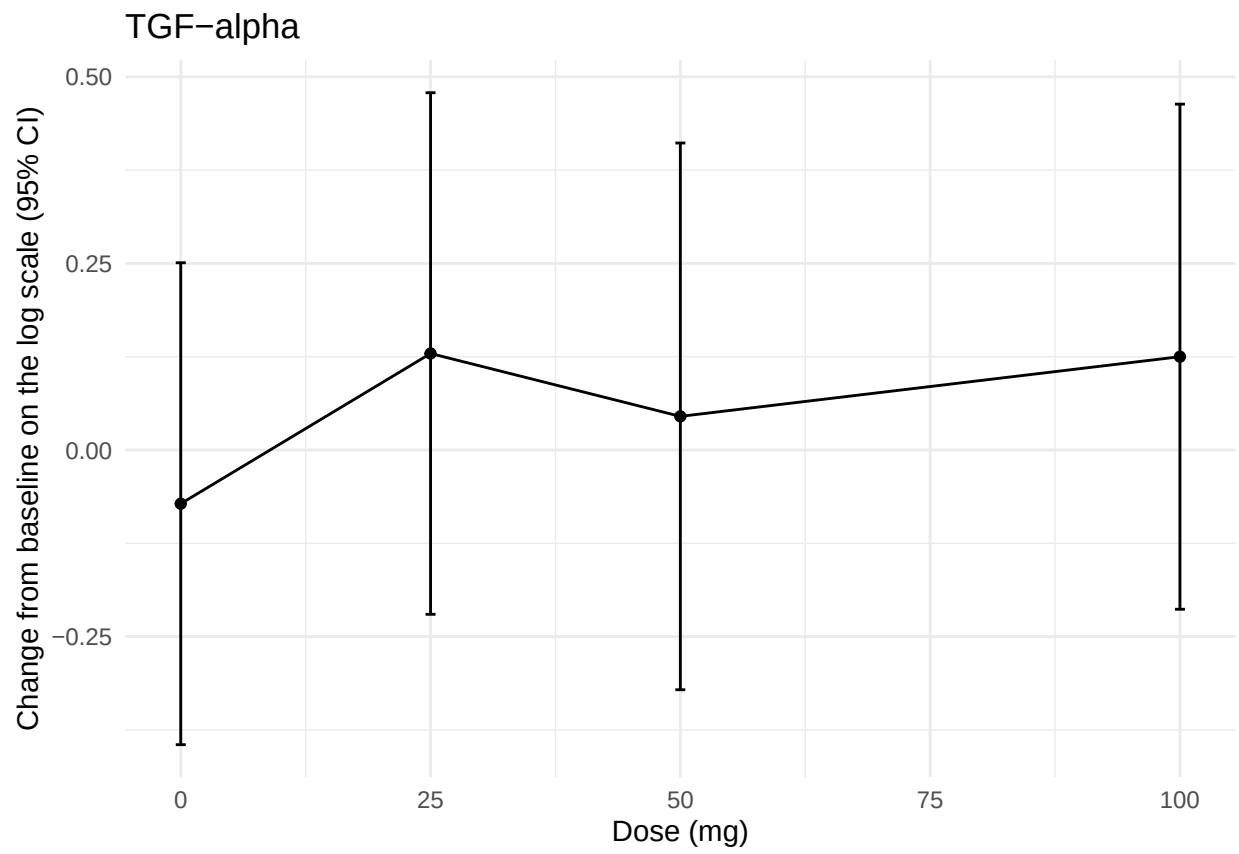
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg	0.20 (-0.39 to 0.79)
losartan 50 mg vs losartan 0 mg	0.12 (-0.48 to 0.71)
losartan 100 mg vs losartan 0 mg	0.20 (-0.38 to 0.78)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	546.5	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	262.3	TRUE

Plot



Thrombospondin-2

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.114 (-0.244 to 0.017)
25 mg	-0.059 (-0.207 to 0.088)
50 mg	-0.111 (-0.264 to 0.042)
100 mg	-0.067 (-0.211 to 0.078)

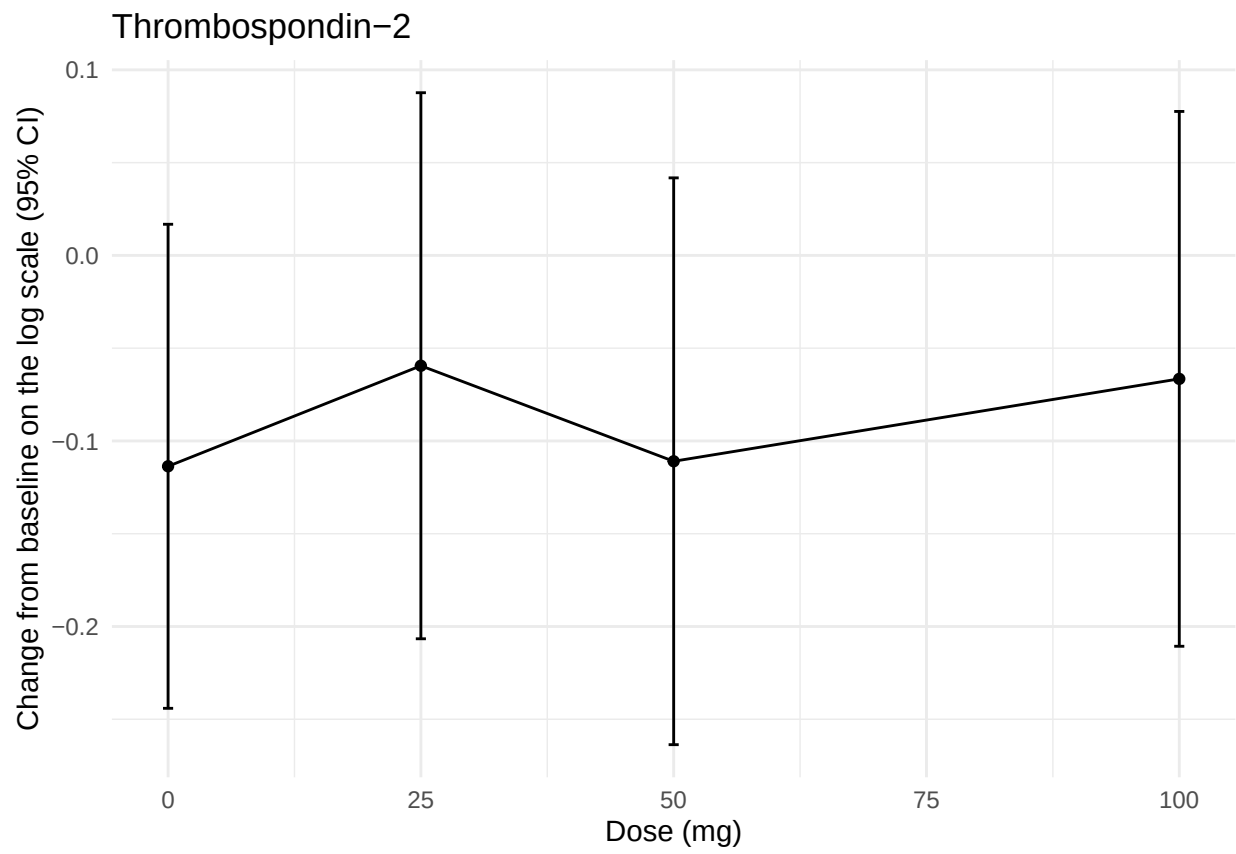
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	0.05 (-0.17 to 0.28)
losartan 50 mg vs losartan 0 mg0	0.00 (-0.22 to 0.23)
losartan 100 mg vs losartan 0 mg0	0.05 (-0.18 to 0.27)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	1882.7	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	84.5	TRUE

Plot



VEGF-A

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.280 (-0.507 to -0.052)
25 mg	-0.470 (-0.720 to -0.219)
50 mg	-0.344 (-0.607 to -0.081)
100 mg	-0.354 (-0.598 to -0.111)

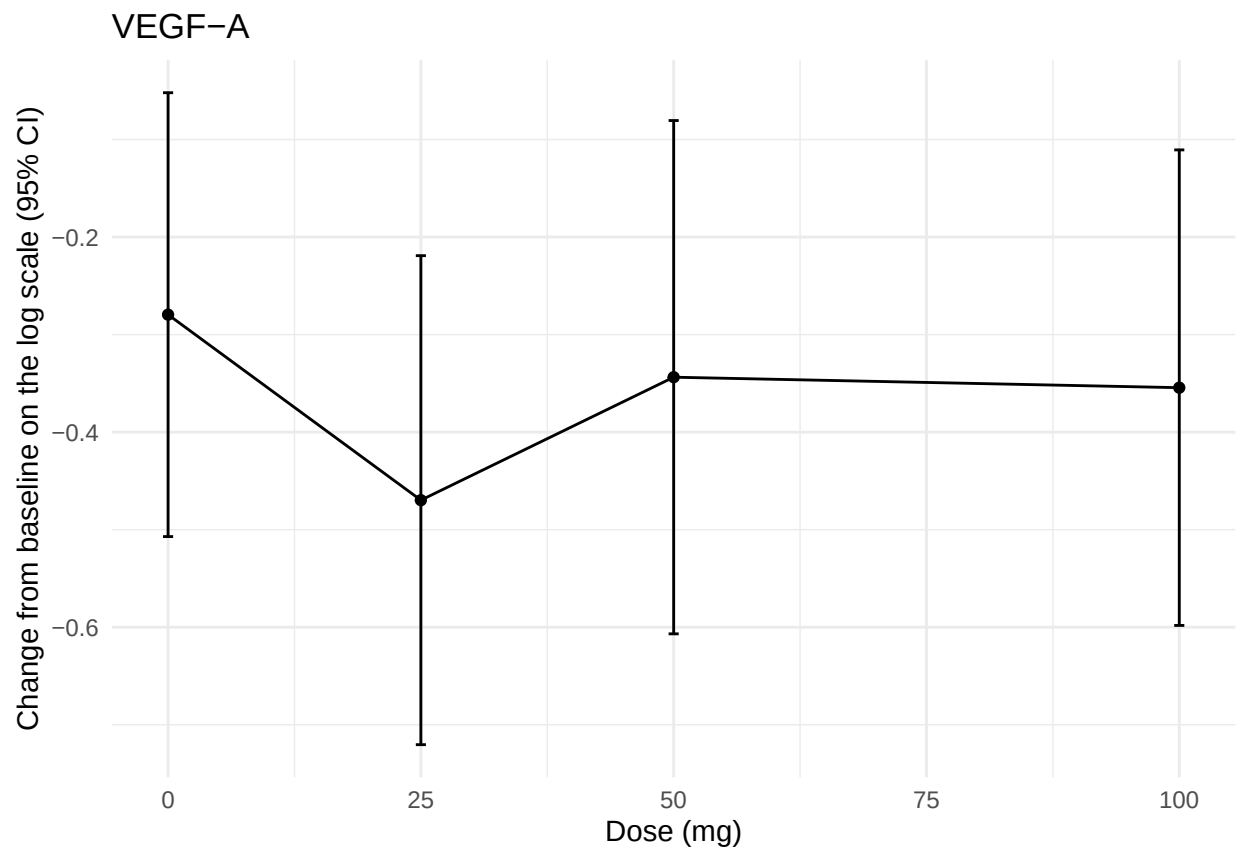
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	-0.19 (-0.60 to 0.22)
losartan 50 mg vs losartan 0 mg0	-0.06 (-0.48 to 0.35)
losartan 100 mg vs losartan 0 mg0	-0.07 (-0.48 to 0.33)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	1079.9	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	193.1	TRUE

Plot



VEGF-C

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.133 (-0.307 to 0.040)
25 mg	-0.101 (-0.299 to 0.096)
50 mg	-0.181 (-0.387 to 0.025)
100 mg	-0.068 (-0.263 to 0.128)

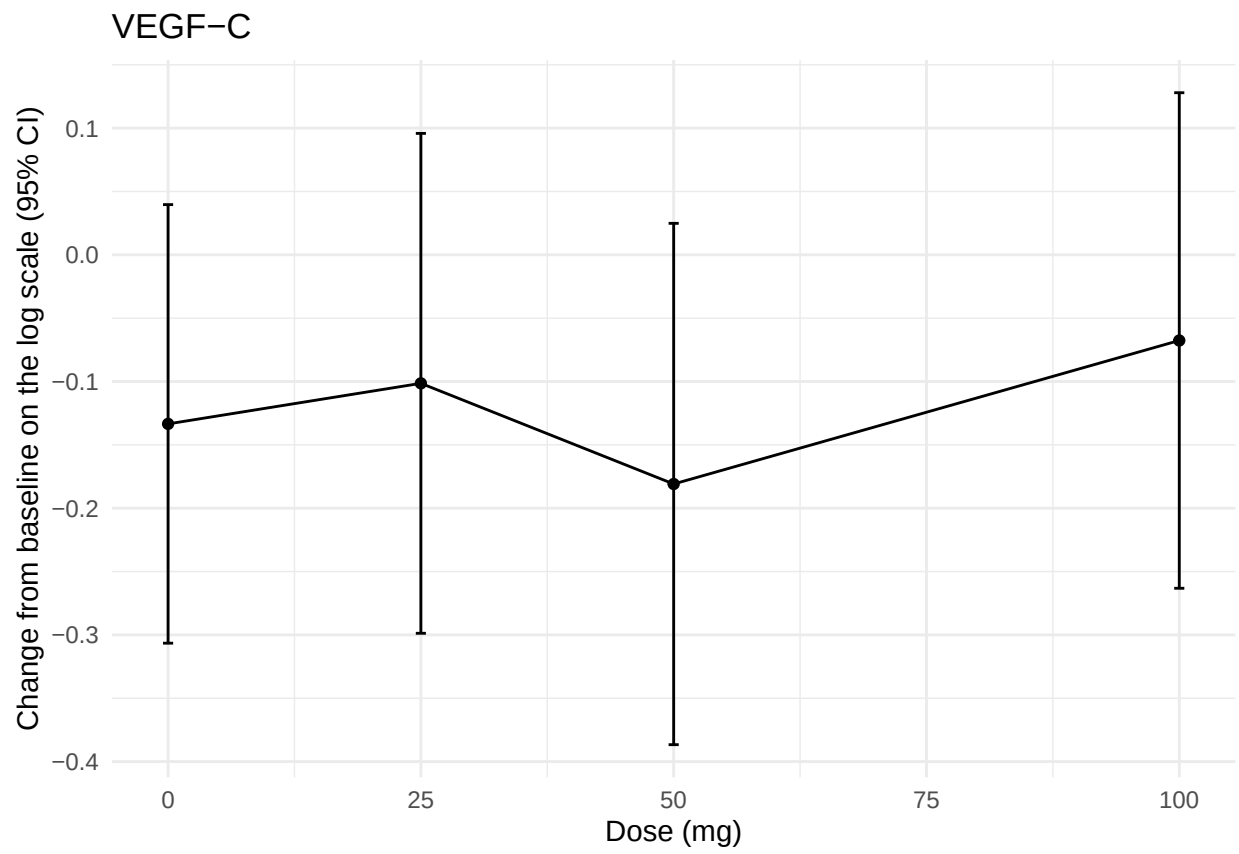
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	0.03 (-0.26 to 0.33)
losartan 50 mg vs losartan 0 mg0	-0.05 (-0.34 to 0.25)
losartan 100 mg vs losartan 0 mg0	0.07 (-0.23 to 0.36)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	1479.2	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	140.2	TRUE

Plot



Angiopoietin-1

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.165 (-0.402 to 0.071)
25 mg	-0.273 (-0.530 to -0.017)
50 mg	-0.137 (-0.410 to 0.135)
100 mg	-0.202 (-0.450 to 0.046)

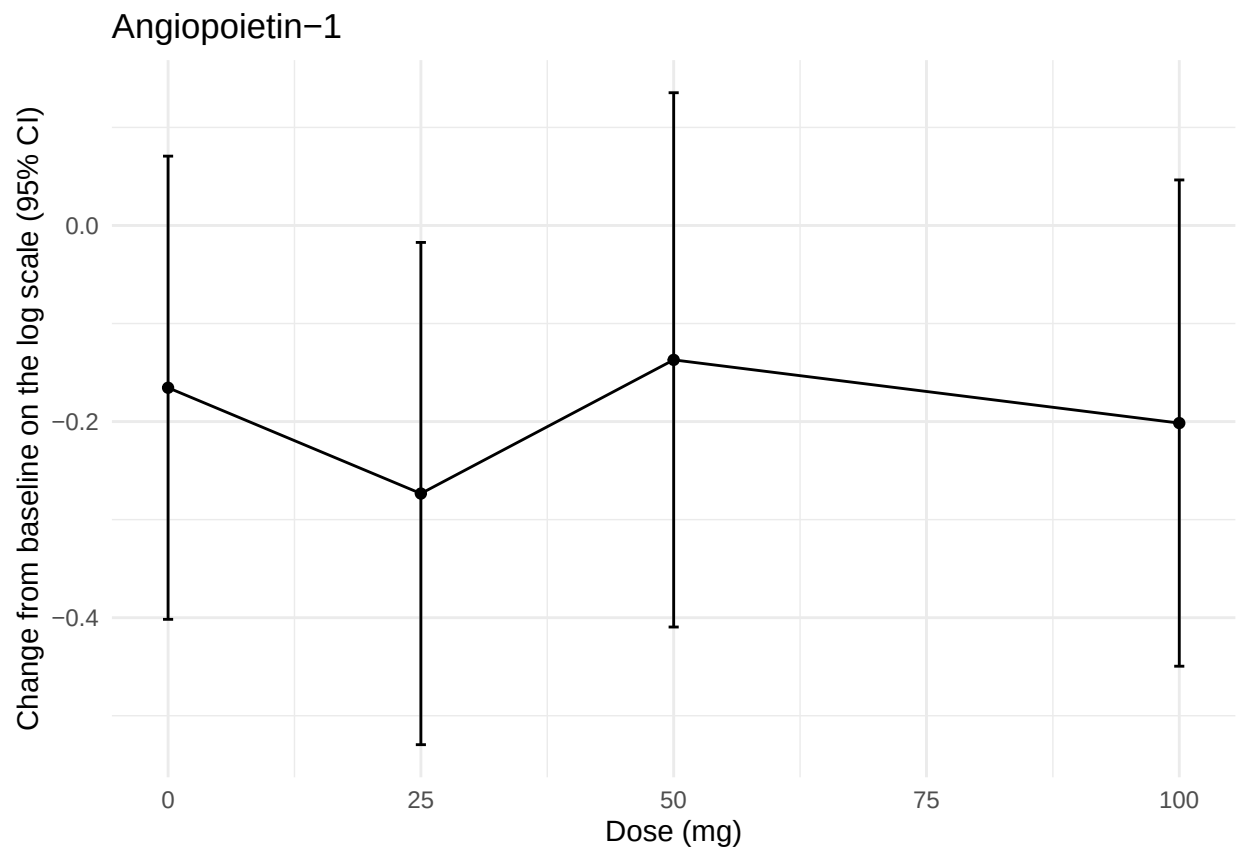
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	-0.11 (-0.54 to 0.32)
losartan 50 mg vs losartan 0 mg0	0.03 (-0.41 to 0.47)
losartan 100 mg vs losartan 0 mg0	-0.04 (-0.45 to 0.38)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	1922.0	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	199.9	TRUE

Plot



Angiopoietin-2

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.044 (-0.124 to 0.036)
25 mg	-0.034 (-0.121 to 0.054)
50 mg	0.013 (-0.079 to 0.105)
100 mg	-0.091 (-0.176 to -0.007)

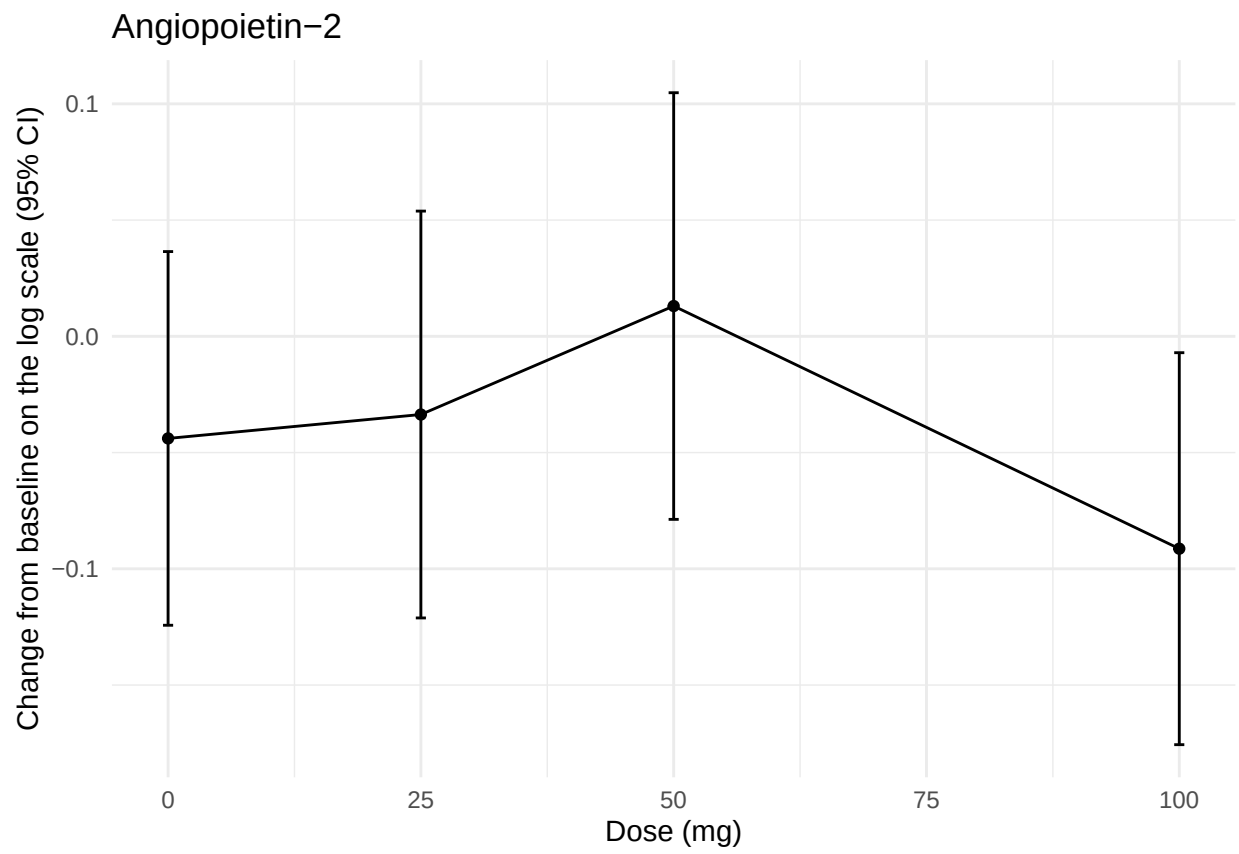
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	0.01 (-0.14 to 0.16)
losartan 50 mg vs losartan 0 mg0	0.06 (-0.09 to 0.20)
losartan 100 mg vs losartan 0 mg0	-0.05 (-0.19 to 0.10)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	1602.8	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	-7.9	TRUE

Plot



ANGPTL4

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.005 (-0.103 to 0.092)
25 mg	-0.118 (-0.228 to -0.009)
50 mg	0.025 (-0.088 to 0.138)
100 mg	-0.081 (-0.187 to 0.025)

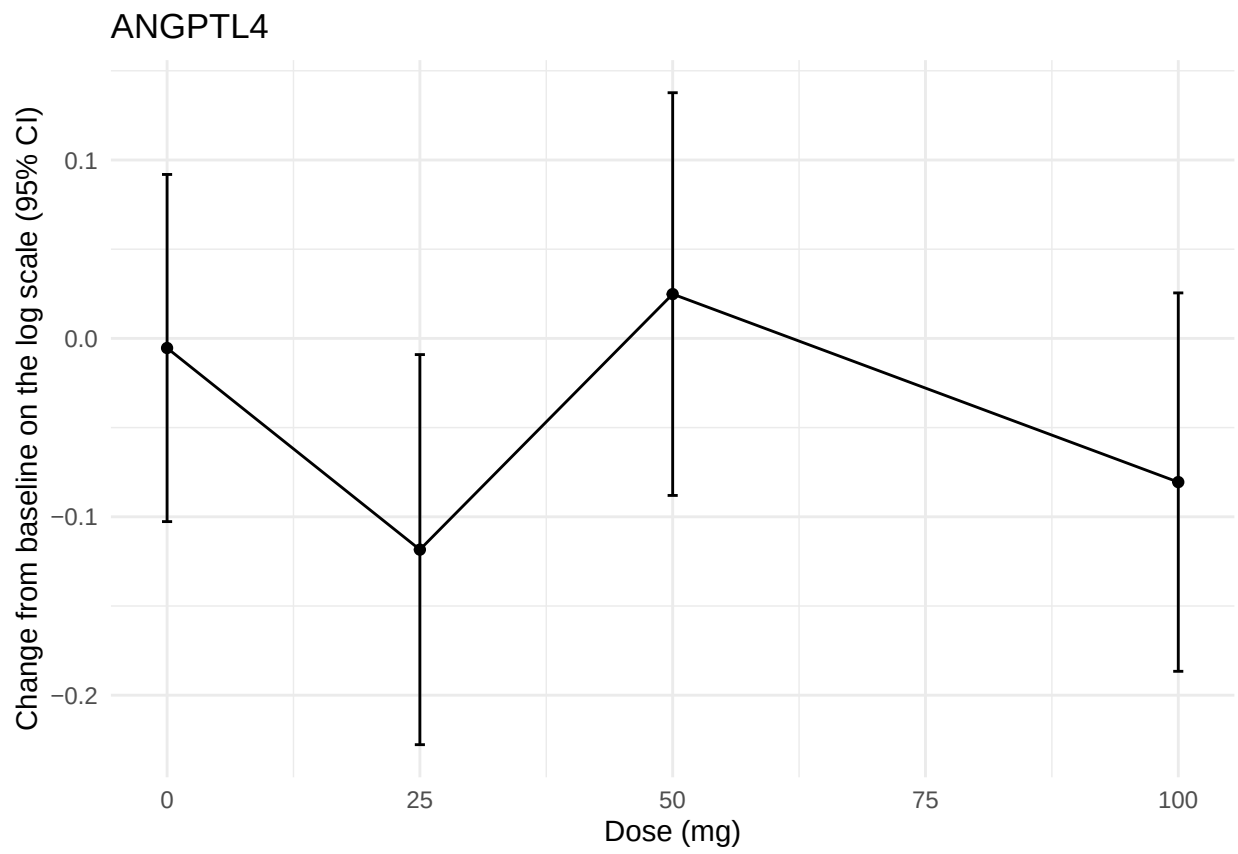
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	-0.11 (-0.29 to 0.06)
losartan 50 mg vs losartan 0 mg0	0.03 (-0.14 to 0.21)
losartan 100 mg vs losartan 0 mg0	-0.08 (-0.25 to 0.10)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	2347.3	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	28.2	TRUE

Plot



CA9

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.024 (-0.302 to 0.255)
25 mg	0.119 (-0.195 to 0.433)
50 mg	0.167 (-0.161 to 0.495)
100 mg	0.038 (-0.268 to 0.343)

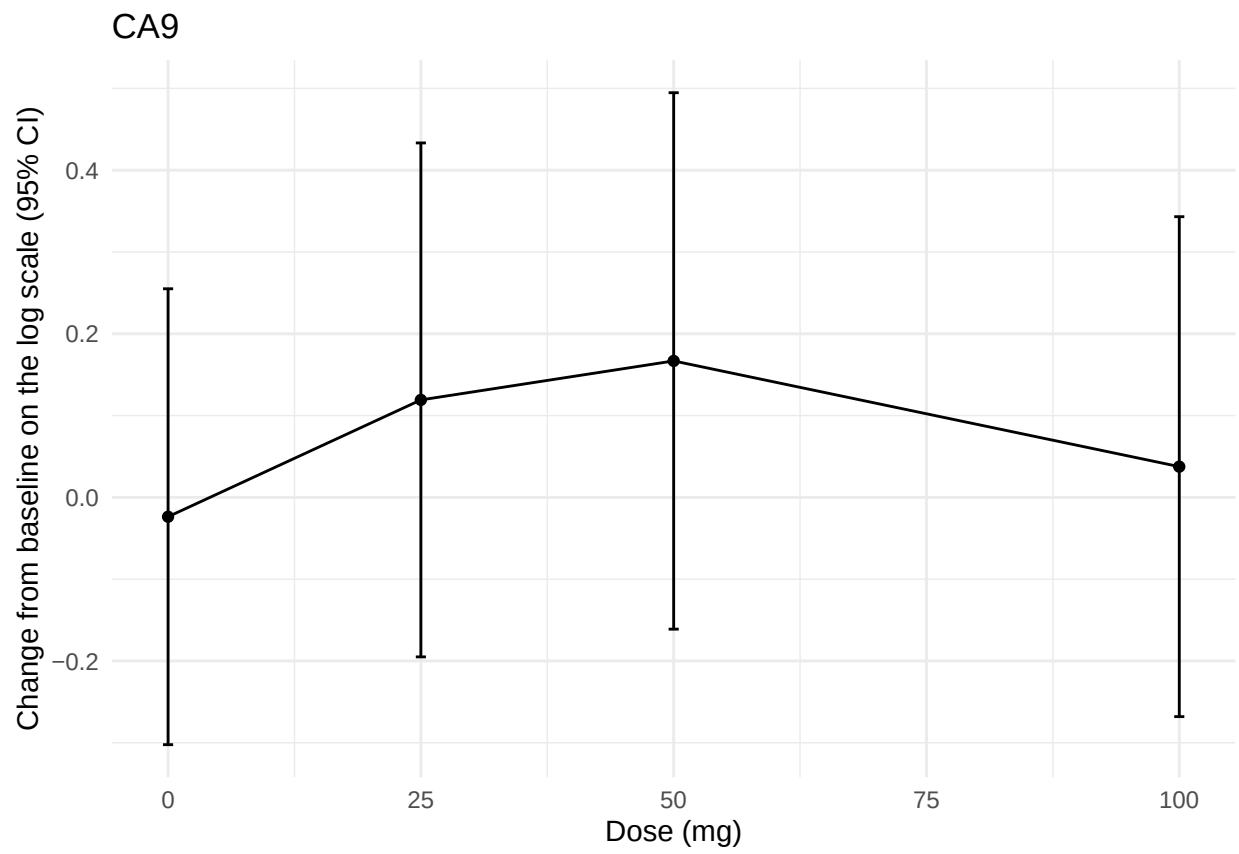
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	0.14 (-0.35 to 0.64)
losartan 50 mg vs losartan 0 mg0	0.19 (-0.31 to 0.69)
losartan 100 mg vs losartan 0 mg0	0.06 (-0.42 to 0.55)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	1073.9	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	233.3	TRUE

Plot



CD163

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	0.095 (0.001 to 0.189)
25 mg	0.137 (0.034 to 0.241)
50 mg	0.117 (0.009 to 0.226)
100 mg	0.038 (-0.063 to 0.139)

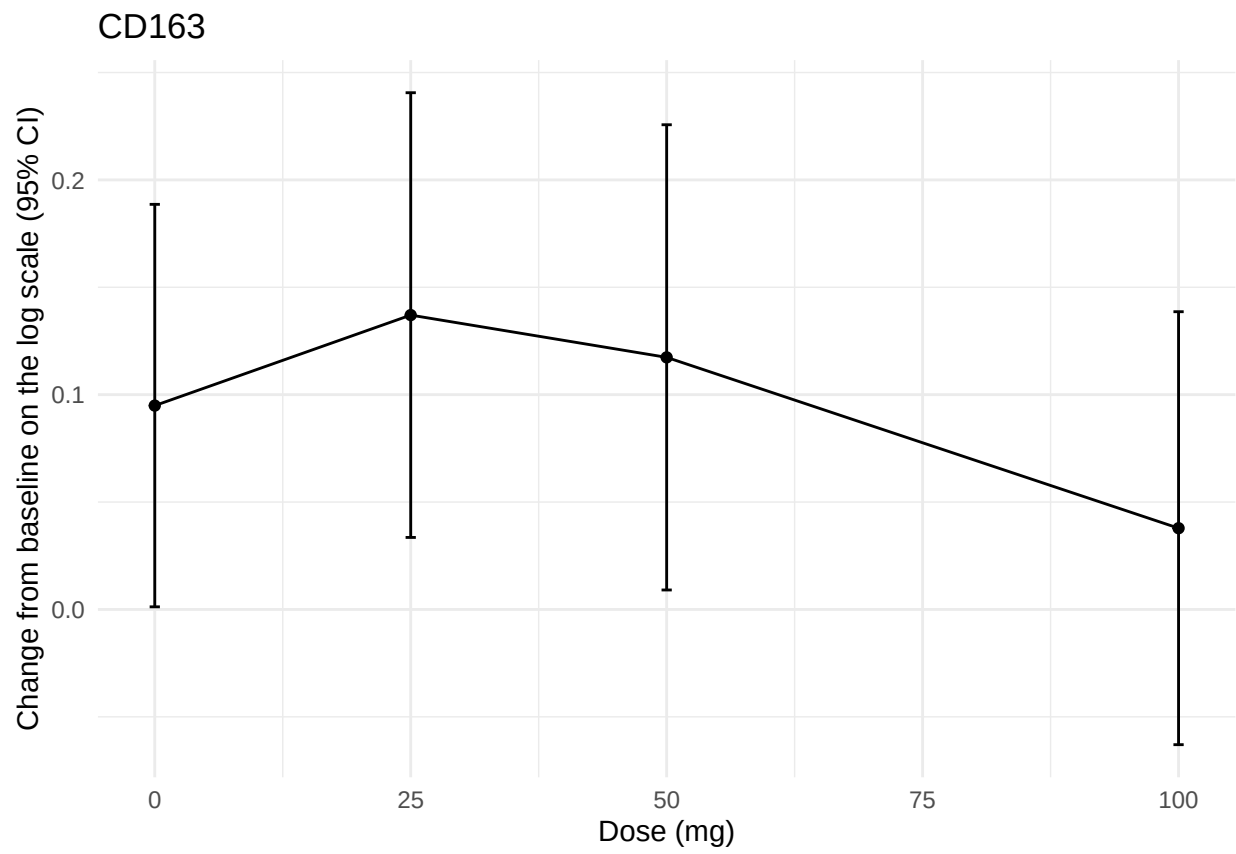
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	0.04 (-0.13 to 0.21)
losartan 50 mg vs losartan 0 mg0	0.02 (-0.15 to 0.19)
losartan 100 mg vs losartan 0 mg0	-0.06 (-0.22 to 0.11)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	2660.3	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	21.8	TRUE

Plot



Collagen IV alpha 1

A total of 52 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	0.127 (0.038 to 0.216)
25 mg	0.088 (-0.012 to 0.189)
50 mg	0.106 (0.002 to 0.209)
100 mg	0.021 (-0.074 to 0.116)

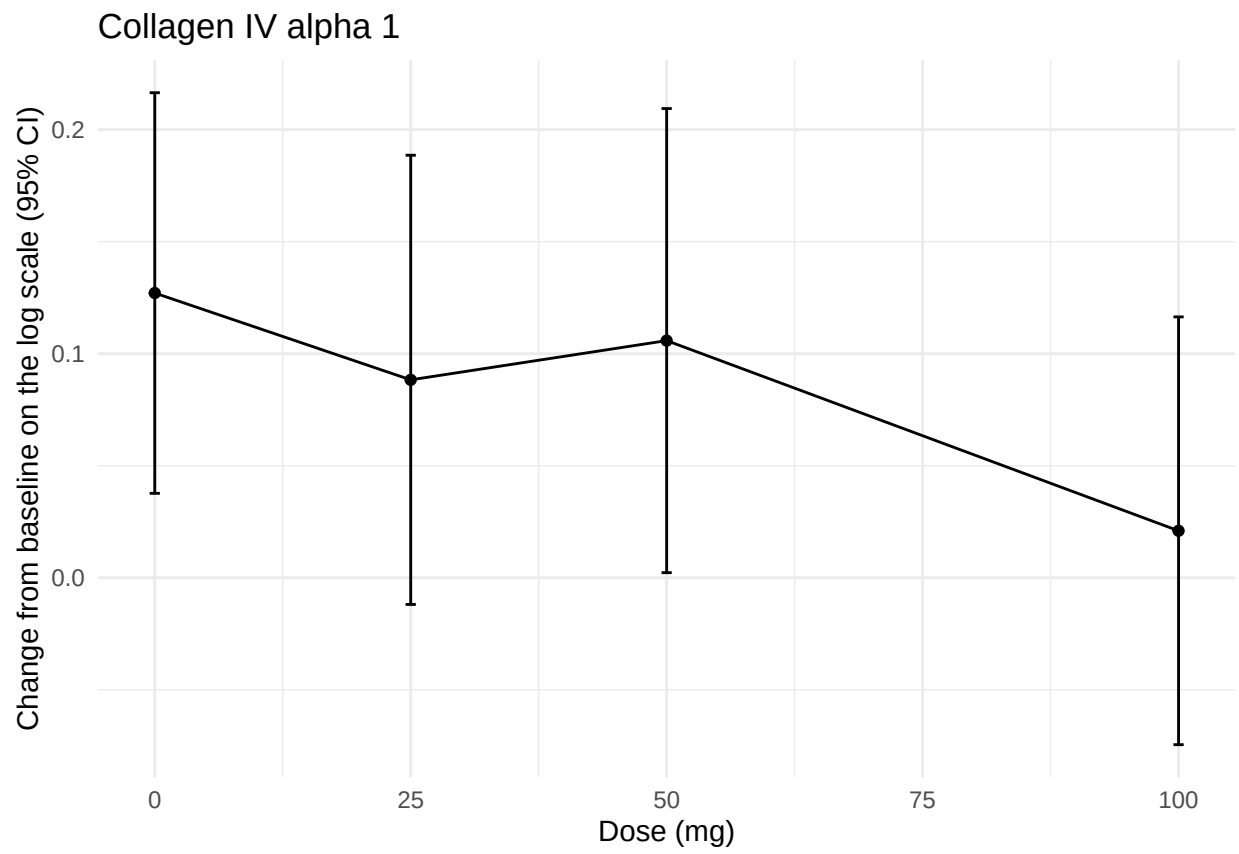
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	-0.04 (-0.20 to 0.12)
losartan 50 mg vs losartan 0 mg0	-0.02 (-0.18 to 0.14)
losartan 100 mg vs losartan 0 mg0	-0.11 (-0.26 to 0.05)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	1327.4	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	10.8	TRUE

Plot



EGF

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.734 (-1.519 to 0.051)
25 mg	-0.718 (-1.573 to 0.137)
50 mg	-0.298 (-1.202 to 0.605)
100 mg	-1.125 (-1.953 to -0.296)

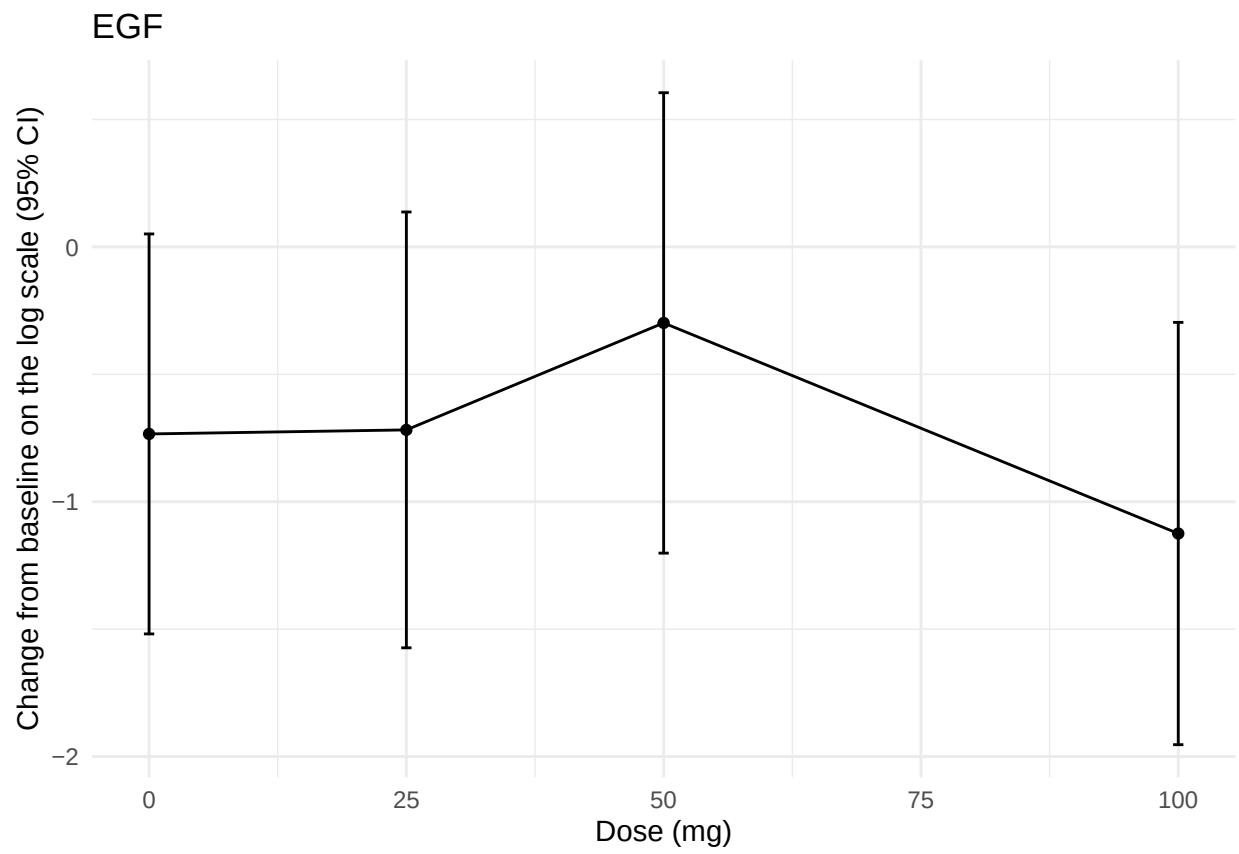
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	0.02 (-1.41 to 1.44)
losartan 50 mg vs losartan 0 mg0	0.44 (-1.01 to 1.88)
losartan 100 mg vs losartan 0 mg0	-0.39 (-1.78 to 1.00)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	871.8	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	432.7	TRUE

Plot



ErbB2

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	0.025 (-0.057 to 0.108)
25 mg	0.047 (-0.046 to 0.140)
50 mg	0.036 (-0.061 to 0.133)
100 mg	0.006 (-0.086 to 0.097)

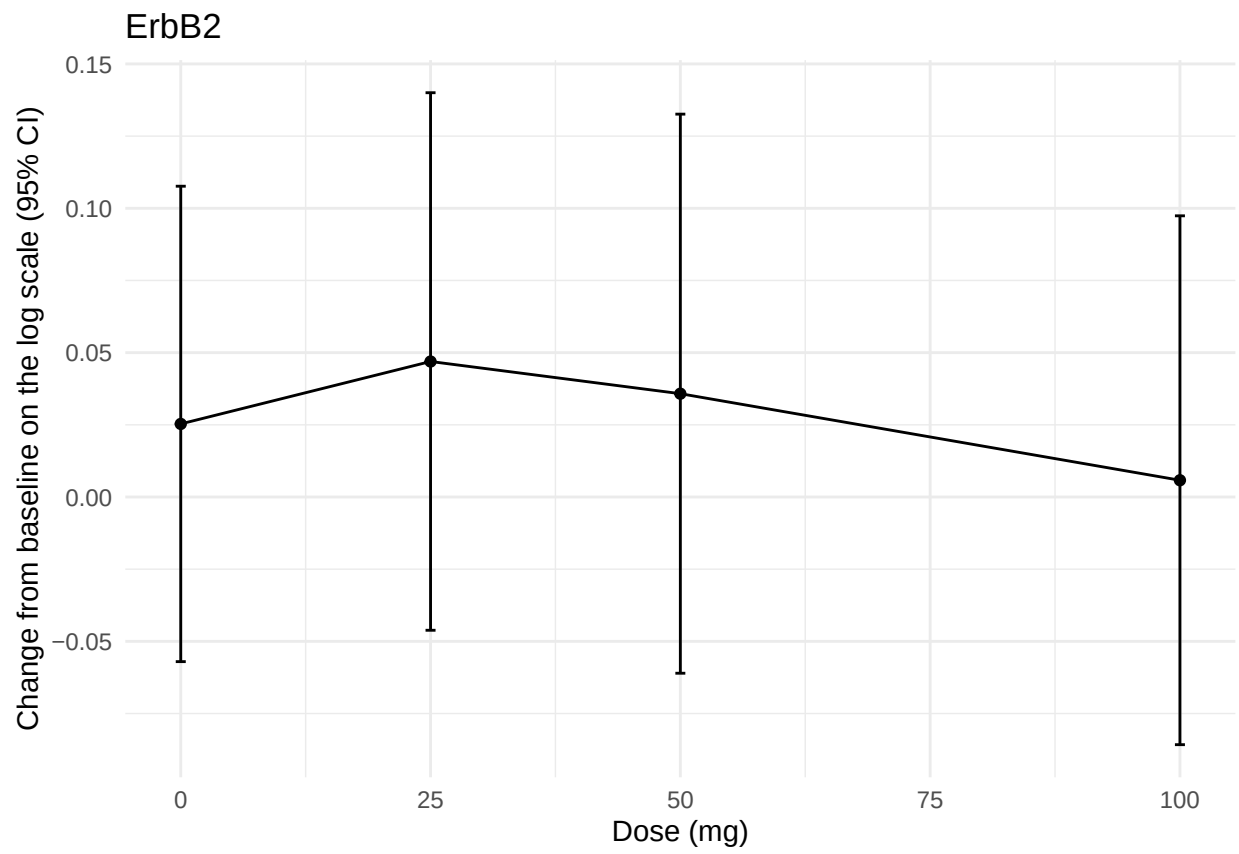
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	0.02 (-0.12 to 0.16)
losartan 50 mg vs losartan 0 mg0	0.01 (-0.13 to 0.15)
losartan 100 mg vs losartan 0 mg0	-0.02 (-0.16 to 0.12)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	1703.6	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	-3.4	TRUE

Plot



FGF Basic

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.255 (-0.502 to -0.007)
25 mg	-0.126 (-0.401 to 0.149)
50 mg	-0.289 (-0.577 to -0.002)
100 mg	-0.259 (-0.527 to 0.010)

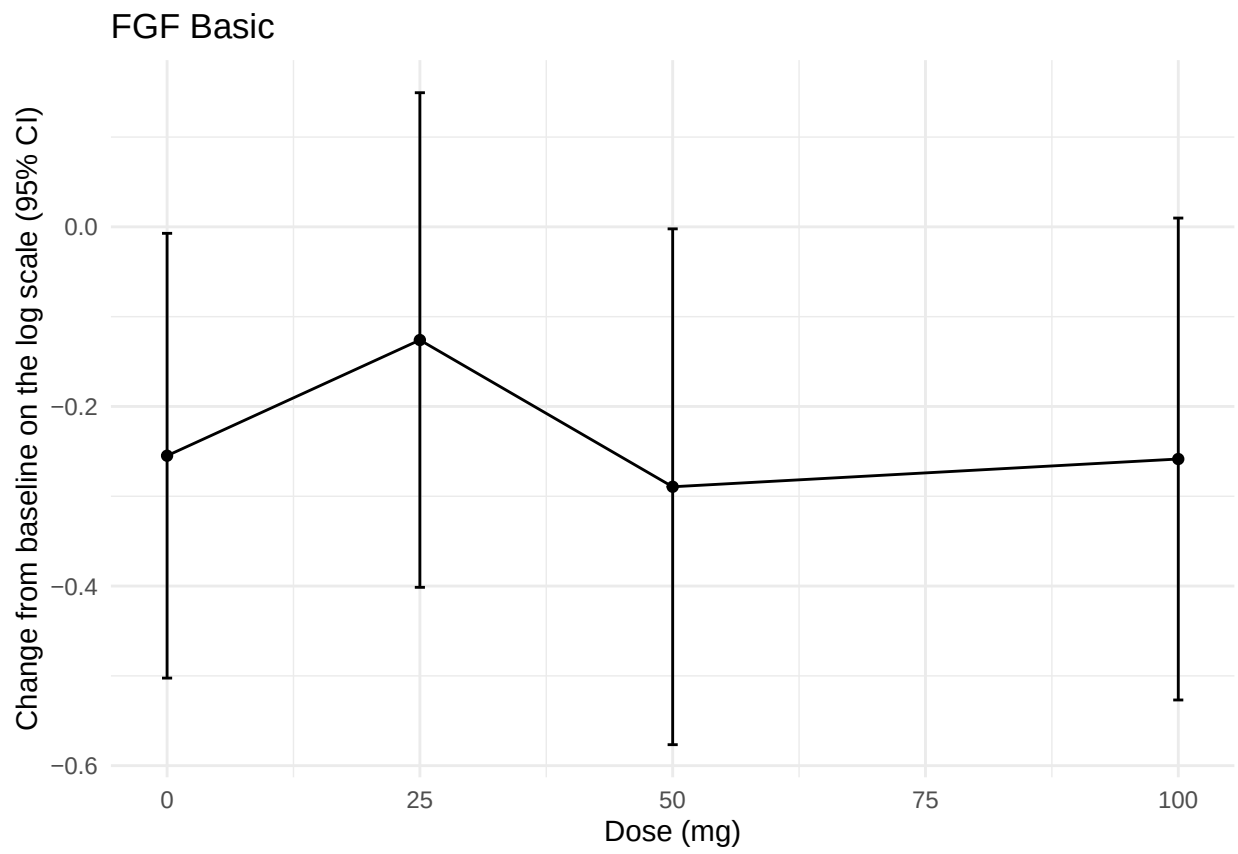
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	0.13 (-0.31 to 0.57)
losartan 50 mg vs losartan 0 mg0	-0.03 (-0.48 to 0.41)
losartan 100 mg vs losartan 0 mg0	0.00 (-0.44 to 0.43)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	648.8	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	209.3	TRUE

Plot



IL-12 p70

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	0.234 (-0.066 to 0.533)
25 mg	0.309 (-0.034 to 0.652)
50 mg	-0.049 (-0.404 to 0.306)
100 mg	0.058 (-0.283 to 0.400)

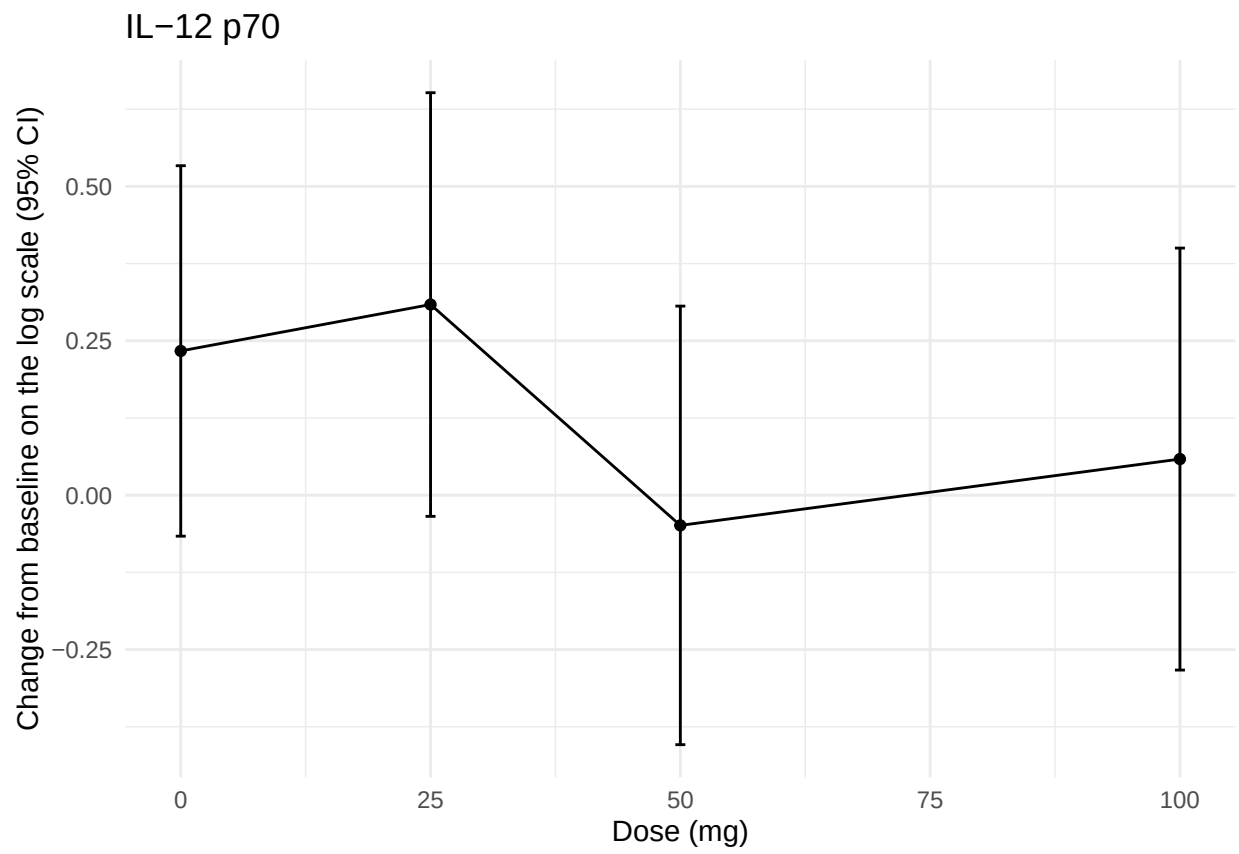
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	0.08 (-0.40 to 0.55)
losartan 50 mg vs losartan 0 mg0	-0.28 (-0.76 to 0.19)
losartan 100 mg vs losartan 0 mg0	-0.18 (-0.65 to 0.30)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	1279.5	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	245.6	TRUE

Plot



IL-2

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.365 (-0.891 to 0.161)
25 mg	-0.095 (-0.658 to 0.467)
50 mg	0.562 (-0.034 to 1.157)
100 mg	-0.396 (-0.934 to 0.143)

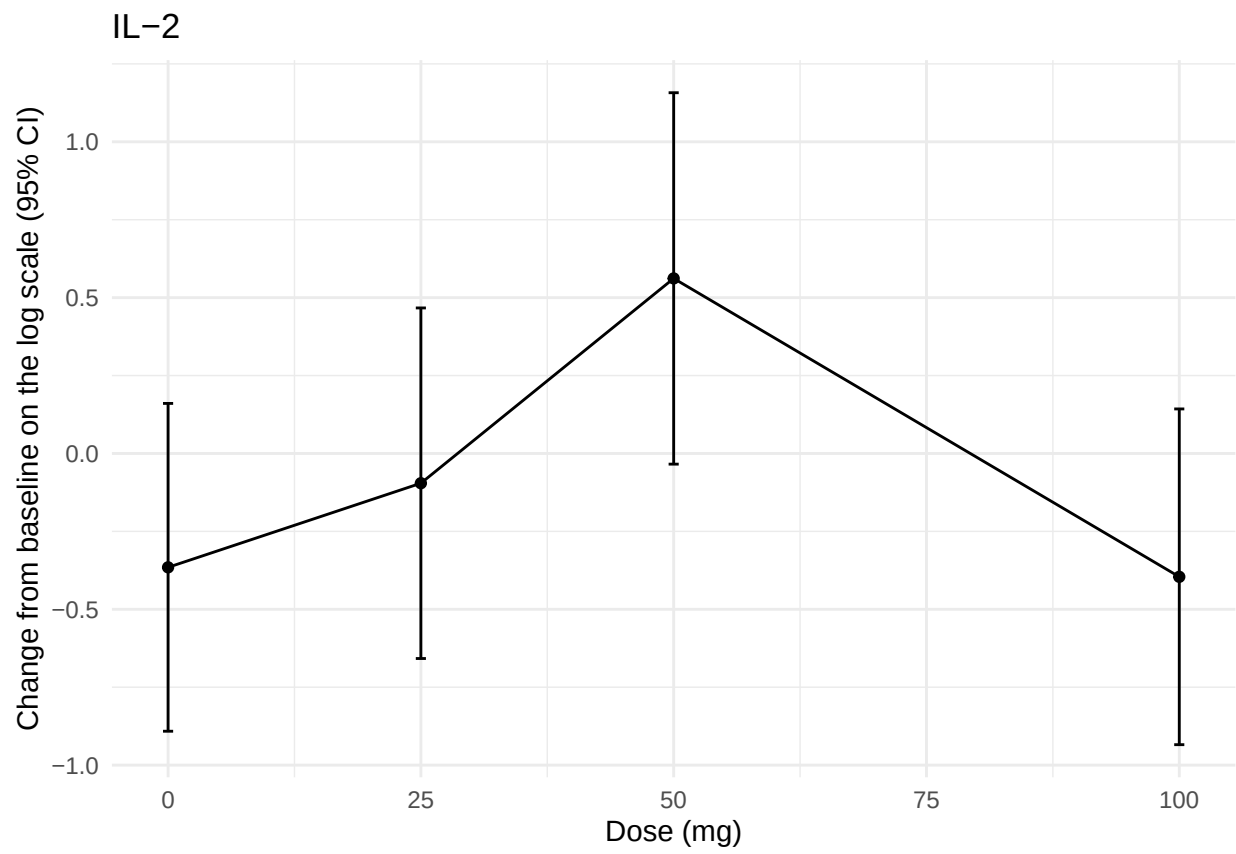
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	0.27 (-0.70 to 1.24)
losartan 50 mg vs losartan 0 mg0	0.93 (-0.05 to 1.91)
losartan 100 mg vs losartan 0 mg0	-0.03 (-0.97 to 0.91)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	783.0	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	356.8	TRUE

Plot



IP-10 (CXCL10)

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	0.229 (0.023 to 0.436)
25 mg	0.181 (-0.045 to 0.407)
50 mg	0.314 (0.077 to 0.550)
100 mg	0.119 (-0.099 to 0.338)

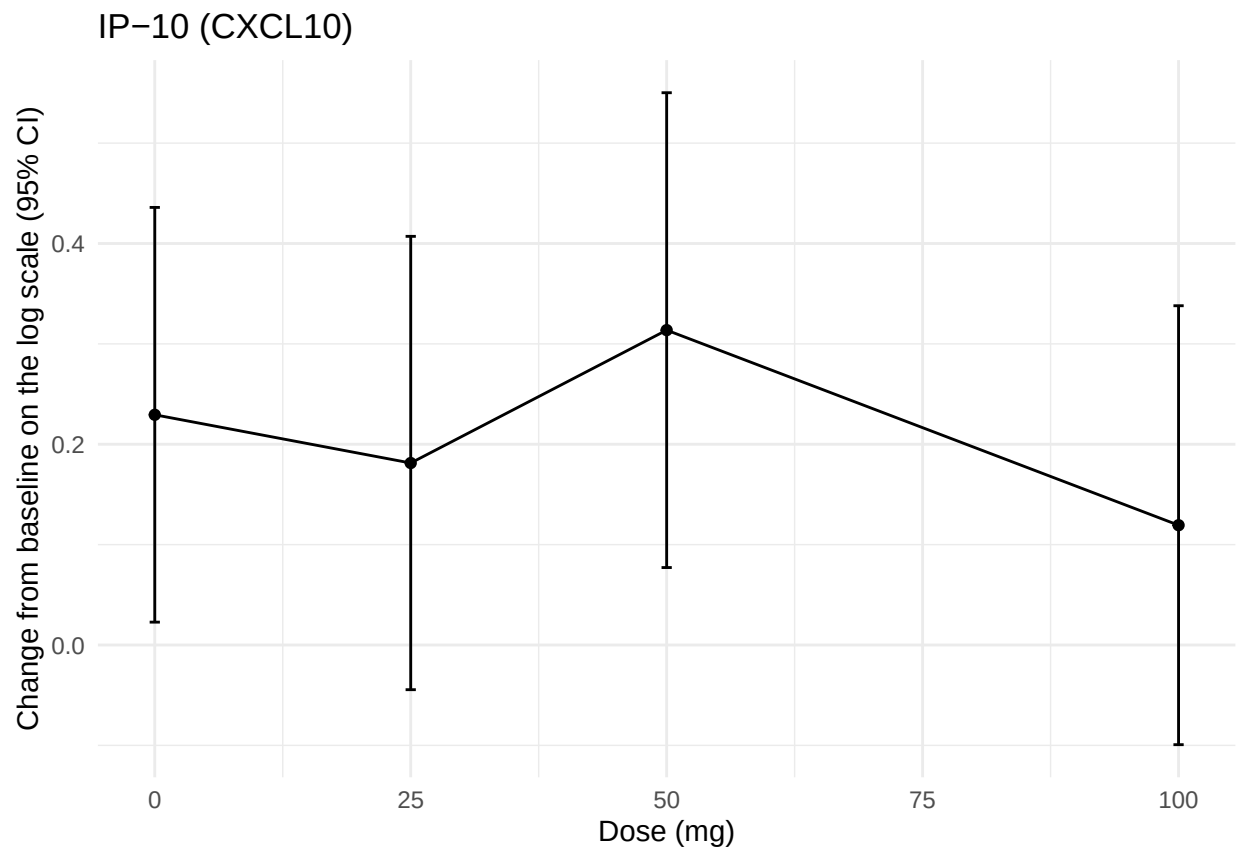
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	-0.05 (-0.42 to 0.33)
losartan 50 mg vs losartan 0 mg0	0.08 (-0.29 to 0.46)
losartan 100 mg vs losartan 0 mg0	-0.11 (-0.48 to 0.26)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	1152.5	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	173.8	TRUE

Plot



I-TAC (CXCL11)

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	0.005 (-0.095 to 0.104)
25 mg	-0.073 (-0.187 to 0.041)
50 mg	0.018 (-0.102 to 0.137)
100 mg	-0.030 (-0.143 to 0.083)

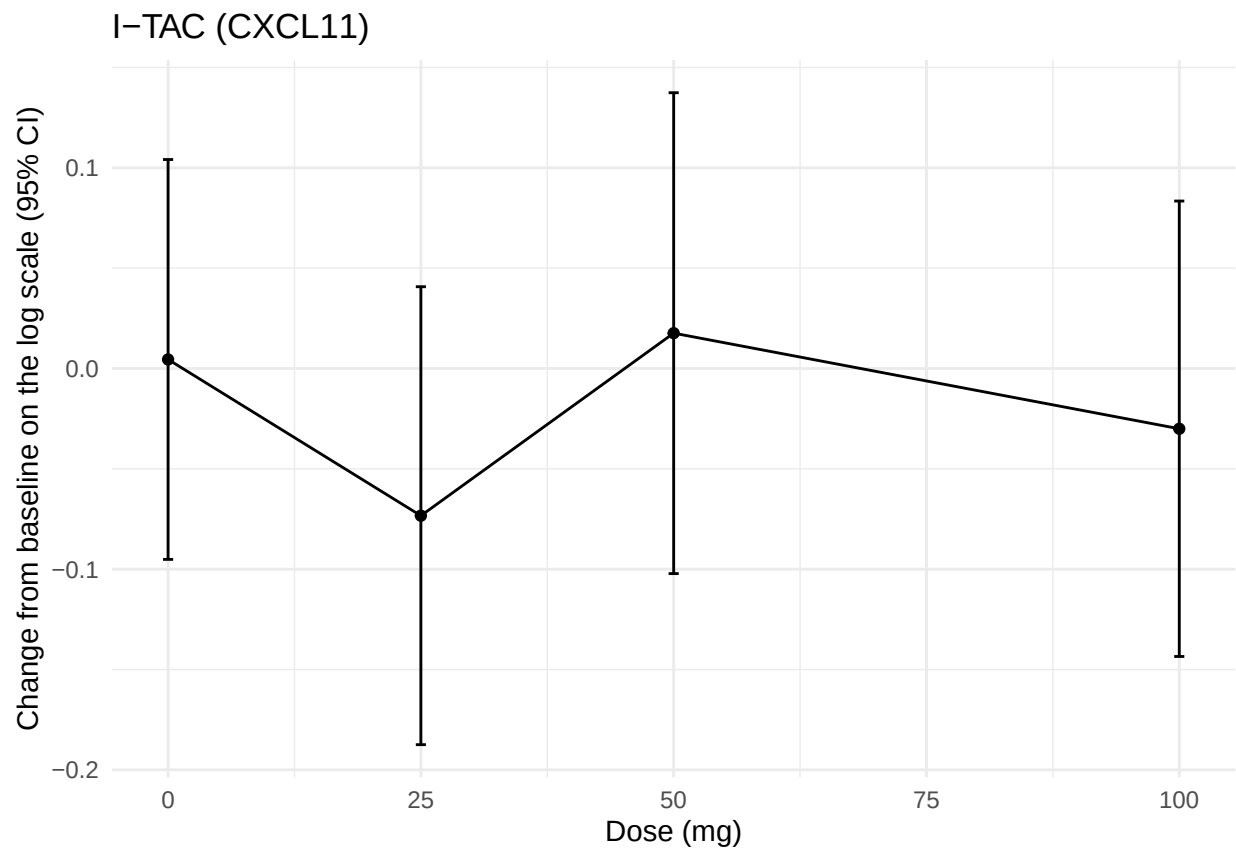
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	-0.08 (-0.24 to 0.08)
losartan 50 mg vs losartan 0 mg0	0.01 (-0.15 to 0.18)
losartan 100 mg vs losartan 0 mg0	-0.03 (-0.20 to 0.13)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	1174.1	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	32.5	TRUE

Plot



MIG (CXCL9)

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	0.203 (-0.277 to 0.684)
25 mg	-0.032 (-0.568 to 0.503)
50 mg	0.047 (-0.512 to 0.607)
100 mg	-0.007 (-0.533 to 0.519)

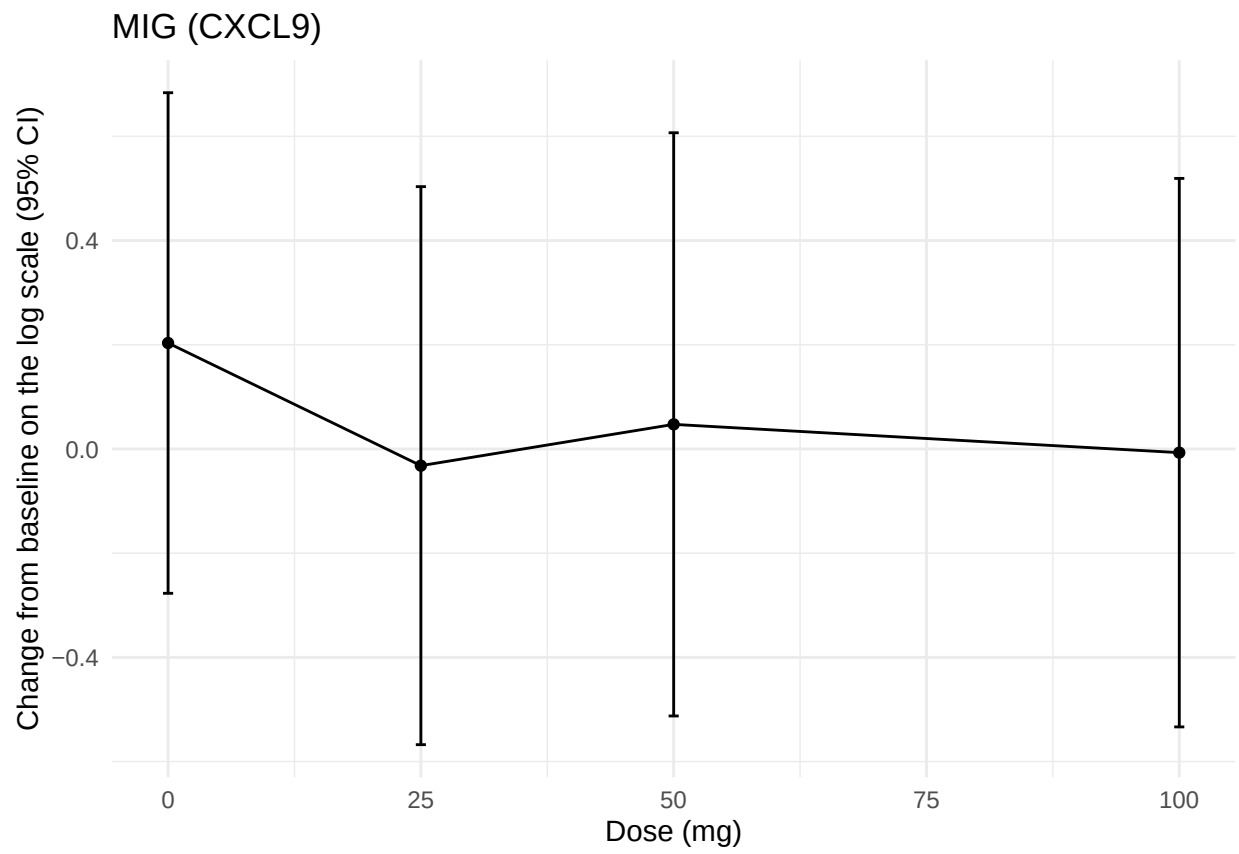
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	-0.24 (-1.09 to 0.62)
losartan 50 mg vs losartan 0 mg0	-0.16 (-1.01 to 0.70)
losartan 100 mg vs losartan 0 mg0	-0.21 (-1.05 to 0.63)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	1452.7	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	338.2	TRUE

Plot



PIGF

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	0.047 (-0.026 to 0.120)
25 mg	0.039 (-0.043 to 0.121)
50 mg	0.051 (-0.035 to 0.136)
100 mg	0.093 (0.012 to 0.173)

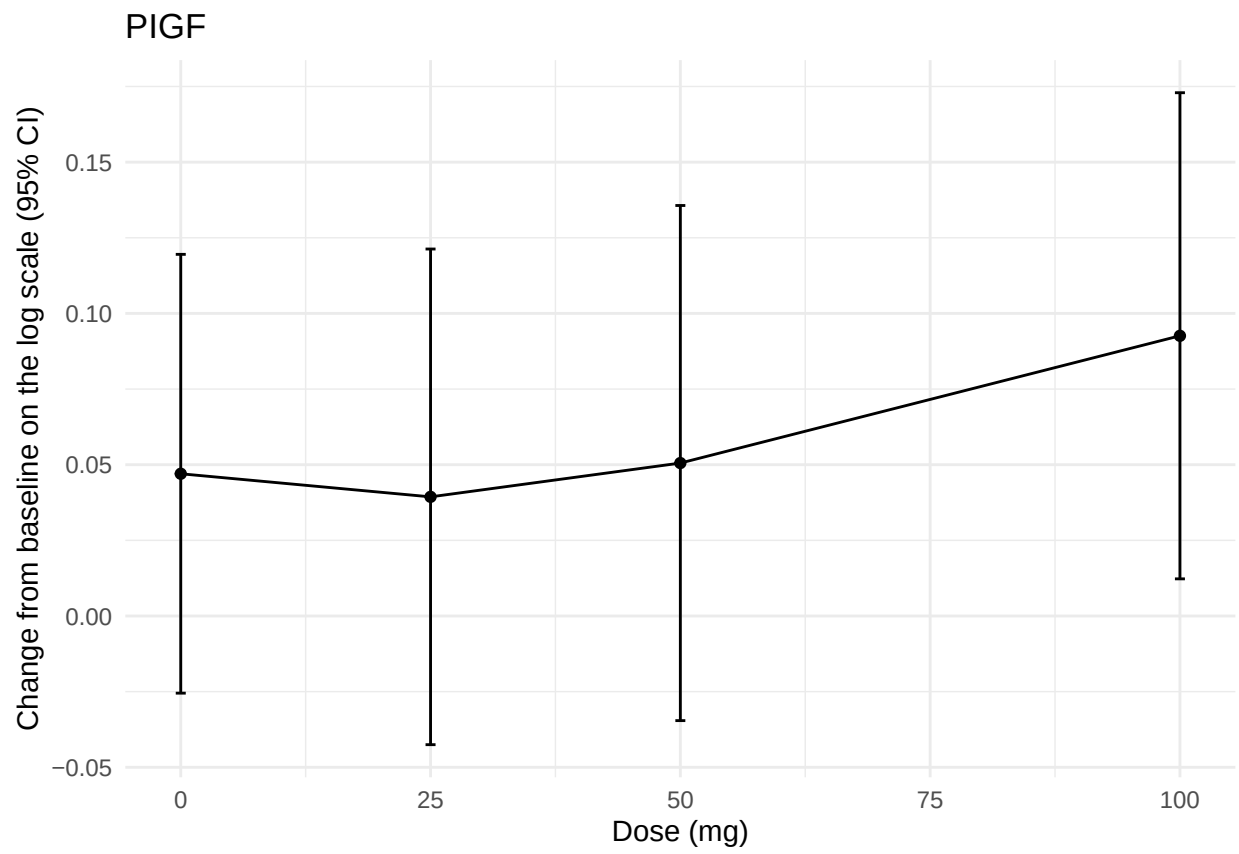
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	-0.01 (-0.13 to 0.12)
losartan 50 mg vs losartan 0 mg0	0.00 (-0.12 to 0.13)
losartan 100 mg vs losartan 0 mg0	0.05 (-0.08 to 0.17)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	788.4	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	-26.8	TRUE

Plot



Tenascin C

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	0.045 (-0.002 to 0.092)
25 mg	0.021 (-0.032 to 0.073)
50 mg	0.021 (-0.034 to 0.076)
100 mg	0.042 (-0.009 to 0.094)

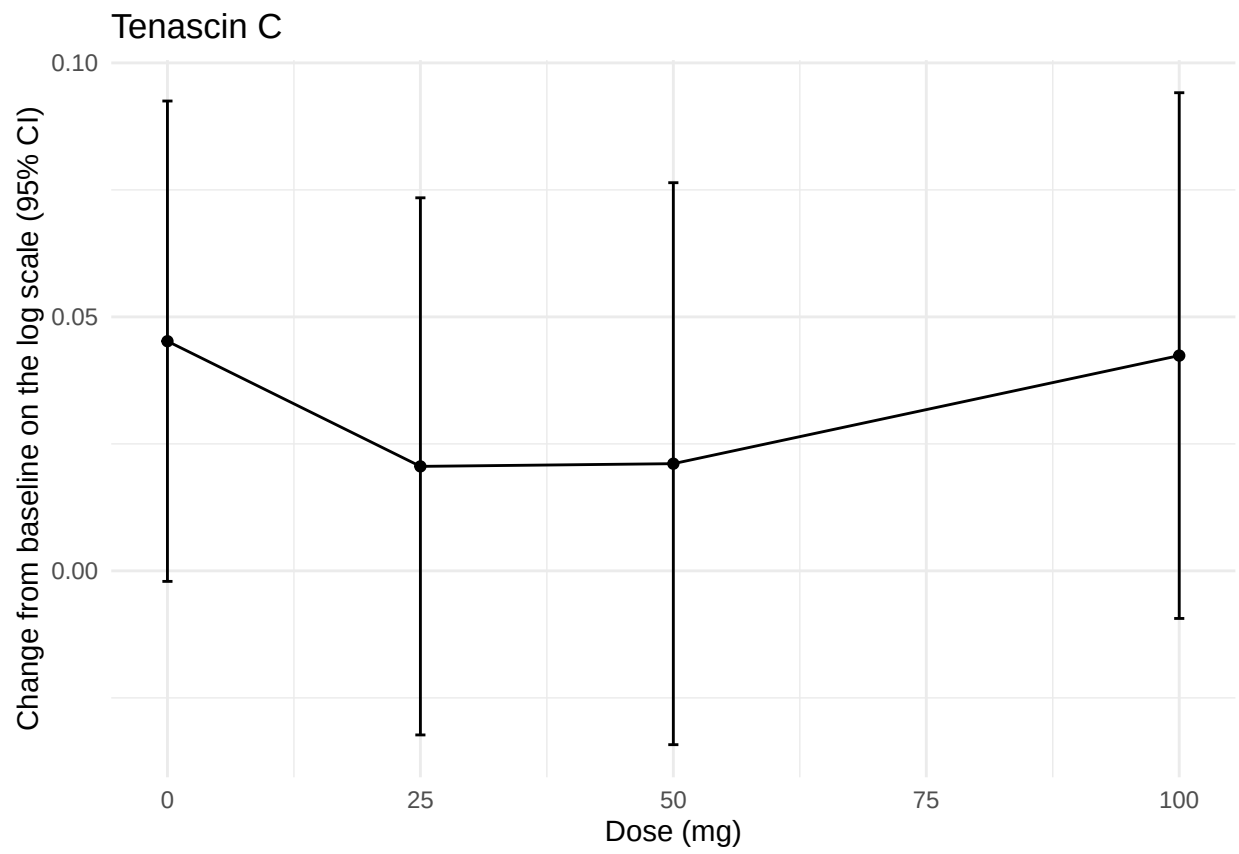
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	-0.02 (-0.11 to 0.06)
losartan 50 mg vs losartan 0 mg0	-0.02 (-0.11 to 0.06)
losartan 100 mg vs losartan 0 mg0	0.00 (-0.09 to 0.08)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	1756.8	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	-111.4	TRUE

Plot



TNFRSF18 (GITR)

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.109 (-0.372 to 0.155)
25 mg	0.039 (-0.259 to 0.336)
50 mg	-0.056 (-0.366 to 0.254)
100 mg	-0.178 (-0.468 to 0.113)

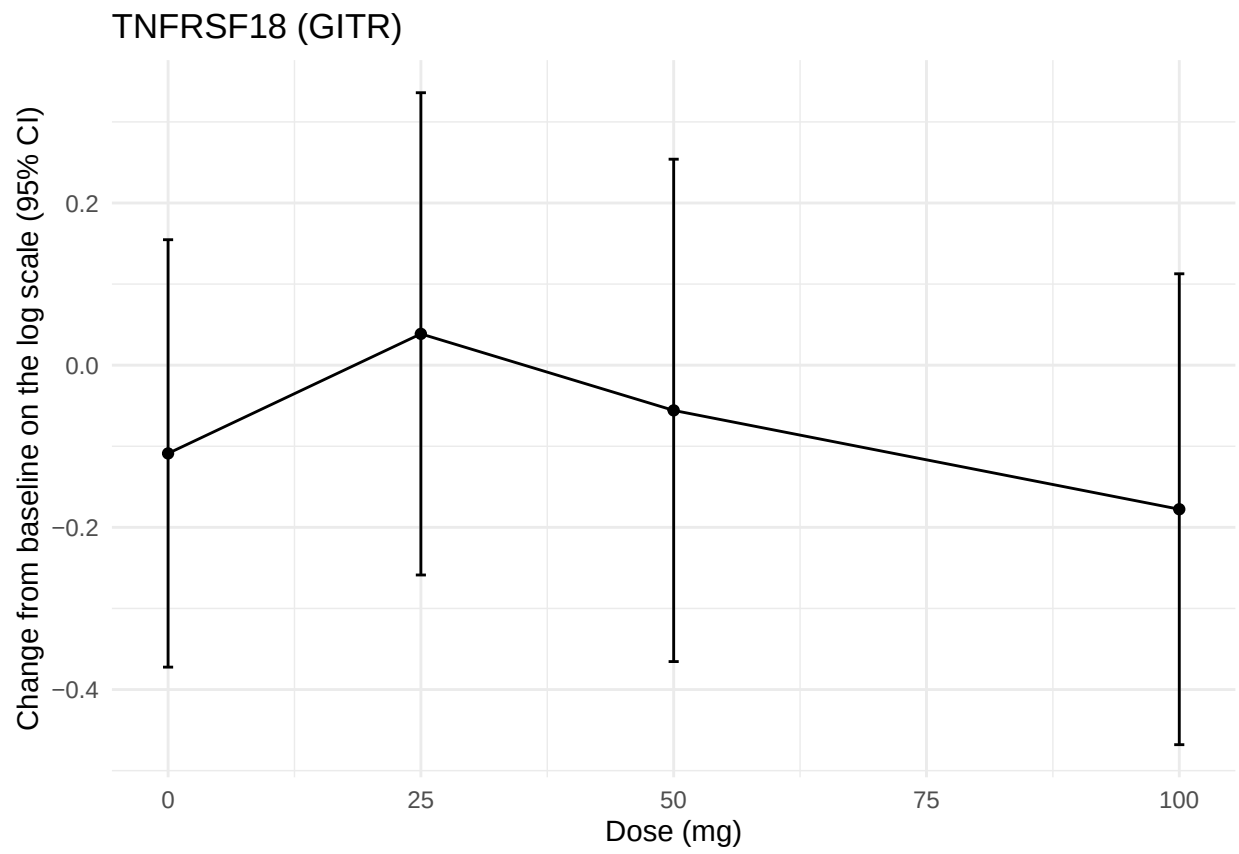
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	0.15 (-0.32 to 0.61)
losartan 50 mg vs losartan 0 mg0	0.05 (-0.41 to 0.52)
losartan 100 mg vs losartan 0 mg0	-0.07 (-0.53 to 0.39)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	1513.7	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	223.0	TRUE

Plot



TNFSF6 (FasL)

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	-0.047 (-0.165 to 0.071)
25 mg	-0.054 (-0.187 to 0.079)
50 mg	-0.001 (-0.140 to 0.137)
100 mg	-0.010 (-0.140 to 0.121)

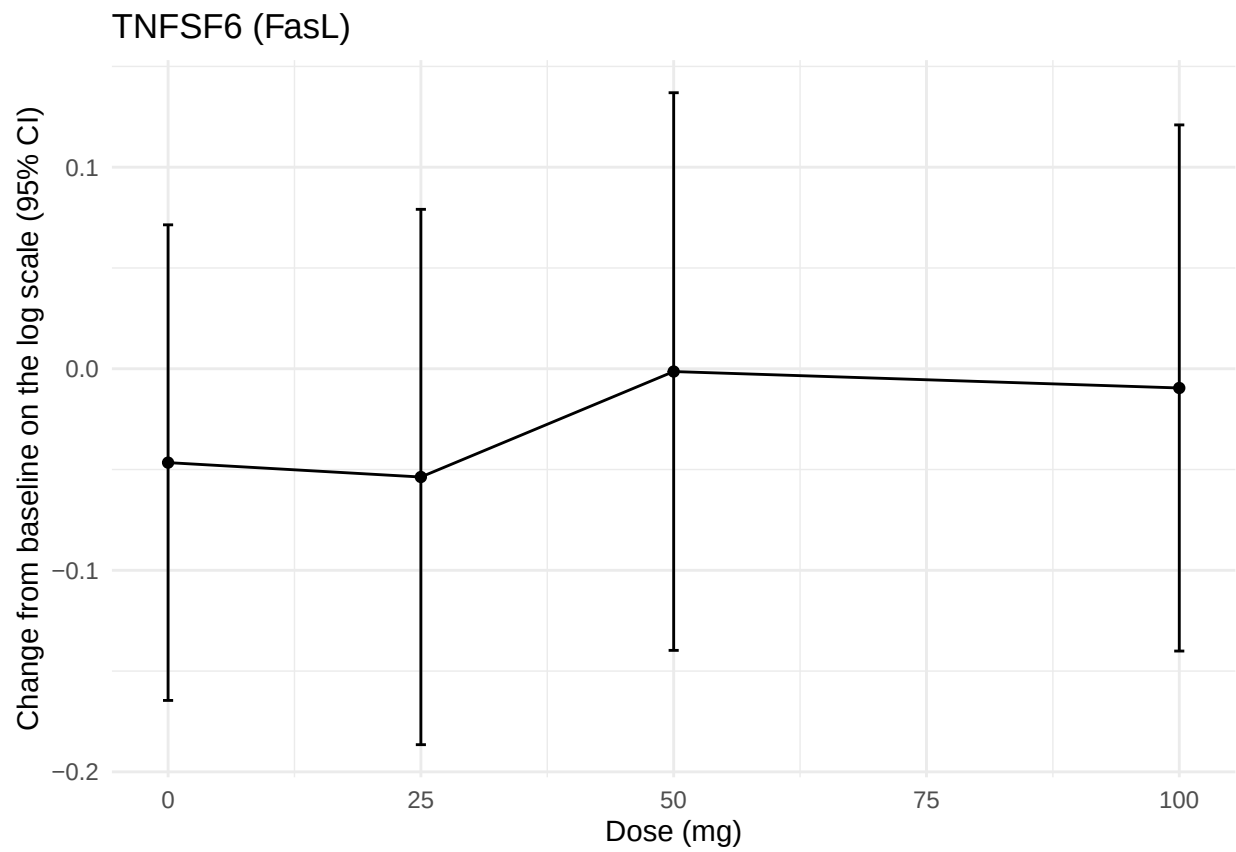
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	-0.01 (-0.21 to 0.20)
losartan 50 mg vs losartan 0 mg0	0.05 (-0.16 to 0.25)
losartan 100 mg vs losartan 0 mg0	0.04 (-0.17 to 0.24)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	878.2	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	66.0	TRUE

Plot



VEGFR2

A total of 53 participants contributed data for this analyte.

Estimated change from baseline by dose

Dose (mg)	Change from baseline on the natural logarithmic scale (95% CI)
0 mg	0.031 (-0.025 to 0.087)
25 mg	0.049 (-0.014 to 0.112)
50 mg	0.017 (-0.049 to 0.083)
100 mg	0.006 (-0.056 to 0.069)

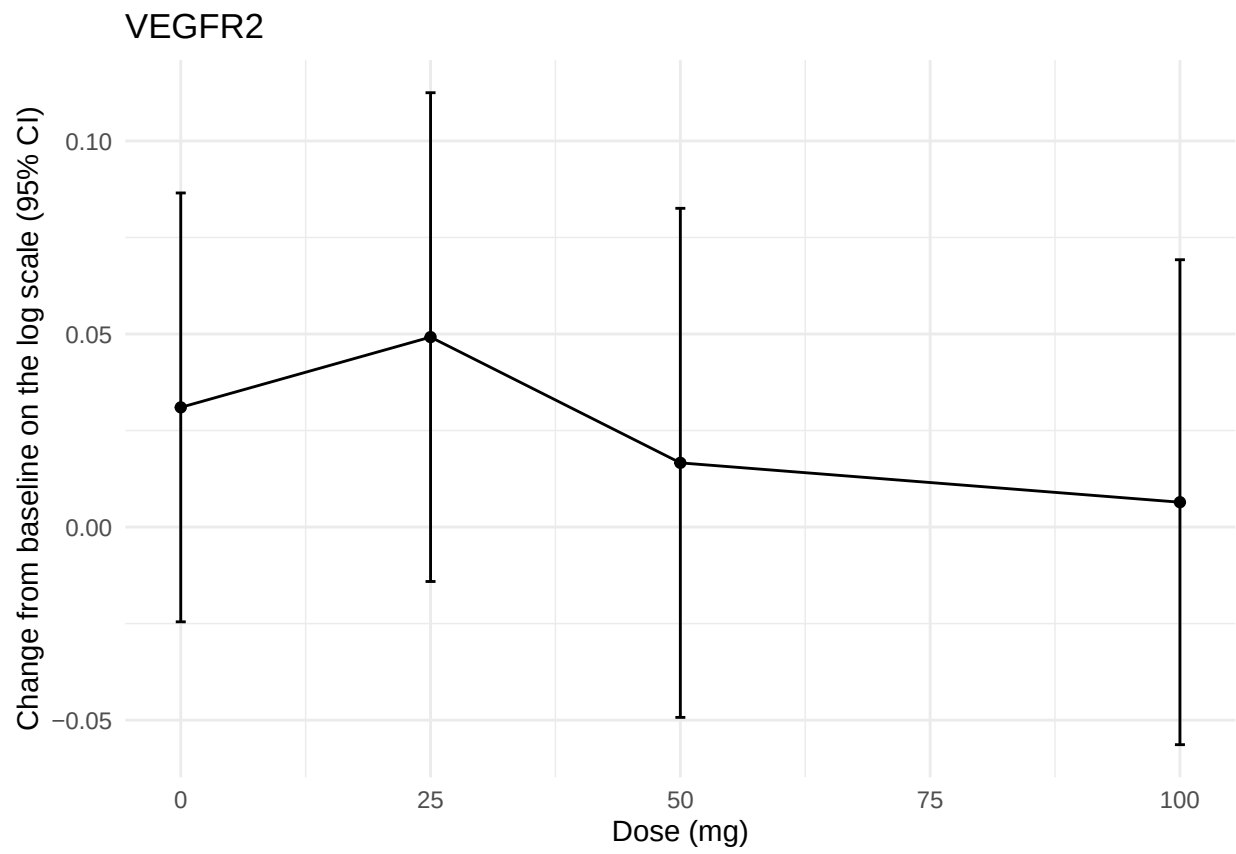
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg vs losartan 0 mg0	0.02 (-0.07 to 0.11)
losartan 50 mg vs losartan 0 mg0	-0.01 (-0.11 to 0.08)
losartan 100 mg vs losartan 0 mg0	-0.02 (-0.12 to 0.07)

Model comparison (AIC)

Model	AIC	Preferred
Linear (chg ~ base + trtcd + avisitn)	1752.4	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trtcd + avisitn)	-80.2	TRUE

Plot



Biomarker Endpoints — Long-term (Cycle 15+)

Model-based estimates of change from baseline at Cycle 15 and later are reported below for analytes with available long-term samples. The same scale (linear or log-transformed) as selected for the early analysis is used for each analyte.

Fractalkine

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	0.514 (-0.062 to 1.090)
losartan 25 mg	0.143 (-0.616 to 0.902)
losartan 50 mg	0.344 (-0.413 to 1.102)
losartan 100 mg	0.035 (-0.671 to 0.741)

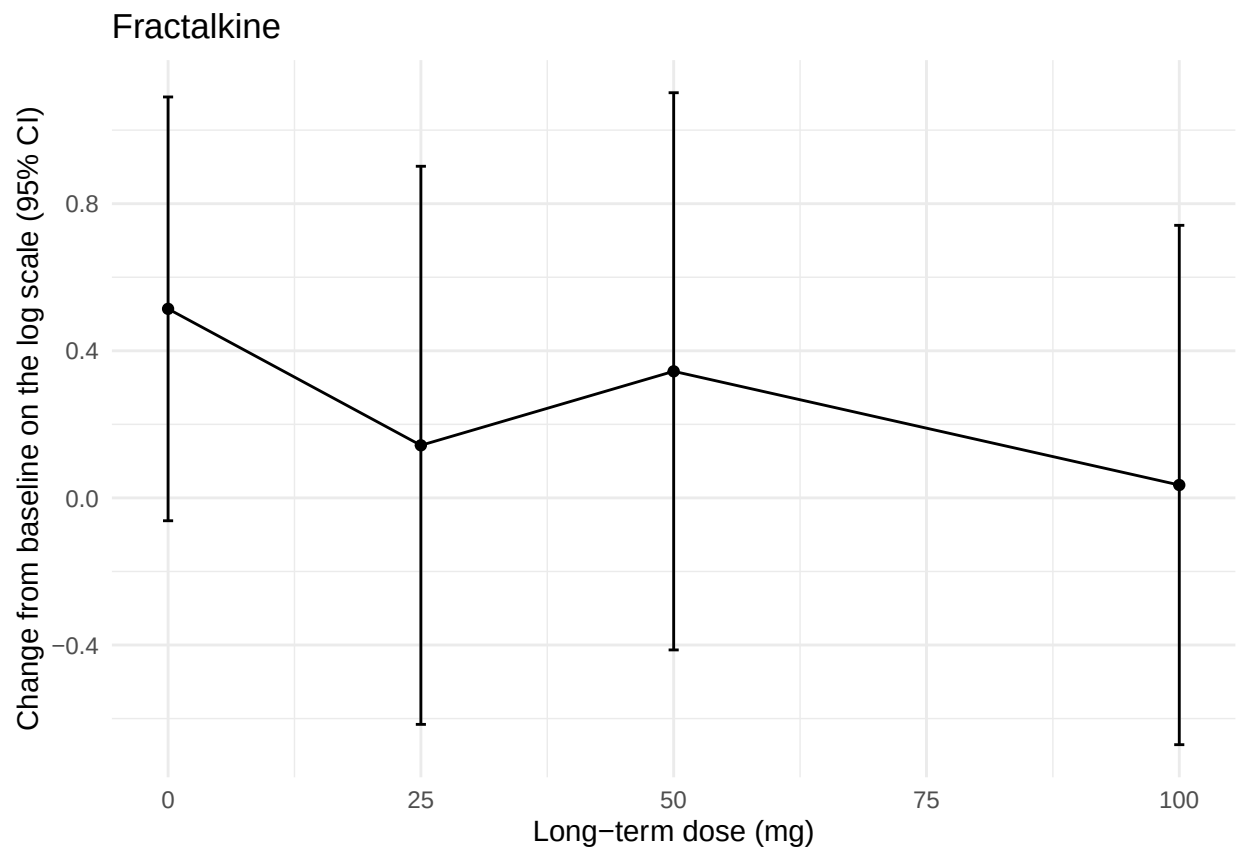
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	-0.37 (-1.55 to 0.81)
losartan 50 mg - losartan 0 mg	-0.17 (-1.34 to 1.00)
losartan 100 mg - losartan 0 mg	-0.48 (-1.58 to 0.62)

Model comparison (AIC)

Model	AIC	Used
Linear ($\text{chg} \sim \text{base} + \text{trt02p}$)	519.7	FALSE
Log-transformed ($\log(\text{value}) - \log(\text{base}) \sim \log(\text{base}) + \text{trt02p}$)	100.6	TRUE

Plot



Granzyme B

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.406 (-1.122 to 0.309)
losartan 25 mg	-0.194 (-1.122 to 0.734)
losartan 50 mg	-1.248 (-2.187 to -0.309)
losartan 100 mg	-0.582 (-1.452 to 0.288)

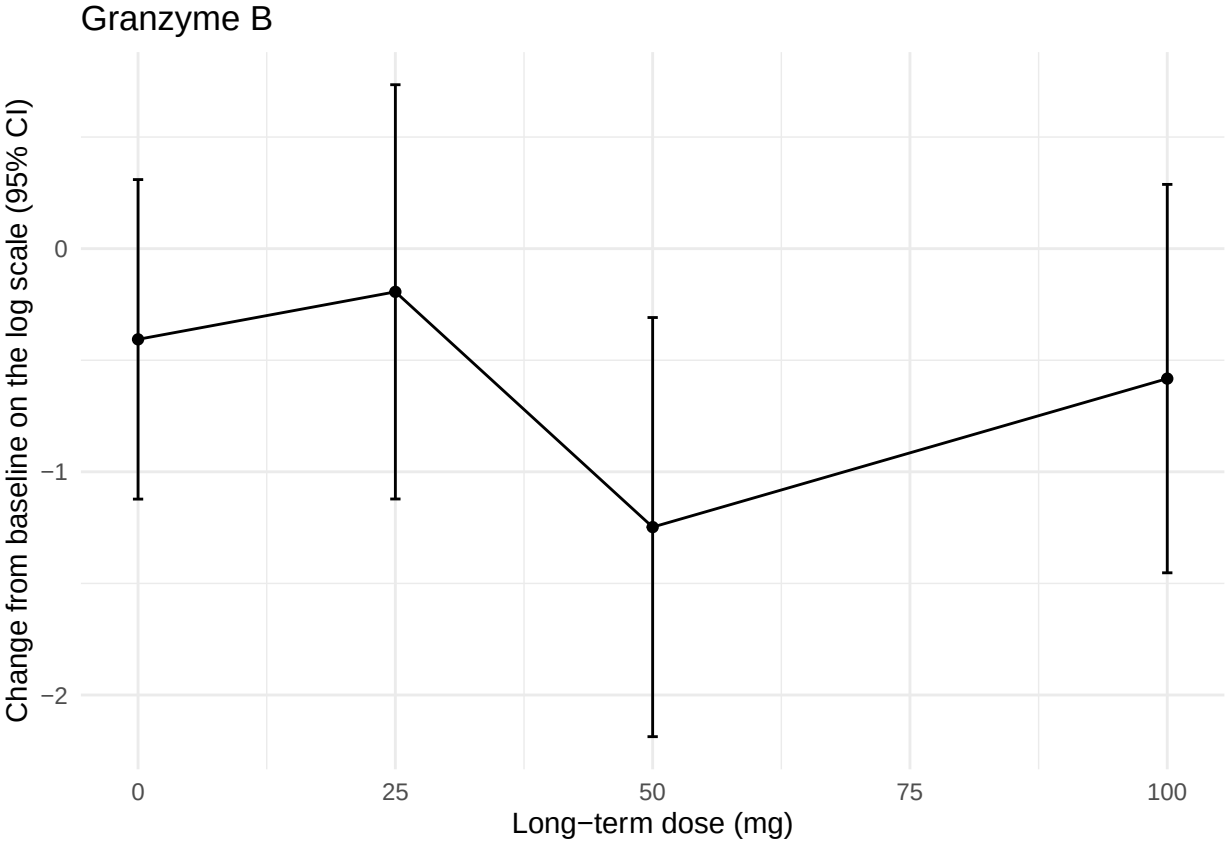
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.21 (-1.21 to 1.63)
losartan 50 mg - losartan 0 mg	-0.84 (-2.30 to 0.62)
losartan 100 mg - losartan 0 mg	-0.18 (-1.56 to 1.21)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	240.0	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	115.4	TRUE

Plot



IFN-g

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.550 (-0.984 to -0.116)
losartan 25 mg	0.295 (-0.265 to 0.856)
losartan 50 mg	-0.310 (-0.871 to 0.251)
losartan 100 mg	-0.259 (-0.775 to 0.258)

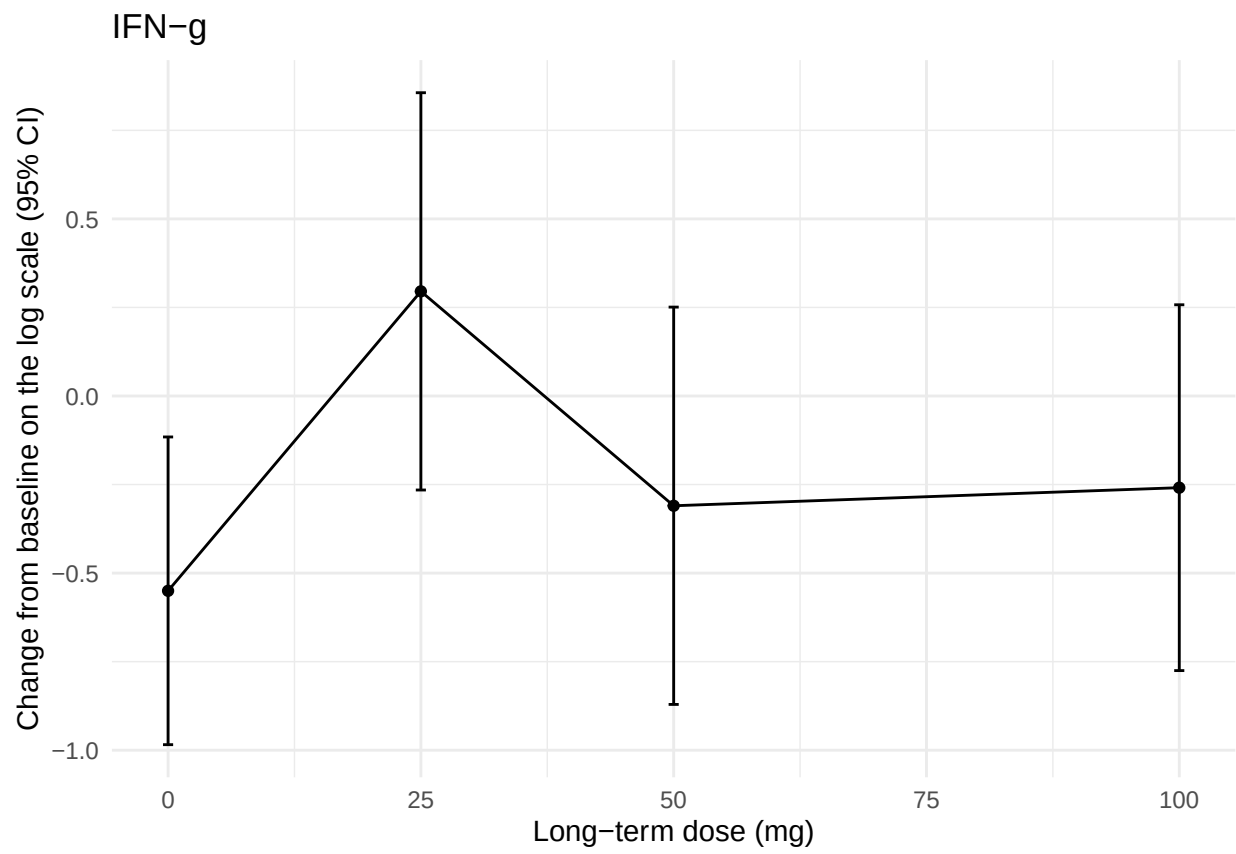
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.85 (-0.04 to 1.73)
losartan 50 mg - losartan 0 mg	0.24 (-0.64 to 1.12)
losartan 100 mg - losartan 0 mg	0.29 (-0.53 to 1.11)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	109.4	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	80.1	TRUE

Plot



IL-1b

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.057 (-0.453 to 0.339)
losartan 25 mg	-0.013 (-0.530 to 0.503)
losartan 50 mg	0.368 (-0.155 to 0.891)
losartan 100 mg	0.025 (-0.458 to 0.508)

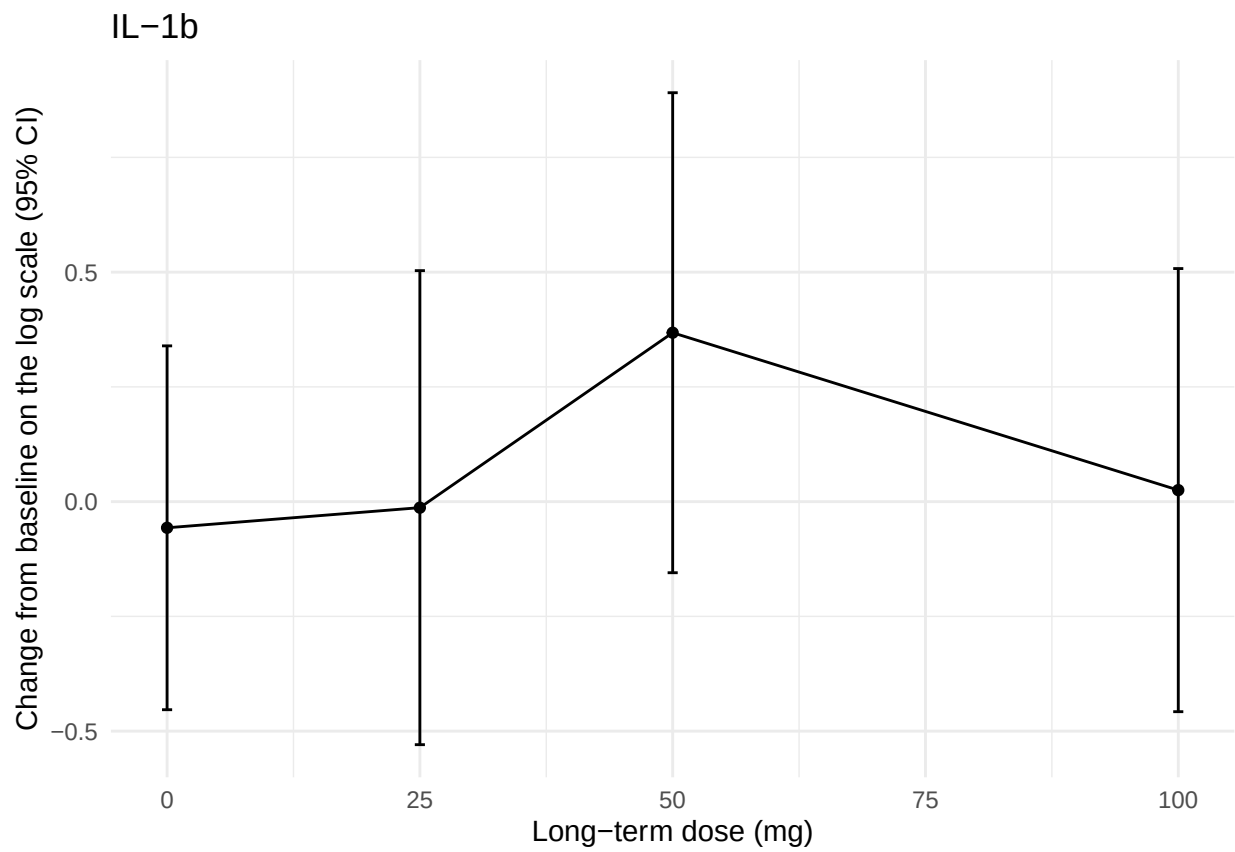
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.04 (-0.75 to 0.84)
losartan 50 mg - losartan 0 mg	0.43 (-0.38 to 1.23)
losartan 100 mg - losartan 0 mg	0.08 (-0.68 to 0.84)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	103.0	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	75.6	TRUE

Plot



IL-6

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.464 (-1.190 to 0.262)
losartan 25 mg	-0.010 (-0.959 to 0.938)
losartan 50 mg	-0.300 (-1.267 to 0.668)
losartan 100 mg	-0.507 (-1.393 to 0.378)

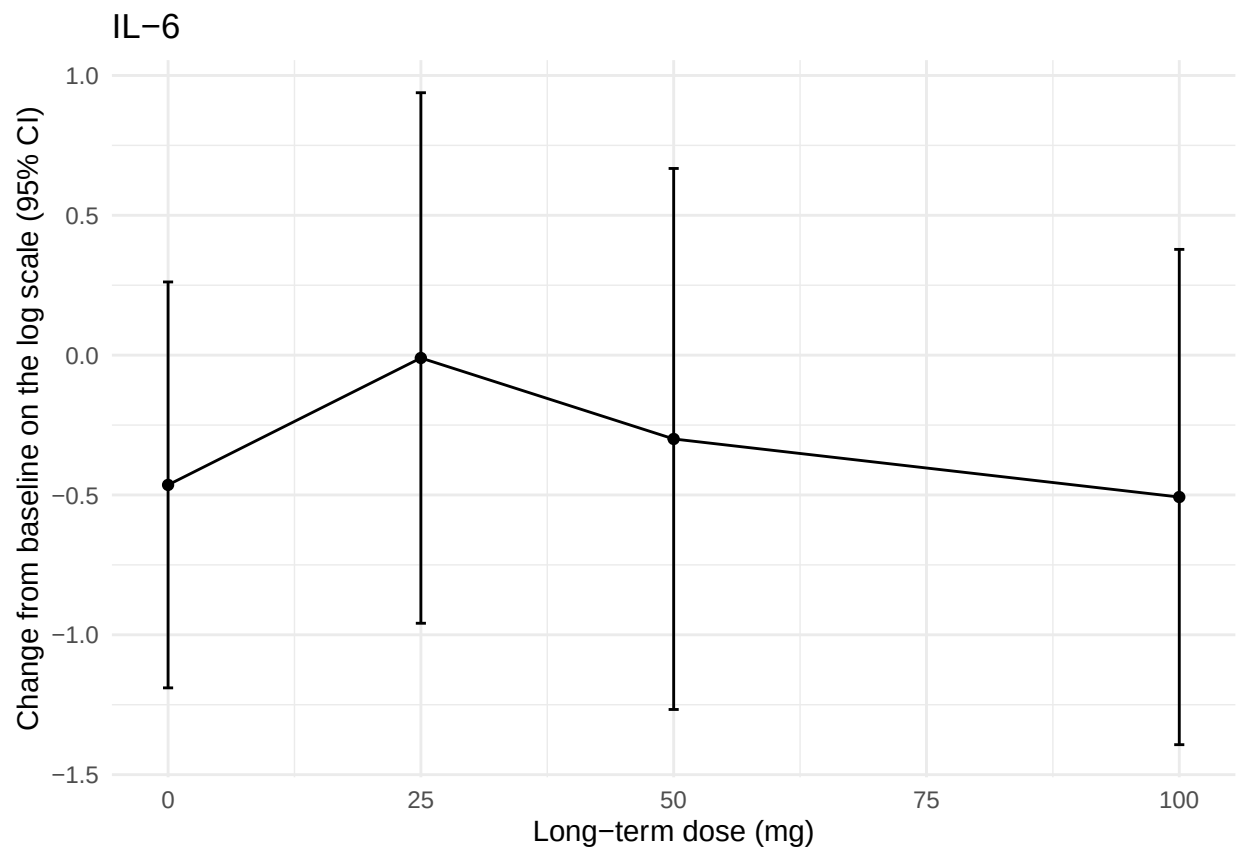
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.45 (-1.00 to 1.91)
losartan 50 mg - losartan 0 mg	0.16 (-1.32 to 1.65)
losartan 100 mg - losartan 0 mg	-0.04 (-1.44 to 1.35)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	110.1	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	116.8	TRUE

Plot



IL-8

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	0.260 (-0.255 to 0.775)
losartan 25 mg	0.058 (-0.616 to 0.732)
losartan 50 mg	-0.016 (-0.692 to 0.660)
losartan 100 mg	-0.322 (-0.960 to 0.315)

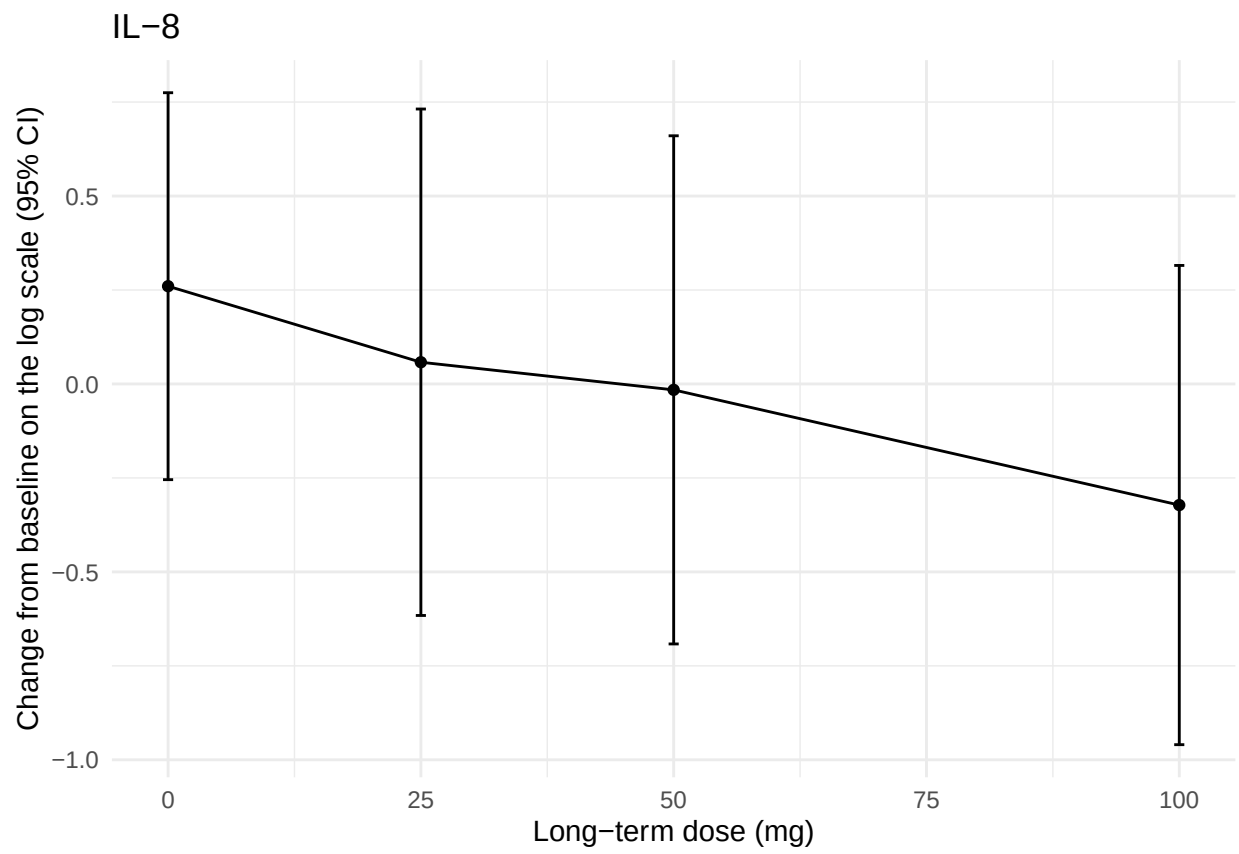
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	-0.20 (-1.24 to 0.83)
losartan 50 mg - losartan 0 mg	-0.28 (-1.31 to 0.76)
losartan 100 mg - losartan 0 mg	-0.58 (-1.58 to 0.42)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	159.0	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	93.7	TRUE

Plot



MCP-1

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	0.005 (-0.185 to 0.195)
losartan 25 mg	-0.022 (-0.271 to 0.226)
losartan 50 mg	-0.017 (-0.270 to 0.237)
losartan 100 mg	-0.110 (-0.345 to 0.125)

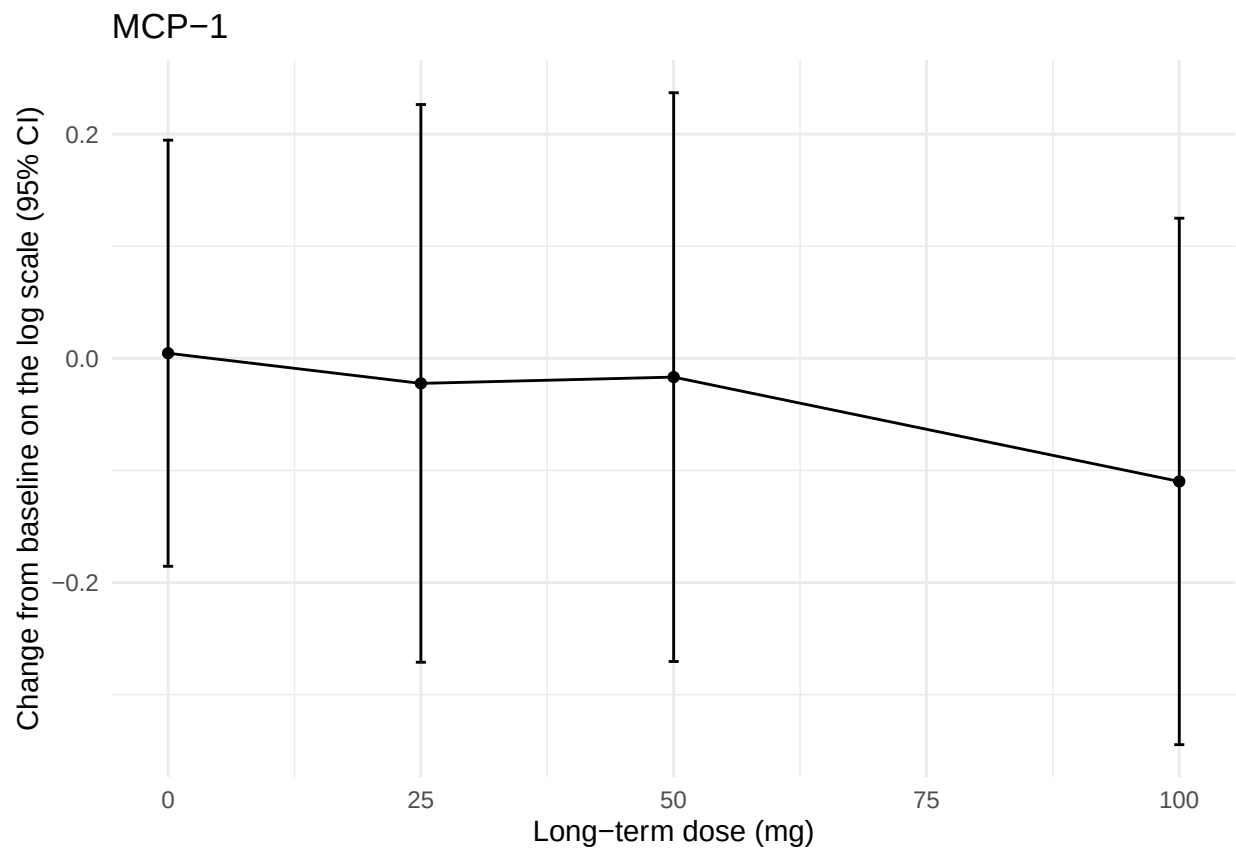
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	-0.03 (-0.41 to 0.35)
losartan 50 mg - losartan 0 mg	-0.02 (-0.41 to 0.37)
losartan 100 mg - losartan 0 mg	-0.11 (-0.48 to 0.25)

Model comparison (AIC)

Model	AIC	Used
Linear ($\text{chg} \sim \text{base} + \text{trt02p}$)	368	FALSE
Log-transformed ($\log(\text{value}) - \log(\text{base}) \sim \log(\text{base}) + \text{trt02p}$)	26	TRUE

Plot



MMP-10

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	0.143 (-0.042 to 0.327)
losartan 25 mg	-0.063 (-0.312 to 0.186)
losartan 50 mg	-0.055 (-0.289 to 0.179)
losartan 100 mg	-0.035 (-0.254 to 0.184)

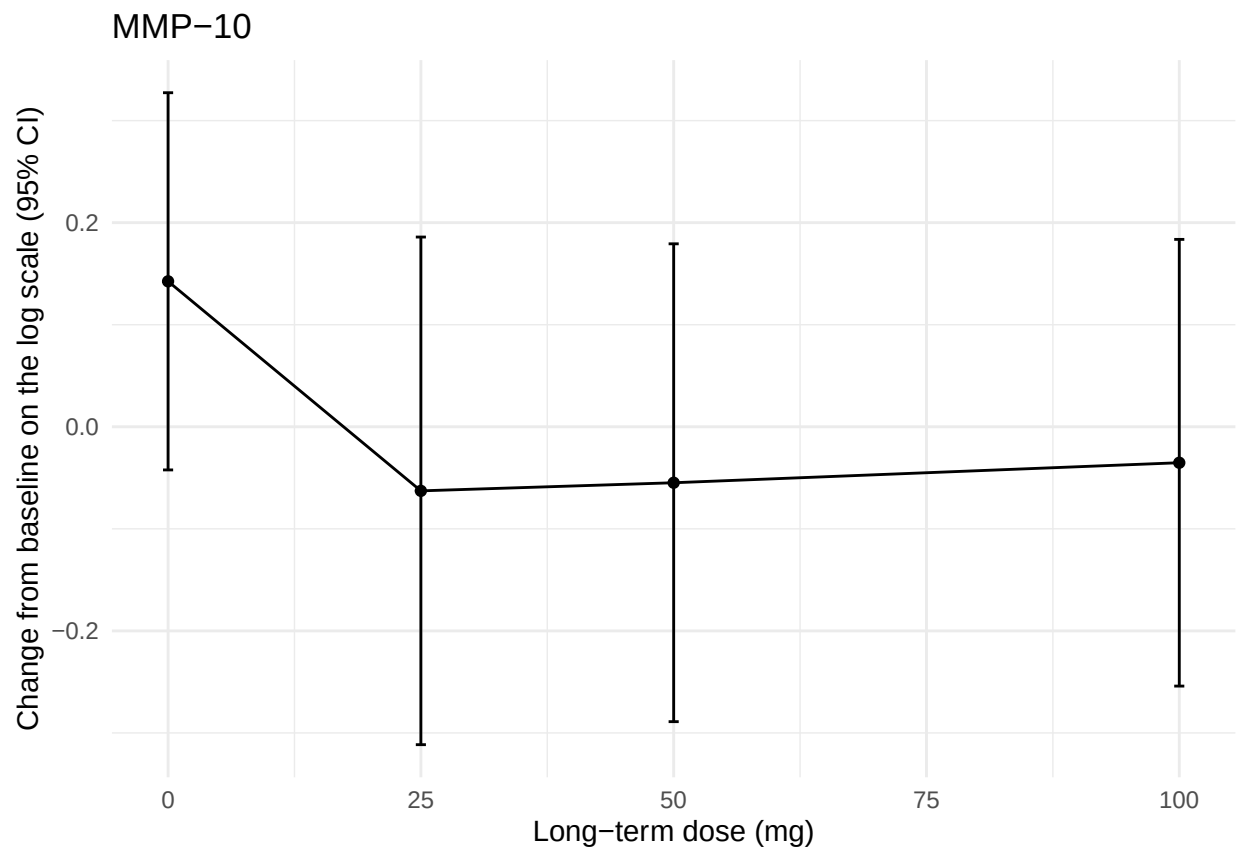
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	-0.21 (-0.60 to 0.19)
losartan 50 mg - losartan 0 mg	-0.20 (-0.56 to 0.16)
losartan 100 mg - losartan 0 mg	-0.18 (-0.53 to 0.17)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	438.3	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	21.8	TRUE

Plot



PDGF-DD

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.055 (-0.397 to 0.286)
losartan 25 mg	-0.284 (-0.732 to 0.163)
losartan 50 mg	-0.441 (-0.891 to 0.008)
losartan 100 mg	-0.242 (-0.659 to 0.175)

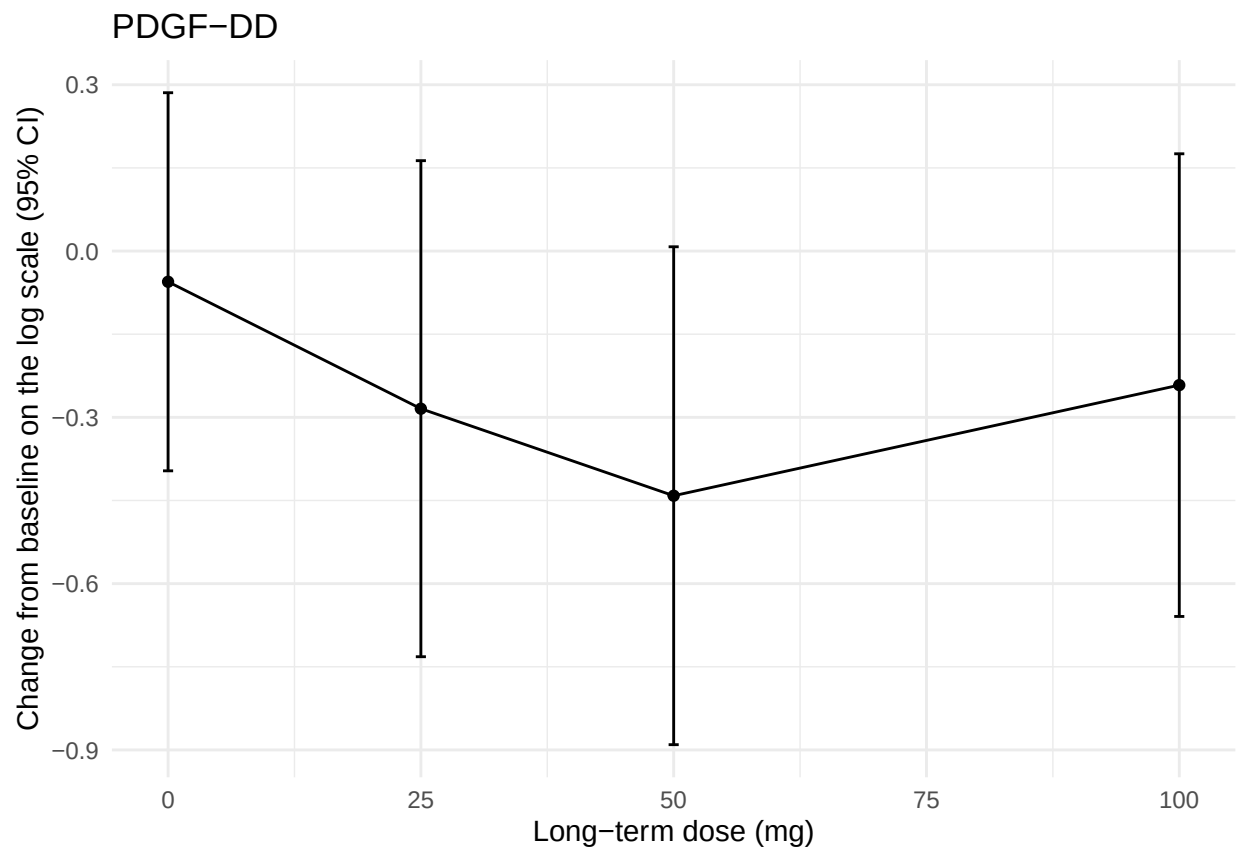
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	-0.23 (-0.91 to 0.46)
losartan 50 mg - losartan 0 mg	-0.39 (-1.08 to 0.30)
losartan 100 mg - losartan 0 mg	-0.19 (-0.84 to 0.47)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	358.4	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	65.7	TRUE

Plot



PECAM-1

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	0.191 (-0.304 to 0.686)
losartan 25 mg	-0.157 (-0.806 to 0.492)
losartan 50 mg	0.215 (-0.431 to 0.861)
losartan 100 mg	0.086 (-0.518 to 0.689)

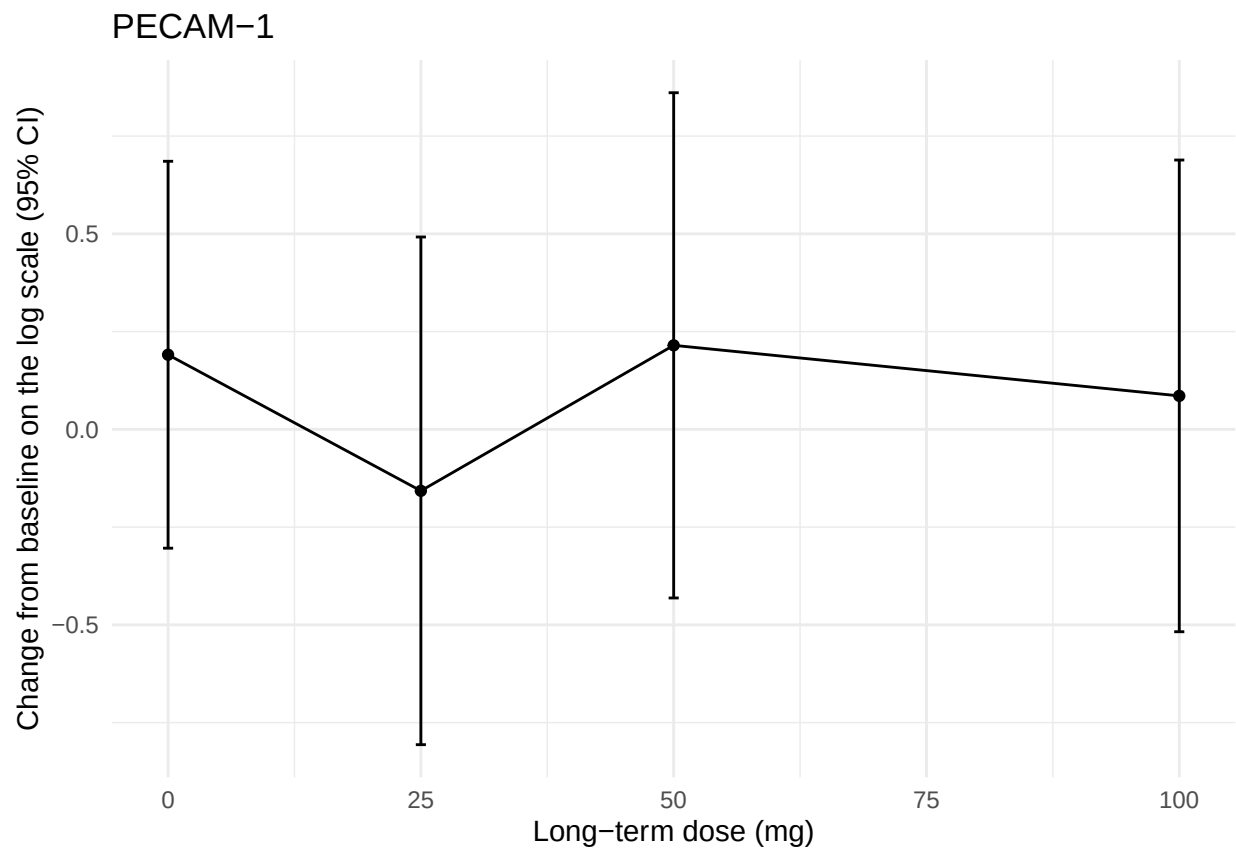
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	-0.35 (-1.35 to 0.65)
losartan 50 mg - losartan 0 mg	0.02 (-0.97 to 1.02)
losartan 100 mg - losartan 0 mg	-0.11 (-1.05 to 0.84)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	833.3	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	90.6	TRUE

Plot



TGF-alpha

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.175 (-0.501 to 0.151)
losartan 25 mg	-0.107 (-0.526 to 0.312)
losartan 50 mg	-0.177 (-0.598 to 0.244)
losartan 100 mg	-0.105 (-0.503 to 0.293)

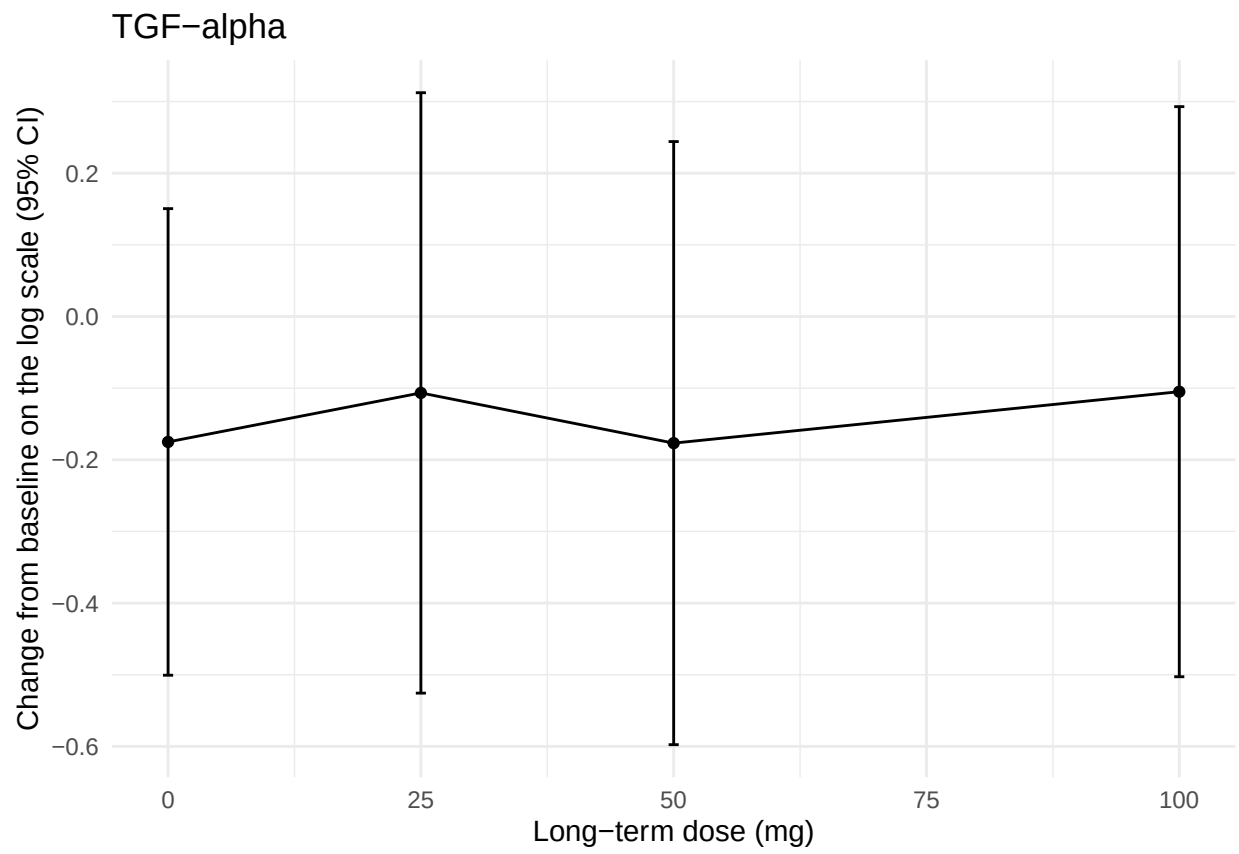
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.07 (-0.58 to 0.71)
losartan 50 mg - losartan 0 mg	0.00 (-0.66 to 0.65)
losartan 100 mg - losartan 0 mg	0.07 (-0.57 to 0.71)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	143.1	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	61.4	TRUE

Plot



Thrombospondin-2

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.133 (-0.317 to 0.051)
losartan 25 mg	0.022 (-0.225 to 0.268)
losartan 50 mg	-0.144 (-0.380 to 0.093)
losartan 100 mg	0.030 (-0.191 to 0.251)

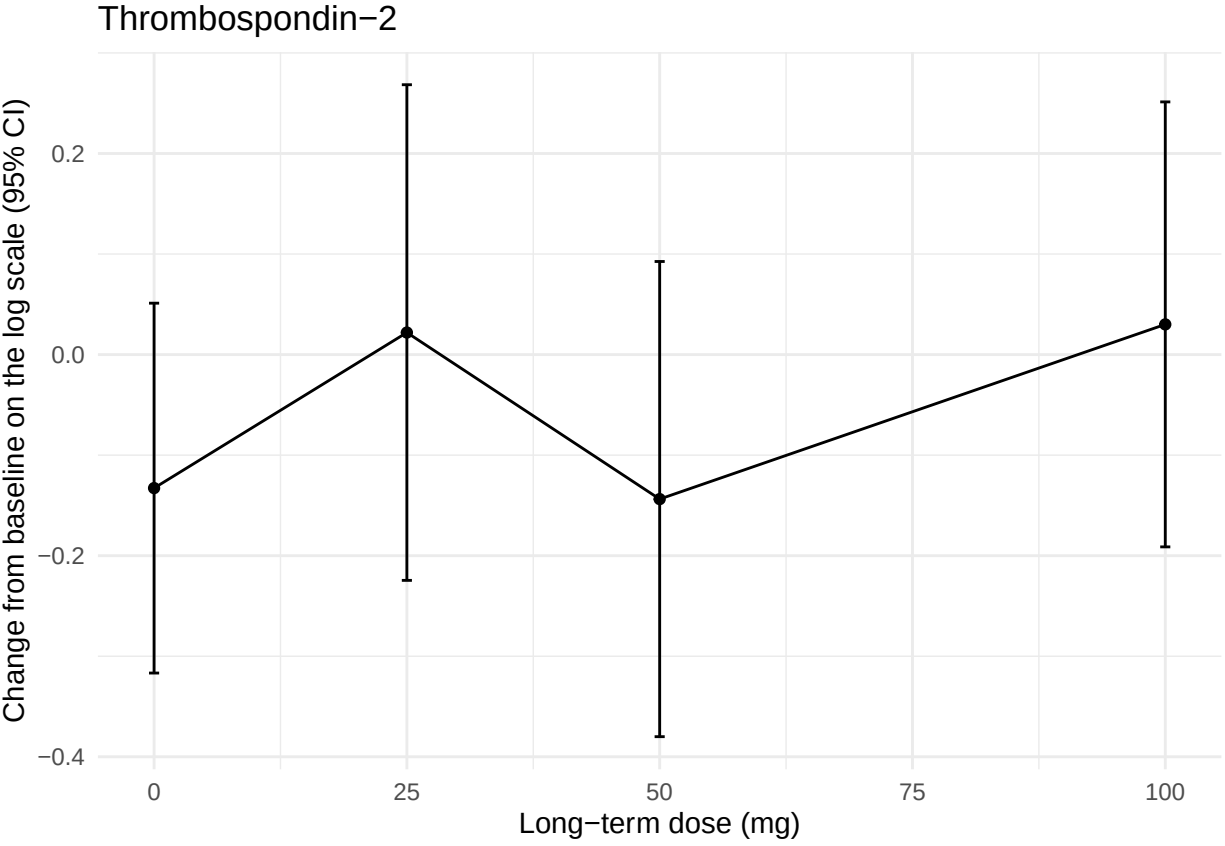
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.15 (-0.23 to 0.54)
losartan 50 mg - losartan 0 mg	-0.01 (-0.38 to 0.36)
losartan 100 mg - losartan 0 mg	0.16 (-0.18 to 0.51)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	660.3	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	22.0	TRUE

Plot



VEGF-A

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.071 (-0.443 to 0.301)
losartan 25 mg	-0.437 (-0.929 to 0.056)
losartan 50 mg	-0.323 (-0.807 to 0.161)
losartan 100 mg	-0.165 (-0.619 to 0.289)

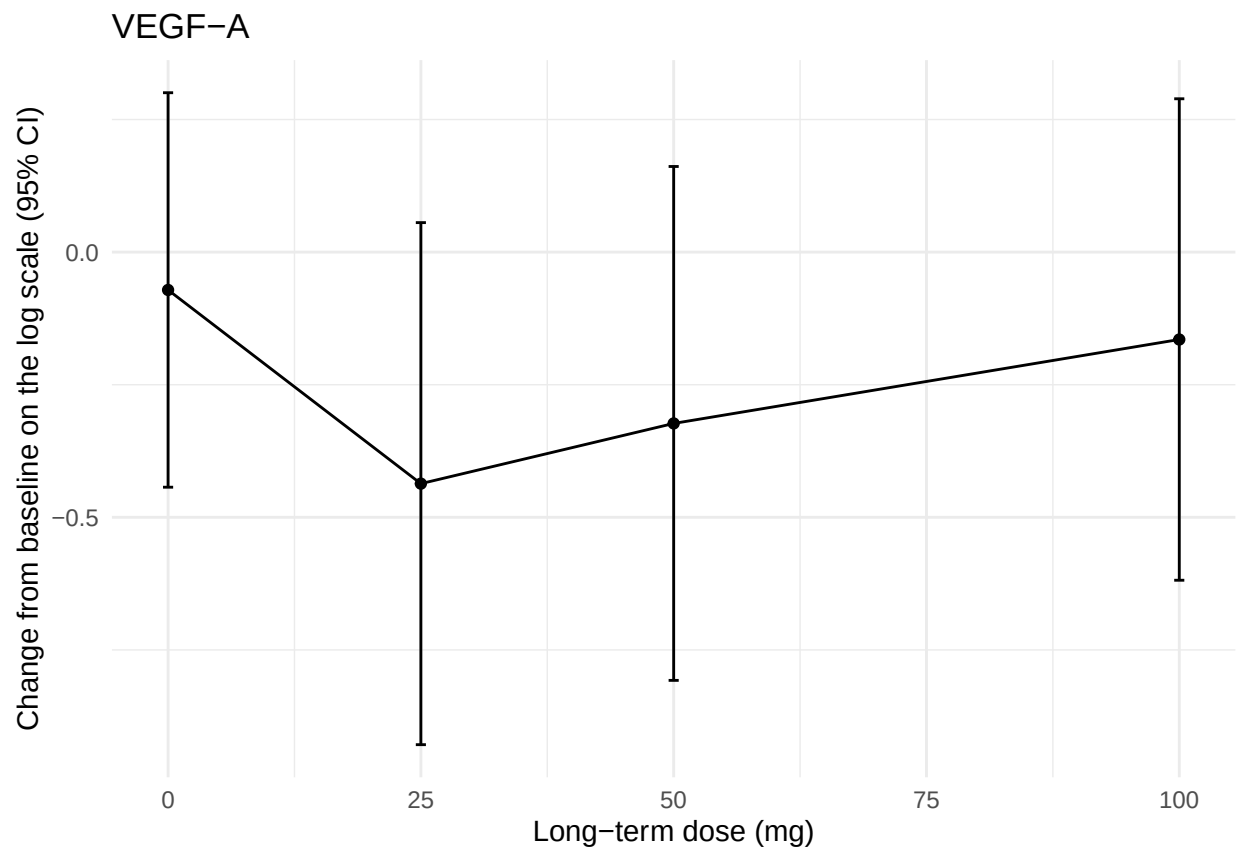
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	-0.37 (-1.12 to 0.39)
losartan 50 mg - losartan 0 mg	-0.25 (-1.00 to 0.49)
losartan 100 mg - losartan 0 mg	-0.09 (-0.81 to 0.62)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	383.8	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	71.3	TRUE

Plot



VEGF-C

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.058 (-0.354 to 0.238)
losartan 25 mg	-0.319 (-0.709 to 0.072)
losartan 50 mg	-0.127 (-0.514 to 0.260)
losartan 100 mg	-0.274 (-0.637 to 0.090)

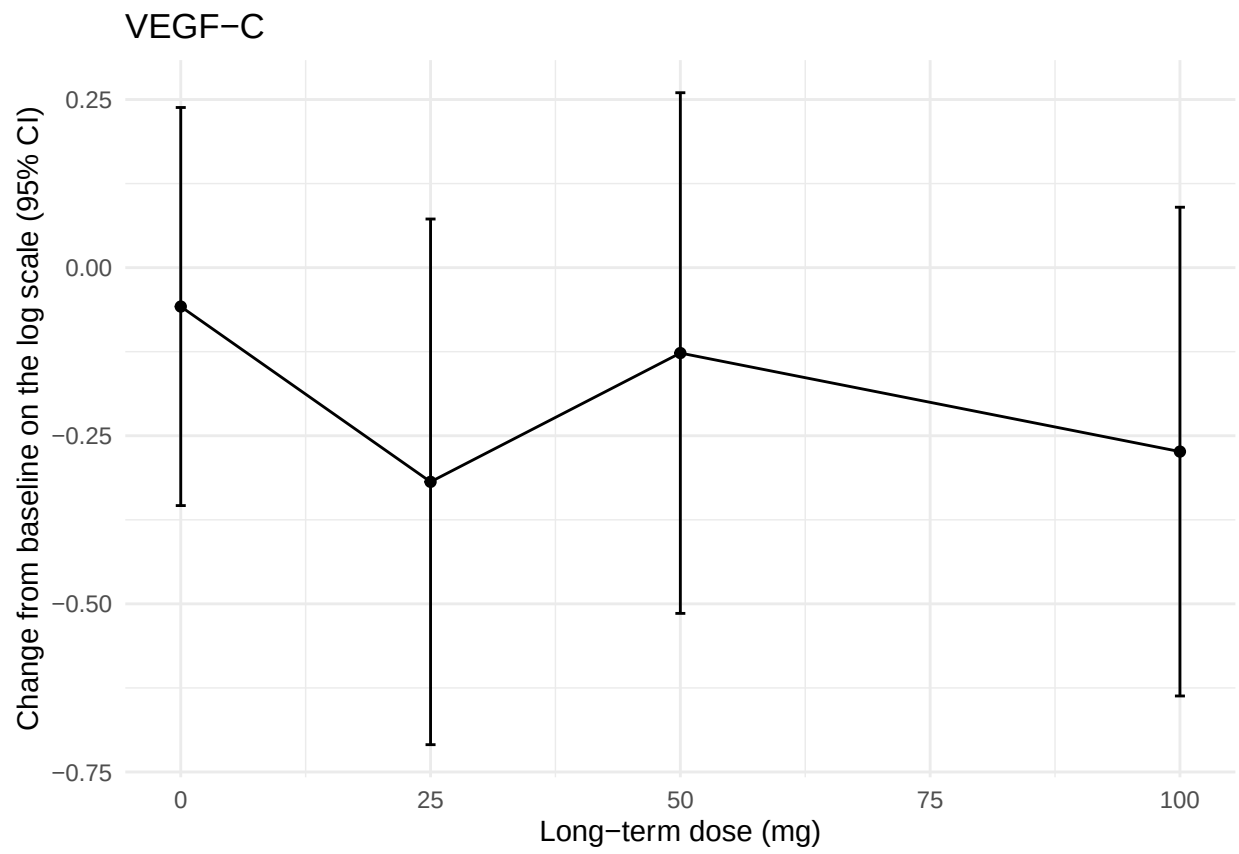
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	-0.26 (-0.86 to 0.34)
losartan 50 mg - losartan 0 mg	-0.07 (-0.66 to 0.52)
losartan 100 mg - losartan 0 mg	-0.22 (-0.79 to 0.35)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	497.5	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	56.1	TRUE

Plot



Angiopoietin-1

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.216 (-0.478 to 0.047)
losartan 25 mg	-0.486 (-0.830 to -0.141)
losartan 50 mg	-0.424 (-0.767 to -0.081)
losartan 100 mg	-0.306 (-0.627 to 0.015)

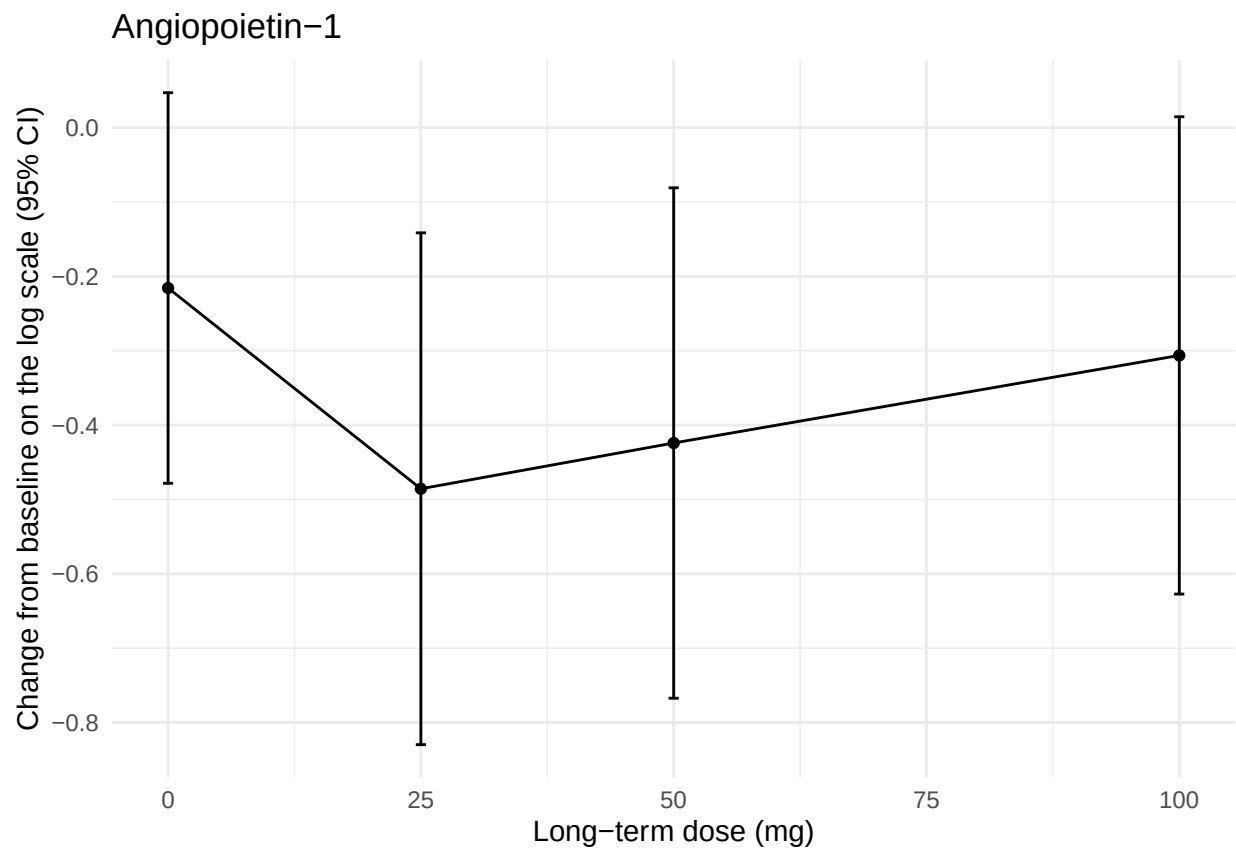
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	-0.27 (-0.80 to 0.26)
losartan 50 mg - losartan 0 mg	-0.21 (-0.74 to 0.32)
losartan 100 mg - losartan 0 mg	-0.09 (-0.60 to 0.41)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	630.3	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	47.8	TRUE

Plot



Angiopoietin-2

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.006 (-0.121 to 0.109)
losartan 25 mg	0.120 (-0.028 to 0.269)
losartan 50 mg	0.004 (-0.144 to 0.152)
losartan 100 mg	0.009 (-0.129 to 0.147)

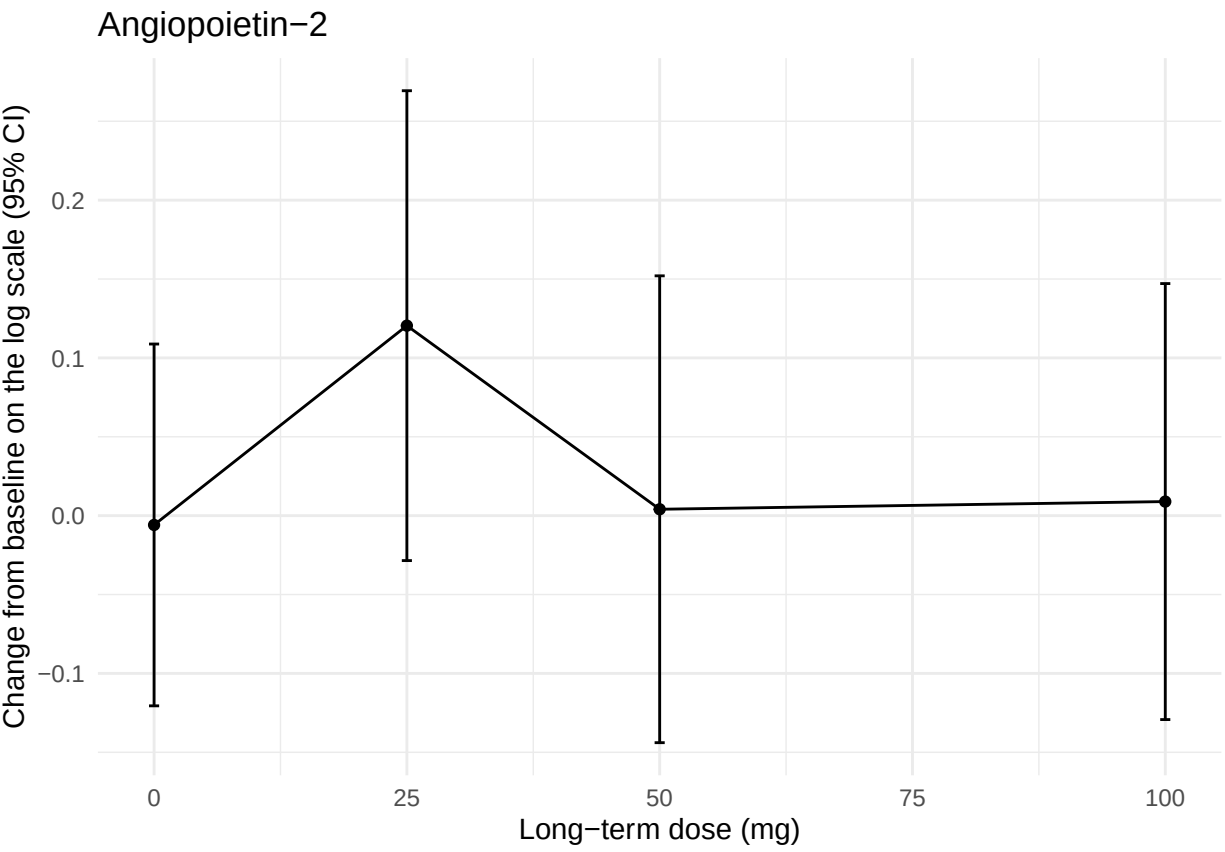
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.13 (-0.11 to 0.36)
losartan 50 mg - losartan 0 mg	0.01 (-0.22 to 0.24)
losartan 100 mg - losartan 0 mg	0.01 (-0.20 to 0.23)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	553.8	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	-9.5	TRUE

Plot



ANGPTL4

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	0.003 (-0.135 to 0.142)
losartan 25 mg	-0.040 (-0.226 to 0.145)
losartan 50 mg	0.024 (-0.157 to 0.204)
losartan 100 mg	-0.063 (-0.233 to 0.107)

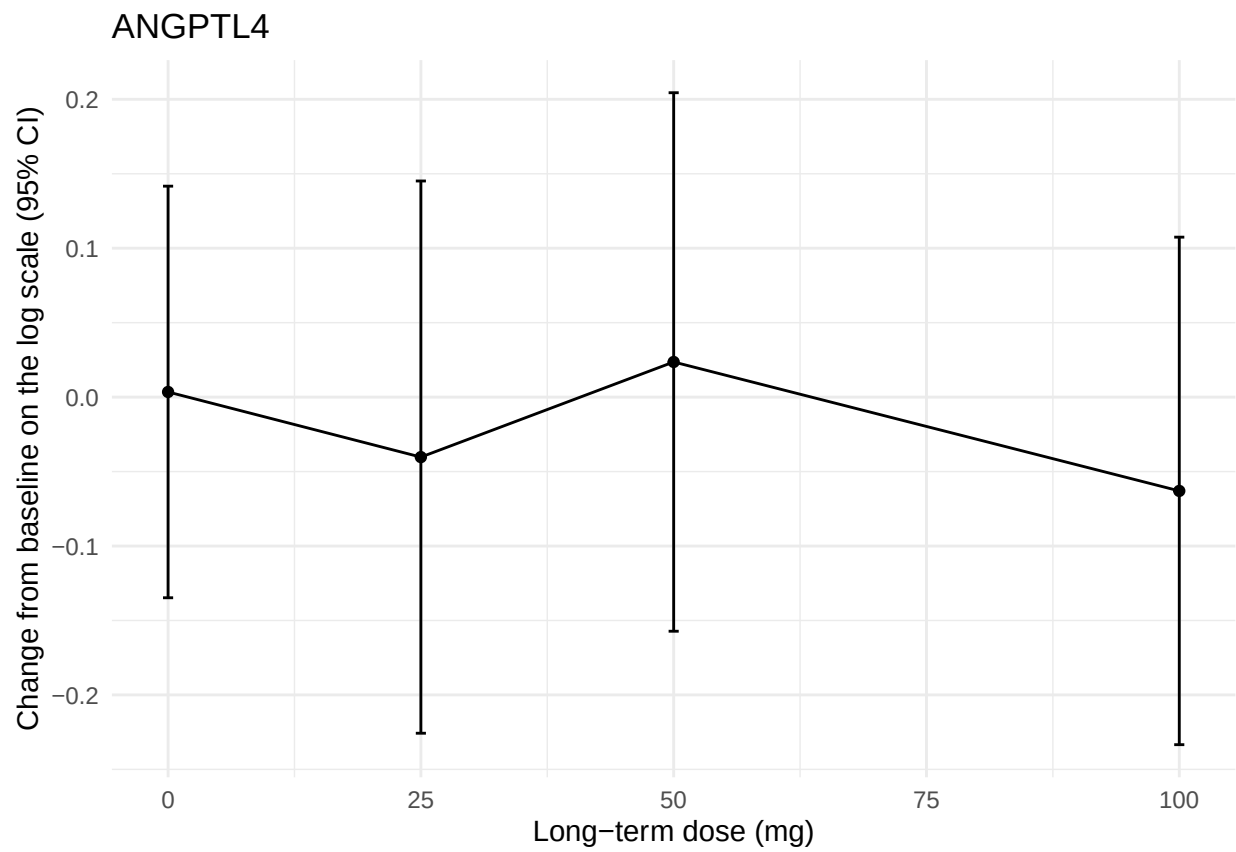
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	-0.04 (-0.33 to 0.24)
losartan 50 mg - losartan 0 mg	0.02 (-0.26 to 0.30)
losartan 100 mg - losartan 0 mg	-0.07 (-0.33 to 0.20)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	823.4	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	3.3	TRUE

Plot



CA9

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	0.073 (-0.520 to 0.667)
losartan 25 mg	0.153 (-0.628 to 0.934)
losartan 50 mg	0.152 (-0.622 to 0.926)
losartan 100 mg	-0.076 (-0.801 to 0.648)

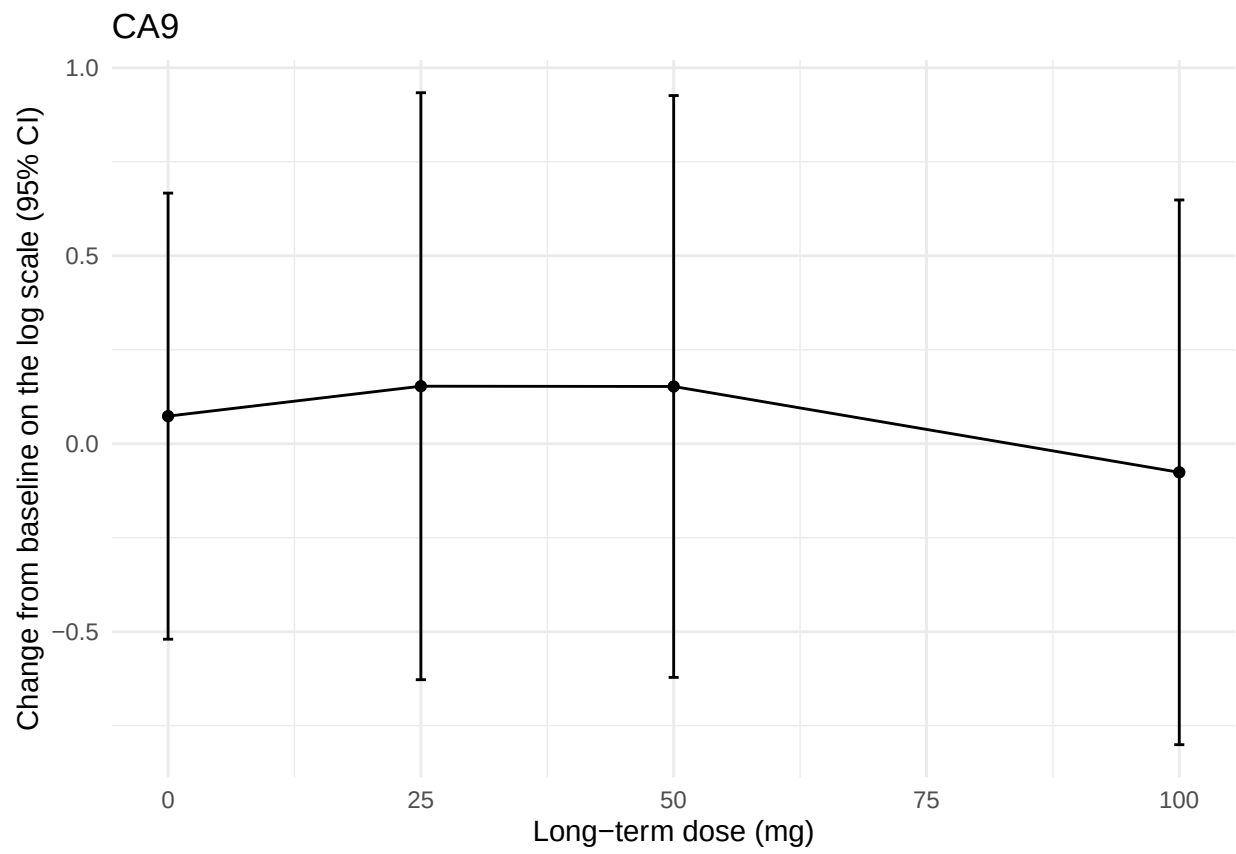
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.08 (-1.13 to 1.29)
losartan 50 mg - losartan 0 mg	0.08 (-1.10 to 1.26)
losartan 100 mg - losartan 0 mg	-0.15 (-1.30 to 1.00)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	353.3	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	102.7	TRUE

Plot



CD163

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.067 (-0.225 to 0.091)
losartan 25 mg	0.315 (0.108 to 0.522)
losartan 50 mg	0.113 (-0.095 to 0.320)
losartan 100 mg	0.139 (-0.055 to 0.333)

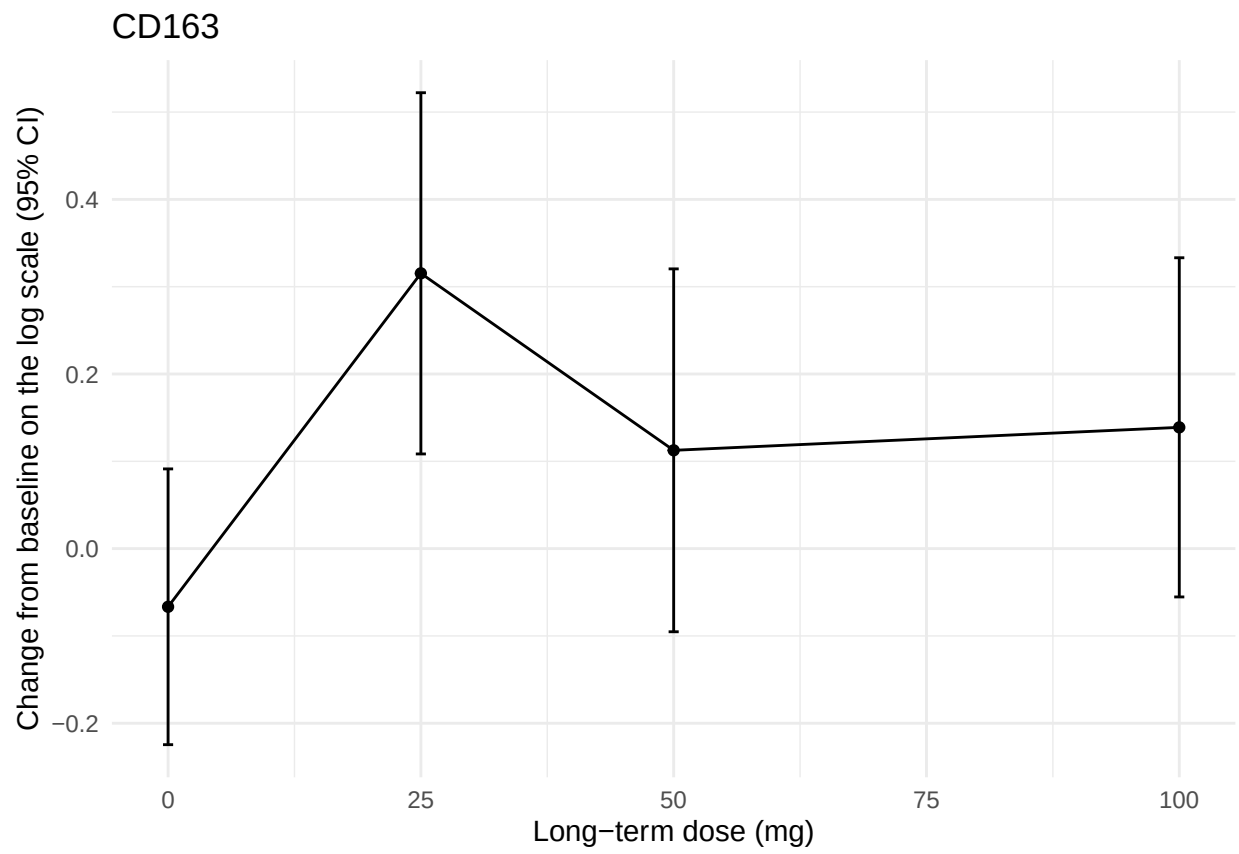
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.38 (0.06 to 0.70)
losartan 50 mg - losartan 0 mg	0.18 (-0.14 to 0.50)
losartan 100 mg - losartan 0 mg	0.21 (-0.10 to 0.51)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	964.0	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	13.4	TRUE

Plot



Collagen IV alpha 1

A total of 33 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	0.087 (-0.051 to 0.225)
losartan 25 mg	0.263 (0.087 to 0.439)
losartan 50 mg	0.130 (-0.047 to 0.308)
losartan 100 mg	-0.040 (-0.201 to 0.122)

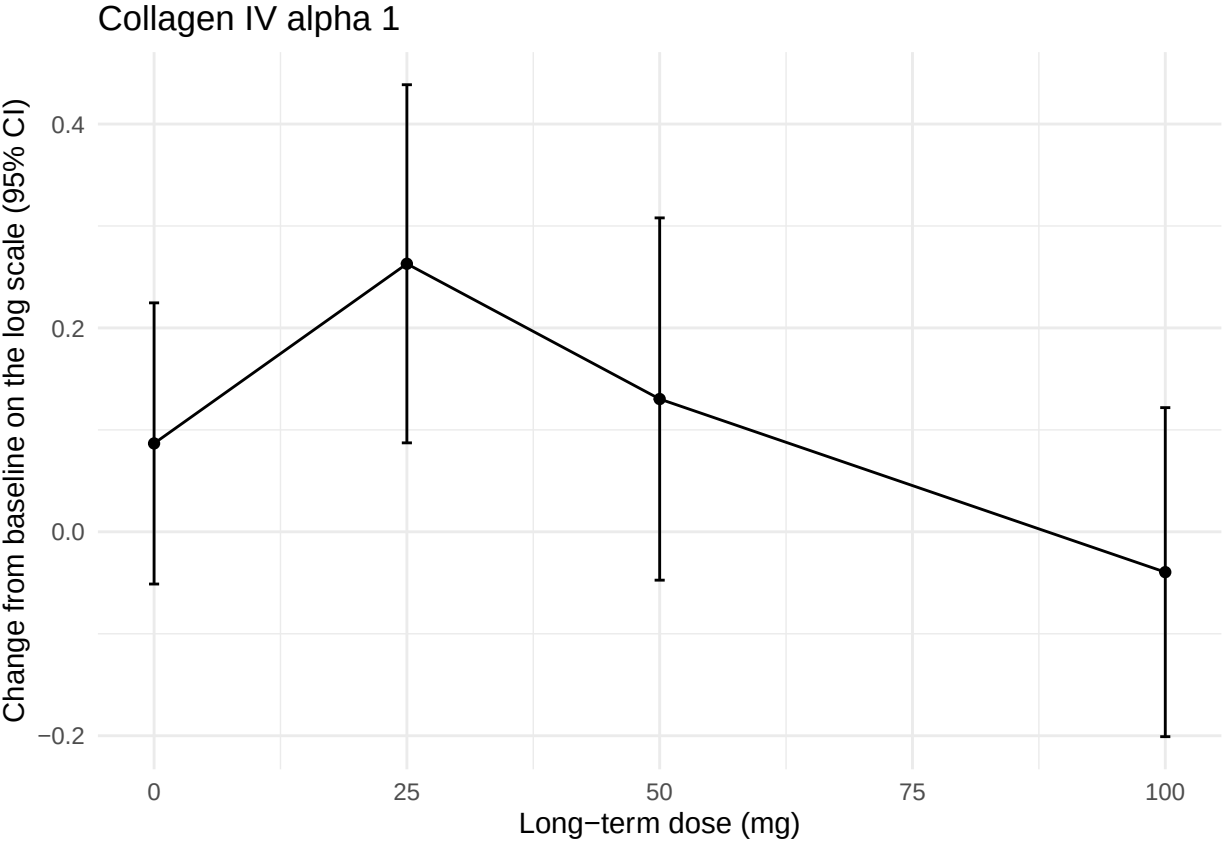
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.18 (-0.09 to 0.45)
losartan 50 mg - losartan 0 mg	0.04 (-0.23 to 0.32)
losartan 100 mg - losartan 0 mg	-0.13 (-0.39 to 0.13)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	455.3	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	1.1	TRUE

Plot



EGF

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.145 (-0.680 to 0.390)
losartan 25 mg	-0.375 (-1.081 to 0.330)
losartan 50 mg	-0.824 (-1.524 to -0.124)
losartan 100 mg	-0.241 (-0.899 to 0.417)

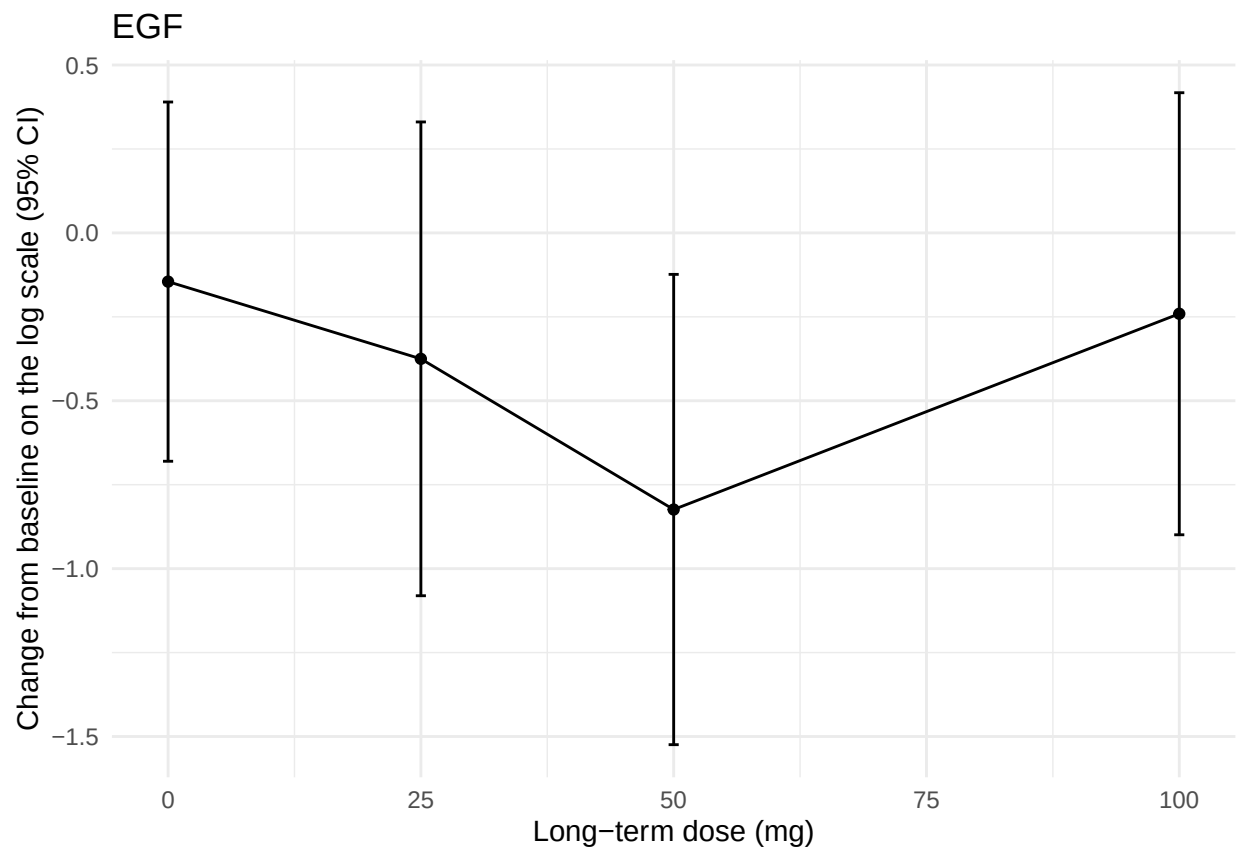
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	-0.23 (-1.31 to 0.85)
losartan 50 mg - losartan 0 mg	-0.68 (-1.75 to 0.40)
losartan 100 mg - losartan 0 mg	-0.10 (-1.13 to 0.94)

Model comparison (AIC)

Model	AIC	Used
Linear ($\text{chg} \sim \text{base} + \text{trt02p}$)	309.7	FALSE
Log-transformed ($\log(\text{value}) - \log(\text{base}) \sim \log(\text{base}) + \text{trt02p}$)	96.4	TRUE

Plot



ErbB2

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	0.035 (-0.094 to 0.165)
losartan 25 mg	0.059 (-0.111 to 0.229)
losartan 50 mg	0.056 (-0.114 to 0.227)
losartan 100 mg	0.140 (-0.020 to 0.301)

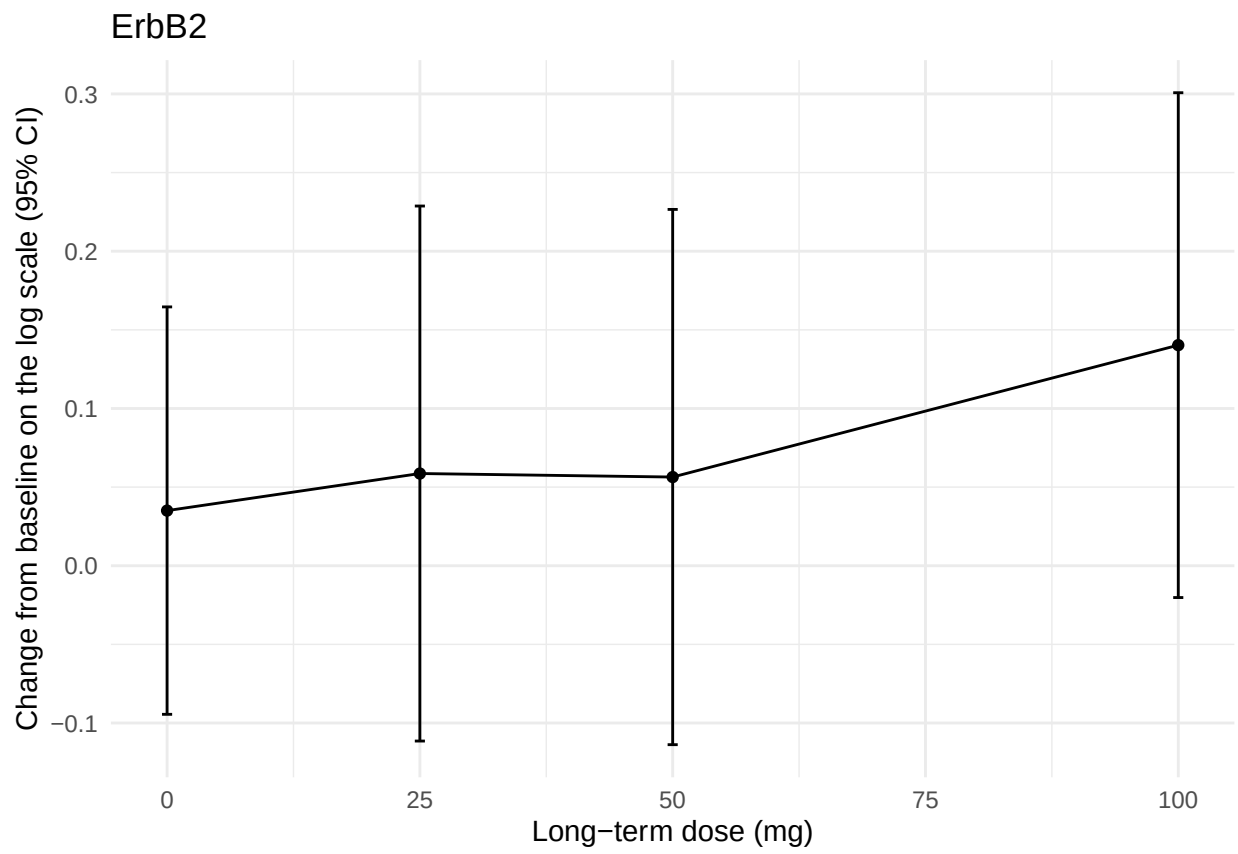
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.02 (-0.24 to 0.28)
losartan 50 mg - losartan 0 mg	0.02 (-0.24 to 0.28)
losartan 100 mg - losartan 0 mg	0.11 (-0.15 to 0.36)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	605.4	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	-0.1	TRUE

Plot



FGF Basic

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	0.038 (-0.369 to 0.445)
losartan 25 mg	0.190 (-0.343 to 0.723)
losartan 50 mg	-0.099 (-0.632 to 0.434)
losartan 100 mg	0.021 (-0.477 to 0.519)

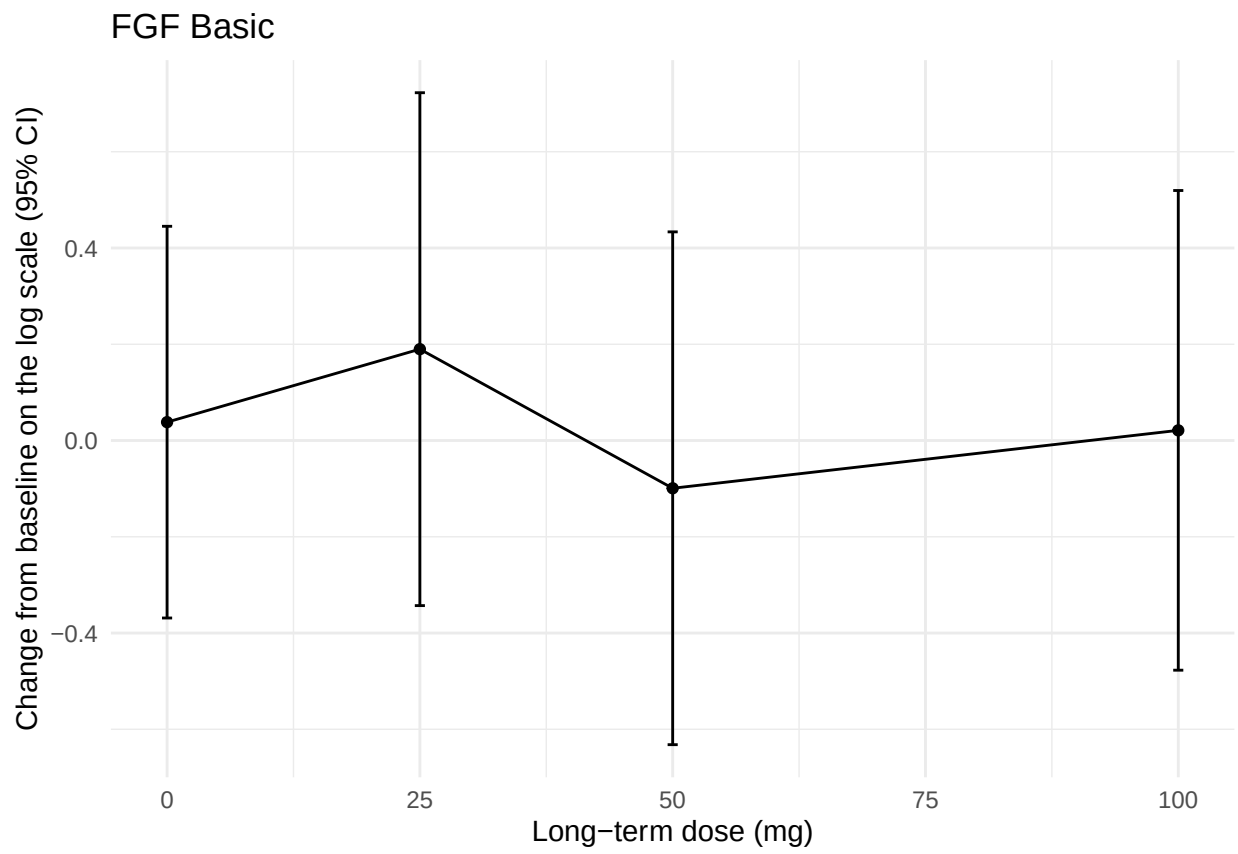
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.15 (-0.67 to 0.97)
losartan 50 mg - losartan 0 mg	-0.14 (-0.95 to 0.68)
losartan 100 mg - losartan 0 mg	-0.02 (-0.80 to 0.77)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	273.4	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	77.8	TRUE

Plot



IL-12 p70

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.114 (-0.753 to 0.524)
losartan 25 mg	0.877 (0.041 to 1.713)
losartan 50 mg	-0.260 (-1.095 to 0.576)
losartan 100 mg	0.091 (-0.692 to 0.874)

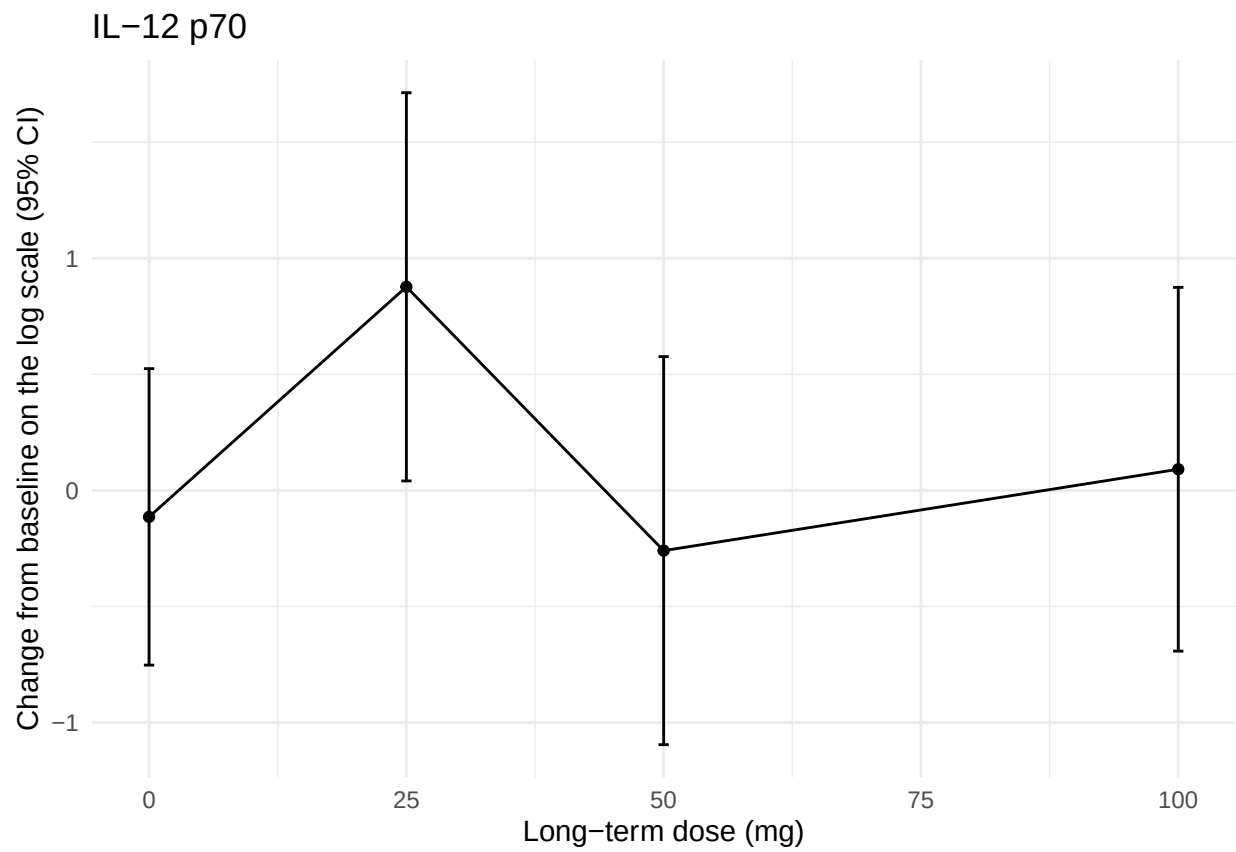
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.99 (-0.29 to 2.27)
losartan 50 mg - losartan 0 mg	-0.15 (-1.43 to 1.14)
losartan 100 mg - losartan 0 mg	0.21 (-1.03 to 1.44)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	444.3	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	108.4	TRUE

Plot



IL-2

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.233 (-1.064 to 0.598)
losartan 25 mg	0.105 (-0.987 to 1.197)
losartan 50 mg	0.239 (-0.861 to 1.339)
losartan 100 mg	-0.411 (-1.433 to 0.611)

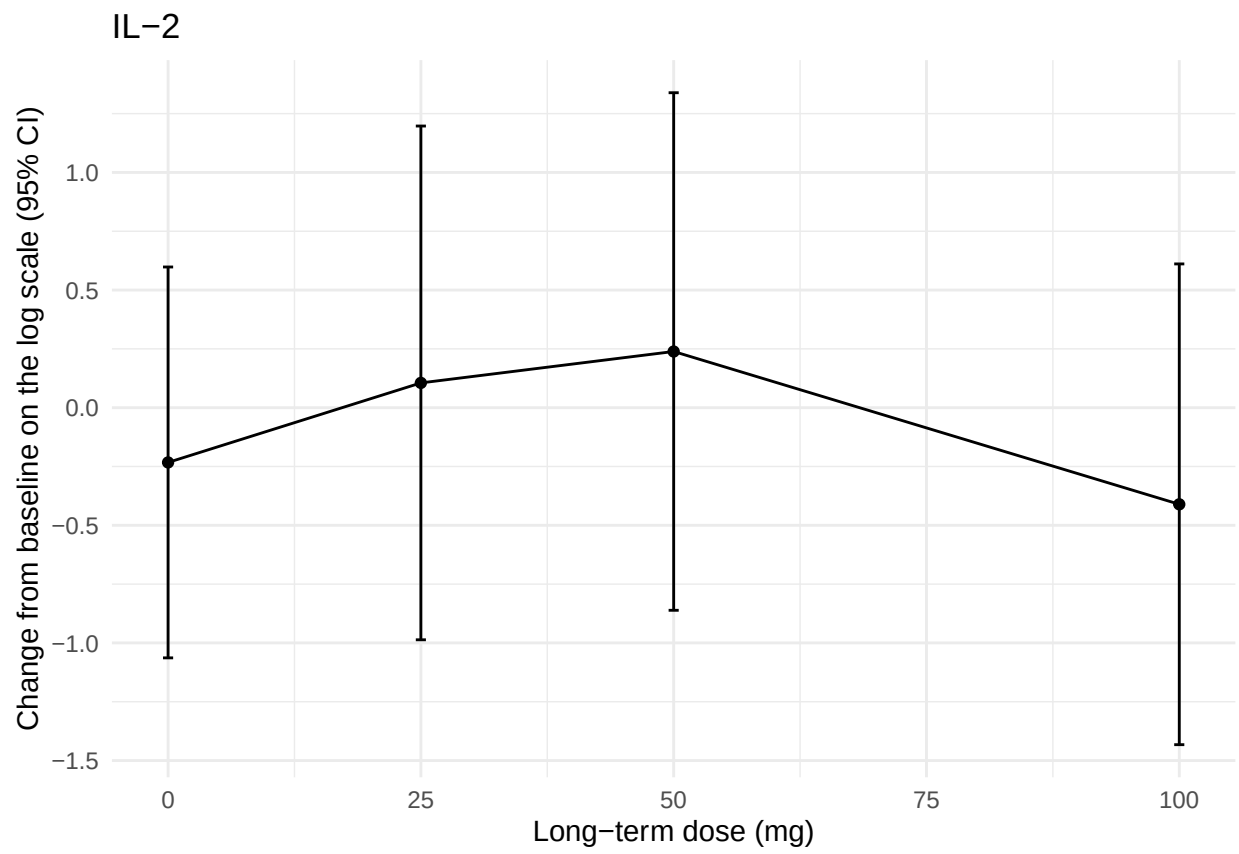
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.34 (-1.34 to 2.01)
losartan 50 mg - losartan 0 mg	0.47 (-1.20 to 2.15)
losartan 100 mg - losartan 0 mg	-0.18 (-1.79 to 1.43)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	348.6	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	126.3	TRUE

Plot



IP-10 (CXCL10)

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.081 (-0.295 to 0.134)
losartan 25 mg	0.234 (-0.047 to 0.515)
losartan 50 mg	0.179 (-0.098 to 0.457)
losartan 100 mg	0.117 (-0.143 to 0.377)

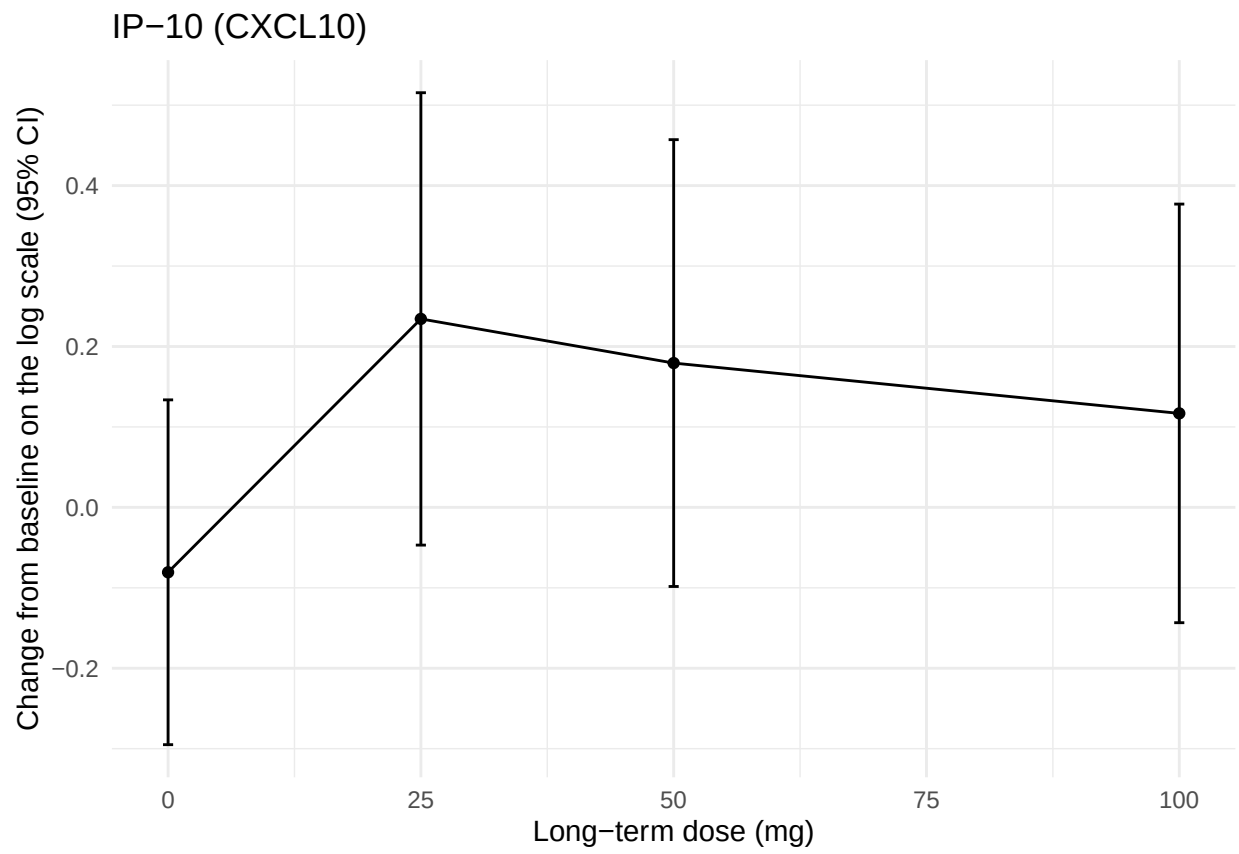
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.31 (-0.12 to 0.75)
losartan 50 mg - losartan 0 mg	0.26 (-0.17 to 0.69)
losartan 100 mg - losartan 0 mg	0.20 (-0.22 to 0.61)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	337.1	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	33.4	TRUE

Plot



I-TAC (CXCL11)

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.092 (-0.294 to 0.110)
losartan 25 mg	-0.100 (-0.364 to 0.165)
losartan 50 mg	-0.177 (-0.453 to 0.100)
losartan 100 mg	-0.070 (-0.320 to 0.179)

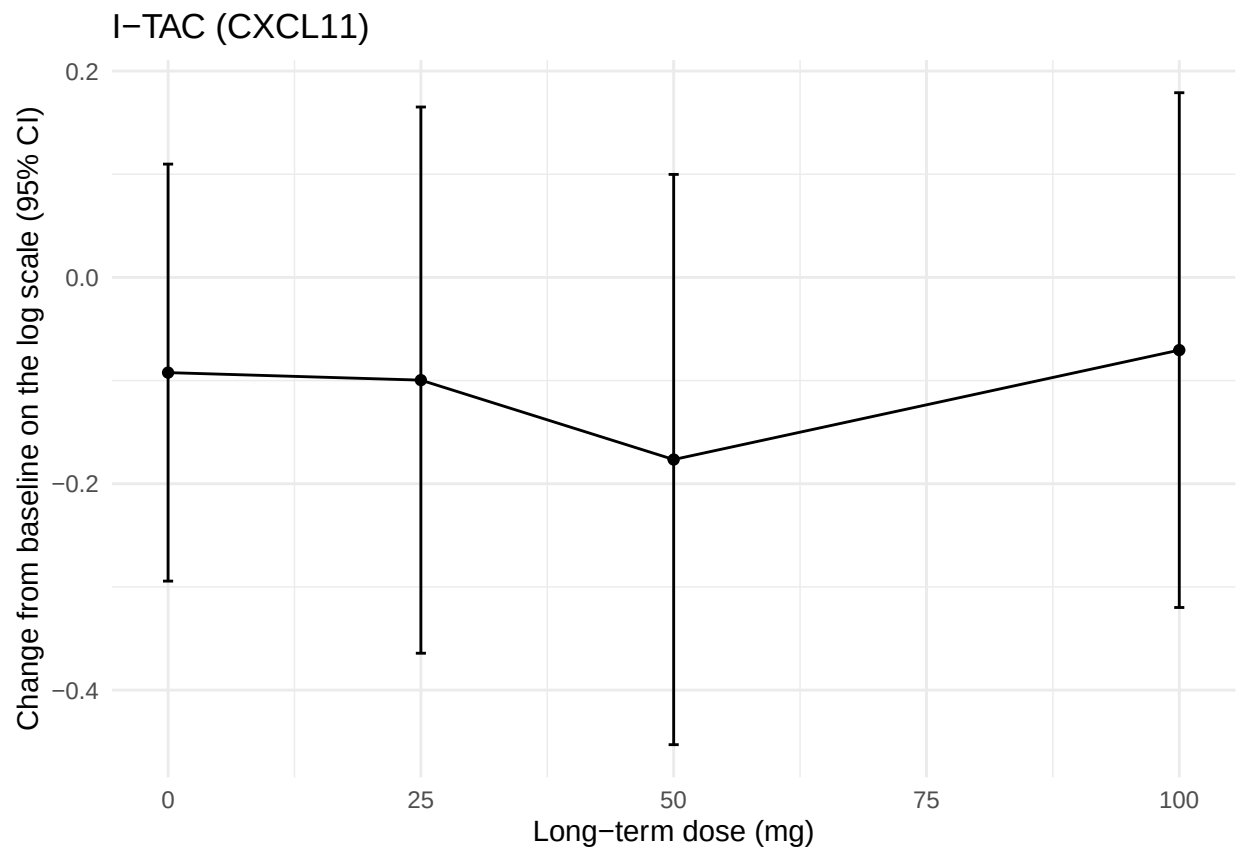
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	-0.01 (-0.41 to 0.40)
losartan 50 mg - losartan 0 mg	-0.08 (-0.51 to 0.34)
losartan 100 mg - losartan 0 mg	0.02 (-0.37 to 0.41)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	421.7	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	30.0	TRUE

Plot



MIG (CXCL9)

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	0.052 (-0.446 to 0.551)
losartan 25 mg	0.343 (-0.309 to 0.996)
losartan 50 mg	-0.421 (-1.074 to 0.233)
losartan 100 mg	-0.356 (-0.968 to 0.257)

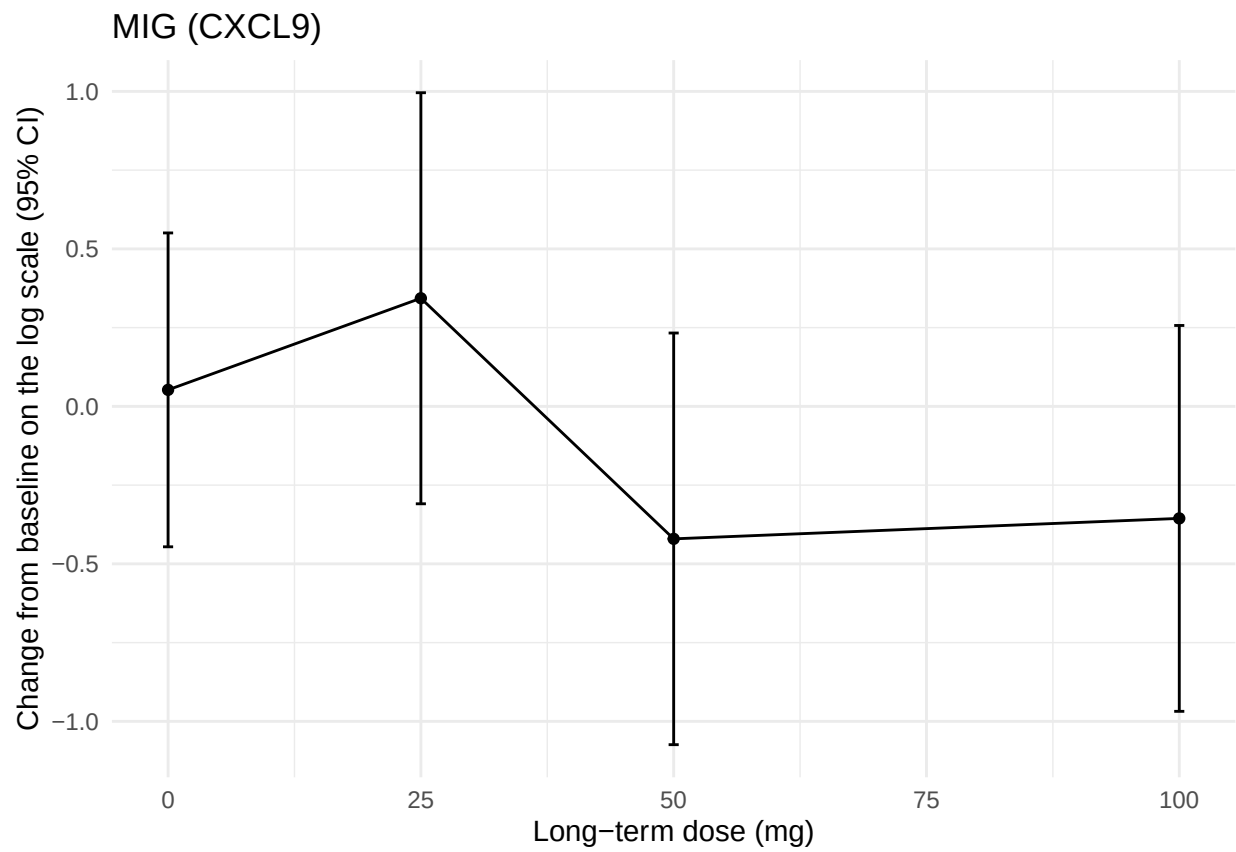
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.29 (-0.71 to 1.30)
losartan 50 mg - losartan 0 mg	-0.47 (-1.47 to 0.52)
losartan 100 mg - losartan 0 mg	-0.41 (-1.38 to 0.56)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	493.5	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	91.2	TRUE

Plot



PIGF

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	0.040 (-0.096 to 0.175)
losartan 25 mg	0.099 (-0.083 to 0.280)
losartan 50 mg	0.059 (-0.121 to 0.239)
losartan 100 mg	0.088 (-0.078 to 0.254)

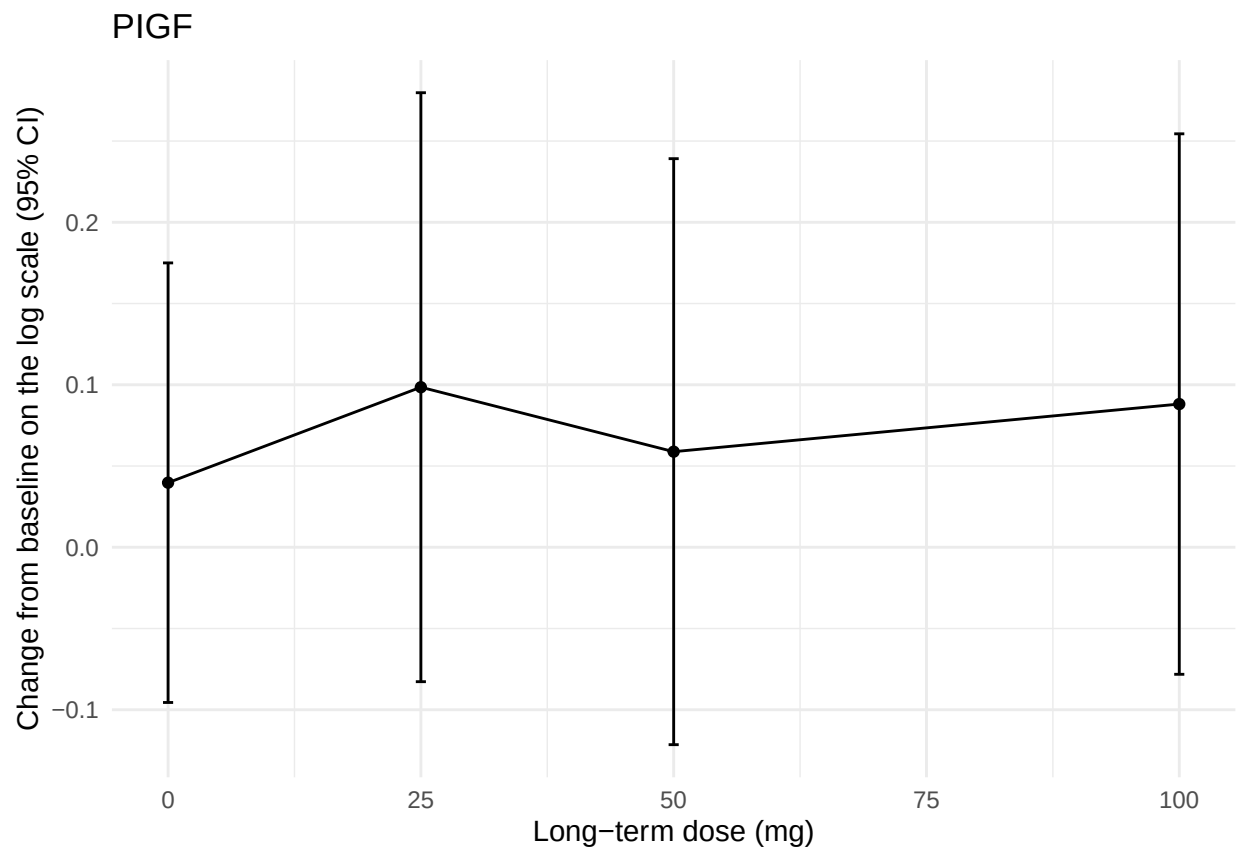
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.06 (-0.21 to 0.33)
losartan 50 mg - losartan 0 mg	0.02 (-0.26 to 0.30)
losartan 100 mg - losartan 0 mg	0.05 (-0.21 to 0.31)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	281.1	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	2.7	TRUE

Plot



Tenascin C

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.020 (-0.103 to 0.064)
losartan 25 mg	0.062 (-0.045 to 0.170)
losartan 50 mg	-0.038 (-0.146 to 0.070)
losartan 100 mg	0.001 (-0.099 to 0.101)

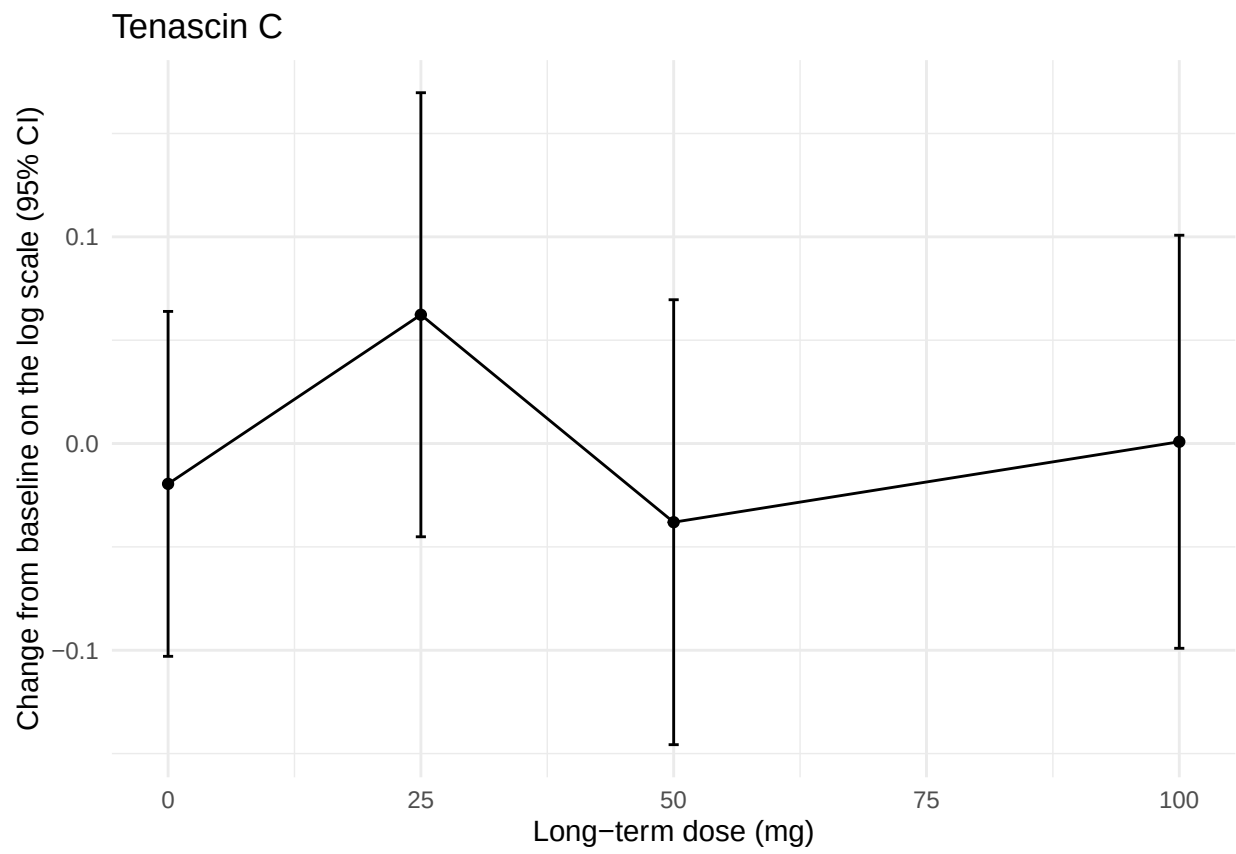
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.08 (-0.09 to 0.25)
losartan 50 mg - losartan 0 mg	-0.02 (-0.19 to 0.15)
losartan 100 mg - losartan 0 mg	0.02 (-0.14 to 0.18)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	620.5	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	-31.6	TRUE

Plot



TNFRSF18 (GITR)

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.691 (-1.480 to 0.099)
losartan 25 mg	0.563 (-0.473 to 1.599)
losartan 50 mg	-0.858 (-1.896 to 0.179)
losartan 100 mg	-0.915 (-1.883 to 0.052)

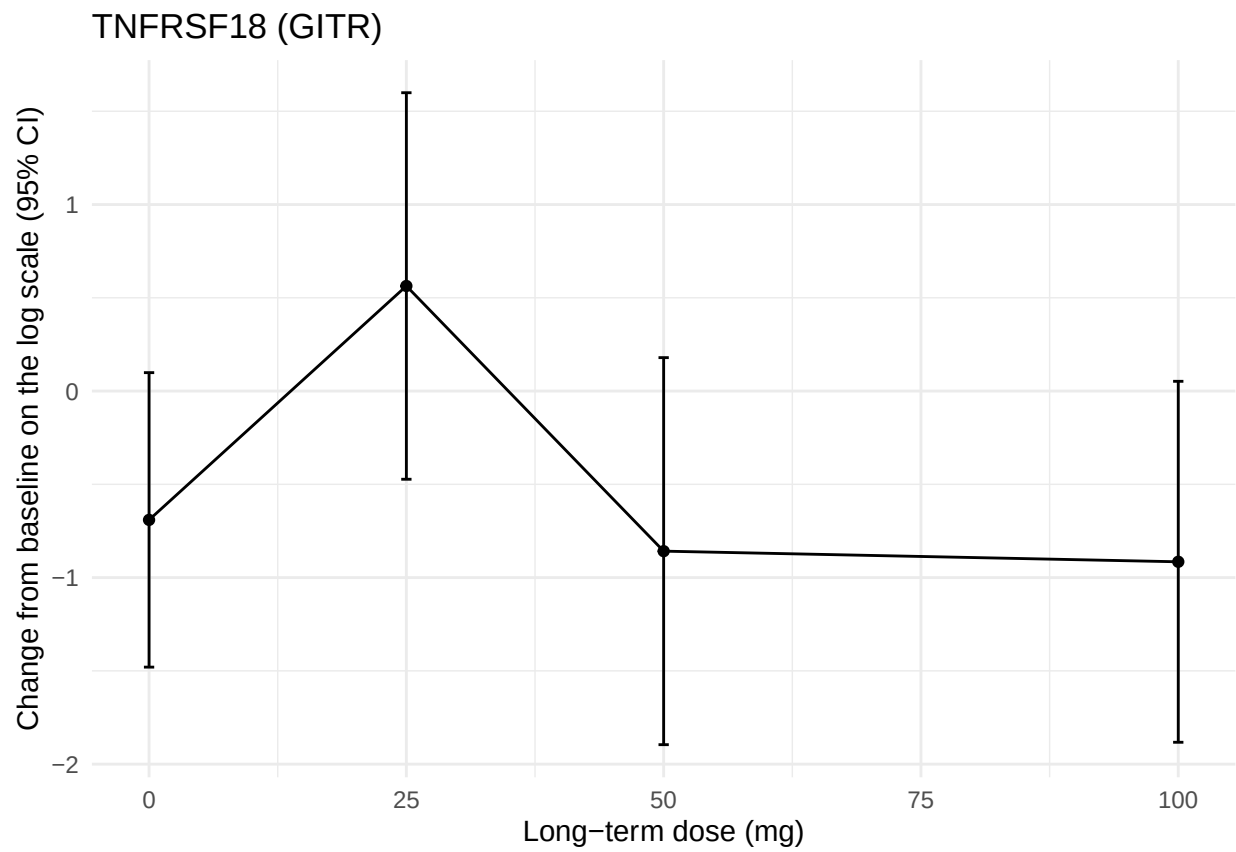
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	1.25 (-0.33 to 2.83)
losartan 50 mg - losartan 0 mg	-0.17 (-1.76 to 1.43)
losartan 100 mg - losartan 0 mg	-0.22 (-1.75 to 1.30)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	496.3	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	122.6	TRUE

Plot



TNFSF6 (FasL)

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.217 (-0.364 to -0.070)
losartan 25 mg	0.087 (-0.105 to 0.280)
losartan 50 mg	-0.057 (-0.250 to 0.136)
losartan 100 mg	-0.021 (-0.201 to 0.158)

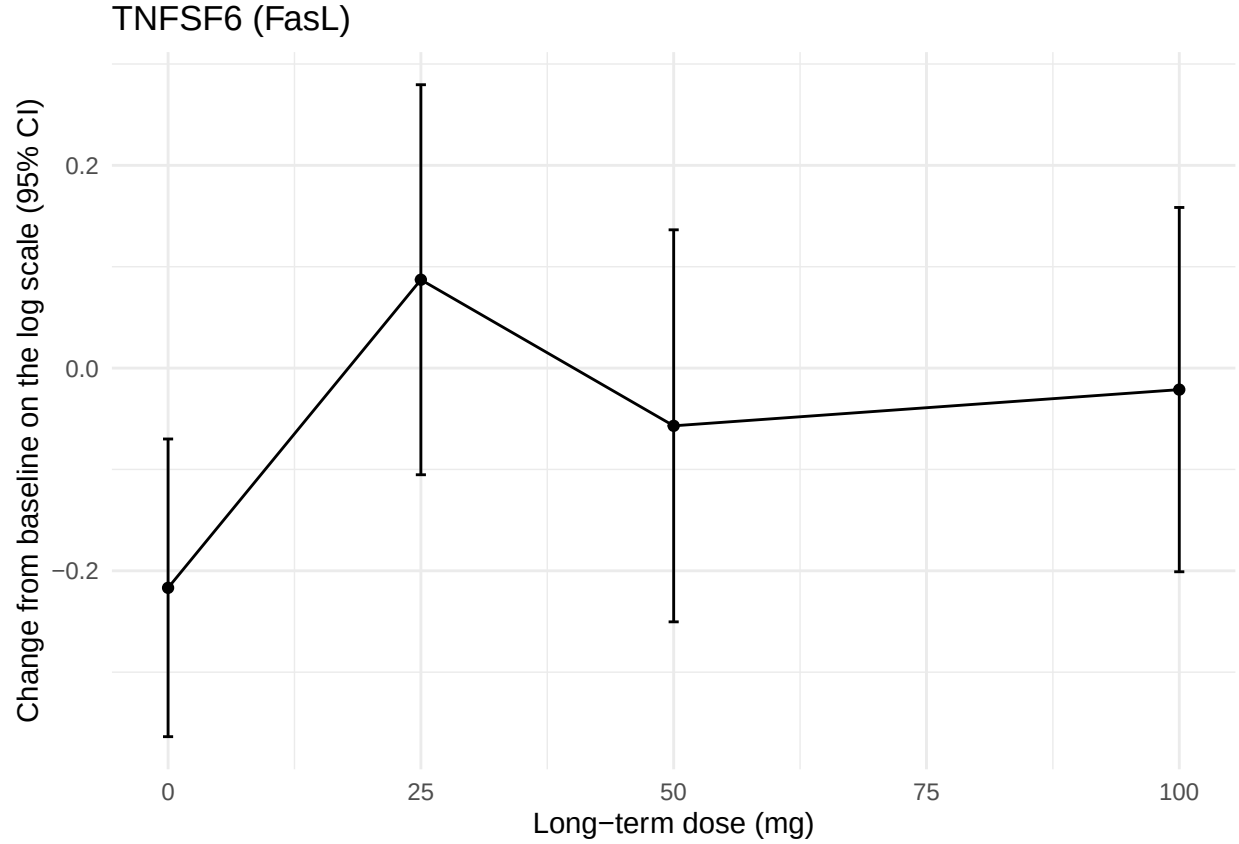
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.30 (0.01 to 0.60)
losartan 50 mg - losartan 0 mg	0.16 (-0.14 to 0.46)
losartan 100 mg - losartan 0 mg	0.20 (-0.09 to 0.48)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	280.8	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	8.4	TRUE

Plot



VEGFR2

A total of 34 participants contributed data for this analyte.

Estimated change from baseline by long-term dose

Long-term dose	Change from baseline on the natural logarithmic scale (95% CI)
losartan 0 mg	-0.008 (-0.119 to 0.102)
losartan 25 mg	0.092 (-0.054 to 0.237)
losartan 50 mg	-0.040 (-0.193 to 0.113)
losartan 100 mg	0.041 (-0.099 to 0.182)

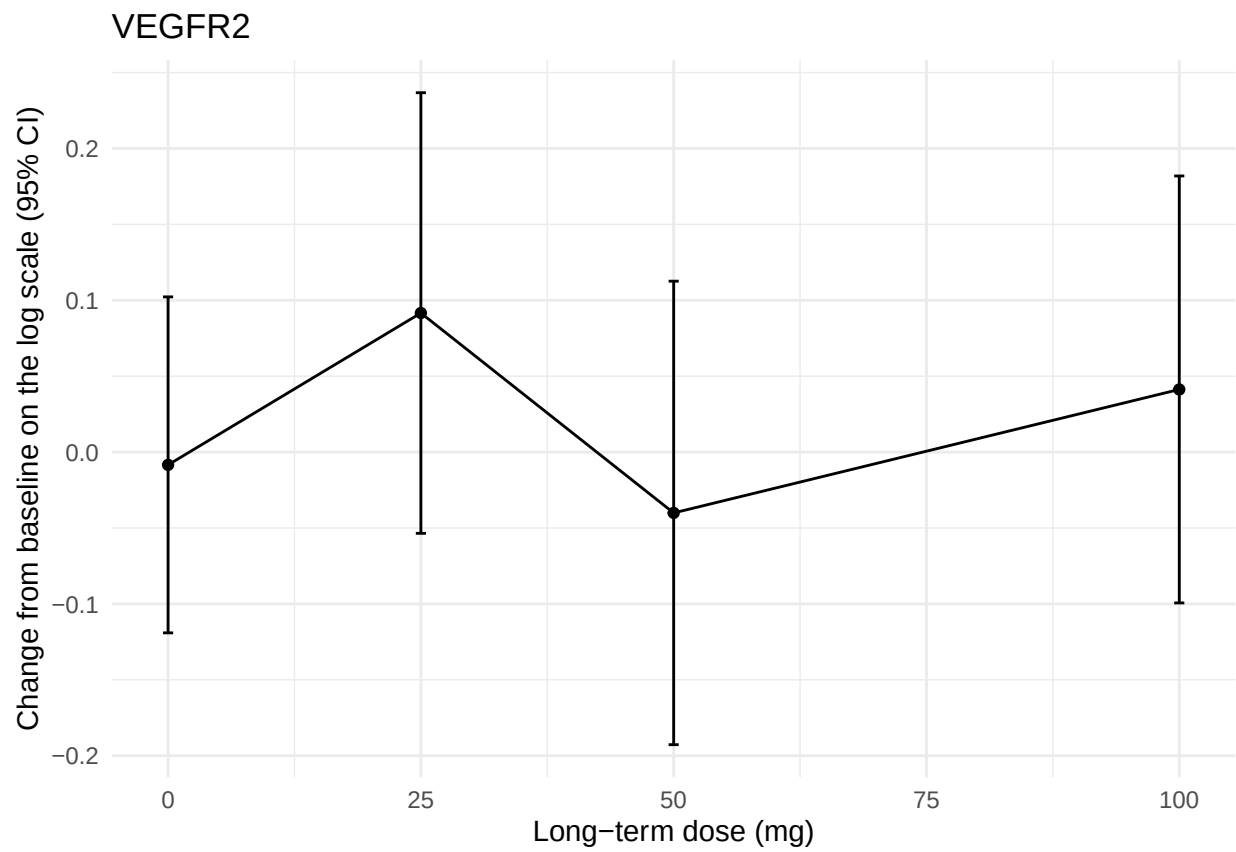
Differences vs 0 mg

Comparison	Difference of the log-transformed values (95% CI)
losartan 25 mg - losartan 0 mg	0.10 (-0.12 to 0.32)
losartan 50 mg - losartan 0 mg	-0.03 (-0.26 to 0.20)
losartan 100 mg - losartan 0 mg	0.05 (-0.17 to 0.27)

Model comparison (AIC)

Model	AIC	Used
Linear (chg ~ base + trt02p)	644.9	FALSE
Log-transformed (log(value) - log(base) ~ log(base) + trt02p)	-10.8	TRUE

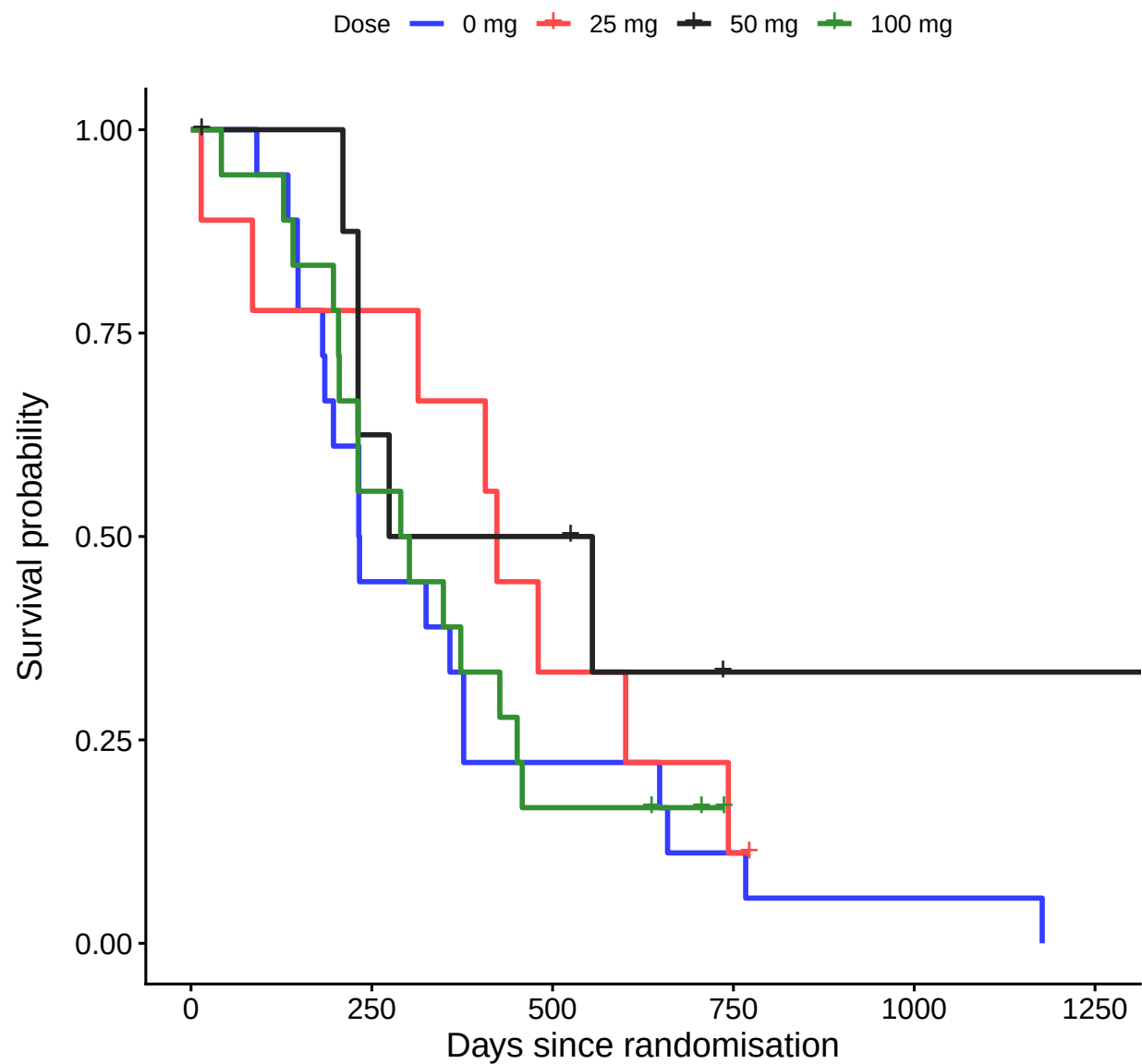
Plot



Time-to-Event Endpoints

Kaplan-Meier curves are shown with the at-risk table below. Overall log-rank p-values are reported for each endpoint, followed by Cox regression hazard ratios versus 0 mg.

Progression-free survival



Number at risk

Dose	0	250	500	750	1000	1250
0 mg	18	8	4	2	1	0
25 mg	9	7	3	1	0	0
50 mg	9	5	4	1	1	1
100 mg	18	10	3	0	0	0

Days since randomisation

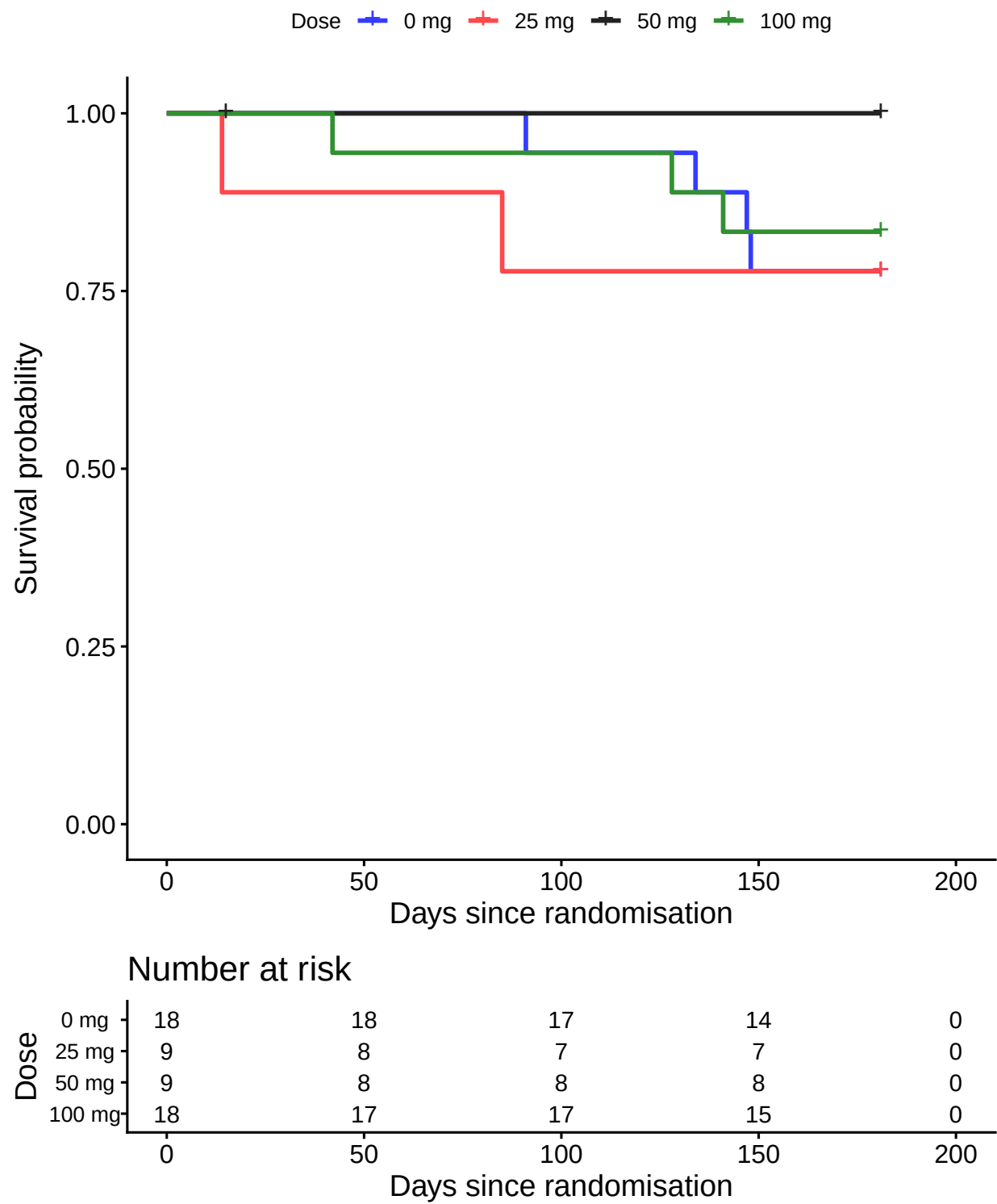
Log-rank p-value

Overall log-rank p-value: 0.363

Cox regression (vs 0 mg)

Comparison	HR (95% CI)	P-value
25 mg vs 0 mg	0.66 (0.28 to 1.52)	0.327
50 mg vs 0 mg	0.45 (0.17 to 1.21)	0.112
100 mg vs 0 mg	0.90 (0.45 to 1.80)	0.761

6-month progression-free survival



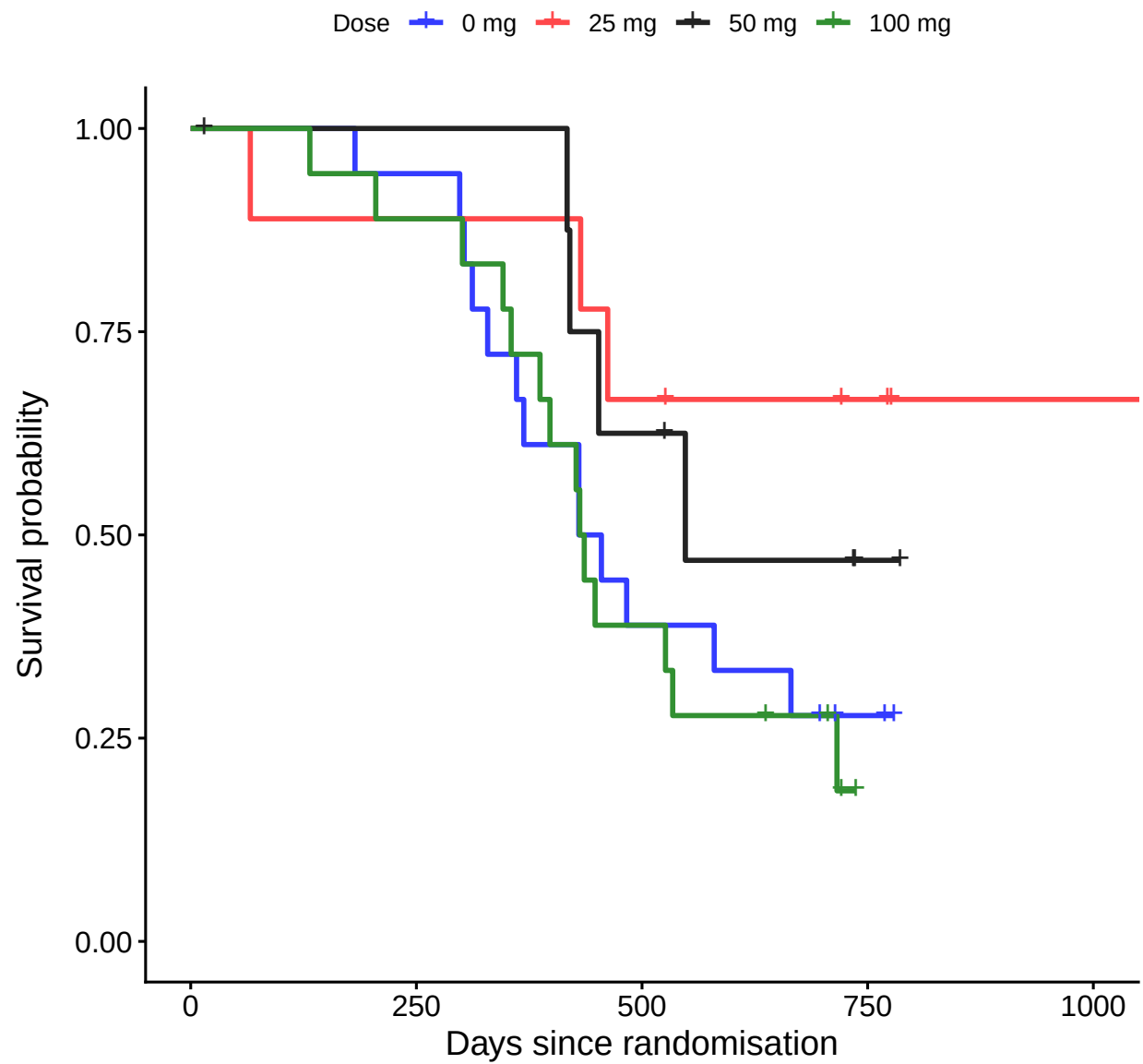
Log-rank p-value

Overall log-rank p-value: 0.564

Cox regression (vs 0 mg)

Comparison	HR (95% CI)	P-value
25 mg vs 0 mg	1.14 (0.21 to 6.24)	0.877
50 mg vs 0 mg	0.00 (0.00 to Inf)	0.999
100 mg vs 0 mg	0.77 (0.17 to 3.44)	0.733

Overall survival



Number at risk

Dose	0	250	500	750	1000
0 mg	18	17	7	3	0
25 mg	9	8	6	4	1
50 mg	9	8	5	1	0
100 mg	18	16	7	0	0

Days since randomisation

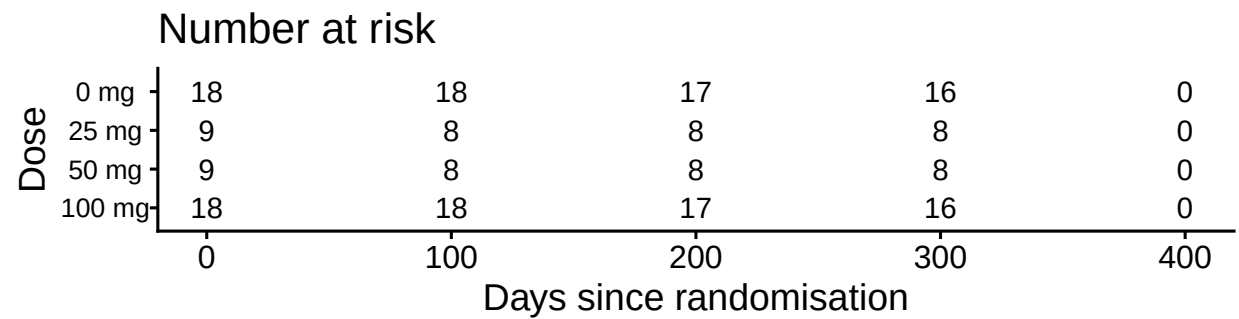
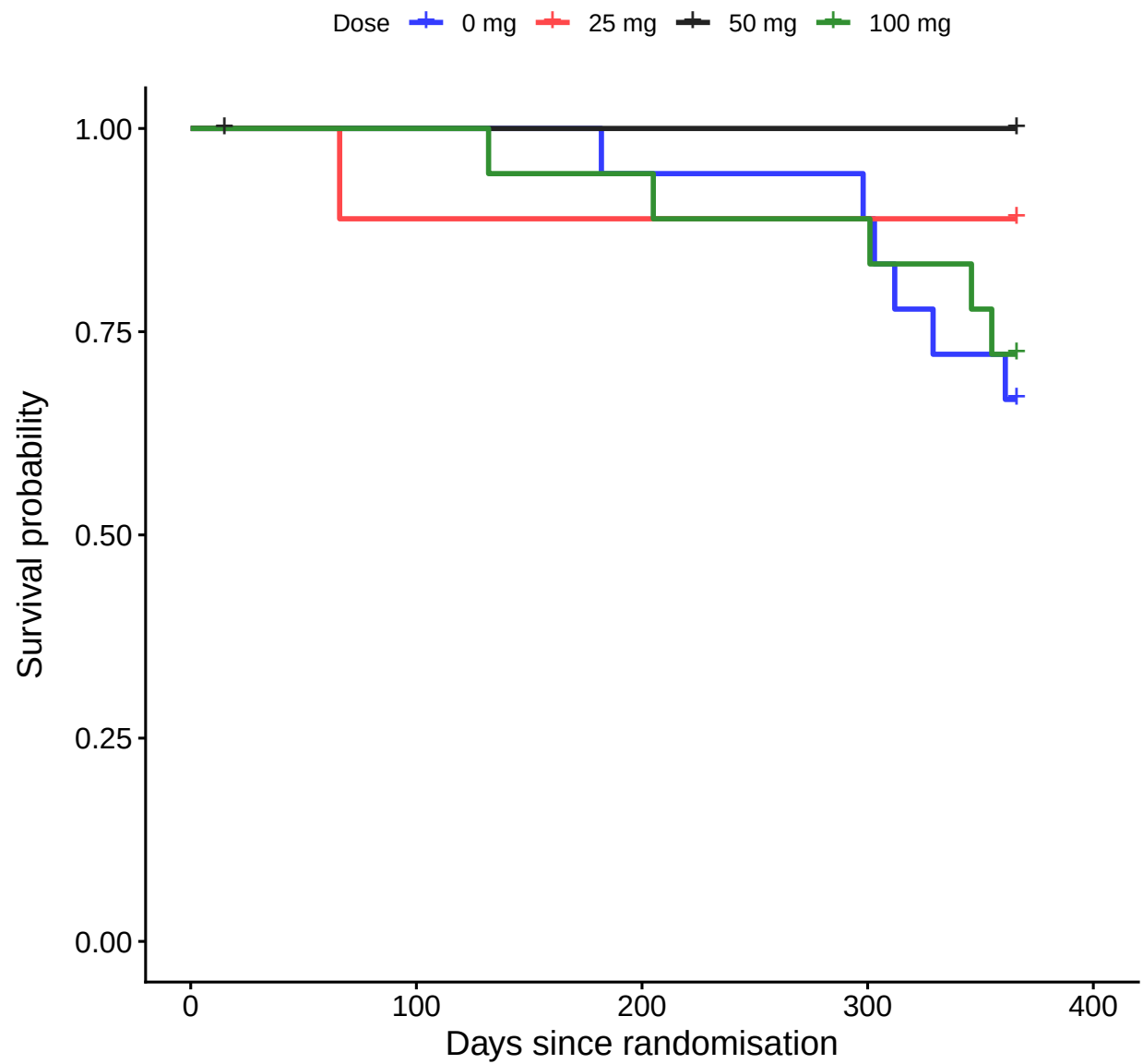
Log-rank p-value

Overall log-rank p-value: 0.139

Cox regression (vs 0 mg)

Comparison	HR (95% CI)	P-value
25 mg vs 0 mg	0.33 (0.09 to 1.16)	0.083
50 mg vs 0 mg	0.52 (0.17 to 1.61)	0.259
100 mg vs 0 mg	1.13 (0.53 to 2.41)	0.752

Overall survival at 12 months



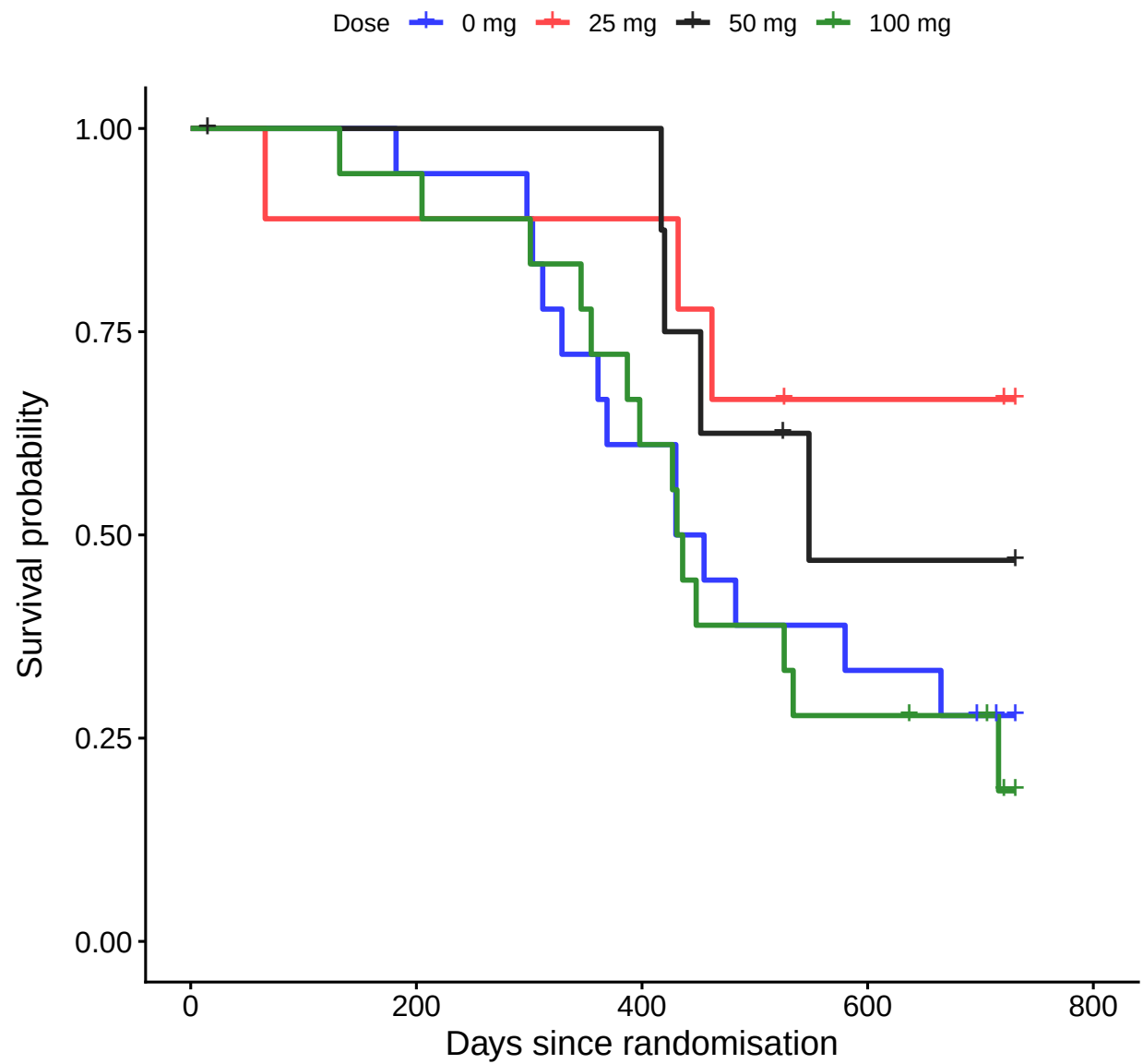
Log-rank p-value

Overall log-rank p-value: 0.271

Cox regression (vs 0 mg)

Comparison	HR (95% CI)	P-value
25 mg vs 0 mg	0.32 (0.04 to 2.66)	0.291
50 mg vs 0 mg	0.00 (0.00 to Inf)	0.998
100 mg vs 0 mg	0.83 (0.25 to 2.71)	0.753

Overall survival at 24 months



Number at risk

Dose	0	200	400	600	800
0 mg	18	17	11	6	0
25 mg	9	8	8	5	0
50 mg	9	8	8	3	0
100 mg	18	17	11	5	0

Days since randomisation

Log-rank p-value

Overall log-rank p-value: 0.139

Cox regression (vs 0 mg)

Comparison	HR (95% CI)	P-value
25 mg vs 0 mg	0.33 (0.09 to 1.16)	0.083
50 mg vs 0 mg	0.52 (0.17 to 1.61)	0.259
100 mg vs 0 mg	1.13 (0.53 to 2.41)	0.752